Notes for Maths 319 — introduction to real analysis

Draft

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Back Matter

Chapter 1

Motivating introduction

Much of mathematics is proving theorems — for example, this polynomial has 2 real roots. But sometimes you need to actually compute[1] something — the roots of this polynomial are …

Back in Calculus-1 you were introduced[2] to a couple of methods for computing the roots[3] of a polynomial: the bisection method and Newton's method. These are both examples of *iterative* methods — you start with a rough approximation to the object you are interested in, and then you do "stuff" to that approximation and turn it into a better approximation. You can then, in turn, do that same "stuff" to that better approximation to make a betterer approximation. And you can keep *iterating* this to get better and better approximations that should get closer and closer to the object you actually want.

Say, for example, we want to compute $\sqrt{2}$ by the bisection method. This is equivalent to finding a zero of the polynomial $f(x) = x^2 - 2$. Since this is a **continuous** function, and we know that $f(0) = -2$ and $f(2) = 2$, the Intermediate Value Theorem tells us that there must be a zero inside the interval $[0, 2]$. And we could, therefore, take the midpoint of that interval, 1, to be an approximation of $\sqrt{2}$.

Of course, this is a pretty crude approximation, since $f(1) = -1$. Indeed, applying the IVT again, we know that f has a zero in the interval $[1, 2]$. So, we can use the midpoint of that interval as a better approximation: 1.5. Now, $f(1.5) = 0.25$, and so the IVT tells us there is a zero inside $[1, 1.5]$.

We can continue to iterate these *bisections* until we are satisfied with the approximation of $\sqrt{2}$.

Of course, there are many assumptions built into this process.

- We need the IVT — what does the IVT need in order to work?

[1] A colleague once told me that this is the difference between a *pure* mathematician and an *applied* mathematician; an applied mathematician sometimes has to compute an actual number.

[2] You should have been introduced?

[3] Well, the real roots, not the complex ones. Let's avoid those complexities (ha!) in this course.

- How do we know that as we keep halving these intervals, we actually end up with an answer?
- How do we know that answer is unique?

All of these things will fail if we consider $f(x) = x^2 - 2$ as a function over the *rationals*. There is some property of the real numbers that makes this actually work. Understanding that property and its consequences is a major goal of this course.

More generally, when we apply iterative methods, that start with an approximate answer, and then iteratively improve it, we need to understand under what conditions they actually work!

- How do I know the approximations get closer to the answer?
- How do I know the answer exists?
- How do I know that a different starting point will not lead to a different answer?
- How do I define "closer" when I'm not dealing with numbers? (eg Taylor polynomial approximations of a function)

We'll examine all of these things in this course. Getting there will take us through a thorough understanding of real numbers, and then extend those ideas to more general **metric spaces** and finally give quite general conditions under which such iterative methods will work.

1.1 Newton iteration and $\sqrt{2}$

In many *real* applications we need to actually compute a quantity or a function. It is not sufficient to just say "it exists". For example, we might need to know π with more accuracy than

$$\pi \approx \frac{22}{7}$$

gives you (it is accurate to about 0.04%).

Rather than starting with π which is a little more involved, let's start by looking at $\sqrt{2}$ which is a bit easier, and we know from our work in a previous course, is not a rational number. One way to compute $\sqrt{2}$ is as the positive root of the polynomial

$$f(x) = x^2 - 2$$

using Newton's method.

Recall [1] that Newton's method starts with an initial rough estimate, say $x_0 = 1$, and then feeds that into a formula to produce a better estimate. That better estimate is then fed into the same formula to make a still better one, etc. In particular,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n}$$

So, if we start with $x_0 = 1$ then we get (iteratively)

$$x_1 = x_0 - \frac{x_0^2 - 2}{2x_0} = 1 - \frac{1-2}{2} = 1 - \frac{-1}{2} = \frac{3}{2} = 1.5$$

$$x_2 = x_1 - \frac{x_1^2 - 2}{2x_1} = \frac{3}{2} - \frac{9/4 - 2}{3} = \frac{3}{2} - \frac{1}{12} = \frac{17}{12} \approx 1.417$$

$$x_3 = x_2 - \frac{x_2^2 - 2}{2x_2} = \frac{17}{12} - \frac{289/144 - 2}{17/6} = \frac{17}{12} - \frac{1}{408} = \frac{577}{408} \approx 1.41422$$

and so on. Continuing in this way, we obtain a sequence of rational numbers that converge to $\sqrt{2}$. Of course, we'd like to be able to prove that this does actually converge and that it converges to the right thing.

There is actually rather a lot going on in the previous paragraphs.

- we are making heavy use of the "fact" that rational numbers are much more concrete objects than objects like "$\sqrt{2}$"; this is not unreasonable — naturals are easier than integers are easier than rationals are easier than reals are easier than complex …
- What does it mean for a sequence to "converge"?
- If we do understand convergence, then does a sequence of rational numbers have to converge?
- If a rational sequence does converge then does it converge to a rational number?

Lots to consider here. The answer to the last one is famously "no"; in fact this follows by considering decimal expansions of $\sqrt{2}$. You can make a sequence of rationals by taking n-digits of the decimal expansion of $\sqrt{2}$. That rational sequence converges to $\sqrt{2}$, but $\sqrt{2} \notin \mathbb{Q}$.

1.2 Picard iteration

A harder example now. Consider the ordinary differential equation (initial value problem really)

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x(1+y) \qquad y(0) = 0.$$

[1] If you don't recall the details, see, for example, CLP-1 Appendix C. As is the case with the naming of many mathematical results, what we know as Newton's method has a long history that involves many people other than Newton. Indeed, the special case of finding square-roots goes back as far as the Babylonians.

What function $y(x)$ satisfies this initial value problem (IVP). One way to approximate a solution is to use something called Picard Iteration[1].

The idea behind this approach is quite similar to that of Newton's method. We start with a rough approximation and then put it into some algorithm to give us a (hopefully) better approximation. Feed that back in to get a still better approximation, and so forth. So, if we have the IVP

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y(x)) \qquad y(0) = y_0$$

then integrating both sides we can restate this as an integral equation

$$y(x) - y(0) = \int_0^x f(t, y(t))\,\mathrm{d}t$$

or

$$y(x) = \int_0^x f(t, y(t))\,\mathrm{d}t + y(0)$$

Much like Newton iteration, we substitute our rough approximation of y into the RHS of this equation and get back a (hopefully) better approximation.

So start with the very crude constant approximation

$$y(x) \approx \varphi_0(x) = y_0.$$

Plug this into the integral equation to get the next approximation

$$\varphi_1(x) = \int_0^x f(t, \varphi_0(t))\,\mathrm{d}t + y_0$$

Take that and plug it back in to get the next one

$$\varphi_2(x) = \int_0^x f(t, \varphi_1(t))\,\mathrm{d}t + y_0$$

and so on. For the example at hand, $f(x, y) = 2x(1 + y)$ and this iteration gives

$$\begin{aligned}
\varphi_0(x) &= 0 \\
\varphi_1(x) &= \int_0^x 2t(1+0)\,\mathrm{d}t + 0 = \int_0^x 2t\,\mathrm{d}t = x^2 \\
\varphi_2(x) &= \int_0^x 2t(1+t^2)\,\mathrm{d}t + 0 = \int_0^x 2t + 2t^3\,\mathrm{d}t = x^2 + \frac{x^4}{2} \\
\varphi_3(x) &= \int_0^x 2t(1+t^2+t^4/2)\,\mathrm{d}t + 0 = \int_0^x (2t + 2t^3 + t^5)\,\mathrm{d}t = x^2 + \frac{x^4}{2} + \frac{x^6}{6} \\
\varphi_4(x) &= x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \frac{x^8}{24} \\
\varphi_5(x) &= x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \frac{x^8}{24} + \frac{x^{10}}{120}
\end{aligned}$$

[1] See, for example, this and this.

and so on. This certainly looks like it is converging to something; we appear to be building Taylor polynomial approximations. Indeed, each iteration seems to build the next polynomial.

To make better sense of this we need to define distances on these objects (ie polynomials). Then we can use that definition to define what it means for objects to be close together or far apart. That, we can then use to define convergence on these objects. This will all happen when we look at **metric spaces**.

We can actually solve this IVP since it is separable:

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} &= 2x(1+y) \\
\frac{1}{1+y}\frac{\mathrm{d}y}{\mathrm{d}x} &= 2x \\
\int \frac{1}{1+y}\frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x &= \int 2x\,\mathrm{d}x \\
\int \frac{1}{1+y}\,\mathrm{d}y &= x^2 + c \\
\log|1+y| &= x^2 + c
\end{aligned}$$

Then since $y(0) = 0$ we must have that $c = 0$ and so

$$y(x) = e^{x^2} - 1.$$

We'd like to be able to *prove* that the Picard iteration does actually converge, and that it converges to the right thing.

1.3 Where are we going?

The above examples are supposed[1] to motivate some of the aims of this course.

- What makes $\mathbb{R}$ different from $\mathbb{Q}$? And what are the consequences of that difference.
- How do we actually define a real number?[2]
- Suprema and their consequences — alternative ways to define $\mathbb{R}$
- Continuity and Derivatives
- Metric spaces
- Contraction mappings
- Compactness (an important mathematical property)

[1] I hope they have

[2] well, actually most of this work is going to get pushed under the proverbial rug (in the appendix)

We are going to start out trying to really deeply understand our number systems – naturals, integers, rationals, reals. The first three of which we will more-or-less take their basic properties for granted (see Section 1.4). Understanding the real numbers on the other hand, is a more complicated task. After all, if you really try to write down what the real numbers are, you start to get to some questions that are difficult to answer, e.g., what is the difference between 0.999999 (repeated) and 1? Unfortunately, we will not have time to go into the full construction of the real numbers in this class, but the interested reader can read about it in the appendix A.

The main differentiating factor between the reals and the rationals, is that the real numbers satisfy the **least upper bound property**. Naturally, this has a great many consequences, which we will explore in the following chapter. We'll then prove some results about continuity and derivatives that you likely already know but haven't really seen *why* those results work. We then extend (some of) these ideas to more general mathematical objects, in order to finally arrive at the idea of a **contraction mapping**. In particular, that allows us to make quite general statements about iterative processes that bring objects closer together.

1.4 Number Systems

Before we get into things, we would like to go into a bit of detail about our basic number systesm: the natural numbers; the integers; and the rational numbers. We know that you're familiar with these already, but we would urge you to try to forget what you know for now, and we will work together to build up our knowledge of how these systems work.

We would also like to acknowledge that we are not going to build up these sets from first principles[1]. If you we're trying to be **really** pedantic, you could go through all the work of defining the natural numbers $\mathbb{N}$ using set theory, the integers $\mathbb{Z}$ by taking a group quotient of $\mathbb{N} \times \mathbb{N}$, and the rational numbers $\mathbb{Q}$ by taking a group quotient of $\mathbb{Z} \times \mathbb{Z}$. Instead, we will assume working knowledge of *some*[2] basic principles of these sets, and build up from there.

Let's get started.

1.4.1 The natural numbers

We begin by defining what we mean by the "natural numbers".

Definition 1.4.1 The Natural Numbers. The natural numbers, denoted $\mathbb{N}$, is the set

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

[1] by "first principles", we really mean the set theory ZFC axioms

[2] not all!

The slightly larger set $\mathbb{N} \cup \{0\}$ is called the "whole numbers" and is equipped with an ordering

$$0 < 1 < 2 < 3 < 4 \ldots$$

and operations of addition $+$ and multiplication $\times$ (or $\cdot$ or just writing nm), which behave as you expect.

More precisely, for all $m, n, k \in \mathbb{N} \cup \{0\}$,

$$m + n = n + m, \quad (m + n) + k = m + (n + k), \quad \text{and} \quad m + 0 = m.$$

In other words, addition is commutative and associative, and 0 is an "additive identity" (meaning adding zero returns the original object, i.e., does nothing).

Similarly for multiplication: For all $m, n, k \in \mathbb{N} \cup \{0\}$,

$$mn = nm \quad (mn)k = m(nk), \quad \text{and} \quad n \cdot 1 = n.$$

In other words, multiplication is commutative, associative, and 1 is a multiplicative identity.

Finally, we assume that *multiplication distributes over addition*, meaning that

$$m(n + k) = mn + mk.$$

$\Diamond$

Of course, you're familiar with these properties. But notice that we have conspicuously left some things off the list, such as things like "1 is the smallest nonzero natural number", for example[3]. We will get to prove some of those properties in just a bit. But first, we actually need to assume one more really fundamental thing about the natural numbers, which you cannot deduce just from the arithmetic properties of the naturals. Indeed, we'll take it as an axiom[4].

Axiom 1.4.2 The well-ordering principle. *Every non-empty subset of the natural numbers has a least element.*

From this we can get a couple of useful generalisations:

Corollary 1.4.3 *Let A be a subset of the integers, and let $k \in \mathbb{Z}$.*

- *if $k \leq a$ for all $a \in A$, then A has a least element, and*
- *if $k \geq a$ for all $a \in A$, then A has a maximal element.*

Proof. In the first case, consider the set $B = \{a - k + 1 \mid a \in A\}$. Since $k \leq a$ for all $a \in A$, $B \subseteq \mathbb{N}$ and thus has a smallest element x. The number $x + k - 1$ is then the smallest element of A.

In the second case, consider the set $C = \{k - a + 1 \mid a \in A\}$. Then since $k \geq a$ for all $a \in A$, the set $C \subseteq \mathbb{N}$ and so has a smallest element y. The number $k - y + 1$ is then the largest element of A. ■

[3] and other things as well!

[4] of course, you could prove it using fundamental ZFC axioms, but that isn't for this class

The well-ordering of the natural numbers is equivalent to the principle of mathematical induction which we'll state as a theorem.

Theorem 1.4.4 Principle of mathematical induction. *Let $P(n)$ be a statement for all $n \in \mathbb{N}$. If*

- *$P(1)$is true, and*
- *for all $k \in \mathbb{N}$,$P(k) \implies P(k+1)$ is true*

then $P(n)$ is true for all natural numbers n.

1.4.2 Induction and well-ordering (optional)

Here we'll show that induction and well-ordering are equivalent. You can prove each from the other.

Well-ordering implies induction. We assume the hypotheses of the principle of induction. That is, $P(n)$ be a statement for all $n \in \mathbb{N}$, so that

- $P(1)$ is true, and
- $P(k) \implies P(k+1)$ is true for all $k \in \mathbb{N}$.

Now construct a set of all the bad values of n, that is the values of n that make $P(n)$ false:

$$B = \{n \in \mathbb{N} \mid P(n) \text{ is false}\} \subseteq \mathbb{N}$$

The set B is empty or non-empty.

- If $B = \varnothing$ then we are done. The law of the excluded middle[a] tells us that $P(n)$ must be true for all $n \in \mathbb{N}$.
- On the other hand, if B is non-empty, then the well-ordering principle tells us that the set B has a minimum element, b. Since $P(1)$ is true, we know that $b \geq 2$, and since b is the smallest value that makes $P(n)$ false, we know that

  $$P(b-1) \text{ is true} \quad \text{and} \quad P(b) \text{ is false}$$

  However this point contradicts our assumption that $P(k) \implies P(k+1)$; if $P(b-1)$ is true, then we must have $P(b)$ is true.

  Thus there is no such number b, and so B must be empty.

■

[a] Given a statement P then either P is true or P is false. It cannot be neither nor can it be both.

Induction implies well-ordering. Assume the principle of induction. Now let B be a bad subset of $\mathbb{N}$, namely one without a minimum element. Let $G = \mathbb{N} - B$

be the complement of B. We will show that B must be empty and $G = \mathbb{N}$.

We know that $1 \notin B$, because otherwise 1 would be the minimum (there is no smaller natural number). Thus $1 \in G$. So we know B is bounded strictly below by 1 — that is for all $b \in B$, we know that $b > 1$ and so $b \geq 2$ (since b must be an integer - see Lemma 1.4.7 below).

Now assume that B is bounded strictly below by k, so that $b > k$ for all $b \in B$. So by the integerness of k (see Lemma 1.4.7 below) this means that all $b \in B$ satisfy $b \geq k+1$. Now, if $k+1 \in B$ then it would have to be the minimum element of B, so we know that $k+1 \notin B$. And thus B is bounded strictly below by $b+1$.

By induction we have proven that for all $n \in \mathbb{N}$ the set B is bounded strictly below by n. Hence there cannot be any $k \in \mathbb{N}$ that also lies in B. So B must be empty. ■

Since we're talking about induction, this feels like a natural spot to put the following warning (though, it really is unrelated to number systems...oh well). We're going to consider the following *false* induction proof. It shows the perils of extrapolating from the finite to the infinite.

Warning 1.4.5 Incorrect induction proof. Every subset of the natural numbers has a maximum.

> Let $P(n)$ be the statement "every set of natural number of cardinality at most n has a largest element". Then
>
> - The set of a single element, $\{a\} \subseteq \mathbb{N}$ trivially has maximum a, so the statement holds when $n = 1$.
> - Now assume that the statement holds for all sets of n elements, and let $B \subseteq \mathbb{N}$ be any $n+1$ element set. Remove any element, ℓ, from B to obtain a set A with n elements. That is
>
> $$B = A \cup \{\ell\}$$
>
> Now we know (by assumption) that A has a maximum, call it k. If $k \geq \ell$, then we know that k is the maximum of B, while if $k < \ell$ then ℓ is the maximum of B. In either case B has a maximum.
>
> So by mathematical induction, all subsets of $\mathbb{N}$ have a maximum.

Can you spot what the issue is in that "proof"? Take a minute and try to spot it for yourself. Once you found it, check that you're correct by opening up the following tab:

Check that you are correct. What we have really shown here is that every *finite* subset of $\mathbb{N}$ has a maximum. The inductive hypothesis fails when we try to jump from a finite set to an infinite one. Consider an *infinite* subset of $\mathbb{N}$, say all the positive odd numbers $\{1, 3, 5, 7, \ldots\}$. Removing a single element from this set does *not* change its cardinality. Similarly, given any *finite* subset of $\mathbb{N}$ we cannot make it infinite by adding an element to it.

This is a surprisingly common induction error.

Notice the error that we made here doesn't invalidate induction. When we use induction to conclude that $P(n)$ is true for all $n \in \mathbb{N}$, it means that no matter what $n \in \mathbb{N}$ we choose the statement $P(n)$ is true. Note that every single $n \in \mathbb{N}$ is *finite*. Induction doesn't tell us what happens for infinite n — you need other techniques to handle such things. ■

1.4.3 The Integers

Now that we've discussed the natural (and whole) numbers a bit, we can expand to include negatives as well. Our definition will seem a bit artificial because we're really not treating these things with the respect they deserve (you should do this in something like a group theory class).

Definition 1.4.6 The *integers*, denoted $\mathbb{Z}$, is the disjoint union

$$\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-n \mid n \in \mathbb{N}\}$$

We also assume that addition and multiplication work just like they do in the definition of the natural numbers 1.4.1, though we won't repeat it all here[5]. ◊

Though we won't do this in too much detail, you can then define the usual ordering on the integers by saying

$$a > b \iff a - b \in \mathbb{N}$$

and

$$a < b \iff a - b \in \{-n \mid n \in \mathbb{N}\}.$$

Then since $a - b \in \mathbb{Z}$, it must be in exactly one of these three sets. Thus we must have exactly one of $a < b, a = b$ or $a > b$.

It doesn't take too much work to see that this ordering is *compatible*, meaning "it preserves", the ordering on $\mathbb{N} \cup \{0\}$, but we won't do that here either.

Now for another little warm-up. Unsurprisingly, there is no integer between 0 and 1 and we prove it using the well-ordering principle.

Lemma 1.4.7 *There is no integer* $0 < n < 1$.

Proof. We prove this by contradiction. So let $A = \{n \in \mathbb{N} \mid 0 < n < 1\}$, and assume, to the contrary, that A is not empty. By the well-ordering principle, A has a least element, a. Since $0 < a < 1$, we can multiply this inequality by a to

[6] for everyone's sanity

get

$$0 < a^2 < a.$$

Since a^2 is a positive integer it is in A, but this contradicts the assumption that a is the smallest element of A. The result follows. ■

1.4.4 The Rationals

Now with the rational numbers, we can start to be a little more formal.

Definition 1.4.8 Consider the set

$$\mathbb{Z} \times \mathbb{N}.$$

Define an equivalence relation $\sim$[6] by

$$(a, b) \sim (c, d) \quad \Longleftrightarrow \quad ad = bc.$$

The **rational numbers** are defined to be the set of equivalence classes

$$\mathbb{Q} = \big(\mathbb{Z} \times \mathbb{N}\big)/\sim .$$

We interpret the equivalence class of (a, b) as the fraction

$$\frac{a}{b},$$

and denote it by $[(a, b)]$. ◊

Theorem 1.4.9 Arithmetic of the rational numbers. *If addition within the rational numbers is defined by*

$$[(a, b)] + [(c, d)] = [(ad + bc, bd)],$$

then this is well-defined *in the sense that*

$$[(a, b)] + [(c, d)] = [(a', b')] + [(c', d')]$$

whenever $[(a, b)] \sim [(a', b')]$ *and* $[(c, d)] \sim [(c', d')]$.

Similarly, multiplication within the rational numbers is defined by

$$[(a, b)][(c, d)] = [(ac, bd)]$$

and is also well-defined.

The integers are naturally identified with a subset of the rational numbers by identifying $a \in \mathbb{Z}$ *with*

$$[(a, 1)] \in \mathbb{Q}.$$

The rational numbers also inherit a natural ordering from the integers, where

$$\frac{a}{b} < \frac{c}{d}$$

[7] hey! It is a great warm-up for you to actually verify that this is an equivalence relation!

is interpreted as meaning that

$$ad < bc \quad \textit{(as integers!)}$$

Proof. We won't get into all these details because it gets really tedious really quickly. Let's just do the claim that addition is well-defined. Start with some equivalent pairs $[(a,b)] \sim [(a',b')]$ and $[(c,d)] \sim [(c',d')]$, meaning that

$$ab' = a'b \quad \text{and} \quad cd' = c'd.$$

We need to prove that $[(ad+bc, bd)]$ and $[(a'd'+b'c', b'd')]$ also define the same equivalence class, which is precisely to say that we need to show that

$$(ad+bc)b'd' = (a'd'+b'c')bd.$$

But wait, this is actually easy[7] because we know that addition and multiplication (let's be careful here; we're talking about addition and multiplication *withhin* $\mathbb{Z}$) behave nicely (associative, commutative, distributive), so we can write

$$\begin{aligned}(ad+bc)b'd' &= (ab')dd' + b(cd')b' \\ &= (a'b)dd' + b(c'd)b' \\ &= (a'd'+b'c')bd.\end{aligned}$$

Wheh, that wasn't *so* bad. But for the sake of everyone's santify, we won't do any more here. You can though! It is kind of fun once you get going. ■

So the above theorem tells us that the rationals are a set that have some addition and multiplication operations, but it *doesn't* say that these things behave like you expect them to. But, of course, they do!

Theorem 1.4.10 The rationals are an ordered field. *With the rational numbers defined as in Definition 1.4.8, the operations of addition $+$ and multiplications $\cdot$ as well as the ordering $\leq$ from Theorem 1.4.9, this makes the rational numbers into an* ***ordered field****. That just means that all the following properties are satisfied for any $x, y, z \in \mathbb{Q}$:*

- *Closed: $x+y$ and $x \cdot y$ are also in $\mathbb{Q}$*
- *Commutative: $x+y = y+x$ and $x \cdot y = y \cdot x$*
- *Associative: $x+(y+z) = (x+y)+z$ and $x \cdot (y \cdot z) = (x \cdot y) \cdot z$*
- *Distributive: $x \cdot (y+z) = (x \cdot y) + (x \cdot z)$*
- *Additive identity: $x + 0 = x$*
- *Additive inverse: $x + (-x) = 0$*
- *Multiplicative identity: $x \cdot 1 = x$ (with $1 \neq 0$)*

[8] Be careful; "easy" is a very dangerous word in mathematics

- *Multiplicative inverse:* $x \cdot (x^{-1}) = 1$ *(provided* $x \neq 0$*)*
- *Trichotomy: Exactly one of* $x < y, x = y, y < x$ *holds.*
- *Transitive:* $(x < y) \wedge (y < z) \implies (x < z)$
- *Order and addition:* $(x < y) \implies (x + z < y + z)$
- *Order and multiplication:* $(0 < x) \wedge (0 < y) \implies (0 < x \cdot y)$

We don't want you to memorize all the above properties in a precise and orderly list. But we *do* want you to be familiar with them and be able to use them and recognize them. We also are not going to delve into the proof of any of these properties, but no individual one of them is overly complicated or difficult. They just require working with the definition of $\mathbb{Q}$ using equivalence relations and the properties of the integers.

We also recognize that we are being a bit uncouth in the above definition because we haven't used the word **field** before. Basically, a field is just a name that we assign to *any* set X equipped with some operations $+$ and $\cdot$ which satisfies the list of requirements:

- Closed: $x + y$ and $x \cdot y$ are real
- Commutative: $x + y = y + x$ and $x \cdot y = y \cdot x$
- Associative: $x + (y + z) = (x + y) + z$ and $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- Distributive: $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$
- Additive identity: $x + 0 = x$
- Additive inverse: $x + (-x) = 0$
- Multiplicative identity: $x \cdot 1 = x$ (with $1 \neq 0$)
- Multiplicative inverse: $x \cdot (x^{-1}) = 1$ (provided $x \neq 0$)

This is all very daunting looking, but you would definitely be familiar with a variety of fields if you saw them. For example, there are **finite fields** which essentially are just the integers mod any prime p, or perhaps the field of all rational functions

$$\left\{ \frac{\text{polynomial}}{\text{(non-zero) polynomial}} \right\},$$

just to name a few. Then the claim that $\mathbb{Q}$ is an *ordered* field is just the extra stipulations that it comes equipped with an ordering operation $<$ which is compatible with the operations of addition and multiplication, as we listed above.

Anyway, all that talk of fields and junk 'n stuff is totally not what this course is about. But you're bound to come across these terms a lot as you progress in your math journey, so you may as well see them here.

Now that we've established the basic properties of the rational numbers, let's delve deeper into some of the more subtle qualities of the rationals. Recall in the definition 1.4.6 of the integers, we defined them to be trichotomous (meaning every element is exactly one of positive, negative, or zero). The same is true for the rationals.

Theorem 1.4.11 *The rational numbers are trichotomous, meaning that for any $p, q \in \mathbb{Q}$, exactly one of*

$$p < q, \quad p > q, \quad p = q$$

are true.

Proof. We will write $p = a/b$ and $q = c/d$ in *lowest form*[8]. Recall, by definition, we have

$$\frac{a}{b} < \frac{c}{d} \quad \iff \quad ad < bc$$

where the latter is using the ordering on the integers. In particular, since the integers are trichotomous, we can say that exactly one of

$$ad < bc, \quad ad > bc, \quad ad = bc$$

is true, which is what we wanted to show. ■

Also, the rationals are **dense**; between any two rationals you can always find another one!

Lemma 1.4.12 *Let $x, y \in \mathbb{Q}$ with $x < y$, then there exists $z \in \mathbb{Q}$ so that $x < z < y$.*

Proof. Let x, y be as stated, and let $z = \frac{x+y}{2}$. Since $x, y \in \mathbb{Q}$ it follows (and we could expand on this) that $z \in \mathbb{Q}$. Then since $x < y$ it follows that $2x < x + y$ and $x + y < 2y$, and so

$$x < \frac{x+y}{2} = z < y$$

as required. ■

As a corollary to the density of the rationals, we also can prove that there is no smallest one.

Corollary 1.4.13 *There is no smallest positive rational number.*

Proof. Assume, to the contrary, that there is such a number; denote it r. Since $0 < r$ and $0, r \in \mathbb{Q}$, Lemma 1.4.12 implies that there is rational s satisfying $0 < s < r$. This contradicts our assumption that r was minimal, and hence there is no such number r. ■

One more property named for Archimedes. He noted (as indeed many other people have) that you can use a small rod to measure a large distance.

[9] meaning, there are no common factors

Theorem 1.4.14 The rationals are Archimedean. *Given any two positive rational numbers p, q one can always find a natural number n so that*

$$np > q$$

Proof. Let $p = \frac{a}{b}$ and $q = \frac{c}{d}$. Now notice that since a, b, c, d are positive, we have

$$q = \frac{c}{d} \leq c$$

and

$$\frac{a}{b} \cdot bc = ac \geq c$$

Combining these inequalities suggests that we can set $n = bc + 1 \in \mathbb{N}$. Then

$$\begin{aligned} np &= (bc+1)p > bcp \\ &= ca \geq c \\ &\geq \frac{c}{d} = q \end{aligned}$$

as required. ■

We have not discussed real numbers yet (only rationals so far!). But we will see later that the rational (and the irrational!) numbers are also dense inside of the reals. We'll get there soon, just not yet.

One may naturally wonder why we can't just stop here. After all, the rational numbers are very nice! They satisfy many of the properties that we want. But the inherit issue with the rational numbers is that they have *holes*[9]. For example, if we would like to write down a solution to the equation $x^2 = 2$, then it is not hard to show that such and x *cannot be rational* (see the following exercise). However, you can get *arbitrarily close* to finding a solution to this equation through the rationals (consider, e.g., the sequence of rational numbers $1, 1.4, 1.41, \ldots,$.) So if we think of $\sqrt{2}$ being the "number"[10] which satisfies the equation $x^2 = 2$, then this number must lie in one of the "holes" of $\mathbb{Q}$. Somehow we need to formally close these holes, so-to-say.

Exercise 1.4.15 Rational numbers have holes. Prove that no rational number $x \in \mathbb{Q}$ satisfies $x^2 = 2$.

Solution. Homework!

1.4.5 The Real Numbers

Oh yea now we're gettin' somewhere! In a perfect world, we would love to give the real numbers all the love and respect they deserve. But, as you might expect given all the work we had to do just to talk about $\mathbb{Q}$, constructing the real

[9] this is not a technical term, and neither is it referencing a movie staring Shia LaBeouf

[10] we haven't defined what real number are yet, so this is really just more of a thought exercise than anything

numbers from the rational numbers is a LOT of work. Unfortunatley we don't have time for that. We'll give just a taste of what one could do in the next section.

1.5 The Real Numbers (a preview)

Now we will go into a little bit of detail about how one **might**[1] go about defining the real numbers from the rationals. Importantly, we will see that the real numbers satisfy, not only all the nice properties that the rationals do, but with the additional property that there are no holes. Let's get into it.

1.5.1 Decimal expansions

Although we've all been working with real numbers for many years, it can be quite surprising just how hard it is to come up with a solid definition of real numbers. A not terrible working definition comes from real numbers are decimal expansions. For example

$$\sqrt{2} = 1.41421356237\ldots$$

and

$$\pi + e = 5.859874482\ldots$$

Note — $\pi + e$ is an interesting example. We know that both π and e are transcendental[2] but no one knows how to prove even that $\pi + e$ is not rational!

Addition of numbers via their decimal expansions, however, does have its problems. For example,

$$0.99999\cdots = 1.0000\ldots$$

These represent the same real number — one — but are two different decimal expansions. In order to take the working definition of "a real number is a decimal expansion" and make it a rigorous piece of mathematics, one has to handle the fact that many numbers have two decimal expansions, while most only have one.

Exercise 1.5.1 It is a fact that the two decimal expansions $0.999999\ldots$ and $1.0000\ldots$ both represent the same number (that is 1). Can you write down other numbers that have two decimal expansions?

Do you think it is possible for a single number to have *more* than 2 decimal expansions? Why or why not? Prove that every number of the form $\frac{a}{2^k 5^k}$ has a decimal expansion (this part is harder, but not terribly so).

Before moving on from decimal expansions, we'd like to point out that even if we did decide to use decimal expansions as a "definition" of real numbers, we

[1] there are quite a few ways, as we will discuss later

[2] The transcendence of e was proved by Hermite in 1873 and the transcendence of π was proved by Lindemann in 1882.

would also have to deal with showing that these satisfy the properties that we want them to (more on exactly what these properties are is coming right up). For example, even showing a simple thing that multiplication behaves well is super messy when working with decimal expansions. Then there is a huge issue of justifying that the real numbers have no "holes", etc. There is a lot of work to do here to justify all these things in a rigorous manner.

Instead of doing all that, we want to give you a taste of a slightly different approach of constructing the reals. The approach that we take here (most details in the appendix) is to define real numbers as a sort of "limit" (of course, we have to be pedantically careful here because we haven't defined what a limit is yet, but ok) of *finite* decimal expansions.

1.5.2 Sequence of intervals (decimal approximations)

We start with something that looks a bit like a decimal expansion, but avoids some of the problems caused by $0.9999\cdots = 1.0000\ldots$. Instead of trying to write down the whole expansion in one go, think about building up a real number by "measuring" it. So, think about measuring $z = \pi + e$ with an instrument that will give us precise upper and lower bounds to as many digits as we want. If we start by a really rough measurement of these numbers, then we know that

$$3 \leq \pi \leq 4$$

and

$$2 \leq e \leq 3$$

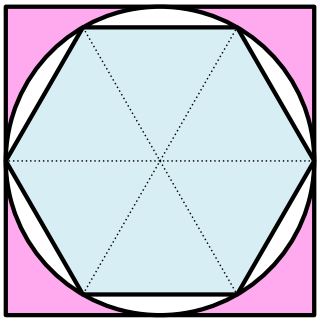
Op! How do we know that? As part of our mathematicians approach, we should have rigorous definitions of these numbers e and π, and not just their decimal expansions (which is exactly what we're trying to construct!). Along those lines, we will use the common definition[3] of $e = \sum_{n=0}^{\infty} 1/n!$ and that $2\pi r = C$, where r, C are the radius and circumference of a circle respectively.

Exercise 1.5.2 Bound: $3 \leq \pi \leq 4$. Prove that $3 \leq \pi \leq 4$ by considering a regular hexagon inscribed[4] within a circle, which is itself inscribed with a square.

Hint. Draw a careful picture in which the hexagon has side-lengths 1, the circle has radius 1 and the square has side-length 2.

Solution. Consider the picture below.

[3] we haven't discussed infinite sums of rational numbers (we will later!), but for now let's just roll with it!



Since the hexagon has side-lengths 1, the circle has radius 1 and the square has side-lengths 2, we know that the circumference of the circle satisfies

$$6 \leq C = 2\pi \leq 8$$

Thus we have

$$3 \leq \pi \leq 4$$

as required.

Exercise 1.5.3 Bound: $2 \leq e \leq 3$. Prove that $2 \leq e \leq 3$ by using the definition

$$e = \sum_{n \geq 0} \frac{1}{n!}$$

Hint. Recall that $0! = 1$. To prove the upper sum, it may be helpful to use the inequality $n! \geq n(n-1)$ when $n \geq 2$.

Solution. Starting from Euler's formula for e, extract out the first two terms of the sum:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \sum_{n=2}^{\infty} \frac{1}{n!}$$

(of course we haven't actually defined what we mean by this infinite sum, but just suspend your scepticism for a little). Since $n! > n(n-1)$ for $n \geq 2$, we can write

$$0 \leq \sum_{n=2}^{\infty} \frac{1}{n!} \leq \sum_{n=2}^{\infty} \frac{1}{n(n-1)}$$

This last sum is telescoping and can be rephrased as

$$\sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = \sum_{n=1}^{\infty} \frac{1}{n} - \sum_{n=2}^{\infty} \frac{1}{n} = 1$$

Hence we have

$$2 \leq e \leq 3$$

as required.

[4] ie drawn inside

So without too much work we know that

$$z = \pi + e \in [5, 7]$$

This is a pretty rough bound on z, but we can improve it. I don't actually want to do the calculations, you should think of this as more of a thought-experiment. We have our imaginary instrument and we turn up the precision a little more, and we arrive at bounds of z to 1 decimal place:

$$z \in [5.8, 5.9] = \left[\frac{58}{10}, \frac{59}{10}\right]$$

Notice that our new measurement $\left[\frac{58}{10}, \frac{59}{10}\right] \subseteq [5, 7]$. If we keep working hard, and turn the precision dial up a little more then we can bound to 2 decimal places[5]

$$z \in [5.85, 5.86]$$

Again, our new bound is nested within the previous one — it improves it. So we have

$$z \in \left[\frac{585}{100}, \frac{586}{100}\right] \subseteq \left[\frac{58}{10}, \frac{59}{10}\right] \subseteq [5, 7]$$

And further still:

$$z \in \left[\frac{5859}{1000}, \frac{5860}{1000}\right] \subseteq \left[\frac{585}{100}, \frac{586}{100}\right] \subseteq \left[\frac{58}{10}, \frac{59}{10}\right] \subseteq [5, 7]$$

Indeed (in principle) we can construct a sequence of intervals

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq I_3 \ldots$$

so that

- for every $n = 0, 1, 2, \ldots$ there are $a_n \leq b_n \in \mathbb{Z}$ so that

$$I_n = \left[\frac{a_n}{10^n}, \frac{b_n}{10^n}\right]$$

  and

- for every $n = 0, 1, 2, \ldots$ we have $z \in I_n$, which implies

$$z \in \bigcap_{n=0}^{\infty} I_n$$

[5]For example, we could use successive partial sums of alternating series expressions for $e^{-1} = \sum(-1)^k/k!$ and $\pi = \sum(-4)^{-k} \cdot (2/(4k+1) + 2/(4k+2) + 1/(4k+3))$ to get easy bounds. This series for π is due to Adamchik and Wagon in 1997.

However this is not enough — we want the intervals to become more and more narrow, and, to be more careful, we need the widths to tend to zero. We haven't defined limits yet, but we can make this idea more precise without having to define limits in all their glory. Consider the intervals defined by

$$I_n = \left[\frac{-1}{n+1}, \frac{n+2}{n+1}\right]$$

and work through this exercise.

Exercise 1.5.4 Want the widths to go to zero. Consider the intervals defined by

$$I_n = \left[\frac{-1}{n+1}, \frac{n+2}{n+1}\right]$$

- Prove that these intervals are nested but that their widths do not go to zero (you don't have to be very rigorous — first year calculus style arguments are enough).
- Prove that $\cap I_n = [0, 1]$ (again, don't be too rigorous)

Solution. To show that the intervals are nested, it suffices to note that

$$\frac{-1}{n+1} < \frac{-1}{n+2}$$

and then

$$\frac{n+3}{n+2} = 1 + \frac{1}{n+2} < 1 + \frac{1}{n+1} = \frac{n+2}{n+1}$$

Now, the width of $I_n = \frac{n+3}{n+1} = 1 + \frac{2}{n+1}$. As n becomes very large, this goes to 1.

To show that these sets are equal, we have to show that each side is a subset of the other.

- Let $x \in [0, 1]$. Then $\frac{-1}{n+1} \leq 0 \leq x \leq 1 < 1 + \frac{1}{n+1}$ and so $x \in I_n$ for all n.
- Now let $x \in \cap I_n$, and so we must have $-\frac{1}{n+1} \leq x \leq 1 + \frac{1}{n+1}$ for all n. Now, assume, to the contrary, that $x < 0$. Set $k = \lceil -1/x \rceil + 1$. Notice that $-1/k > x$ and so $x \notin I_k$ which gives a contradiction. If we assume that $x > 1$ we can arrive at a similar contradiction. Hence $x \in [0, 1]$.

Hence there are uncountably infinite real numbers that lie in the intersection of all these intersections. We don't want that; we would like to have exactly one number lying in the intersection of all these sets — the number we are trying to measure.

Borrowing ideas from limits, we require that the widths of the intervals become arbitrarily small. That is, no matter what small width ε you choose, we can make n large enough so that $|I_n| < \varepsilon$. Writing this more mathematically[6],

[6] i.e., in a pedantically careful manner.

we need

$$\forall \varepsilon > 0, \exists N \ \mid \ (n > N) \implies |I_n| < \varepsilon.$$

We are going to use this ideas to define real numbers. In particular, we are going to associate each real number with such a sequence of intervals. That is (and abusing the notation "$\cong$" for the moment)

$$z \cong \left([5,7], \left[\frac{58}{10}, \frac{59}{10}\right], \left[\frac{585}{100}, \frac{586}{100}\right], \left[\frac{5859}{1000}, \frac{5860}{1000}\right], \dots \right)$$

Of course, this looks an awful lot like associating the real number z with something like its decimal expansion. This is no coincidence. Part of what we are trying to do here is to write z as its decimal expansion while avoiding many of the issues that arise from the fact that $0.999\dots$ and $1.000\dots$ both represent the real number 1.

Exercise 1.5.5 A sequence of nested intervals whose intersection contains 1. Construct a sequence of nested intervals (I_n) so that their widths go to zero and so that $1 \in \cap I_n$.

Solution. There are lots of possibilities here, for example:

$$I_n = [0.\underbrace{99\dots9}_{n}, 1.\underbrace{00\dots0}_{n}]$$
$$I_n = [1,1]$$

Or many others!

There is still quite a bit of work to do. To start, notice that the sequence of intervals about is not as "tight" as it could be. In particular, we know (from bounding further significant figures) that

$$5 \leq z = \pi + e = 5.859874482\cdots \leq 6$$

and so we could improve our sequence of intervals to

$$z \cong \left([5,6], \left[\frac{58}{10}, \frac{59}{10}\right], \left[\frac{585}{100}, \frac{586}{100}\right], \left[\frac{5859}{1000}, \frac{5860}{1000}\right], \dots \right)$$

and then at each depth, the interval is as narrow as possible. Hence we should really prove (among other things) that given any sequence of nested intervals

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$$

that

- there exists an "equivalent" sequence of intervals that is as narrow as possible,
- that this narrowest sequence is unique, and

- that equivalence of sequences of intervals is well defined.

Not to mention that along the way we would also have to understand how we can compare and order sequences of intervals, how we can define addition and the other usual arithmetic operations, that the real numbers really do contain the rationals (in some formalized sense), that this set really does not have any "holes", etc. That is, we need to show that these sequences of intervals do the things that real numbers do. That is all a looooooot of work (like 3 weeks of class time). For a proper treatment of these things, we direct the reader to the appendix A.

1.5.3 So what can we assume about the real numbers then?

Well, the real numbers, which we will denote by $\mathbb{R}$, should be an *extension* of the rational numbers, in the sense that they should satisfy all the properties we know and love from the rationals, but also more! So let's start by saying that much.

Theorem 1.5.6 The Real Numbers are an Ordered Field. *The real numbers are an ordered field, meaning that for any $x, y, z \in \mathbb{R}$, all the following are satisfied*[7]*:*

- *Closed: $x + y$ and $x \cdot y$ are real*
- *Commutative: $x + y = y + x$ and $x \cdot y = y \cdot x$*
- *Associative: $x + (y + z) = (x + y) + z$ and $x \cdot (y \cdot z) = (x \cdot y) \cdot z$*
- *Distributive: $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$*
- *Additive identity: $x + 0 = x$*
- *Additive inverse: $x + (-x) = 0$*
- *Multiplicative identity: $x \cdot 1 = x$ (with $1 \neq 0$)*
- *Multiplicative inverse: $x \cdot (x^{-1}) = 1$ (provided $x \neq 0$)*
- *Trichotomy: Exactly one of $x < y, x = y, y < x$ holds.*
- *Transitive: $(x < y) \wedge (y < z) \implies (x < z)$*
- *Order and addition: $(x < y) \implies (x + z < y + z)$*
- *Order and multiplication: $(0 < x) \wedge (0 < y) \implies (0 < x \cdot y)$*

Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as an ordered subfield *in the sense that there is some injective function*

$$\varphi : \mathbb{Q} \to \mathbb{R}$$

which respects the operations $+, x$ and the ordering $<$. In otherwords, $+, x, <$ mean the same thing that they did in $\mathbb{Q}$ as they do here in $\mathbb{R}$.

So far though, the previous theorem 1.5.6 does not say anything about $\mathbb{R}$ filling the holes of the rational numbers. In order to make that precise however, we need a definition.

Definition 1.5.7 Upper Bounds and Least Upper Bounds. Let $A \subseteq \mathbb{R}$ be any of real numbers. We say that $b \in \mathbb{R}$ is an **upper bound** of A if

$$a \leq b \quad \text{for all } a \in A.$$

In this case, we say that A is **bounded above**

We say that $\alpha \in \mathbb{R}$ is the **Least Upper Bound** if A is bounded above, and if

$$\alpha \leq b$$

for *every* upper bound b. In this case, we denote $\alpha = \sup A$, and is called the "supremum of A". ◊

[7] hah! we get to through this list at you again! muah ha ha

A Least Upper Bound is a concept that you are surely casually familiar with; it is a type of "maximum" of a set, even if the maximum is not necessarily contained in the set. The following exercise demonstrates this.

Exercise 1.5.8 A concrete LUB example. We will discuss what exactly the "Least Upper Bound Property" means for the real numbers later, but for now, let's start examine what this property means for a set of real numbers that we understand better, that is $\mathbb{Q}$. First, consider the set

$$A = \{x \in \mathbb{Q} : x^2 < 1\}.$$

In other words, A is the set of all rational numbers x such that $x^2 < 1$. Convince yourself that $y = 1$ is the *least upper bound* for the set A. To do this you must

1. show that is an *upper bound* for the set, meaning that $x \leq y$ for every $x \in A$, and

2. it is the *least* such bound, meaning that if $z < y$, then z is *not* an upper bound for A.

You'll do this on your homework.

As it turns out, *every* bounded set of real numbers has a least upper bound, which we will call a *supremum*. Let's formally state this as a theorem, then deomstrate with an example why this should convince you that there are no "holes" in $\mathbb{R}$.

Theorem 1.5.9 Least Upper Bound Property for the Real Numbers. *Let $A \subseteq \mathbb{R}$ be any non-empty set of real numbers that is bounded above. Then* $\sup(A)$ *exists as a real number.*

Hopefully the following example helps to convince you that this "least upper bound property" is exactly the trick that we needed to fill the holes in the rational numbers.

Exercise 1.5.10 Filling the hole — A definition for the square root of two. Consider the set

$$A = \{x \in \mathbb{R} : x^2 < 2\}.$$

Note that this set is, in fact, bounded above because $b = 10^8$ is an upper bound. Therefore, the Theorem 1.5.9 tells us that $\alpha := \sup(A)$ exists as a real number! Let's show that $\alpha^2 = 2$, in other words, this α could be the definition of $\sqrt{2}$

Oof, wait! We probably shouldn't actually complete this exericse here because you'll do it on your homework!

To round out this section, we would like to state one more result about the real numbers, and we hope you find it kind of surprising and also very pleasing. It is not a result that we prove (even in the appendix), but that's ok.

[8] who cares what upper bound we take at this point! All that we need to know is that it is bounded above by *something*

Theorem 1.5.11 Uniqueness of the Real Numbers. *Suppose* $(X, +, \times, <)$ *is* any *ordered field (in the sense of the Theorem 1.5.6), and also assume that* $(X, <)$ *satisfies the least upper bound property (in the same sense as in the above theorem 1.5.9, but with the general set* X *replacing* $\mathbb{R}$*). Then* X *is* precisely *the set* $\mathbb{R}$ *in the sense that there exists a bijection*

$$\varphi : X \to \mathbb{R}$$

which respects the addition, multiplication, and ordering of both X *and* $\mathbb{R}$*. In other words,* $(X, +, x, <)$ *is* indistinguishable *from* $(\mathbb{R}, +, x, <)$ *as ordered fields.*

To rephrase the above slightly, the real numbers are the only possible way that the rational numbers can be extended while still preserving all the structure $\mathbb{Q}$ and while also filling all the holes! Of course, you can call your dog by a different name, but at the end of the day, it is still your same old wet dog[9].

1.6 Useful resources

While writing this document the authors made heavy use of the resources below. Most of these (but perhaps note all) should also prove to be useful resources for you.

References

[1] N.H. Katz *Calculus for cranks* Link to full text[1]
This is a nice resource for someone looking for a text that focuses on a rigorous treatment of calculus on the real numbers, which is fairly easy to read, and contains many examples and exercises.
Sections 1.4, 2.1 and 3.1-3.5 largely mirrors the material of our chapters 3,4,5. *Cranks* provides an alternate construction of the real number than the one we provide in chapter 2, though they do not go into full detail regarding some of the properties of the real numbers that we desire. The material in after section 3.5 in *Cranks* is not applicable to our course. Conversely, *Cranks* focuses entirely on standard calculus on the real line, and does not consider metric spaces, which is the topic of our Chapter 6.

[2] Jiří Lebl *Basic Analysis I & II* Link to full online text[2]
This resource reads more like a series of lecture notes, rather than a text-book. Its language is clear and easy to follow, and it features nice visuals when appropriate.
Volume 1, Chapters 1 - 4 and 7 (omit 2.5, 2.6, 4.4) in *Basic* covers roughly the same material as our Chapters 2 - 6. Of particular note, *Basic* takes a different approach to the discussion of the real numbers, by first discussing

[9] something something rose by any other name something something
[1] www.math.caltech.edu/~nets/cranks.pdf
[2] www.jirka.org/ra

and proving some properties of $\mathbb{R}$, and then after (in section 1.5) outlining the construction, which is done there using digit strings (as opposed to intervals like we do).

[3] K.R. Davidson and A.P.Donsig *Real analysis and applications — Theory in practice* Text homepage[3]
This is a good resource for someone looking for a text that situates calculus within broader mathematics. It is a great supplementary text, though not recommended as a primary source.
This is a textbook that reaches far beyond our course material, but most of what we will learn is contained in its depths. You should have free access in Springer via your CWL. The relevant sections in *Analysis* for our purposes are 2.1-2.9, 4.1-4.3, 5.1-6.2 (omit 5.5) and chapter 9.

[4] W.F. Trench *Introduction to real analysis* Link to full text[4]
This book "is designed to fill the gaps left in the development of calculus". As such, more general analysis topics (like metric spaces, and compactness, for example) aren't touched on until the end of the book (chapter 8). But it reads easily, is approachable, and has a plethora of exercises.
Chapter 1 focuses on properties of the real numbers (though not their construction) and chapter 2 covers the majority of the topics from our chapters 3-5. Chapters 3-7 discuss integral and multivariable calculus, which we won't discuss in this course.

[5] W. Rudin *Principles of mathematical analysis* Link to text[5]
Often referred to as *Baby Rudin*, this work is a classic for aspiring mathematicians. It's known for its high level of rigor, terse prose, and challenging problem sets. *Principles* is ideal for those aiming for deep theoretical understanding, but less accessible for beginners or interdisciplinary students. The layout of *Rudin* is different than what we will follow. While it does start with a construction of the real numbers (which is very terse and not very easy to follow by the way), *Rudin* jumps right into metric spaces and topology, which we don't discuss until the end of the course. Much of the rest of the text depends on this introduced technical material, and therefore this may not be the best accompanying source for your for this course, though it really is a nice short classic.

[6] G.H. Hardy *A course of pure mathematics* link to full text[6]
This is a classic mathematical text, but can feel dated compared to modern texts. Good for readers interested in traditional, theoretically precise presentation.
The majority of chapters 1,2,4-6 are applicable to this course, though you

[3] www.math.uwaterloo.ca/~krdavids/RAA/real.html
[4] digitalcommons.trinity.edu/mono/7/
[5] david92jackson.neocities.org/images/Principles_of_Mathematical_Analysis-Rudin.pdf
[6] www.gutenberg.org/ebooks/38769

may have to apply a bit of a sieve in your reading. Moreover, this text does not even mention what a **metric space** is.

[7] M. Bridger *Real Analysis: A Constructive Approach Through Interval Arithmetic* link to full text[7]
This is a very readable book that takes a somewhat different approach to introducing analysis. We recommend skimming this one just to get some ideas as to how we could be doing all this stuff differently. For instance, immediately after constructing the real numbers (which is done with a similar — still using interval arithmetic —, though slightly different method than the one we present), Bridger jumps into a long section on the inverse function theorem (which we don't choose to address at all). Also, Bridger elects to focus only on *uniformly* continuous functions, rather than simply continuous ones. This is somewhat unique among mathematical texts. Bridger sees this as a practical approach, due to the fact that in most applications, and for most people who are not going to necessarily pursue mathematics as a career, they care about a continuous function on a closed bounded interval $[a, b]$ (when, integrating, for instance). And in this setting, the fact that the function is continuous implies it actually satisfies this stronger notion of *uniform* continuity. We will touch on this application in Chapter 2.

[8] B. Green *Metric spaces* Lecture notes[8] Videos of lectures[9]
Maybe not so useful as a text, but it was a useful reference for the authors.

[9] D.J. Eck *Metric spaces* Lecture notes[10]
These lecture notes cover roughly the same material as our chapter 6, though in a slightly different order (they discuss compactness before completeness, for example). This is a great recourse for you to follow along with while we are learning about metric spaces for the first time because it should give you a slightly different perspective and an extra supply of exercises at an attainable level.

[10] M. Wheeler *Advanced real analysis* Lecture notes[11]
Maybe not so useful as a text, but it was a useful reference for the authors.

[11] J.B. Conway *A course in point set topology* Text on google[12]
Maybe not so useful as a text, but it was a useful reference for the authors.

[12] J.R. Munkres *Topology* Text on amazon[13]
Maybe not so useful as a text, but it was a useful reference for the authors.

[7] www.gutenberg.org/ebooks/38769
[8] courses.maths.ox.ac.uk/mod/resource/view.php?id=2115
[9] youtube.com
[10] math.hws.edu/eck/metric-spaces/index.html
[11] www.mileshwheeler.com/ma30252/
[12] books.google.ca
[13] amazon.ca

Chapter 2

Consequences of the Supremum

The formalization of the supremum (A.K.K., the least upper bound propery 1.5.9) leads to many of the properties of the real numbers that you are already familiar with. But we do want to caution the reader that we really need to build all these things from the ground up, and not rely on our intuition or on the things that you already "konw" to be true. A good practice is to justify every logical implication in your work with a previously known theorem, axiom, or definition. Take nothing for granted!

We will also show that the existence of suprema is equivalent to two other properties, namely the nested interval property and the completeness of the reals. These, in turn, allow us to prove many results about convergence and continuity and …

But first, take a moment to recall the triangle inequality and its reversed friend. These inequalities will be crucial in our coming work.

Theorem 2.0.1 The triangle inequality. *Let $x, y \in \mathbb{R}$. The* ***triangle inequality*** *states that*

$$|x + y| \leq |x| + |y|$$

while the ***reverse triangle inequality*** *says*

$$|x - y| \geq \big||x| - |y|\big|$$

Proof. The proof of the triangle inequality requires some careful case-analysis.

There reverse inequality follows from some careful substitutions. Starting with the triangle inequality, set $y = b - a$ and $x = a$, then we have

$$|a + (b - a)| \leq |a| + |b - a| \qquad \text{and so} \qquad -|b - a| \leq |a| - |b|$$

Similarly, setting $x = a - b$ and $y = b$ gives

$$|(a - b) + b| \leq |a - b| + |b| \qquad \text{and so} \qquad |a| - |b| \leq |b - a|$$

Putting these together gives

$$-|b - a| \leq |a| - |b| \leq |b - a| \qquad \text{and so} \qquad \big||b| - |a|\big| \leq |b - a|$$

where we have used $(-x \leq y \leq x) \iff (|y| \leq x)$. ■

2.1 Archimedes

Using the Theorem 1.5.9, we can immediately get the Archimedean property of the reals.

Theorem 2.1.1 Archimedean Property for the Reals. *For each real $x > 0$, the set $\{nx \mid n \in \mathbb{N}\}$ has no upper bound. Consequently, if $y \in \mathbb{R}$ then there exists $n \in \mathbb{N}$ so that $nx > y$.*

Proof. Let $x > 0$ be given. Assume, to the contrary, that the set $A = \{nx \mid n \in \mathbb{N}\}$ has an upper bound. In this case the real number $u = \sup A$ exists, by the Least Upper Bound property 1.5.9.

Now since $x > 0$, we have $u - x < u = \sup A$, so there must be some $a \in A$ so that $u - x < a$. And since $a \in A$ we know that $a = kx$ for some $k \in \mathbb{N}$. This gives

$$u - x < kx.$$

Adding x to both sides gives

$$u < (k + 1)x.$$

This gives us the desired contradiction: clearly $(k + 1)x \in A$, so we must have $u \geq (k + 1)x$, but this is the opposite of the inequality above.

Thus the set A can have no upper bound. In particular, if some real number y is given, it cannot be an upper bound for A. By definition, there must be some $a \in A$ satisfying $y < a$. Recalling the definition of the set A proves that some $n \in \mathbb{N}$ must satisfy $y < nx$, as required. ■

Corollary 2.1.2 *The following three statements are logically equivalent*

- *For every positive $x, y \in \mathbb{R}$ there is some $n \in \mathbb{N}$ so that $nx > y$*
- *For every positive $x \in \mathbb{R}$ there is some $n \in \mathbb{N}$ so that $n > x$.*
- *For every positive $x \in \mathbb{R}$ there is some $n \in \mathbb{N}$ so that $0 < \frac{1}{n} < x$*

Proof. We prove the first implies the second, the second implies the third and then the third implies the first.

- Assume the first statement is true. Then it is true for $x = 1$ and any y, giving us $n > y$ as required.
- Assume the second is true. Let positive $x \in \mathbb{R}$ and set $y = \frac{1}{x} > 0$. Then we must have n so that $n > y = \frac{1}{x} > 0$. Multiply this by $\frac{x}{n} > 0$ to get $x > \frac{1}{n} > 0$ as required.
- Assume the last is true. Let $x, y > 0$, then $\frac{x}{y} > 0$ and so there is some n so that $\frac{1}{n} < \frac{x}{y}$. Multiply this by $ny > 0$ to get $y < nx$ as required.

■

From this we can then get the **density** of the rationals inside the reals. This is quite a bit more technically difficult than showing we can always find a rational between any two distinct rationals. Given two reals x, y we can't just set $z = \frac{x+y}{2}$ because we need *rational* z.

But first we need[1] what might feel like a really obvious result. Namely if two reals are more than one apart then there is an integer between them.

Lemma 2.1.3 *Let $x, y \in \mathbb{R}$ so that $1 < y - x$. Then there exists $n \in \mathbb{Z}$ so that $x < n < y$.*

Proof. A good homework problem! ■

Corollary 2.1.4 Density of $\mathbb{Q}$ inside $\mathbb{R}$. *Let $x, y \in \mathbb{R}$ with $x < y$. Then there exists $q \in \mathbb{Q}$ so that $x < q < y$.*

Proof. Another good homework problem! ■

The density of rationals inside the reals tells us that we can always use a rational number to make a good approximation of any given real number. To be slightly more precise, given any real number x and any $\varepsilon > 0$, we can always find a rational number q so that $x - \varepsilon < q < x + \varepsilon$.

The interested reader should search-engine their way to a discussion of Diophantine approximation, which is, among other things, the approximation of real numbers by rational numbers. In particular, a theorem by Legendre proves that "best" approximations of real numbers by rational numbers are achieved by continued fractions.

Of course, you're probably wondering about the *irrational*[2] numbers. And yes, we ought to mention that they are dense too I guess[3]. In the proof, we will use the fact that $\sqrt{2}$ is irrational, as you will prove in your homework.

Result 2.1.5 *Let $x, y \in \mathbb{Q}$ with $x < y$. Then there exists irrational z so that $x < z < y$.*

Proof. Wait! This is one of the exercises in this section! ■

Notice that while the Corollary 2.1.2 may feel fairly obvious, one can find some quite natural examples in which it does not hold.

Example 2.1.6 No Archimedes for rational functions. Consider the set of rational functions with rational coefficients. That is, all functions

$$f(x) = \frac{a_n x^n + \cdots + a_0}{b_k x^k + \cdots + b_0}$$

where all the $a_i, b_i \in \mathbb{Q}$ and $b_k \neq 0$. We add, subtract, multiply and divide these functions in the usual way. Notice that this set contains $\mathbb{N}$ and $\mathbb{Q}$.

We can define an order on this set by saying that $f > 0$ when $\frac{a_n}{b_k} > 0$. This

[1] Actually — you need to! It will be on your homework.

[2] you know this already, but we can formally define an "irrational number" to be a real number that is not rational

[3] though really, no one ever wanted to approximate a real number with an irrational number

then naturally extends to $f > g$ when $(f - g) > 0$.

To show that this field is not Archimedean, consider the functions $f(x) = x$ and $g(x) = 1$. Since $f - g = x - 1$ we know that $f > g$. However, for any natural number n we have

$$f - n \cdot g = x - n > 0.$$

and so no matter how large we take n we cannot make $n \cdot g$ larger than f. □

Exercises

1. No such thing as infinitesimals. Use the Archimedean property to show there is no smallest positive real number.

Background: Back when calculus was first being invented (discovered?), one approach was to work with *infinitesimals*, thought of as a kind of "smallest number"[4]. We know better now though, there is no such thing. Instead, we work with limits (as we will see in the proceeding sections/chapters).

2. Trapped between two integers. Given any $x \in \mathbb{R}$, show that there is a *unique* $n \in \mathbb{Z}$ such that $n \leq x < n + 1$.

We would like to emphasize the importance of "uniqueness" here! Make sure you prove that *any* such integer satisfying that property is equal to any other. It is often the case that we want to say something like "let n be *the* integer with $n \leq x < n + 1$", where the "the" there is doing some pretty heavy lifting! Be careful to use the right determiner when writing mathemtaical proofs! There is a big difference between "an integer n" and "the integer n".

3. A useful thing for proving inequalities. Suppose $x, y \in \mathbb{R}$ are such that $x \leq y + \varepsilon$ for every $\varepsilon > 0$. Show that $x \leq y$.

Note: On the other hand, if $x < y + \varepsilon$ for all $\varepsilon > 0$, this does *not* imply that $x < y$, because they may actually be equal.

4. Irrationals are dense too. Show that $\mathbb{R} \setminus \mathbb{Q}$ is also dense inside $\mathbb{R}$.

You may use the fact that $\sqrt{2}$ is not rational, where this number is defined as

$$\sqrt{2} := \sup\{x : x^2 < 2\}.$$

Note that $\sqrt{2}$ is well defined since the above set is non-empty (it contains $0, 1$) and bounded above (say by 2) and so the supremum exists.

Hint. Use the irrationality of $\sqrt{2}$ and the density of the rational numbers in $\mathbb{R}$.

[4]The interested reader should search-engine their way to more information on infinitesimals, surreal numbers, and non-standard analysis. We'll stick to *standard* analysis in this course.

2.2 Nested intervals

2.2.1 The nested interval property

Another useful property of the real numbers is the nested interval property 2.2.2. Roughly speaking, this states that if we take a sequence of intervals, $I_1 \supseteq I_2 \supseteq I_3 \supseteq \ldots$, then (with a few conditions) there is always a unique real number that belongs to *all* the intervals. We will see later that the nested interval property is actually equivalent to the least upper bound property for the real numbers, but we don't need that yet.

Just to make sure we all understand *precisely* what we mean by an interval, let's write down the definition.

Definition 2.2.1 Intervals. Let $a, b \in \mathbb{R}$ with $a < b$. The **closed interval** of all reals between a and b is

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

while the **open interval** is

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$$

Let the **length** of an interval I be $|I| = |b - a|$, so that $(a, b), [a, b]$ both have finite lengths.

We also consider intervals that have infinite length. The intervals

$$(-\infty, b) = \{x \in \mathbb{R} \mid x < b\} \quad \text{and} \quad (a, +\infty) = \{x \in \mathbb{R} \mid x > a\}$$

are open infinite intervals, while

$$(-\infty, b] = \{x \in \mathbb{R} \mid x \leq b\} \quad \text{and} \quad [a, +\infty) = \{x \in \mathbb{R} \mid x \geq a\}$$

are closed infinite intervals.

We also sometimes write $(-\infty, \infty)$, the bi-infinite interval, to represent all real numbers. It is both open and closed — a fact we will return to later in this section. ♢

With that out of the way, consider the following procedure for computing (numerically) the square-root of (say) 17. Since $17 > 1$ we know that $1 < \sqrt{17} < 17$. So

$$\sqrt{17} \in [1, 17] = I_1$$

Now look at the midpoint of this interval I_1. The midpoint of this is $\frac{17+1}{2} = 9$, and $9^2 = 81 > 17$, so we know that

$$\sqrt{17} \in [1, 9] = I_2$$

Repeat — look at the midpoint $\frac{1+9}{2} = 5$, since $5^2 = 25 > 17$, we know that

$$\sqrt{17} \in [1, 5] = I_3$$

Again, midpoint is $\frac{1+5}{2} = 3$, and $3^2 = 9 < 17$, so

$$\sqrt{17} \in [3, 5] = I_4$$

Repeat, midpoint is 4 and $4^2 = 16 < 17$, so

$$\sqrt{17} \in [4, 5] = I_5$$

Once more, midpoint is 4.5 and $4.5^2 = 20.25 > 17$, so

$$\sqrt{17} \in [4, 4.5] = I_6$$

and so on. This is very easy to code and the distance between our upper and lower bounds on $\sqrt{17}$ halves each iteration.

So we start with an interval I_1, from which we obtain a sequence of **nested intervals**, meaning that

$$I_{n+1} \subseteq I_n$$

so that the length of each interval gets shorter and shorter — informally, the lengths of the intervals go to zero. This gives a sequence of improving upper and lower bounds on the quantity $\sqrt{17}$.

Of course, in order for this to work, we need to be sure that

- there actually is something in all of those intervals, and
- it is unique.

The existence of suprema is sufficient to give us this result, and so validate the algorithm above. Namely if we can keep improving our upper and lower bounds, then they will converge to a unique (correct) real number. Let us formalise this.

Theorem 2.2.2 Nested interval property. *Let $\{I_n \subseteq \mathbb{R} \mid n \in \mathbb{N}\}$ be a set of closed non-empty intervals of the real line so that*

- *the intervals are nested, $I_{n+1} \subseteq I_n$, and*
- *the lengths of the intervals become arbitrarily short, that is for any $\varepsilon > 0$ there exists N so that for any $k > N$, the length $|I_k| = |b_k - a_k| < \varepsilon$.*

then the set $\bigcap_n I_n$ consists of exactly one real number.

Proof. Let a_n, b_n be the lower and upper terminals of the interval I_n. First notice that since the intervals are nested we have $a_k \leq b_\ell$ for all k, ℓ .

Now consider the set $A = \{a_n \mid n \in \mathbb{N}\} \subset \mathbb{R}$. It is bounded above by b_1 and so has a supremum $\alpha = \sup A$. Hence we have $\alpha \geq a_n$ for all n.

Then notice that since α is the supremum, we know that for any $x < \alpha$ there is $a_k \in A$ so that $x < a_k$. This tells us that $\alpha \leq b_n$ for all $n \in \mathbb{N}$.

Hence $a_n \leq \alpha \leq b_n$ for all n and so $\alpha \in \bigcap_n I_n$.

Now say there were a distinct real number $\beta \in \cap_n I_n$ with $\alpha < \beta$ (of course β cannot be smaller than α since it is the supremum of the a_n). Then since $\alpha, \beta \in I_n$

for any n, we must have that $b_n - a_n > \beta - \alpha$ for any n. However, this contradicts the assumption that for any $\varepsilon > 0$, there exists k so that $|I_k| = |b_k - a_k| < \varepsilon$.

Hence α is the unique real in $\bigcap_n I_n$. ■

Notice that it is not sufficient just that the intervals get shorter. Consider

$$I_n = \left[-\frac{1}{n}, 1 + \frac{1}{n}\right]$$

this forms a sequence of nested intervals, but (informally) the lengths of these intervals go to 1, not 0.

Further, it is vitally important that the intervals are closed. The same idea definitely does not work with open intervals, as illustrated by Exercise 2.2.3.1.

Thus the Theorem 1.5.9 implies the nested interval property. Interestingly, the converse is also true! That is to say, if it is known that an ordered field satisfies the nested interval property, then it also satisfies the least upper bound property! We won't prove that here, but will save it for later this chapter 2.4 (we will have by then an additional property that is also equivalent.)

2.2.2 Open and closed

In the section above we told you that certain intervals were **open** and others were **closed**, without really telling you what it means for a set of real numbers to be open or closed. We will do this in more detail in Chapter 5 but we'll do a little bit more now. It isn't a bad place to practice some nested quantifiers.

Definition 2.2.3 Open and closed sets of reals. Let $A, B \subseteq \mathbb{R}$. We say that A is **open** when for any $x \in A$ there exists $r > 0$ so that the open interval $(x - r, x + r)$ is contained in A. We say that the set B is **closed** when its complement $\mathbb{R} - B$ is open. ◊

Notice that there are sets that are neither open nor closed. For example $A = (0, 1]$ — you should prove this! See Exercise 2.2.3.2.

For work that is coming soon, it is helpful for us to have an idea of what it means for a set of reals to be **bounded**. Indeed, we'll see later in this chapter 2.5 that closed bounded sets are particularly "nice" and have good properties.

Definition 2.2.4 Bounded. A subset $A \subseteq \mathbb{R}$ is **bounded** when there exists $a, b \in \mathbb{R}$ so that $A \subseteq [a, b]$. ◊

See Exercise 2.2.3.3 for an equivalent definition of bounededness.

The authors apologize to the readers about defining what a bounded function means here when we really won't need to use this definition for several more sections. But alas, this is just the most convenient place for it.

Definition 2.2.5 Bounded Function. Given any set A (not necessarily a subset of $\mathbb{R}$), we say that a function $f : A \to \mathbb{R}$ is **bounded** if its image

$$\{f(x) \mid x \in A\}$$

is a bounded subset of $\mathbb{R}$. ◊

Openness and closeness interact nicely with unions and intersections (and certain functions). For example, you will show in Exercise 2.2.3.4, any intersection of closed sets is closed, and any union of open sets is open. Unfortunately, infinite unions of closed sets can be non-closed and infinite intersections of open sets can be non-open (see Exercise 2.2.3.5). Finite unions and intersections behave as expected, see Exercise 2.2.3.6.

We will continue to explore these phenomena as we continue to build up our toolboox (for example, we will revisit open/closed sets in the context of a metric spaces 5).

2.2.3 Exercises

1. **Nested open intervals don't work.** Show that the nested interval property can fail if the intervals are not closed by considering the intervals $I_n = (0, 1/n)$. You should verify that the intervals are indeed nested with widths going to zero, but that their intersection is problematic.

   Hint. The Archimedean property might help you prove your answer (not just argue it).

2. **Neither open nor closed.** Prove that the set $A = (0, 1]$ is not open and prove that is not closed.

3. **Alternative definition of bounded.** Show that the set $A \subseteq \mathbb{R}$ is bounded if and only if there exists $M > 0$ so that $|x| \leq M$ for every $x \in A$.

   Remark: Feel free to use this alternate defninition of bounded in your work (and tests!) if you prefer.

4. **Open, closed, union, intersection.**

   Hint. The intersection of closed sets can be written as the complement of a union of open sets.

5. **Infinity does not play nicely.** Show that

   (a) $\bigcup_{n\in\mathbb{N}} [1/n, 1 - 1/n] = (0, 1)$, and

   (b) $\bigcap_{n\in\mathbb{N}} \left(-\frac{1}{n}, 1 + \frac{1}{n}\right) = [0, 1]$

6. **Finite unions and intersections are nice.** Show that the finite intersection of open sets is open. By taking complements, show that the finite union of closed sets is closed.

   Hint. A finite set of positive reals has a minimum.

7. **Open is open and closed is closed.** Let $a, b \in \mathbb{R}$ with $a < b$. Prove that (a, b) is open and $[a, b]$ is closed$\{a\}$

8. Both open and closed. Let $A \subseteq \mathbb{R}$. If A is both open and closed then $A = \emptyset$ or $A = \mathbb{R}$

Hint. Use a contradiction so that you can find $x \in A$ and $y \in \mathbb{R} - A$, so that (without loss of generality) $x < y$. Then use the supremum to find $x \leq z \leq y$ so that $z \notin A$ and $z \notin (\mathbb{R} - A)$ due to needing both these sets to be open.

2.3 Cauchy and Completeness

The existence of suprema also implies the convergence of a certain family of limits. First, let us talk about convergence of sequences (which you should have seen before, though likely not in the level of rigor that we will use).

2.3.1 Convergence, Bolzano and Weierstrass

Definition 2.3.1 Convergence of a sequence. Let $(a_1, a_2, \dots) = (a_n)_n$ be a sequence of real numbers. We say that the sequence **converges** to the **limit** $L \in \mathbb{R}$ when for any $\varepsilon > 0$ there exists an integer N so that if $n > N$ then $|a_n - L| < \varepsilon$.

In this case we write $a_n \to L$. ◊

This rigorous definition of convergence is due to Bolzano. He laid the foundations for the rigorous treatment of Calculus by introducing the $\varepsilon - \delta$ definition of limits of functions (similar to this $\varepsilon - N$ definition for sequences). He was also the first to prove the lower bound property of the reals — the interested reader can find a careful exploration of Bolzano's original proof here[1]. Indeed, we recommend the TRIUMPHS[2] project to the interested reader. It contains many interesting articles that teach mathematics by exploring the original sources.

This definition lets us prove that a sequence converges to a given number L. We will need more machinery to prove convergence without knowing the limit.

Example 2.3.2 Convergence of $a_n = \frac{3n}{n+1}$. The sequence $a_n = \frac{3n}{n+1}$ converges to 3; we know this from all our calculusing.

We need to show that no matter which ε we pick we can always make n large enough so that $|a_n - 3| < \varepsilon$. As is pretty normal for inequality proofs, we start from the inequality we want to prove and then work backwards. So

$$\begin{aligned} \varepsilon > |a_n - 3| &= \left| \frac{3n}{n+1} - 3 \right| \\ &= \left| \frac{3n - 3n - 3}{n+1} \right| = \frac{3}{n+1} \end{aligned}$$

So we need $\frac{3}{n+1} < \varepsilon$, or $n + 1 > \frac{3}{\varepsilon}$. So if we set $N = \lceil \frac{3}{\varepsilon} \rceil$ then we know that when

[1] digitalcommons.ursinus.edu/triumphs_analysis/14/

[2] blogs.ursinus.edu/triumphs

$n > N$, we get $\frac{3}{n+1} < \varepsilon$. Let's write this up nicely.

Let $\varepsilon > 0$ and then set $N = \lceil 3/\varepsilon \rceil$. Now for any $n > N$ we have that $\frac{3}{n+1} < \varepsilon$. Then

$$|a_n - 3| = \frac{3}{n+1} < \varepsilon$$

as required. □

Convergent sequences play nicely with one another[3] in the sense that if you know that two sequences are convergent and they stay close together, then they actually approach the same value. *If you and a friend stick together, and one of you is going to Cleveland, then you're both going to Cleveland*

Lemma 2.3.3 *Let a_n and b_n be convergent sequences so that $(a_n - b_n) \to 0$. Then a_n and b_n converge to the same limit.*

Proof. A good homework problem! ■

Another nice property of convergent sequences is that they are *always* bounded. Roughly speaking, for large n, the sequence must be close to its limit, and there are only a finite number of early terms. Essentially, you can find an upper bound by taking a number a little larger than the limit, and then checking the size of the early terms. *If you never leave your city after high school, then you can't have travelled everywhere.*

Lemma 2.3.4 A convergent sequence is bounded. *Let (a_n) be a convergent sequence. Then $\{a_n \mid n \in \mathbb{N}\}$ is bounded.*

Proof. Let L be the limit of the sequence and let $\varepsilon = 1$. Then there is some N so that for all $n > N$ we know that

$$|a_n - L| < 1$$

and thus $|a_n| < |L| + 1$. Then the set

$$\{|a_1|, |a_2|, \ldots, |a_N|, |L| + 1\}$$

is finite and so has a maximum M. The terms of the sequence are bounded by M. ■

The converse of this is false. You can take a sequence like $a_n = (-1)^n$; it is bounded but does not converge. This is, roughly speaking, because it bounces up and down forever. However, if we concentrate on **monotonic** sequences then we get a nice result.

Lemma 2.3.5 Bounded monotone sequence converges. *Let (a_n) be a **monotonically** increasing sequence of reals; that is $a_n \leq a_{n+1}$ for all n. If (a_n) is bounded above then it converges to the supremum of its terms.*

A similar result holds for monotonically decreasing sequences.

[3] unlike my kids

Proof. Let a_n be as stated and let $A = \{a_1, a_2, \ldots\}$. The set A is bounded above (by assumption) and so let $L = \sup A$.

Now let $\varepsilon > 0$, then we know that $L - \varepsilon$ is not an upper bound for the set and so there is some $a_k \in A$ so that

$$L - \varepsilon < a_k < L$$

Since the terms of the sequence are increasing, we know that $a_n \geq a_k$ and so

$$L - \varepsilon < a_k \leq a_n < L$$

Now set $N = k - 1$, then for any $n > N$ we have

$$|a_n - L| < \varepsilon$$

and so $a_n \to L$ as required. ■

Example 2.3.6 Showing convergence without knowing the limit. Consider the sequence $a_n = \frac{3n-2}{n+7}$. Notice that

$$a_{n+1} - a_n = \frac{3n+1}{n+8} - \frac{3n-2}{n+7} = \frac{23}{(n+7)(n+8)} > 0$$

and so is monotonically increasing. Also notice that

$$\frac{3n-2}{n+7} < \frac{3n-2}{n} < \frac{3n}{n} = 3$$

and so is bounded above. Hence the sequence converges.

By the way, this is a common trick when working with sequences. First show that the limit converges to *something*, then you can use some other techniques which may be predicated on knowing that the limit exists! to decude the actual limiting value. See, for example, Exercise 2.3.3.10. □

Consider another related non-convergent sequence $b_n = (-1)^n \cdot \frac{3n-1}{n}$; it does not converge, it is comprised of two **subsequences** that do converge. All the even terms give $3 - 1/n$ and all the odd terms give $1/n - 3$; those subsequences *are* monotonic. Now this is a pretty simple example, but it demonstrates a more widespread phenomenon. First, let's sure up the definition of subsequence.

Definition 2.3.7 Subsequences. Let $(a_n)_n$ be a sequence of reals. A **subsequence** of (a_n) is a new sequence $(a_{n_k}) = a_{n_1}, a_{n_2}, a_{n_3}, \ldots$ with $n_1 < n_2 < n_3 < \ldots$ ◊

We can show that every sequence has some monotone subsequence by studying the **peaks** of the sequence. That is, start scanning the sequence and record when you find a term which is larger than all the terms that come after it, i.e., find n so that $a_k < a_n$ for all $k > n$. Notice that each peak must be larger than all future peaks; so if there is an infinite number of peaks, they form a decreasing subsequence. If there is a finite number of peaks we have a little more work to

do.

Lemma 2.3.8 Always monotone subsequences. *Every sequence (a_n) has a monotonic subsequence.*

Proof. Call the location of the first peak n_1, the second n_2, and so on. Notice that we must have

$$a_{n_1} > a_{n_2} > a_{n_3} > \cdots$$

- If there are an infinite number of peaks then we have found an monotonically decreasing subsequence.
- If there are no peaks at all, then let $k_1 = 1$. While if there are finitely many (but not zero) peaks, let a_N be the last; set $k_1 = N + 1$. Since a_{k_1} is not a peak, there must be some later $a_\ell > a_{k_1}$; set $k_2 = \ell$. Since a_{k_2} is not a peak, there is some later $a_{k_3} > a_{k_2}$. Iterate this process to get a monotonically increasing subsequence

  $$a_{k_1} < a_{k_2} < a_{k_3} < \ldots$$

  This must be infinite, otherwise there would be a peak after a_N.

In either case we have found a monotonic subsequence. ■

Example 2.3.9 Monotone subsequences of $\sin(n)$. Consider the sequence $a_n = \sin n$ — note the argument is not an integer multiple of π, just the integer.

By Lemma 2.3.8 it has a monotone subsequence. Indeed, with a little bit of computer code you can show that if we let (b_n) be the sequence

$$1, 45, 89, 133, 177, 221, 265, 309, 353, 397, 441, 485, \ldots$$

then a_{b_n} forms the start of an increasing subsequence. While if we set (c_n) to be the sequence

$$1, 21, 65, 442, 819, 1196, 1573, 1950, 2288, 2327, \ldots$$

then a_{c_n} forms the start of a decreasing subsequence.

Of course, *proving* that one can continue these subsequences to infinity is much more delicate[4]. But we *know* that (a_n) contains at least one monotone subsequence, even if we cannot nail it down explicitly. □

Putting these together gives us a very famous result. It was first proved by Bolzano in 1817 (on his way to proving the Intermediate Value Theorem); however Bolzano and his work was relatively isolated at the time. It was later reproved by Weierstrass around 50 years later and was actually known as Weierstrass' theorem until Bolzano's work was more widely known.

[4]There are no simple answers here. Though the sine function is periodic with period 2π the sequence is not periodic. This is because π is irrational — proved first by Lambert in 1760.

Theorem 2.3.10 Bolzano-Weierstrass. *Every bounded sequence has a convergent subsequence.*

More precisely, if (x_n) is a sequence contained in $[a, b]$ then there exists $c \in [a, b]$ and a subsequence (x_{n_k}) so that $\lim_{k\to\infty} x_{n_k} = c$.

Using least upper bound property. Let (x_n) be a bounded sequence. By Lemma 2.3.8 we know that there is a monotone increasing or monotone decreasing subsequence of (x_n). Since the original sequence is bounded, the subsequence is also bounded. Then by Lemma 2.3.5 we know that the subsequence converges to some real number x. The fact that $x \in [a, b]$ is proved in Lemma 2.3.13 below. This completes the proof. ■

Alternative proof using nested intervals. A good homework problem! ■

So for any bounded sequence $(x_n)_n$, Theorem 2.3.10 guarantees the existence of *some* subsequence of it which converges. But the question is: "How do you know *which* subsequence converges?" Indeed, this phenomenon is very common in mathematics; we may know that something exists but it is a totally different thing to actually write down what it is. The following example illustrates this.

Example 2.3.11 It might be (really) hard to explicitly write down a converging subsequence. Consider the sequences

$$a_n = \sin(n) \qquad b_n = \operatorname{sgn}(\sin(n))$$

where $\operatorname{sgn}(x)$ is the sign-function

$$\operatorname{sgn}(x) = \begin{cases} +1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

In both cases the sequences are bounded $-1 \leq a_n, b_n \leq +1$, so by Theorem 2.3.10 we know that both have convergent subsequences. In one case, it is very easy to write down one such subsequence. In the other case, however, it is *much* more difficult (impossible?). □

Remark 2.3.12 Be careful with limits and open intervals. Consider the sequence $(a_n) = (1/n)$ that lies in the interval $(0, 1)$. Every term of the sequence lies in $(0, 1)$ however the limit of the sequence is $0 \notin (0, 1)$. Indeed, we must be careful with open intervals, we can (as we have shown above) define sequences whose limits lie just outside the set.

To complete the proof above we need one more little technical result. Namely, if a sequence in a closed bounded interval $[a, b]$ converges, then its limit is in the same interval. You will prove this more generally in Exercise 2.5.3.

Lemma 2.3.13 Closed bounded intervals contain their limits. *Let $(x_n)_n$ be a sequence in the closed interval $[a, b]$. That is, $a \leq x_n \leq b$ for all $n \in \mathbb{N}$. If the sequence converges to a limit $L \in \mathbb{R}$, then $L \in [a, b]$.*

Proof. Let $x_n \to L$, and assume, to the contrary that $L \notin [a,b]$. Then either $L < a$ or $L > b$. WLOG, assume that $L < a$. Then set $\varepsilon = (a-L)/2$. Since $x_n \to L$, we know that there is N so that for any $n > N$ we have $|x_n - L| < \varepsilon$. However, this means that

$$\begin{aligned} x_n - L < \varepsilon &= \frac{a}{2} - \frac{L}{2} \\ x_n < \frac{a}{2} + \frac{L}{2} &< \frac{a}{2} + \frac{a}{2} \\ &= a \end{aligned}$$

However this contradicts our assumption that the sequence is contained in $[a,b]$.

The argument when $L > b$ is similar. Thus we must have the limit $L \in [a,b]$. ■

Remark 2.3.14 Extend B-W to closed and bounded sets. In Section 2.5 we'll show how Theorem 2.3.10 can be extended to any closed and bounded set of reals. We'll also touch on how useful closed and bounded sets of reals are.

2.3.2 Convergence, Cauchy and Complete

The usual definition of convergence requires us to know the limit L. But what if we don't know the number but still want to show that a sequence converges. Well, it is not too hard to show that if it converges to L then the terms of the sequence must get arbitrarily close together. In fact, this notion of "*the terms get arbitrarily close together*", turns out to be a mathemtically significant one. So we'll just do the thing that mathemticians do and make an a priori unmotivated definition so that we have a name for this.

Definition 2.3.15 Cauchy sequences. A sequence $(a_n)_n$ is called **Cauchy** when for any positive ε, there is some integer N so that for all $k, \ell > N$ we have

$$|a_\ell - a_k| < \varepsilon.$$

♢

Let's see an example before proving anything.

Example 2.3.16 Example of a Cauchy sequence. Consider the sequence $a_n = \frac{3n}{n+1}$. We know (from all our calculusing) that this converges to three. But let us prove that it is also Cauchy.

First of all, consider the difference of terms

$$\begin{aligned} |a_k - a_\ell| &= \left| \frac{3k}{k+1} - \frac{3\ell}{\ell+1} \right| \\ &= \left| \frac{3k(\ell+1) - 3\ell(k+1)}{(k+1)(\ell+1)} \right| \\ &= \left| 3\frac{k-\ell}{(k+1)(\ell+1)} \right| \end{aligned}$$

$$\begin{aligned} &< 3\left|\frac{k-\ell}{k\ell}\right| = 3\left|\frac{1}{\ell}-\frac{1}{k}\right| \\ &\leq \left|\frac{3}{k}\right| + \left|\frac{3}{\ell}\right| && \text{triangle inequality} \\ &\leq \frac{6}{\min\{k,\ell\}} \end{aligned}$$

So if we wish to make the difference between a_k, a_ℓ small, then it suffices to make k and ℓ both large. In particular, if we want

$$|a_k - a_\ell| < \varepsilon$$

then we can require that

$$\frac{6}{\min\{k,\ell\}} < \varepsilon$$

or equivalently

$$\min\{k,\ell\} > \frac{6}{\varepsilon}$$

So let $\varepsilon > 0$, then let $N = \lceil \frac{6}{\varepsilon} \rceil$. Now pick $k, \ell > N \geq \frac{6}{\varepsilon}$ then

$$\begin{aligned} |a_k - a_\ell| &= 3\left|\frac{k-\ell}{(k+1)(\ell+1)}\right| \\ &\leq 3\left|\frac{\ell-k}{k\ell}\right| = 3\left|\frac{1}{\ell}-\frac{1}{k}\right| \\ &\leq \frac{3}{k}+\frac{3}{\ell} \leq \frac{6}{\min\{k,\ell\}} \\ &\leq \frac{6}{6/\varepsilon} = \varepsilon \end{aligned}$$

as required. □

Before we prove the general statement that a convergent sequence is Cauchy, we (the authors) urge you to compare the methods used in the previous example with Example 2.3.2 (where we proved that same sequence converges). Notice the similarities, in particular with the way that N is chosen in regards to ε. Notice that if you replaced the ε in the convergent proof with $\varepsilon/2$, then essentially the N is the same. That is basically all we need to do to prove the following theorem.

Lemma 2.3.17 Convergent Implies Cauchy. *A convergent sequence is Cauchy.*

Proof. Let $(x_n)_n$ be a convergent sequence, with limit $x \in \mathbb{R}$ and let $\varepsilon > 0$. Since we know that $x_n \to x$, we can find some $N \in \mathbb{N}$ such that whenever $n \geq N$, we have $|x_n - x| < \varepsilon/2$. In particular, whenever $n, m \geq N$, we have

$$|x_n - x_m| \leq |x_n - x| + |x_m - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This shows that $(x_n)_n$ is Cauchy. ■

So *if* a sequence converges *then* its coefficients become arbitrarily close to each-other. It is *not* obvious that all Cauchy sequences converge — that *would* be the converse (if it were true) of Lemma 2.3.17.

In other words, a convergent sequence is "nicer"[5] than a Cauchy sequence (always). One "nice" property of convergent sequences is that they are necessarily bounded, as we saw in Lemma 2.3.4. It turns out that Cauchy sequences are still "nice" enough to retain this property.

Lemma 2.3.18 *Let (a_n) be a Cauchy sequence. Then $\{a_n \mid n \in \mathbb{N}\}$ is bounded.*

Proof. Let $\varepsilon > 1$ then we know that there is some N so that for any $n, k > N$ we have $|a_n - a_k| < 1$. So set $\ell = N + 1$, then we know that

$$|a_\ell - a_k| \leq 1$$

and so $|a_k| < |a_\ell| + 1$. Form the finite set

$$\{|a_1|, |a_2|, \ldots, |a_\ell|, |a_\ell| + 1\}.$$

This set has a maximum M and the sequence is bounded by M. ■

Another way in which Cauchy sequences are *almost* as nice as convergent ones is that if *some* subsequence of a Cauchy sequence converges, then the *entire* sequence must as well. *If a group of you stick together, and some of you go to Cleveland, then you all go to Cleveland.*

Lemma 2.3.19 Convergent subsequence of Cauchy sequence. *Let (a_n) be a Cauchy sequence and let (a_{n_k}) be a subsequence. If (a_{n_k}) converges to α then (a_n) converges to α.*

Proof. Another good homework problem. ■

The question still remains as to how "nice" Cauchy sequences are in general. In hindsight, the reader is probably aware that "convergent" and "Cauchy" probably do not always mean the same thing (otherwise, why would we need a different name for them). But still, it would be nice to be able to point to a concrete sense in which "Cauchy" is **not** as nice as "convergent". The following example does just that.

Example 2.3.20 Example of non-convergence of Cauchy sequence in $\mathbb{Q}$. Let a_n be the sequence of rationals given by

$$a_n = 10^{-n} \lfloor 10^n \sqrt{2} \rfloor$$

that is, the first n digits of the decimal expansion of $\sqrt{2}$.

$$1, 1.4, 1.41, 1.4142, 1.41421, 1.414213, \ldots$$

[5] what a mathematician thinks is "nice" usually means that it satisfies many properties that are useful

We need to show that this is Cauchy.

So let $k < \ell$ and consider $a_\ell - a_k$:

$$\begin{aligned} &1.41421\ldots b_k \ldots b_\ell - 1.41421\ldots b_k \\ &= 0.00\ldots b_{k+1}\ldots b_\ell \\ &< 0.00\ldots 9\ldots 9 < 10^{-k} \end{aligned}$$

So if we want $|a_\ell - a_k| < \varepsilon$ then we need to choose $N = \min\{k, \ell\}$ so that $10^{-N} < \varepsilon$. Hence the sequence is Cauchy.

However, since this sequence converges to the *irrational* number $\sqrt{2}$, it does not converge in the set of *rational* numbers. □

The previous example is a critical one. It shows that one can have a sequence a_n in a set A that is Cauchy but does not converge to any $\alpha \in A$. This does happen in the rationals, but does not happen in the reals. We say that the reals are **complete**.

Definition 2.3.21 Let $A \subseteq \mathbb{R}$. We say that A is **complete** when every Cauchy sequence $(a_n) \in A$ converges to some $\alpha \in A$. ◊

You're already (intuitively) familiar with some sets that are complete and some are not. Let's make that more concrete with an example.

Remark 2.3.22 Complete, open, and closed. As noted above, any open interval (a, b) is not complete. Take, for example, the sequence

$$x_n = a + \frac{b-a}{2n}$$

We know that $a < x_n < b$ but the limit

$$\lim_{n\to\infty} x_n = a \notin (a, b).$$

On the other hand, if we take any sequence $y_n \in [a, b]$ it cannot converge to point outside $[a, b]$. Say, to the contrary that it did converge to (without loss of generality) $c < a$. Then let $\varepsilon = \frac{a-c}{2}$. Since y_n converges to c we know that there is N so that for any $n > N$ we have

$$|y_n - c| < \varepsilon = \frac{a-c}{2}$$

But this means that

$$y_n - c < \frac{a-c}{2} = \frac{a}{2} - \frac{c}{2}$$

and so

$$y_n < \frac{a}{2} + \frac{c}{2} < a$$

But this contradicts our assumption that $y_n \in [a, b]$.

You will explore these ideas more in Exercise 2.5.3, Exercise 2.5.4, and Exercise 2.5.5.

Also, as we've seen in Example 2.3.20, the rational numbers are not complete (that is, "incomplete"), but as it turns out, the rigid structure of the real numbers (in particular, the existence of a supremum) is enough to show that the reals are, in fact, complete. In other words, in the real numbers, Cauchy sequences are **just as nice** as convergent ones. That is one of the miracles of the real[6] numbers.

The proof is actually very straight-forward with all the work that we have done to get here.

Theorem 2.3.23 The reals are complete. *The set of real numbers is complete. That is, any Cauchy sequence in $\mathbb{R}$ converges to some limit in $\mathbb{R}$.*

Proof. So finally we can stitch together the proof. Let (a_n) be a Cauchy sequence in the reals. Then we know by Lemma 2.3.18 that it must be bounded. By Bolzano-Weierstrass 2.3.10 it has a convergent subsequence whose limit is a real number. And finally by Lemma 2.3.19 the full sequence must converge. ■

But one can also prove the least upper bound property from the completeness of the reals — see Section 2.4. Indeed, many texts take this approach. They essentially start with the **field** properties of the rational numbers (see Theorem 1.4.10) and then take the real numbers to be the **completion**[7] (we will visit this topic in Chapter 5), from which Theorem 2.3.23 follows by definition. From that they deduce the existence of the supremum A.7.3, and, if desired, the nested interval property 2.2.2.

For the sake of reference later, we can combine the results in this section to state the *Cauchy criterion for sequences*:

Corollary 2.3.24 Cauchy criterion for sequences. *A sequences of real numbers converges if and only if it is Cauchy.*

2.3.3 Exercises

1. **Seems obvious, but you should check.** Show that a constant sequence converges.
2. **Is it complete?** For each of the following sets, decide whether it is a complete subset of $\mathbb{R}$. Justify your answers.
   - $(0, 1) \cup [2, 4]$.
   - A finite set.
   - $\{1/n : n \in \mathbb{N}\} \cup \{0\}$.
3. **Non-convergence.** Write down (in logical quantifiers) what it means for a sequence $(x_n)_n$ to *not* converge.

[6] or the complex!

[7] One way to think of the completion of a set is to imagine that you are can just "fill in" all the holes in the set

4. **A simple construction.** Construct a sequence $(x_n)_n$ which does *not* converge, but has at least 2 converging subsequences.

5. **2 converging subsequences.** Show that if a sequence $(x_n)_n$ is bounded and does not converge, then there exists two subsequences of $(x_n)_n$ that converge to *different* values.

   Hint. Consider breaking up a large interval (containing the whole sequence) into an equal number of "bins" of the same size, and take the size of the bins to zero. Argue that, eventually, you must have at least two bins that have infinitely many values.

6. **Convergence and subsequences.** Let (x_n) be a sequence of reals. Prove that (x_n) converges to x if and only if every subsequence of it converges to x.

   Hint. The "only if" direction should be straight-forward. For the "if" direction, use a proof by contradiction.

7. **Unbounded sequences don't converge**[8]**.** Let (x_n) be a sequence an **unbounded** sequence, i.e., it is not a bounded sequence. Show that it does not converge.

   Hint. Recall the definition that the sequence is bounded 2.2.4.

8. **Convergence and Subsequences 2.** Prove that a sequence $(x_n)_n$ converges if and only if every subsequence of it converges.

   Note: This is different than Exercise 2.3.3.6 because here we are not assuming that every subsequence converges to the same value x.

   Hint. For the "if" direction, consider breaking into cases of whether or not the sequence is bounded (you may want to use Exercise 2.3.3.7)

9. **Arithmetic of limits.** Suppose that (x_n) and $\{y_n\}$ are converging sequences with $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$. Prove, directly from the definition of converging sequences, the following:

   - (Sum Rule) $\lim_{n\to\infty}(x_n + y_n) = x + y$.
   - (Product Rule) $\lim_{n\to\infty}(x_n y_n) = xy$.
   - (Quotient Rule) If $y \neq 0$, then $\lim_{n\to\infty}\left(\frac{x_n}{y_n}\right) = \frac{x}{y}$.

   Takeaway: Limits behave the way we expect them to with respect to sums, products, and quotients.

   Hint. For the product rule, use the fact that a converging sequence is bounded. While for the quotient rule, notice that when n is large enough, you know that y_n is bounded away from 0.

10. **Recursive sequence.** Define a sequence of real numbers $(x_n)_n$ recursively by $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{x_n + 2}$ for $n \geq 1$.

[8] to a real number

- Prove that $x_n < 2$ for every $n \in \mathbb{N}$
- Prove that $(x_n)_n$ is an increasing sequence
- Conclude that the sequence converges and that $\lim_{n\to\infty} x_n = 2$.

Hint. You can assume that the square-root function is increasing (but you should be able to prove this if you really have to). For the last task you may use the fact that the square-root function is *continuous* and thus respects limits (a fact we will prove as Corollary 3.2.10 in Chapter 4).

11. Now you do it. Define a sequence of real numbers (x_n) recursively by $x_1 = 2$ and $x_{n+1} = \frac{x_n+5}{3}$ for $n \geq 1$. Prove that $(x_n)_n$ converges, and find the limit L.

Hint. The process is similar to the previous exercise.

12. Know the Cauchy condition precisely. Let $x_n = \sqrt{n}$. Prove that

$$\lim_{n\to\infty} (x_{n+1} - x_n) = 0$$

but the sequence (x_n) is not Cauchy.

Takeaway: The Cauchy condition requires that $|x_n - x_m|$ is small for *any* choices of n, m sufficiently large. It does *not* require that n and m are themselves close.

13. Infinite (improper) limits. So far, when we have written $\lim_{n\to\infty} x_n = a$, it is assumed that a is a real number. Often, however, mathematicians will use the notation

$$\lim_{n\to\infty} x_n = +\infty \qquad (\text{resp. } = -\infty)$$

to mean that the terms of the sequence become arbitrarily large and positive (resp. large and negative). This notion is made more precise by stating that for all $M > 0$, there exists $N \in \mathbb{N}$ such that whenever $n \geq N$, there holds $x_n \geq M$ (resp. $x_n \leq -M$). We will sometimes use the words **diverges to infinity** to mean that the limit is either $\pm\infty$ according to the sense above.

Using this definition, prove the following.

- If $x_n \to \infty$, then the limit $\lim_{n\to\infty} x_n$ does not exist in the usual (finite) sense.
- The sequence $x_n = \sqrt{n}$ diverges to infinity.
- If $x_n \to \infty$ and y_n is a sequence with $y_n \geq x_n$ for all sufficiently large n, then $y_n \to \infty$.

2.4 Three equivalent properties

In the previous two sections, we have used the Least Upper Bound Property 1.5.9 to prove the the nested interval property 2.2.2 and also the completeness of hte real numbers 2.3.23. But as it turns out, all three of these properties are logically equivalent, in the sense that any one of them implies the other two.

We developed our understanding of the real numbers by starting with digit-strings and then working to the least-upper bound property and from that we proved these other two properties. Many other texts instead start from either the nested-interval property or completeness and infer these other properties from there. The interested reader should also search-engine their way to Dedekind cuts[1] which give another way of constructing the real numbers[2].

We will prove the equivalence by showing that each of the the nested interval property 2.2.2 and the completeness of hte real numbers 2.3.23 implies, in turn, the existence of the supremum.

Theorem 2.4.1 Suprema, nested intervals, and completeness. *The following three properties of the reals are logically equivalent*

- *the least upper bound property 1.5.9*
- *the nested interval property 2.2.2*
- *completeness of the real numbers 2.3.23*

Nested interval property implies least upper bound property. Assume that the nested interval property holds, and let $A \subseteq \mathbb{R}$ be non-empty and bounded above by b_1. Let c be any element of A (we know it exists because the set is non-empty). Then set $a_1 = c - 1$.

Now define $I_1 = [a_1, b_1]$. We know that a_1 is not an upper bound because of c, while b_1 is an upper bound (by assumption). Now let

$$m_1 = \frac{a_1 + b_1}{2}$$

be the midpoint of the interval.

- If m_1 is an upper bound, set $I_2 = [a_1, m_1]$ or
- If m_1 is not an upper bound set $I_2 = [m_1, b_1]$

Let a_2, b_2 be the ends of I_2. Again, notice that a_2 is not an upper bound, but b_2 is an upper bound. Also notice that $|I_2| = |I_1|/2$ and $I_2 \subset I_1$.

Keep repeating this to obtain a sequence of nested intervals $I_1 \supset I_2 \supset I_3 \supset \ldots$ whose lengths halve each time. This satisfies the nested interval property and so there is a unique real number $q \in \cap I_n$.

[1] wikipedia.org/wiki/Dedekind_cut

[2] wikipedia.org/wiki/Construction_of_the_real_numbers#Construction_by_Dedekind_cuts

Notice that since the widths of the intervals become arbitrarily small, for any $\varepsilon > 0$, there is always an interval I_k so that

$$a_k \leq q \leq b_k \qquad \text{and } b_k - a_k < \varepsilon.$$

Notice that we are using the Archimedean property here to tell us that we can make k large enough so that

$$|I_k| = 2^{1-k}|I_1| < \varepsilon.$$

Thus for any ε, you can find an upper bound b_ℓ so that $b_\ell < q + \varepsilon$. And similarly, you can find a_ℓ, which is not an upper bound by construction, so that $a_\ell > q - \varepsilon$.

We now prove that q is the supremum of A.

- Assume, to the contrary, that q is not an upper bound. Then this means that there is some $x \in A$ so that $x > q$. But, by the reasoning above, there is some b_ℓ with $q \leq b_\ell < x$ and b_ℓ is an upper bound for A. Hence no such x exists.

- Now let $z < q$. Then by the reasoning above, there is some a_ℓ so that $z < a_\ell \leq q$. By construction we know that a_ℓ is not an upper bound and so we know that z is not an upper bound.

Thus $q = \sup A$ as required. ■

Completeness implies least upper bound property. The argument required here is very similar to that used to prove that the nested-interval property implies the supremum.

Start by assuming that $\mathbb{R}$ is complete; all Cauchy sequences converge in $\mathbb{R}$. Then let A be a non-empty subset of $\mathbb{R}$ with an upper bound u_1. Since A is non-empty, there exists a real number ℓ_1 that is *not* an upper bound of A (one can take $\ell_1 = x - 1$ for any $x \in A$).

At this point we have u_1 which is an upper bound, and ℓ_1 which is not. From this consider the midpoint $m_1 = \frac{u_1+\ell_1}{2}$.

- if m_1 is an upper bound, then set $\ell_2 = \ell_1, u_2 = m_1$, otherwise

- if m_1 is not an upper bound, then set $\ell_2 = m_1, u_2 = u_1$.

This gives u_2 which is an upper bound and ℓ_2 which is not, so that $|u_2 - \ell_2| = \frac{1}{2}|u_1 - \ell_1|$.

Iterating this gives sequences $\ell_1 \leq \ell_2 \leq \ell_3 \leq \cdots \leq u_3 \leq u_2 \leq u_1$, so that $u_n - \ell_n \to 0$. We next show that each of these sequences are Cauchy.

So, for any $\varepsilon > 0$ there exists N so that for any $n > N$ we have $|u_n - \ell_n| < \varepsilon/2$. Then for any $j, k > n$ we also have that

$$|u_j - u_k| = |(u_j - \ell_n) + (\ell_n - u_k)| \leq |u_j - \ell_n| + |\ell_n - u_k|$$

Now since $\ell_n \leq u_j, u_k \leq u_n$, we know that $|u_j - \ell_n|, |u_k - \ell_n| \leq |u_n - \ell_n| \leq \varepsilon/2$, and thus

$$|u_j - u_k| \leq \varepsilon$$

as required. Hence the u-sequence is Cauchy. A similar argument implies that the ℓ-sequence is Cauchy. By completeness both converge, and since $u_n - \ell_n \to 0$ they both converge to the same limit (by Lemma 2.3.3) — call this limit α.

It remains to show that $\alpha = \sup A$.

- Assume that α is not an upper bound. Then there is some $x \in A$ so that $\alpha < x$. Now set $\varepsilon = \frac{x-\alpha}{2}$. Since $u_n \to \alpha$, there is some u_k with $\alpha < u_k < \alpha + \varepsilon < x$. But, by construction we know that u_k is an upper bound, so no such x can exist. Thus α must be an upper bound.

- Let $\varepsilon > 0$, then since $\ell_n \to \alpha$, we know that there is some k so that $|\alpha - a_k| < \varepsilon$. That is, we know there is some $\alpha - \varepsilon < \ell_k < \alpha$ and by construction ℓ_k is not an upper bound. Hence $\alpha - \varepsilon$ is not an upper bound. So there is no upper bound smaller than α.

Thus α is the supremum of A as required. ■

2.5 An important aside to closed and bounded sets of reals

So far, we have discussed at length a variety of categorizations of types of sets of real numbers: open, closed, bounded, complete. We have seen how these properties interact with each other.

Now there is one more major category of sets that we have yet to discuss, which mathematicians care about *a lot* — **compact**. Now we don't want to get into precisely what this means yet because the precise definition of this is quite cumbersome and unnecessary for us at this point (we will save that for later in the text 6). In the special case of the real numbers, however, the notion that a set K is **compact** is actually *equivalent* to the condition that it is closed *and* bounded. Importantly, we need both this conditions to hold *together at the same time!*

Theorem 2.5.1 Closed and bounded sets are nice. *The following conditions are equivalent for a set $A \subseteq \mathbb{R}$:*

- *A is closed* and *bounded,*

- *Every sequence $(x_n)_n$ contained in A has a subsequence $(x_{n_k})_k$ that converges to a point $a \in A$.*

Proof. Homework! ■

We are obligated to mention[1], though you don't need to remember these words yet, that the subsequence condition given in the above theorem for a set K is referred to as the set being **sequentially compact**. For now, however, you only need to remember that this condition is equivalent to the set being a closed and bounded subset of the real numbers.

As you may imagine, the equivalent categorisation given by Theorem 2.5.1 means that closed and bounded subsets of $\mathbb{R}$ satisfy many nice properties. You will explore these in the following exercises.

Exercises

1. **You really neeed both.** Show that Theorem 2.5.1 is false if you replace "closed and bounded" with just "closed" or just "bounded".

2. **Unbounded function and a bad point.** Suppose that $K \subseteq \mathbb{R}$ is closed and bounded and $f : K \to \mathbb{R}$ is an *unbounded* function, i.e., it is *not* a bounded function 2.2.5. Show that there exists some $x \in K$ such that $f(N_r)$ is unbounded for every $r > 0$, where $N_r = (x - r, x + r) \cap K$.

   Hint. First show there is a sequence of points $(x_n)_n$ with $|f(x_n)| \to \infty$.

3. **Closed sets contain their limits.** Fix some closed set $K \subseteq \mathbb{R}$, and let $(x_n)_n$ be a sequence contained in K, which converges to some real number x. Prove that $x \in K$ as well.

   Takeaway: Closed sets are "closed" under sequences, i.e., they contain all their limiting points. This is an extension of Lemma 2.3.13.

   Hint. Recall that K being closed means that $\mathbb{R} \setminus K$ is open. Use this to show that $x \notin \mathbb{R} \setminus K$ (by contradiction, for instance).

4. **Limits contained implies closed.** Prove that the converse to the previous exercise is true as well. That is, let $A \subseteq \mathbb{R}$ be some set that satisfies the following peculiar property:

   "For every converging sequence of real numbers $(x_n)_n$ in A, the limiting value x is contained in A as well."

   Show that this property implies that A is closed.

   Hint. Use a proof by contradiction by assuming that A is open. Now first show there is some $x \in \mathbb{R} \setminus A$ for which for every $r >$, it is true that

   $$[(x - r, 0) \cup (0, x + r)] \bigcap A \neq \emptyset.$$

   Since this condition must hold *for every* $r > 0$, you can apply this with a sequence of $r \to 0$ to obtain a "bad" sequence.

5. **Closed same as Complete (in $\mathbb{R}$).** Use the previous two exercises to show that a set $A \subseteq \mathbb{R}$ is complete if and only if it is closed.

   Hint. Use the completeness of the real numbers.

[1] and we would like to apologise to the reader for throwing even more terminology

6. **Nested closed bounded sets — loosen the nested interval property.** Suppose that $(A_n)_n$ is a sequence of closed, bounded, and non-empty sets in $\mathbb{R}$. Further assume that the sets are **nested** so that for all $n \in \mathbb{N}$

$$A_n \supseteq A_{n+1}.$$

Prove that the intersection

$$\bigcap_{n\in\mathbb{N}} A_n$$

is non-empty.

In fact, if one assumes that the *diameters*[2] of the $(A_n)_n$ are decreasing to zero as $n \to \infty$, then the intersection is composed of precisely one real number. This is therefore a generalisation of Theorem 2.2.2.

The following exercises involve continuity which is a topic we haven't covered yet — you'll do it in Section 3.2 and Section 3.3. We have placed them here because they involve properties of closed bounded sets of reals. We recommend that you come back to them after you've worked through the material in Section 3.2 and Section 3.3.

7. **Generalised Extreme Value Theorem.** Prove Corollary 3.3.7 by first showing that for any closed and bounded set $K \subset \mathbb{R}$, if $f : K \to \mathbb{R}$ is continuous, then the image

$$f(K) = \{f(x) \mid x \in K\},$$

is *also* a closed and bounded set of real numbers.

Hint. Start by considering a sequence of points in the image $f(K)$ and use the equivalence of Theorem 2.5.1 (twice!).

8. **Uniformly continuous is easy if you're closed and bounded.** Suppose that $f : K \to \mathbb{R}$ is continuous, where K is a closed and bounded set. Prove that f is actually **uniformly continuous** on K, i.e., that for any $\varepsilon > 0$, there exists $\delta > 0$ (not depending on x!) such that

$$|f(x) - f(y)| < \varepsilon$$

whenever $x, y \in K$ such that $|x - y| < \delta$.

Hint. Prove by contradiction so that for some given $\varepsilon > 0$, you can find sequences of points $x_n \in K$ and $y_n \in K$ such that $|x_n - y_n| < 1/n$ but for which $|f(x_n) - f(y_n)| \geq \varepsilon$. Now use the characterization of closed and bounded subsets of $\mathbb{R}$ that such sets always have converging subsequences and the fact that f is continuous to derive a contraction.

[2] in this setting, the diameter of the set is just maximum value minus the minimum value of the set

2.6 Series

So far, we have studied sequences:

$$a_1, a_2, a_3, \ldots$$

where we ask whether the individual terms approach some limiting value.

Now we turn to a related but substantially more subtle object: *infinite sums*. The basic question is deceptively simple: How do we interpret the infinite sum

$$a_1 + a_2 + a_3 + \cdots ?$$

What could this possibly mean? After all, we cannot literally add infinitely many numbers one at a time and "finish." The key idea is that we interpret an infinite sum through its *partial sums*. In other words, series are secretly sequences in disguise.

At this point, we do not have any sort rigorous notion for what we precisely mean by "convergent", or even a proper definition for what a series even is! But we want to give you a few examples of series to whet your appetite.

Example 2.6.1 Convergent series - Geometric series. A classic example of a convergent series is the **geometric series** (we will visit this again in Example 2.6.6). The most well-known one is the series described by Zeno in the popular *Zeno's Paradox*.

Imagine trying to walk across a room. Before reaching the other side, you must first travel halfway. Then you must travel half of the remaining distance, then half again, and so on forever:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

At first glance, this seems impossible: if there are infinitely many steps, how could you ever arrive? This is where the supposed "paradox" pops up; it is not actually a paradox, of course. We all know that it is, in fact, possible to cross a room. We will develop the tools to properly address this.

(spoiler alert: The infinite sum equals 1) □

Example 2.6.2 Another convergent series. Another famous convergent series is

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$$

This is sometimes called the **Basel series**. In the 1700s, mathematicians tried for decades to determine its exact value. The problem was finally "solved"[1] by Euler, who showed that the sum is

$$\frac{\pi^2}{6}.$$

This answer seemed astonishing: why should the number π, which comes from geometry, appear in a problem about fractions and infinite sums? Unfortunately,

however, we will not have time[2] in this course to see why this equality is true, but we will be able to prove that this sum does, in fact, converge. □

Example 2.6.3 A (very important) divergent series. Not every infinite series settles down to a finite value. A famous example is the **harmonic series**:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$$

The terms get smaller and smaller, so one might reasonably guess that the series should converge. Surprisingly, it does not. The sum grows without bound, although it does so extremely slowly.

The divergence of the harmonic series was first written down[3] by a French scholar *Oresme* in the 12th century. The fact that this series really is divergent should serve as a reminder as we progress that series are inherently more complicated than sequences, and that "terms going to zero" is not enough to guarantee convergence.

By the way, the reason this series is called the **harmonic series** is derived from music theory; the amplitude of the waves produced by vibrating strings are inversely proportional to the frequency. □

Historically, infinite series caused an astonishing amount of confusion. People in the 17th and 18th centuries manipulated divergent series with alarming confidence and occasionally obtained correct answers for completely wrong reasons. Modern analysis exists partly to clean up this mess. In this class, we will rigorously cover some of the basics of infinite series.

Example 2.6.4 You've seen this before, but not rigorously. You've seen in first-year calculus a bunch of Taylor series, for example

$$\sin(x) = 1 - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$$

Taylor series are a special type of series, which are more generally referred to as **power series**. Essentially, the idea here is that for each *fixed* value of x, you can consider the series give for that value of x. The natural follow-up question is: *Which values of x make the series converge?* Then one can define the **radius of convergence** and deduce nice properties of such series within its radius of convergence. We will discuss some of these properties within the homework. □

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[1] perhaps not as rigorously as we would require nowadays

[2] or the tools!

[3] that we know of!

2.6.1 Definition and Basic Notation

Definition 2.6.5 Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. The **series**

$$\sum_{n=1}^{\infty} a_n$$

is the sequence of partial sums

$$s_N = \sum_{n=1}^{N} a_n = a_1 + a_2 + \cdots + a_N.$$

We say that the series **converges** if the sequence $\{s_N\}$ converges.

If the sequence $\{s_N\}$ converges, i.e., if

$$\lim_{N\to\infty} s_N = s,$$

for some $s \in \mathbb{R}$, then we write

$$\sum_{n=1}^{\infty} a_n = s.$$

If the sequence $\{s_N\}$ does not converge, the series is said to **diverge**. ◊

The sigma notation

$$\sum_{n=1}^{\infty} a_n$$

is compact and convenient, but it is important not to lose sight of what it actually means:

$$\sum_{n=1}^{\infty} a_n \quad \text{really means studying the sequence} \quad \left\{ s_N = \sum_{n=1}^{N} a_n \right\}_{N=1}^{\infty}.$$

So every theorem about sequences immediately becomes a theorem about series once we apply it to the sequence of partial sums.

Example 2.6.6 Geometric Series - Revisited. Consider the **geometric series**

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots.$$

To understand whether this series converges, we study its partial sums:

$$s_N = 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^N}.$$

At first glance, finding a formula for s_N may seem difficult, but there is a clever trick that works for geometric series.

Multiply both sides by $\frac{1}{2}$:

$$\frac{1}{2}s_N = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^{N+1}}.$$

Now subtract the two equations:

$$s_N - \frac{1}{2}s_N = \left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^N}\right) - \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{N+1}}\right).$$

Almost every term cancels. This is the whole point of the trick. We are left with

$$\frac{1}{2}s_N = 1 - \frac{1}{2^{N+1}},$$

or equivalently,

$$s_N = 2 - \frac{1}{2^N}.$$

Now we can compute the limit:

$$\lim_{N\to\infty} s_N = \lim_{N\to\infty}\left(2 - \frac{1}{2^N}\right) = 2.$$

Therefore the series converges, and

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = 2.$$

You'll have a chance to practice this same method in Exercise Exercise 2.6.8.1. □

Notice something slightly philosophical in the above example: we are declaring that an infinite sum "equals" 2 because the *approximations* approach 2. Infinite series are fundamentally limiting processes.

Example 2.6.7 Some (easy) divergent series. Not every infinite series converges. Here are two simple examples of divergent series.

First, consider

$$\sum_{n=1}^{\infty} n = 1 + 2 + 3 + 4 + \cdots$$

The sequence of partial sums are

$$1, \quad 1 + 2 = 3, \quad 1 + 2 + 3 = 6, \quad 1 + 2 + 3 + 4 = 10, \quad \ldots$$

These sums grow larger and larger without bound. Hence, the series diverges.

Next, consider

$$\sum_{n=0}^{\infty} (-1)^n = 1 - 1 + 1 - 1 + \cdots$$

The partial sums are

$$1, \quad 0, \quad 1, \quad 0, \quad \dots$$

Since the partial sums do not approach a single real number, this series also diverges. □

Example 2.6.8 A warning about assuming convergence. We want to stress the importance of *not assuming*[4] that a series converges before before you actually confirm that it does. For example, the series $\sum n$ is somewhat well-known[5] in the pop-sci community; *if you assume* that this series converges (it doesn't!), then you can algebraically manipulate the sum to obtain

$$\sum_{n=1}^{\infty} = -\frac{1}{12}$$

This "equality" (it's not actually an equality because the limit of partial sums does not exist!) is actually somewhat important in some fields of physics, e.g., string theory, and mathematics, e.g., the analytic continuation of the Riemann-Zeta function, but we won't worry about those things in this class. The interested reader can go jumping down rabbit holes. □

Example 2.6.9 Divergence of the Harmonic Series. We prove that the **harmonic series**

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

diverges. Note that the individual terms $a_n = \frac{1}{n}$ do indeed go to zero, but the series does *not* converge.

The key idea is to group terms in a way that reveals a uniform lower bound.

We write the series as

$$\begin{aligned} & 1 \\ & + \frac{1}{2} \\ & + \left(\frac{1}{3} + \frac{1}{4}\right) \\ & + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \\ & + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) + \cdots \end{aligned}$$

Each successive group doubles in size. We now estimate each group from below.

The first two terms are

$$1 \quad \text{and} \quad \frac{1}{2}.$$

[4] y' know what they say about assuming

[5] to nerds anyway

For the third group,

$$\frac{1}{3} + \frac{1}{4} \geq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

For the fourth group,

$$\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \geq 4 \cdot \frac{1}{8} = \frac{1}{2}.$$

Similarly, for the block from $2^{k-1} + 1$ to 2^k, each term satisfies

$$\frac{1}{n} \geq \frac{1}{2^k},$$

and there are 2^{k-1} such terms. Hence the entire block satisfies

$$\sum_{n=2^{k-1}+1}^{2^k} \frac{1}{n} \geq 2^{k-1} \cdot \frac{1}{2^k} = \frac{1}{2}.$$

Therefore the partial sums s_N are unbounded and thus do not converge. □

The previous example surprises many people the first time they see it. The terms do go to zero, but they do so *too slowly*. In fact, one can show (not in this class!) that the harmonic series grows basically like $\log(n)$.

Remark 2.6.10 Growth of the Harmonic Series. Let

$$H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

Then it is a fact (we won't prove) that

$$H_n \approx \log n.$$

More precisely, there exists a constant γ (the Euler-Mascheroni constant[6], $\gamma \approx 0.57721...$) such that

$$H_n = \log n + \gamma + c(n),$$

where $\{c_n\}$ is a sequence of real numbers such that $\lim_{n\to\infty} c(n) = 0$.

All of the previous examples are series with only positive terms. Of course, this is not a necessary condition. Let's check out a slight modification of the harmonic series, where you take the exact same terms but just alternate the sign of the terms.

Example 2.6.11 Convergence of the alternating harmonic series. Con-

[6]Euler did a lot of the proverbial heavy lifting regarding the discovery and properties of this constant, but Mascheroni is credited with actually computing the constant

sider the **alternating harmonic series**

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$$

We will show that this series converges without using any special convergence tests. Let

$$s_n = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k}$$

denote the sequence of partial sums.

First, consider the even partial sums:

$$s_{2n} = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n}\right).$$

Each bracket is positive, and therefore the sequence (s_{2n}) is increasing. On the other hand,

$$s_{2n} = 1 - \left(\frac{1}{2} - \frac{1}{3}\right) - \left(\frac{1}{4} - \frac{1}{5}\right) - \cdots - \frac{1}{2n} < 1.$$

Since the sequence (s_{2n}) is increasing and bounded above, it converges.

Considering the odd partial sums, we know that

$$s_{2n+1} = s_{2n} + \frac{1}{2n+1}.$$

We may take the limit on the RHS above (because both sequences converge) to see that the

$$\lim_{n\to\infty} s_{2n+1} = \lim_{n\to\infty} s_{2n},$$

and thus the overall sequence of partial sums converges.

Note: the convergence of the alternating harmonic series is *painfully* slow! Because the error of the n'th partial sum (in regards to $S = \log(2)$) is almost the size of the next term. One can quantify the slowness of the convergence: You need roughly 10^n terms to ensure that the n'th partial sum is accurate to n decimal places. There are acceleration methods for computing these types of sums[7], but we won't discuss those here. □

The previous two examples raises an interesting question. Since the two series (harmonic vs alternating harmonic) only differ based on the sign of the terms within the sum, and because one of them converges and one does not, it is natural to want to distinguish a series for which the sum of the **absolute value** of its terms converges, rather than "ordinary" convergence. We will discuss this concept more later, but for now we will just state the definition.

[7] credit Euler again for this one

Definition 2.6.12 A series $\sum_{n=1}^{\infty} a_n$ is said to **converge absolutely** if

$$\sum_{n=1}^{\infty} |a_n|$$

converges. That is, the series of absolute values converges. ◊

We will see later (see Theorem 2.6.17) that absolute convergence implies regular convergence, as one might expect. We'll also see that sequences like the alternating harmonic series, which converge (as we've seen) but are not absolutely convergent, are extremely delicate mathematical objects and need to be treated carefully (see the section on rearrangements later in chapter).

2.6.2 The Cauchy Criterion for Series

In practice, we often do not know what the limit of a series is. In many important examples, there may not even be a closed formula for the partial sums. The Cauchy criterion is valuable because it allows us to determine convergence *without already knowing the limit.*

Let's revisit an example of a series that we have seen that converges, but determining the limit is difficult.

Example 2.6.13 Basel Series Revisited. Let's turn back to a previous example 2.6.2, the Basel series:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

If we define the partial sums by

$$s_N = \sum_{n=1}^{N} \frac{1}{n^2},$$

then it is far from obvious how to find a simple formula for s_N. In fact, there is no elementary closed form for these partial sums.

Nevertheless, we can still prove that the series converges. For instance,

$$0 \leq \frac{1}{n^2} \leq \frac{1}{n(n-1)} \qquad (n \geq 2),$$

and the series

$$\sum_{n=2}^{\infty} \frac{1}{n(n-1)}$$

is a "telescoping"[8] one, meaning that

$$\frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}.$$

Hence

$$\sum_{n=2}^{N} \frac{1}{n(n-1)} = 1 - \frac{1}{N} \leq 1.$$

By comparison (see the comparison test 2.6.25[9], which basically just says that if one series has smaller terms than one that we know converges, then we can conclude the convergence of our original series),

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges as well.

In fact, a much deeper result shows that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.644934\ldots.$$

The first few partial sums illustrate the slow convergence:

$$1, \quad 1.25, \quad 1.3611, \quad 1.4236, \quad \ldots$$

This example is useful because it separates two ideas:

- proving that a series converges, and
- determining the exact value of the sum.

The first is often achievable with elementary tools; the second may require much deeper ideas. □

Remark 2.6.14 One natural question at this point is: how might one even *guess* that this sum should involve π? In practice, one can numerically approximate partial sums and then apply various acceleration techniques to improve convergence and reveal hidden structure much more quickly. Such experiments often suggest the appearance of familiar constants (like π) before any rigorous proof is known, and this "constant-guessing" perspective plays an important role in the historical development of series analysis.

Theorem 2.6.15 Cauchy Criterion for Series. *The series*

$$\sum_{n=1}^{\infty} a_n$$

[8]We use the term "telescoping" to mean that each subsequent term can partially cancel part of the previous one, thus simplifying the overall sum. More generally if we can write $a_n = b_n - b_{n+1}$ for some other sequence (b_n), then $\sum_{n=1}^{N} a_n = \sum_{n=1}^{N}(b_n - b_{n+1}) = b_1 - b_{N+1}$. This is a useful trick and generalises into the WZ-method. The interested student is invited to search-engine their way to more details.

[9]oof, we haven't actually proved this yet. Oh well, we're going to haphazardly use it anyway

converges if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\left| \sum_{k=n}^{m} a_k \right| < \varepsilon$$

whenever $m \geq n \geq N$.

Proof. Let

$$s_N = \sum_{k=1}^{N} a_k$$

be the sequence of partial sums.

The series converges if and only if the sequence $\{s_N\}$ converges. By the Corollary 2.3.24, this occurs if and only if for every $\varepsilon > 0$ there exists N such that

$$|s_m - s_{n-1}| < \varepsilon$$

whenever $m \geq n \geq N$.

But

$$s_m - s_{n-1} = \sum_{k=n}^{m} a_k.$$

Hence the result follows. ■

The interpretation is important. The quantity

$$\sum_{k=n}^{m} a_k$$

is part of the "tail" of the series. The term **tail** is not really one that we ever rigorously define, but it is a common one in math; it just is referencing the *sum of all remaining terms*, where the sum does not have to start at any given point (we could start at the 10th term, the 100th term, or the 10^{100}'th term. Using this meaning of `tail', the theorem says:

> A series converges precisely when its tails eventually become arbitrarily small.

That is exactly what we should expect from a convergent infinite sum: once we are sufficiently far out in the series, adding more terms should not substantially change the value.

It is important to note in the Cauchy Criterion, that the terms are added up *in order*! For example, consider the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n},$$

which is the so-called "alternating harmonic series". It converges, but you can't just add up the terms in a willy-nilly order and still expect convergence. For

example, if you just took the even terms above before taking the odd terms [10]

$$\sum_{n=1}^{\infty} \frac{1}{2n},$$

then you can see that this is precisely half the value of the harmonic series, and is therefore infinite (diverges). Thus, you should not expect that you should be allowed to sum up the terms in a different order, without changing the result. We will revisit this topic later. In fact, the following example gives an explicit example of what can go wrong when you rearrange the order of terms.

Example 2.6.16 A rearrangement paradox for the alternating harmonic series. Consider the alternating harmonic series

$$S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots .$$

Remember, we've already seen this series converges, but not absolutely.

We first group terms in pairs:

$$S = (1 - \tfrac{1}{2}) + (\tfrac{1}{3} - \tfrac{1}{4}) + (\tfrac{1}{5} - \tfrac{1}{6}) + \cdots .$$

Each pair simplifies as

$$\frac{1}{2n-1} - \frac{1}{2n} = \frac{1}{2n(2n-1)},$$

so

$$S = \sum_{n=1}^{\infty} \frac{1}{2n(2n-1)}.$$

Note that each term in the sum is positive, and in particular, $s_n \geq s_n = \frac{1}{2}$. Because S is the limit of the partial sums, and because we know that this limit exists, we can certainly say that S is *strictly* positive (this will be important later).

Now we regroup the original alternating series in a different way, grouping as

$$(1 - \tfrac{1}{2} - \tfrac{1}{3}) + (\tfrac{1}{4} - \tfrac{1}{5} - \tfrac{1}{6}) + (\tfrac{1}{7} - \tfrac{1}{8} - \tfrac{1}{9}) + \cdots .$$

Each block simplifies to

$$\frac{1}{2n-1} - \frac{1}{2n} - \frac{1}{2n+1} = \frac{1}{4n(2n-1)}.$$

Hence we obtain

$$S = \sum_{n=1}^{\infty} \frac{1}{4n(2n-1)} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{2n(2n-1)} = \frac{1}{2} S.$$

[10] Strictly speaking, this is not actually a "rearrangement", but it works well enough to serve as a pedagogical example

This leads to the formal identity $S = \frac{1}{2}S$. But wait a second! We know that this identity implies, in particular, that $S = 0$. But this directly contradicts our comment earlier that $S > 0$. Clearly something has broken in our manipulations, and indeed, one must be extremely careful about what "rearranging" or "regrouping" an infinite series means. This example shows that "convergence" alone is not a sufficiently strong condition for a series to be unchanged by rearranging its terms. □

As we will see at the end of this chapter, all this discussion about rearrangement of series is intimately related to the distinction between absolute and regular (or "conditional") convergence. As the names suggest, absolute convergence is stronger than conditional convergence.

Theorem 2.6.17 Absolute convergence implies convergence. *If the series*

$$\sum_{n=1}^{\infty} |a_n|$$

converges, then the series

$$\sum_{n=1}^{\infty} a_n$$

also converges. Moreover,

$$\left| \sum_{n=1}^{\infty} a_n \right| \leq \sum_{n=1}^{\infty} |a_n|$$

In other words, absolute convergence implies convergence.

Note: The converse of this statement is false. More precisely, there exists series that converge but do not *converge absolutely (the alternating harmonic series is the classic example).*

Proof. In the proof we will freely use the Corollary 2.3.24 without explicitly saying so.

$$\sum_{n=1}^{\infty} |a_n|$$

converges. Let

$$s_n = \sum_{k=1}^{n} a_k$$

be the sequence of partial sums of $\sum a_n$.

We will show that (s_n) *is a Cauchy sequence. Let* $\varepsilon > 0$. *Since*

$$\sum_{n=1}^{\infty} |a_n|$$

converges, its sequence of partial sums is Cauchy. Hence there exists N *such that whenever* $m > n \geq N$,

$$\sum_{k=n+1}^{m} |a_k| < \varepsilon.$$

Now, for $m > n \geq N$,

$$|s_m - s_n| = \left| \sum_{k=n+1}^{m} a_k \right| \leq \sum_{k=n+1}^{m} |a_k| < \varepsilon.$$

Thus (s_n) *is Cauchy. Since* $\mathbb{R}$ *is complete, the sequence* (s_n) *converges. Therefore,*

$$\sum a_n$$

converges.

Note, the claim that

$$\left| \sum a_n \right| \leq \sum |a_n|$$

just follows from the finite case

$$|s_N| = \left| \sum_{n=1}^{N} a_n \right| \leq \sum_{n=1}^{N} |a_n|$$

and taking the limit $N \to \infty$. ■

Since absolute convergence is a stronger notion than "regular" convergence, we would like a word to describe a series that converges, but *not* absolutely.

Definition 2.6.18 We say a series $\sum a_n$ is **conditionally convergent** if it converges, but not absolutely. ◊

Note that Example 2.6.9 and Example 2.6.11 shows the alternating harmonic

series converges conditionally (it converges, but not absolutely).

2.6.3 A Necessary Condition for Convergence

The next result is extremely simple, but it is probably the single most commonly misused fact about series by students seeing the subject for the first time.

Corollary 2.6.19 *If*

$$\sum_{n=1}^{\infty} a_n$$

converges, then

$$\lim_{n\to\infty} a_n = 0.$$

Proof. Let

$$s_n = \sum_{k=1}^{n} a_k.$$

If the series converges, then the sequence $\{s_n\}$ converges.

Observe that

$$a_n = s_n - s_{n-1}.$$

By the sum rule for sequences (this was explored in a previous exercise 2.3.3.9) and the limit of subsequence theorem (well, this wasn't formally a theorem either, but was also explored in a previous exercise 2.3.3.6), we have

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (s_n - s_{n-1}) = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} \to s - s = 0.$$

■

Actually, the above theorem statement is probably more useful if we use the contrapositive instead:

Corollary 2.6.20 *If the terms of a series do not approach zero, then the series cannot converge.*

Example 2.6.21 The series

$$\sum_{n=1}^{\infty} 1 = 1 + 1 + 1 + \cdots$$

clearly diverges because the terms do not go to zero. □

Example 2.6.22 Now we have a shortcut. We already saw in Example 2.6.7 that the series

$$\sum_{n=1}^{\infty} n$$

diverges. But now we have a shortcut! We could just say *the terms do not go to zero, so it does not converge*, instead of going through all the trouble of showing

divergence from scratch.

Aside: We would like to remind the reader that this series DEFINNITELY does not converge to $-1/12$, as some online sources say (for the clicks, I assume). □

However — and this is the important warning — the converse of Corollary 2.6.19 is false. We have already seen an example of this in the Harmonic series! Thus, it is possible for the terms of a power series to go to zero, but too slowly to allow convergence.

2.6.4 Comparing and Positive Series

Life becomes much simpler when all terms are nonnegative. Remember, the convergence (or non convergence) of a series is just the same thing as asking whether or not the *sequence* of partial sums s_N converges. When all the terms are nonnegative (or nonpositive), this is equivalent to the partial sums forming an increasing sequence. We have already developed a bunch of tools to address such sequences. Let's take advantage of that.

If every term satisfies $a_n \geq 0$, then each new term can only increase the partial sums:

$$s_{N+1} = s_N + a_{N+1} \geq s_N.$$

More precisely, the sequence $\{s_n\}_n$ is a *monotone increasing* sequence, i.e., there is no possible cancellation happening in each summation. Remember, a monotone increasing sequence converges if and only if it remains bounded. Let's state this precisely.

Corollary 2.6.23 *Suppose $a_n \geq 0$ for all n. Then*

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if the sequence of partial sums

$$s_N = \sum_{n=1}^{N} a_n$$

is bounded.

Proof. This is a direct consequence of Lemma 2.3.4 and Lemma 2.3.5 and the fact that $\{s_n\}_n$ forms a monotone increasing sequence. ■

This theorem is extremely important conceptually. Let's again reword it using the negation.

Corollary 2.6.24 *A nonnegative series diverges if and only if the partial sums are unbounded.*

Also, nonnegative series are really nice when it comes to comparing. Phrasing things in terms of two monotone increasing sequences, if it is known that one

sequence is smaller than another, that the latter is bounded, then so is the former (and thus converges!). We can state this formally as the "Comparison Test". In practice, this is often how convergence proofs actually work: compare a complicated series to a simpler "benchmark" series whose behaviour we already understand.

Theorem 2.6.25 Comparison Test. *Suppose*

$$0 \leq a_n \leq b_n$$

for all [11] *sufficiently large* n*.*

1. *If* $\sum_{n=1}^{\infty} b_n$ *converges, then* $\sum_{n=1}^{\infty} a_n$ *converges.*
2. *If* $\sum_{n=1}^{\infty} a_n$ *diverges, then* $\sum_{n=1}^{\infty} b_n$ *diverges.*

Proof. Since $0 \leq a_n \leq b_n$ for all "sufficiently large n", this means that there is some $N \in \mathbb{N}$ such that this condition holds for all $n \geq N$. So, we can take advantage of this fact, and write

$$s_n = \sum_{n=N}^{N+n} a_n, \qquad t_n = \sum_{n=N}^{N+n} b_n.$$

These partial sums of the tails thus satisfy

$$0 \leq s_n \leq t_n$$

for all $n \geq 0$.

Note that

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{N} b_n + \sum_{n=N+1}^{\infty} b_n = \sum_{n=1}^{N} b_n + \lim_{n\to\infty} t_n$$

So if $\sum b_n$ converges, then $\{t_n\}$ is bounded. Hence $\{s_N\}$ is also bounded and also convergent because it is increasing. Therefore, we have

$$\begin{aligned}\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n &= \lim_{n\to\infty} \left[\sum_{n=1}^{N} a_n + \sum_{n=N+1}^{\infty} a_n\right] \\ &= \lim_{n\to\infty} \sum_{n=1}^{N} a_n + \lim_{n\to\infty} \sum_{n=N+1}^{\infty} a_n.\end{aligned}$$

Both of the above limits exist, and therefore $\sum a_n$ converges.

The second statement follows from the contrapositive of the first. ■

Note: In applying this result, we will often only require the hypotheses to hold eventually, that is, on the "tail" of the sequence (note the requirement is "for sufficiently large n"). This is because adding or removing finitely many terms at the beginning of a series does not affect convergence, so the initial segment can be treated separately.

This same idea appears in the proof that every convergent sequence is bounded: the finitely many "head" terms are automatically bounded since there are only finitely many of them, while the "tail" requires the actual convergence argument to show that the infinite sum, i.e., the limit of the finite partial sums, converges.

For this reason, we state the theorem in terms of eventual behavior, as this is both sufficient and more convenient for applications.

The comparison test is only as useful as the series you already know. The

[11] This is somewhat informal language that is pretty common in math. It's nice for you to see it and get used to understanding its meaning, but we expect a bit more rigor from you in your homework and tests.

most important benchmark examples are:

$$\sum \frac{1}{n^p} \quad \text{and} \quad \sum r^n.$$

The reader is likely familiar with these series from first-year calculus, and also hopefully recalls that the conditions

$$p > 1 \quad \text{and} \quad |r| < 1$$

respectively are precisely the conditions required for convergence.

Lemma 2.6.26 Geometric series and p-series. *The geometric series*

$$\sum_{n=1}^{\infty} r^n,$$

where r is a constant[12]*, converges if and only if $|r| < 1$.*

The p-series

$$\sum_{n=1}^{\infty} \frac{1}{p^n},$$

where p is a real number, converges if and only if $p > 1$.

Proof. Homework! ■

These appear constantly throughout analysis. In some sense, much of the theory of series amounts to determining whether a given series behaves more like a geometric series or more like the harmonic series.

2.6.5 Root Test and Ratio Test

At this point, we have a few basic tools for proving convergence: directly proving a limit exists using the sequence of partial sums; comparison test; and the Cauchy criterion. Now we continue to develop our toolbox by proving two tests that you are likely familiar with from first-year calculus.

The first tool that we state is the root test, and is especially useful when the terms of a series behave like something raised to the power n.

Theorem 2.6.27 Root Test. *Let $\sum a_n$ be a series and define*

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|}.$$

1. *If $L < 1$, then $\sum a_n$ converges absolutely.*
2. *If $L > 1$, then $\sum a_n$ diverges.*
3. *If $L = 1$, the test is inconclusive; the series could be convergent or divergent.*

[12]We are using the word "constant" here because it works, not only for any real number, but more generally for any complex number. We won't worry about it for complex numbers here though.

Proof. Suppose $L < 1$. Choose r such that $L < r < 1$ We would like to emphasize here that this choice of r is henceforth *fixed*, and is not variable; we really need to be able to say there is some fixed distance between r and L and also between r and 1. By the definition of the limit, there exists N such that for all $n \geq N$,

$$\sqrt[n]{|a_n|} \leq r,$$

so

$$|a_n| \leq r^n.$$

Since the geometric series $\sum r^n$ converges, comparison implies $\sum |a_n|$ converges.

Suppose now that $L > 1$. Then, similarly to before, we can fix some $1 < r < L$ so that we can find some N such that for all $n \geq N$, we have $\sqrt[n]{|a_n|} \geq r$, and thus

$$|a_n| \geq r^n.$$

In particular, the sequence $\{a_n\}_n$ does not even go to zero, so the series must diverge by Corollary 2.6.20.

Finally, we the harmonic series (divergent) and the alternating harmonic series (converges) are both series that satisfy $L = 1$, and thus shows this boundary case is inconclusive. We will go into more detail on such examples later. ■

As demonstrated in the proof, the root test is essentially taking the given series and trying to compare it to a geometric series. This works well a lot of the time, but usually does not usually work well when factorials are involved. In such settings, the ratio test is much more effective.

Theorem 2.6.28 Ratio Test. *Suppose each $a_n \neq 0$ and that*

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L.$$

1. *If $L < 1$, then $\sum a_n$ converges absolutely.*
2. *If $L > 1$, then $\sum a_n$ diverges.*
3. *If $L = 1$, the test is inconclusive; the series could be convergent or divergent.*

Proof. Homework! ■

Remark 2.6.29 The ratio test is especially natural for expressions involving powers and factorials, such as

$$\frac{n!}{r^n}, \quad \frac{r^n}{n!}, \quad \frac{n!}{n^n}, \quad \frac{n^n}{n!}, \quad \frac{1}{n!}, \quad \text{etc.}$$

because powers and factorials behave very predictably under quotients.

Both the root and ratio tests have a common weakness: they can fail to give any information. So $L = 1$ is genuinely a "boundary case", not a failure of the test. In fact, one could argue that this boundary case of $L = 1$ is really

where basically *all* the interesting mathematics comes up. For example, we have already mentioned that the harmonic (and alternating) harmonic series are boundary cases ($L = 1$), which are interesting. Or consider a couple more:

$$\sum \frac{1}{n \log(n)} \quad \text{and} \quad \sum \frac{1}{n \log(n)^2}.$$

It is a fact (you can trust us for now; we will explore these later in the exercises) the former series diverges but the latter converges. Yet, both are boundary $L = 1$ cases for both the root and ratio tests.

Example 2.6.30 Divergence when $L = 1$. Consider the harmonic series

$$a_n = \frac{1}{n}.$$

Then

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \to 1$$

and

$$\sqrt[n]{a_n} \to 1,$$

so that both the ratio and the root test are inconclusive, even though the series diverges. □

Example 2.6.31 Convergence when $L = 1$. Consider

$$a_n = \frac{1}{n^2}.$$

Then

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \to 1$$

and

$$\sqrt[n]{a_n} \to 1,$$

so that both the ratio and the root test are inconclusive, even though the series converges as seen in Example 2.6.13. □

Oof! Wait a second! In both of the examples above, we have used the fact that

$$\sqrt[n]{a_n} \to 1$$

even though this is far from obvious! Oh dear... how awkward. The problem is that in order to prove this limit[13] is to use L'Hopital's Rule, which we have not developed yet. But the authors think that you can trust us for the time being. The trick is to take the logarithm of a_n, which then looks like an indeterminate form and L'Hopital's Rule can be applied, but we won't worry about that yet because we haven't even defined what a derivative is!

[13] a relatively painless way anyway!

Given the similarity between the root and ratio test, a very natural question is whether or not one of these tests is *better* than the other? Well, as usual, it depends on what you mean by "better". On the one hand, the ratio test is usually *easier* to apply, but it turns out that the root test is *always* better from a mathematical perspective. What we mean by that is that it always gives more information, and applies to more general circumstances.

Theorem 2.6.32 Root test is stronger than ratio test. *Suppose that*

$$\sum a_n$$

is a series with $a_n \neq 0$ for all n. Let

$$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

and

$$R = \lim_{n\to\infty} |a_n|^{1/n}.$$

Assume that both limits above exist. Then

$$R \leq L.$$

Consequently, whenever the ratio test applies, the root test also applies and gives the same conclusion. Thus, the root test is stronger than the ratio test.

Proof. Define

$$r_n = \left| \frac{a_{n+1}}{a_n} \right|.$$

Then for every $n \geq 1$,

$$|a_n| = |a_1| r_1 r_2 \cdots r_{n-1}.$$

Taking n-th roots gives

$$|a_n|^{1/n} = |a_1|^{1/n} (r_1 r_2 \cdots r_{n-1})^{1/n}.$$

Fix $\varepsilon > 0$. By the definition of the limit, there exists N such that

$$r_n \leq L + \varepsilon$$

for all $n \geq N$. So for $n > N$,

$$r_1 r_2 \cdots r_{n-1} \leq C(L+\varepsilon)^n,$$

where

$$C = \frac{r_1 r_2 \cdots r_{N-1}}{(L+\varepsilon)^N}.$$

Notice that C does not depend on n![14] Therefore,

$$|a_n|^{1/n} \leq |a_1|^{1/n} C^{1/n} (L+\varepsilon).$$

Taking the limit[15] yields

$$\lim_{n\to\infty} |a_n|^{1/n} \leq L + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary,

$$R = \lim_{n\to\infty} |a_n|^{1/n} \leq \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L.$$

■

The above Theorem raises the natural follow-up question as to whether or not the root test is *strictly* stronger than the ratio test, or if they actually always just give the same information. The following example answers that question.

Example 2.6.33 Root test is strictly stronger than ratio test. The root test can succeed even when the ratio test fails. Consider the series

$$\sum_{n=1}^{\infty} 3^{-n-(-1)^n}.$$

and let

$$a_n = 3^{-n-(-1)^n}.$$

First compute the quantity appearing in the root test:

$$(a_n)^{1/n} = 3^{-1-\frac{(-1)^n}{n}}.$$

Since

$$\frac{(-1)^n}{n} \to 0,$$

we obtain

$$\lim_{n\to\infty} (a_n)^{1/n} = 3^{-1} = \frac{1}{3} < 1.$$

Hence, by the root test, the series converges.

Now consider the ratio test:

$$\frac{a_{n+1}}{a_n} = 3^{-(n+1)-(-1)^{n+1}+n+(-1)^n}.$$

If n is even, then $(-1)^n = 1$ and $(-1)^{n+1} = -1$, so

$$\frac{a_{n+1}}{a_n} = 3^{-1-(-1)+1} = 3.$$

[14]This is an excited explanation point, not the quantity n factorial.

[15]there is also a bit of shenanigans happening here; you can convince yourself that the limit of $C^{1/n}$ is 1 by taking the logarithm then undoing that operation. The subtle issue here is that when you undo such an operation, you are implicitly using the **continuity** of the exponential function, which is something that you of course know and are familiar with, but we just haven't proven it yet...oh well

On the other hand, if n is odd, then $(-1)^n = -1$ and $(-1)^{n+1} = 1$, so

$$\frac{a_{n+1}}{a_n} = 3^{-1-1-1} = \frac{1}{27}.$$

Thus the sequence

$$\left(\frac{a_{n+1}}{a_n}\right)$$

alternates between 3 and 1/27, so the limit does not exist. Even if you don't worry about the actual limit existing, you have the bigger problem that the ratios alternate between being larger and smaller than 1. Therefore, the ratio test fails.

This demonstrates that the root test is strictly stronger than the ratio test. □

2.6.6 Arithmetic of Series

We now discuss how series behave under basic algebraic operations. First, addition and subtraction behave as expected.

Theorem 2.6.34 Series Summation. *For convergent series $\sum a_n$ and $\sum b_n$,*

$$\sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} (a_n + b_n).$$

In particular, the series on the RHS is convergent as well.

Proof. For partial sums, there are no problems:

$$\sum_{n=1}^{N}(a_n + b_n) = \sum_{n=1}^{N} a_n + \sum_{n=1}^{N} b_n.$$

Now taking the limit $N \to \infty$, we apply the sum rule for limits 2.3.3.9[16] (this is valid because both series on the RHS converge, i.e., the limits exist) so that

$$\begin{aligned}
\sum_{n=1}^{\infty}(a_n + b_n) &= \lim_{N\to\infty} \sum_{n=1}^{N}(a_n + b_n) \\
&= \lim_{N\to\infty} \left[\sum_{n=1}^{N} a_n + \sum_{n=1}^{N} b_n\right] \\
&= \lim_{N\to\infty} \sum_{n=1}^{N} a_n + \lim_{N\to\infty} \sum_{n=1}^{N} b_n \\
&= \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.
\end{aligned}$$

■

Remark 2.6.35 In other words, addition and subtraction of series are not mysterious operations: they are inherited directly from addition of real numbers applied to partial sums.

Multiplication is more subtle; given two series $\sum a_n$ and $\sum b_n$, it is not immediately obvious that there is a *natural* way to compute the product

$$\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right)$$

One way that you might think to do this is index in terms of the a_n terms:

$$a_0\left(\sum_{n=0}^{\infty} b_n\right) + a_1\left(\sum_{n=0}^{\infty} b_n\right) + a_2\left(\sum_{n=0}^{\infty} b_n\right)\ldots$$

But this isn't really *natural* in the sense that the terms $a_0 b_n$ are going to be extremely small for very large n (assuming that the series' both converge, of course!). So if you chose to expand the multiplication in this way, you're doing a lot of summation work that doesn't do a lot of the "heavy lifting" before you add something like $a_1 b_1$, which could be very large! With that in mind, it makes more sense to add up all the "large" terms first.

This is actually best demonstrated with **power series**[17]:

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right)$$

[16] hopefully you did this exercise!

[17] a topic that you'll explore more in homework

In this case, we would like to express the above product *also* as a power series, i.e., in the form of

$$\sum_{n=0}^{\infty} c_n x^n$$

In other words, we'd like to group the terms in the product *in terms of the degree of* x! Circirling back to the language of the previous paragraph, this is akin to adding together all the "big" terms first. If you play around with this, you'll see that coefficient in front of the x^n term is precisely

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Of course, the question of whether or not the resulting product series actually converges is quite subtle. We'll explore that shortly. For now, let's just formally state what the series product is.

Definition 2.6.36 Cauchy Product. Given two series $\sum a_n$ and $\sum b_n$, their **product** is formally defined by

$$\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty} c_n,$$

where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

$\Diamond$

We will not prove general product convergence results here; this is a delicate topic. The following gives and explicit example where the Cauchy product of two convergent series is not necessarily convergent!

Example 2.6.37 Two convergent series with divergent Cauchy product. Let

$$a_n = b_n = \frac{(-1)^{n-1}}{\sqrt{n}}.$$

Then both series

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} b_n$$

converge (we have not proven this, but we may explore it on homework). However, their Cauchy product has general term

$$c_n = \sum_{k=1}^{n} \frac{(-1)^{k-1}}{\sqrt{k}} \cdot \frac{(-1)^{n-k}}{\sqrt{n-k+1}}.$$

These terms do not even decay to zero! - and therefore the product series certainly does not converge. $\square$

Not all is bad news however. If we increase the strength of our hypotheses slightly to assume that at least one of the series is *absolutely* convergent, then we *do* get convergence of the product.

Theorem 2.6.38 Merten's[18] Product Theorem. *Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be two series such that $\sum_{n=0}^{\infty} a_n$ converges absolutely and $\sum_{n=0}^{\infty} b_n$ converges (not necessarily absolutely).*

Define the Cauchy product series (c_n) by

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Then the series $\sum_{n=0}^{\infty} c_n$ converges, and moreover

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right).$$

Proof. We omit the proof for now. Perhaps we will explore this on homework.... ■

Note the subtlety of what Merten's theorem is saying: the left-hand side

$$\sum_{n=0}^{\infty}\sum_{k=0}^{n} a_k b_{n-k}$$

is an infinite sum, and is therefore a limit; the right-hand side

$$\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right)$$

is a product of two numbers, each of which in turn is defined as a limit. So really, this result gives a method of splitting a limit of a particular form into two new simpler limits. Just some food for thought.

2.6.7 Rearrangement of series

We now reach one of the most conceptually important (and slightly dangerous) topics in the subject.

At the level of finite sums, there is no ambiguity: addition is commutative and associative, so the order of summation does not matter. This basic fact might suggest that the same should remain true for infinite sums. And indeed, there is certainly no issue if you just rearrange *finitely many* of the terms in an infinite sum. For example,

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{1}{4} + 1 + \frac{1}{9} + \frac{1}{16} + \ldots$$

[18]Definitely not the same as "Martens", as in one of the authors. Although, I would like to point out here that there is also no "Rechnitzer's Theorem". And I would like to point out there is no "Rechnitzer Theorem" yet.

But an issue arises when you start rearranging *infinitely many* terms. We have already seen an example of this 2.6.16. One has to be really careful when it comes to rearranging infinite series.

Not all is lost, however. In the case where your series converges absolutely, then we will see that, in fact, any rearrangement is equal 2.6.39. So absolute convergence is certainly *sufficient* to rearrange things. It turns out that this condition is also *necessary*; we prove in this section that any *conditionally convergent* series can be rearranged, not only to a different sum, but to equal *any* value that you want! 2.6.40

In summary: When a series converges absolutely, its sum is stable under reordering of terms. If a series is only conditionally convergent, however, rearrangements are very, very, very bad.

Theorem 2.6.39 Absolute Convergence and Rearrangements. *If $\sum a_n$ converges absolutely, then any rearrangement of $\{a_n\}$ converges to the same sum.*

Proof. Assume $\sum_{n=1}^{\infty} a_n$ converges absolutely, so

$$\sum_{n=1}^{\infty} |a_n| < \infty.$$

Let

$$S = \sum_{n=1}^{\infty} a_n.$$

Let (b_n) be any rearrangement of (a_n), i.e., $b_n = a_{\pi(n)}$ for some bijection $\pi : \mathbb{N} \to \mathbb{N}$. Define the partial sums

$$s_n = \sum_{k=1}^{n} a_k, \qquad p_n = \sum_{k=1}^{n} b_k.$$

Fix $\varepsilon > 0$. Since $\sum |a_n|$ converges, there exists N such that

$$\sum_{k>N} |a_k| < \varepsilon.$$

Because π is a bijection, there exists M such that for all $m \geq M$, the set

$$\{\pi(1), \ldots, \pi(m)\}$$

contains $\{1, \ldots, N\}$.

Fix $m \geq M$. Then we can decompose

$$p_m = \sum_{k=1}^{N} a_k + \sum_{k \in A_m} a_k,$$

where $A_m \subset \{N+1, N+2, \ldots\}$ is the set of indices that have appeared among $\pi(1), \ldots, \pi(m)$ and are greater than N.

Now write

$$S - p_m = \sum_{k>N} a_k - \sum_{k \in A_m} a_k = \sum_{k>N,\, k \notin A_m} a_k.$$

Now to this infinite series, we may apply the latter inequality in this theorem 2.6.17, to obtain

$$|S - p_m| \leq \sum_{k>N,\, k \notin A_m} |a_k| \leq \sum_{k>N} |a_k| < \varepsilon.$$

Thus $p_m \to S$, so the rearranged series converges to the same sum.

Since the rearrangement was arbitrary, every rearrangement converges to S. ■

We now turn to the more surprising case, which is that if you *don't* have absolute convergence, then *literally anything can happen when you rearrange!* Note, the following theorem is certainly cool, and we are including its proof in full precisely for that reason, but its certainly not something you'd be expected to reproduce in full on an exam!

Theorem 2.6.40 Riemann Rearrangement Theorem, Conditional Case. *Let $\sum a_n$ be a conditionally convergent series of real numbers. Then by rearranging the terms of the series, one can make the resulting series converge to any prescribed real number, or even diverge to $\pm\infty$.*

Before jumping into the proof of this, we first give a lemma which will be necessary in the proof. It tells you, essentially, that if you have a conditionally convergent series and throw away all the negative (or positive!) terms, then what you're left with must diverge.

Lemma 2.6.41 *Let $\sum a_n$ be a conditionally convergent series of real numbers. Define*

$$a_n^+ = \max(a_n, 0), \qquad a_n^- = \max(-a_n, 0).$$

Notice that if $a_n < 0$ then $a_n^+ = 0$ and similarly, if $a_n > 0$ then $a_n^- = 0$. Then

$$\sum a_n^+ = \infty \quad \textit{and} \quad \sum a_n^- = \infty.$$

In particular, both the positive and negative parts have divergent (infinite) sums.

Remark 2.6.42 In the statement of the above lemma, we were faced with a notational choice to make. To represent the negative terms, we could have kept them negative (in which case, we would always have to be conscience that a_n^- is actually a negative number). We chose instead to conform to the mathematical community on this matter, which is to flip the sign of the negative terms in advance, i.e., so that a_n^- is actually a *positive* number. This has its drawbacks as well, of course, but is the more standard choice.

Proof. (Lemma 2.6.41) Since $\sum a_n$ converges, we have $a_n \to 0$. But because the

convergence is not absolute, we also have

$$\sum |a_n| = \infty.$$

Now observe that

$$|a_n| = a_n^+ + a_n^-, \text{ and } a_n = a_n^+ - a_n^-.$$

If both $\sum a_n^+$ and $\sum a_n^-$ were finite, then

$$\sum |a_n| = \sum (a_n^+ + a_n^-) < \infty,$$

contradicting conditional convergence.

So at least one of $\sum a_n^+$ or $\sum a_n^-$ must diverge to $+\infty$.

We now show both diverge. Suppose instead that $\sum a_n^+ < \infty$ but $\sum a_n^- = \infty$. Then

$$\sum a_n - \sum a_n^+ = -\sum a_n^-.$$

The LHS above would be a number (the sum of two convergence series) whereas the RHS is infinite, which is a contradiction. The other case of $\sum a_n^+ = \infty$ and $\sum a_n^- < \infty$ is similar.

Note in the above work, we have been freely using series summation 2.6.34 without saying so, but we think you're OK with that at this point. ■

Before we leap into the proof, let's consider how we can use the previous lemma to get the result. To make this concrete, let's think of taking the alternating harmonic series $\sum (-1)^n/n$ (which is conditionally - not absolutely! - convergent) and let's see how to rearrange it to make it add up to $\sqrt{2}$ (which is *not* the value of the alternating harmonic series, recall). Notice that all the even terms are positive and all the odd terms are negative.

- Since $\sqrt{2} > 0$ we start by taking positive terms until the partial sum is *just* bigger than $\sqrt{2}$:

$$1/2 + 1/4 + \cdots + 1/18 \approx 1.414484127 > \sqrt{2}$$

- Now since we have gone over, start adding in negative terms (from the beginning of the list) until the partial sum drops below $\sqrt{2}$. In this case, we only need the first negative term, because it's pretty big (compared to how far above $\sqrt{2}$, that is).

$$1/2 + 1/4 + \cdots + 1/18 - 1 \approx 0.414484127 < \sqrt{2}$$

- The partial sum is too low, so we have to add (unused) positive terms until it exceeds $\sqrt{2}$ again. We know that we can do this because, as per the previous lemma, the sum of all the positive terms diverges. In this case

$$1/2 + 1/4 + \cdots + 1/18 - 1 + 1/20 + \cdots + 1/140 \approx 1.416418379 > \sqrt{2}$$

- Wow, we had to use a lot of positive terms to do that. But that's OK because we have an infinite supply of them![19] Now that we have just gone over our desired sum, let's add in some more negative terms (way back near the beginning where we left them!) until we dip below $\sqrt{2}$:

  $$1/2+1/4+\cdots+1/18-1+1/20+\cdots+1/140-1/3 \approx 1.083085046 < \sqrt{2}$$

- this is now too negative, so we add in positive terms until it goes over $\sqrt{2}$.
- Keep iterating over, under, over, under \dots

This is essentially the proof. It's not too bad to wrap your head around for this particular choice of example. But, as you well know, an example doesn't constitute a general proof. Let's do it.

Proof. (Theorem 2.6.40) By the previous lemma 2.6.41, if $\sum a_n$ is conditionally convergent, then

$$\sum a_n^+ = \infty \qquad \text{and} \qquad \sum a_n^- = \infty.$$

We will first deal with the *any real number case*, i.e., we let $L \in \mathbb{R}$ be arbitrary, and we will construct a rearrangement of the series whose partial sums converge to L.

List the positive terms of the series in their original order:

$$p_1, p_2, p_3, \dots,$$

and the negative terms in their original order:

$$q_1, q_2, q_3, \dots.$$

Since

$$\sum p_k = \infty,$$

we may choose m_1 minimal such that

$$p_1 + \cdots + p_{m_1} > L.$$

Next, since

$$\sum q_k = -\infty,$$

we may choose n_1 minimal such that

$$p_1 + \cdots + p_{m_1} + q_1 + \cdots + q_{n_1} < L.$$

[19] You should, of course, be concerned if you see a marketing pitch offering higher payouts to an infinite number of people. That's called a pyramid scheme. In a similar way to what we're doing, the math checks out, but the number of people willing to buy in doesn't! See also Ponzi scheme.

Note here we're always working with *finite* sums, and in particular, these finite sums can be rearranged to be in the (convenient) order presented above, instead of in the order in which they appeared in the original series.

Continue inductively. Having chosen m_j and n_j, choose $m_{j+1} > m_j$ minimal such that

$$p_1 + \cdots + p_{m_{j+1}} + q_1 + \cdots + q_{n_j} > L,$$

and then choose $n_{j+1} > n_j$ minimal such that

$$p_1 + \cdots + p_{m_{j+1}} + q_1 + \cdots + q_{n_{j+1}} < L.$$

This defines a rearrangement of the original series, since every positive term and every negative term appears exactly once.

Now (by slight abuse of notation), let's use (s_n) to denote the sequence of partial sums of this *rearranged* series (not the original series). We need to show that these partial sums *actually converge* to L. This is a bit delicate.

Define

$$u_j = p_1 + \cdots + p_{m_j} + q_1 + \cdots + q_{n_{j-1}},$$

so that $u_j > L$, and

$$v_j = p_1 + \cdots + p_{m_j} + q_1 + \cdots + q_{n_j},$$

so that $v_j < L$. In other words, u_j (resp. v_j) is the partial sum in which we cross above (resp. below) L for the j'th time.

Because m_j was chosen minimally, i.e., the *first* time that we cross L after adding positive terms, if we remove the last term from the partial sum u_j, then it is too small. More precisely,

$$u_j - p_{m_j} \leq L.$$

Similarly, the same is true for the n_j. If we remove the last term from the partial sum v_j then it is too big:

$$L \leq v_j + |q_{n_j}|.$$

That is

$$u_j - L \leq p_{m_j} \qquad \text{and } L - v_j \leq |q_{n_j}|$$

Since the original series converges, we have $a_n \to 0$. Hence these positive and negative summands in our series must also converge to zero:

$$p_{m_j} \to 0 \qquad \text{and} \qquad q_{n_j} \to 0.$$

Therefore,

$$u_j \to L \qquad \text{and} \qquad v_j \to L.$$

Now every partial sum s_n lies between some consecutive pair u_j and v_j, because during each stage we add terms of only one sign and cross from one side of L to the other exactly once. We leave it to the reader to finish off the proof precisely, using a Squeeze Theorem-esque argument (see also Lemma 2.3.3).

To obtain divergence to $+\infty$, construct a rearrangement by adding enough positive terms to make the partial sum exceed 1, then add one negative term, then enough positive terms to exceed 2, then one negative term, and so on. Since the negative corrections can be made arbitrarily small (because $a_n \to 0$), the partial sums tend to $+\infty$. The $-\infty$ case is similar. We leave the precise proof as an exercise. ■

2.6.8 Exercises

1. **Now you do it (the "hard" way).** Work just as in Example 2.6.6 to show that

$$\sum_{n=0}^{\infty} \frac{1}{3^n}$$

converges and deduce a formula for the value of the series.

2. e **is irrational.** The number e is classically defined as

$$\sum_{n=0}^{\infty} \frac{1}{n!}$$

a Why does the series defining e converge?

b With s_n denoting the partial sums of the series, show that

$$0 < e - s_n < \frac{1}{n!n}$$

c If e were rational, then we could write $e = p/q$ for integers p, q. Apply part (b) with $n = q$ and deduce the existence of an *integer* that lies strictly between 0 and 1, which is obvious nonsense (wait, why is this "obvious"?)

3. **A term-by-term product.** Suppose that $\sum a_n$ converges and that $(b_n)_n$ is a monotonic bounded sequence. Prove that $\sum a_n b_n$ converges. Is the result true if we do not assume that the sequence b_n is monotone?

Hint. The easiest way to prove this is likely to break into cases depending on what b_n converges to and to apply the Cauchy criterion for convergence.

4. **Alternating series test.** In this question, you will prove the *Alternating Series Test*, which is a test that can be applied to conclude the convergence of the alternating harmonic series for example.

Consider a series of the form

$$S = \sum_{n=1}^{\infty} (-1)^{n+1} a_n,$$

where the a_n are all nonnegative, monotone decreasing, i.e., $a_{n+1} \leq a_n$ for all n, and go to zero $a_n \to 0$. We will show that the series S converges.

a Denote by S_n the partial sums. Show that the sequence of even partial sums

$$S_2, S_4, S_6, \ldots$$

is increasing, and likewise the odd partial sums are decreasing. Conclude that both of these subsequences converge converge.

b Consider the difference

$$S_{2k-1} - S_{2k}$$

to conclude that the even partial sums and odd partial sums converge to the same limit. Conclude that the overall series S converge.

5. Some series. In each of the following, determine whether the series converges or diverges. If it converges, determine if it converges absolutely.

a

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^{3/2}}$$

b

$$\sum_{n=2}^{\infty} \frac{(-1)^{n}}{\ln(n)}$$

c

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n+1}$$

6. Dyadic comparison test. Let (a_n) be a sequence of nonnegative real numbers which is monotone decreasing. In this question, you will prove that the series

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if the dyadic series

$$\sum_{k=0}^{\infty} 2^k a_{2^k}$$

converges.

a For each $k \geq 0$, compare the block

$$a_{2^k} + a_{2^k+1} + \cdots + a_{2^{k+1}-1}$$

with $2^k a_{2^{k+1}}$ and $2^k a_{2^k}$. Conclude that

$$2^k a_{2^{k+1}} \leq \sum_{n=2^k}^{2^{k+1}-1} a_n \leq 2^k a_{2^k}.$$

b Use the comparison test for series to show that the convergence of the original series is equivalent to the convergence of the dyadic one.

Remark: This criterion is called the *Dyadic Comparison Test* or *Cauchy's Condensation Test*.

7. ***p*-series.** Use the previous exercise to conclude that

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

8. **A very fine line.** In this question, we will consider more series that look similar to the harmonic one, in particular seeing that the line between convergence and divergence is much more subtle than we may have thought.

a Prove that

$$\sum_{n=2}^{\infty} \frac{1}{n \log(n)^p}$$

converges if and only if $p > 1$.

Hint: This is similar to the previous exercise.

b Prove that

$$\sum_{n=3}^{\infty} \frac{1}{n \log(n)(\log(\log(n)))^p}$$

converges if and only if $p > 1$.

Hint: After dyadic comparison, you will have to find a way to deal with the $\log(k \log(2))$ term. Try to compare this to just $\log(k)$ (up to a constant). If you're having trouble with this step, first just solve the problem by replacing the $\log(k \log(2))$ with $\log(k)$.

c Use a computer (spreadsheet or otherwise) to calculate a portion of the tail of the sum, say something like 100 terms:

$$\sum_{n=N}^{N+100} a_n$$

(where N can be taken arbitrarily large) for both of the series

$$\sum_{n=3}^{\infty} \frac{1}{n \log(n) \log(\log(n))}$$

and

$$\sum_{n=3}^{\infty} \frac{1}{n \log(n)(\log(\log(n)))^2}.$$

Compare their relative size and in particular, how slowly they grow to be different! Remember, one of these series converges and one does not!

If you're having trouble building such a spreadsheet (this is a useful skill for you to develop), we have drafted one up for you: at this link[20]. The cell A1 is the N above, and the $K2$ is the size of one partial sum divided by the other one. Change N to be larger and larger to see how slowly the quotient grows.

Remark: Evidently, you could continue this pattern indefinitely (try writing down a general statement!). One might be led to conjecture that there is some "boundary" for which all series on one side converge and all on the other diverge. Unfortunately (or fortunately, depending on your worldview), such a result is impossible, though this is beyond the scope of this course to investigate.

9. **Mertens'[21] Theorem.** Suppose that $A = \sum_{n=0}^{\infty} a_n$ and $B = \sum_{n=0}^{\infty} b_n$ are both convergent and (at least) one of them converges absolutely. Show that the Cauchy product series converges to the series product. That is to say,

$$\sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) = AB$$

a Write A_n, B_n for the partial sums of the A and B series respectively. Likewise, write

$$C_n = \sum_{j=0}^{n} \left(\sum_{k=0}^{j} a_k b_{j-k} \right)$$

Furthermore, without loss of generality, assume it is A that converges absolutely. Write $\beta_n = B_n - B$.

Prove that

$$C_n = A_n B + a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$$

Remark: This seems like an unmotivated thing to do. It also seems a little daunting, but we promise it isn't *too* bad if you just work your way through the definitions. Bear with us.

b Show that – in order to prove the desired result – it suffices to show that

$$\gamma_n := a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$$

converges to zero.

[20] docs.google.com/spreadsheets/d/1TPExwxkrPHWl6i3I5wYsz3Z_e7zG-p6EZs_ _ZUHqiy4/edit?usp=sharing

c Split up the sum defining γ_n into two pieces: one of the pieces will vanish as $n \to \infty$ because the $a_n \to 0$; try to use the fact that each $\beta_k \to 0$ as well as the absolute converge of the A series to show that the other sum can be made arbitrarily small.

10. Convergence practice. For each of the following, determine the convergence or divergence of the series $\sum a_n$. Fully justify your answers (like you would on a midterm... cough cough)

a $a_n = \dfrac{1}{n^2+1}$

b $a_n = \dfrac{\sin(n)}{n^2}$

c $a_n = \dfrac{(-3)^n}{n!}$

d $a_n = \dfrac{1}{n^{3/4}}$

e $a_n = \dfrac{1}{n \ln n}$

f $a_n = (-1)^n \dfrac{n}{n^2+1}$

g $a_n = (-1)^{n+1} \dfrac{1}{\sqrt{n}}$

h $a_n = \dfrac{3^n}{5^n+n}$

11. An important trick. Suppose that $\sum_{n=1}^{\infty} a_n$ is a convergent series of nonnegative terms. Show also that

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n}$$

converges.

Hint: You could apply the Cauchy-Schwarz inequality (Google!) or, more simply, the inequality $\sqrt{xy} \leq \frac{x+y}{2}$ which holds for all nonnegative real numbers x, y (why?).

[21] Not Martens' (unfortunately)

Chapter 3

Continuity

In this section we'll revisit some of the ideas from differential calculus on continuity and now *prove* them! We'll prove a bunch of other things along the way of course.

We will assume that you recall the definition of a function, and start from there. We will also assume that you have understood the $\varepsilon - N$ proof work from the previous chapter.

Many of the definitions and results in the coming sections rely on functions having nice properties. Typically we don't need those properties to hold everywhere on the real line, but we also need them to hold at more than just a single point. We usually need them to hold for all points that are "close" to a given point on the real line. To this end we define the following.

Definition 3.0.1 Balls and Neighbourhoods. Let $\varepsilon > 0$. The **ε-ball** of a point $q \in \mathbb{R}$ is the open interval

$$(q-\varepsilon, q+\varepsilon) = \{x \in \mathbb{R} \mid |q-x| < \varepsilon\}$$

The **punctured ε-ball** of q is

$$(q-\varepsilon, q+\varepsilon) - \{q\} = (q-\varepsilon, q) \bigcup (q, q+\varepsilon) = \{x \in \mathbb{R} \mid 0 < |q-x| < \varepsilon\}$$

A set $A \subseteq \mathbb{R}$ is a **neighbourhood** of $q \in \mathbb{R}$ when there exists an $\varepsilon > 0$ so that A contains an ε-ball of q. We similarly define a **punctured neighbourhood** of a point. $\Diamond$

Remark 3.0.2 Precise and loose. Note that one reason that we define both **balls** and **neighbourhoods** is that they allow us to do slightly different things. A ball is very useful when we need to be very precise. A ball knows exactly which elements are included and which are not; we'll need this to define limits, for example. On the other hand, sometimes, it is useful to say "there is some set around this point" without having to go to the extra work of specifying exactly what it is. For example, to ensure certain limits actually exist, we need that a given function is continuous in some non-trivial region around the given

point. Now rather than writing "there exists some $\varepsilon > 0$ so that the function is continuous on the ball of radius ε around the point", we can write "the function is continuous in a neighbourhood of the point". Much easier.

Also notice that the definition of neighbourhood allows us to simplify our earlier definition of an open set of reals, Definition 2.2.3. A set of reals, A, is open if for every point there is a neighbourhood about that point that is contained in A. Actually, we can go further, the set A is open when it is a neighbourhood for all its elements.

3.1 Limits of functions

We saw in Chapter 2 a notion of convergence of a sequence $x_n \to a$. In this section we will define what it means for a *function* to converge. You have doubtless seen this in a first course in differential calculus, but here we will formalise this notion, as well as see how this is related to sequence convergence.

Let us start with the rigorous definition of the limit of a function. This goes back initially to Bolzano around 1817, and then Cauchy in his 1821 text *Cours d'analyse*, and finally in 1861 Weierstrass introduced the $\varepsilon - \delta$ notation[1]. The interested reader should look at this excellent article by JV Grabiner[2] and this other excellent article by GI Sinkevich[3] for more history of this topic.

Definition 3.1.1 Limit. Let f be a function on the reals defined on some punctured neighbourhood of the point $a \in \mathbb{R}$. That is, there is some $\zeta > 0$ so that f is defined on at least the set of points $(a - \zeta, a) \cup (a, a + \zeta)$; it need not be defined at a .

Then we say that the **limit** of f as x approaches a is L, which we denote as

$$\lim_{x \to a} f(x) = L \qquad \text{or} \qquad f(x) \to L \text{ as } x \to a$$

when the following holds

- for every real $\varepsilon > 0$, there exists real $\delta > 0$ so that $0 < |x - a| < \delta$ implies that $|f(x) - L| < \varepsilon$.

$\Diamond$

Remark 3.1.2 Informally, the definition of limits says if $f(x) \to L$ as $x \to a$, then we can make $f(x)$ arbitrarily close to L by taking x closer and closer to a .

Note that we do not require that $f(a)$ exists, so it is important that the condition that "x is close to a" is written as $0 < |x - a| < \delta$ and not as just $|x - a| < \delta$.

This definition can be re-expressed quite concisely in terms of neighbour-

[1] To be a little more careful, we have lecture notes written in 1861 by a student of Weierstrass.

[2] core.ac.uk/download/pdf/148357947.pdf

[3] arxiv.org/pdf/1502.06942.pdf

hoods 3.0.1: for any $\varepsilon > 0$, there is some $\delta > 0$ so that when x is in the punctured δ-ball of a, we know that $f(x)$ is in the ε-ball of L.

Example 3.1.3 A first limit. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = 3x + 2$. Then

$$\lim_{x \to 1} g(x) = 5$$

Hint. So given any $\varepsilon > 0$ we have to find δ so that

$$|x - 1| < \delta \quad \Longrightarrow \quad |(3x + 2) - 5| < \varepsilon.$$

As is often the case with inequalities, our scratch work starts with the inequality we want to prove, and then works backwards. So we want

$$|(3x + 2) - 5| < \varepsilon.$$

So, clean things up a bit $|3x - 3| = 3|x - 1|$ — hiding in there is the term we need to make small! This tends to be how these proofs work. So

$$\varepsilon > |(3x + 2) - 5| = 3|x - 1|$$

So, if we make $|x-1| < \varepsilon/3$ then we can ensure that we get what we want. That is, set $\delta = \varepsilon/3$ (or anything smaller).

Running this logic backwards gives the proof.

Solution. Let $\varepsilon > 0$.[4] Take $\delta = \varepsilon/3$ and let x be a real number so that $|x - 1| < \delta$. Then

$$|(3x + 2) - 5| = 3|x - 1| < 3\delta = \varepsilon$$

and so $\lim_{x \to 1}(3x + 2) = 5$ as required. □

Example 3.1.4 Another limit. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Then

$$\lim_{x \to 2} f(x) = 4$$

Hint. Given any $\varepsilon > 0$ we need to find δ so that when $|x - 2| < \delta$ we know that $|x^2 - 4| < \varepsilon$. Again, start with the conclusion and work backwards — we still have to make sure our logic is sound when we get to the actual proof.

$$\varepsilon > |x^2 - 4| = |x - 2||x + 2|$$

Now, we know that we are going to make x really close to 2 and so we expect that the term $x + 2$ is going to be pretty close to 4. Of course we can't say "its close to 4" in a proof, we need to make that rigorous. So lets assume that $|x - 2| < 1$; since we have control over δ, this is just making sure that $\delta \leq 1$. Juggling this inequality around:

$$|x - 2| < 1$$

[4]It is kind of a joke in analysis that all proofs start with "Let $\varepsilon > 0$." So much so, in fact, that one of the authors has a t-shirt with that sentence printed on it

$$-1 < x - 2 < 1$$
$$3 < x + 2 < 5$$

And thus we know that the $|x + 2| < 5$ and

$$\varepsilon > 5|x - 2| > |x - 2||x + 2|$$

We need to make sure that $|x - 2| < \frac{1}{5}\varepsilon$. That is, we need $\delta < \frac{\varepsilon}{5}$.

Running this reasoning backwards gives us our proof.

Solution. Let $\varepsilon > 0$ and set $\delta = \min\{1, \varepsilon/5\}$. Now let x be a real number so that $|x - 2| < \delta$.

Since $|x - 2| \leq 1$, we know that $3 < x + 2 < 5$, and thus

$$|x - 2||x + 2| < 5|x - 2| < 5\delta$$

and since $\delta \leq \varepsilon/5$ we get the required inequality

$$|x^2 - 4| = |x - 2||x + 2| < \varepsilon.$$

□

Example 3.1.5 A less simple limit. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \sqrt{x}$. Then

$$\lim_{x \to 9} f(x) = 3$$

Hint. Given any $\varepsilon > 0$ we need to find δ so that when $|x - 9| < \delta$ we know that $|\sqrt{x} - 3| < \varepsilon$. Again, let's start with what we want and work backwards

$$|\sqrt{x} - 3| < \varepsilon$$

and thus

$$3 - \varepsilon < \sqrt{x} < 3 + \varepsilon$$

Squaring both sides gives

$$(9 - 6\varepsilon + \varepsilon^2) < x < (9 + 6\varepsilon + \varepsilon)^2$$

Rearrange a bit more

$$-6\varepsilon + \varepsilon^2 < x - 9 < 6\varepsilon + \varepsilon^2$$

Notice that this is of the form $-a < x - 9 < b$ and we want an expression of the form $-c < x - 9 < c$ so that we can rewrite it as $|x - 9| < c$. Consequently we should use the *minimum* of either side. Hence we want

$$|x - 9| < \min\left\{6\varepsilon - \varepsilon^2, 6\varepsilon + \varepsilon^2\right\} = 6\varepsilon - \varepsilon^2$$

(of course, this makes no sense if $\varepsilon > 6$ since it will become negative).

Running this reasoning backwards gives us our proof.

Hold on, we can do better than that using difference of squares. Notice that

$$|x-9| = |(\sqrt{x}-3)(\sqrt{x}+3)| = |\sqrt{x}-3| \cdot |\sqrt{x}+3|$$

Now, since $\sqrt{x} \geq 0$, we know that $|\sqrt{x}+3| \geq 3$. And thus

$$|x-9| \geq 3|\sqrt{x}-3|$$

So, if we make $|x-9|$ nice and small, then $|\sqrt{x}-3|$ will be small too.

$$|\sqrt{x}-3| \leq \frac{1}{3}|x-9| \leq \varepsilon$$

Hence, we need to make $|x-9| \leq 3\varepsilon$. That is $\delta = 3\varepsilon$.

Solution. Let $\varepsilon > 0$ and set $\delta = 3\varepsilon$. Now let x be a real number so that $|x-9| < \delta$. Then

$$\begin{aligned} |\sqrt{x}-3| &= \frac{|x-9|}{\sqrt{x}+3} \\ &\leq \frac{1}{3}|x-9| && \text{since } |\sqrt{x}+3| \geq 3 \\ &\leq \frac{\delta}{3} = \varepsilon \end{aligned}$$

Hence the limit is 3 as required. □

Notice how in each of the previous examples, the finalized polished solution starts with an arbitrary $\varepsilon > 0$ and then magically pulls a corresponding δ out of the hat that just works. Of course, all that "magic" has to be thought through ahead of time. That is how these proofs usually go; first you work through the problem backwards to discover the "proper" value of δ, then when you're ready to write the good-copy proof, you start by defining that value of δ and reveal the trick.

3.1.1 Limits and sequences

We have now seen two different notations of convergence:

- that of converging sequences of numbers $x_n \to a$ as $n \to \infty$ in Section 2.3; and
- that of convergence of functions $f(x) \to L$ as $x \to a$.

Of course, these two notions are not unrelated! For instance, imagine you have a converging sequence of points $x_n \to a$ and a function f defined in a neighborhood of a such that $\lim_{x \to a} f(x) = L$. Now you can define a *new* sequence of points by taking the images of the x_n under the function f, i.e.,

$$y_n = f(x_n).$$

The natural question to ask is whether or not it is true that this sequence of points satisfies

$$\lim_{n\to\infty} y_n = \lim_{n\to\infty} f(x_n) \overset{???}{=} L.$$

In other words, we are asking whether or not the two notions of convergence that we have defined agree?

Thankfully, the answer is **yes**. In fact, these two notions of convergence are *equivalent*. The following results make this more precise.

Lemma 3.1.6 *Let (x_n) be a sequence that converges to a but for which $x_n \neq a$ for every $n \in \mathbb{N}$. Let f be a function for which $\lim_{x\to a} f(x) = L$. Then*

$$\lim_{n\to\infty} f(x_n) = L$$

Note: *the function f may not even be defined at a for this to work! Of course, the most useful application of this result will be when f is* ***continuous*** *at a, i.e., when $L = f(a)$. See the next section 3.2 for this discussion.*

Proof. Let (x_n) and f be as stated. Let $\varepsilon > 0$, we need to show that there exists N so that for all $n > N$ we have $0 < |f(x_n) - L| < \varepsilon$.

Since $\lim_{x\to a} f(x) = L$, there exists $\delta > 0$ so that when $0 < |x - a| < \delta$ we have $|f(x) - L| < \varepsilon$. Then since $x_n \to a$ (and $x_n \neq a$) we know that there is N so that when $n > N$, $0 < |x_n - a| < \delta$.

So let $n > N$, then we know that $0 < |x_n - a| < \delta$ and thus $|f(x_n) - L| < \varepsilon$. Hence $f(x_n) \to L$ as required. ■

The converse of this is also true. Namely

Lemma 3.1.7 *Let f be a function. If for all sequences (x_n) with $x_n \neq a$ for every $n \in \mathbb{N}$,*

$$\text{if } \lim_{n\to\infty} x_n = a \text{ then } \lim_{n\to\infty} f(x_n) = L,$$

then $\lim_{x\to a} f(x) = L$.

Proof. We prove the contrapositive of this statement. That is, if $\lim_{x\to a} f(x) \neq L$, then there exists a sequence (x_n) with $x_n \neq a$ so that $x_n \to a$, but $f(x_n) \not\to L$. So from the non-existence of the f-limit we need to construct a sequence x_n that does what we need.

So assume that $\lim_{x\to a} f(x) \neq L$. This means that there is some $\varepsilon > 0$ so that for any $\delta > 0$ there is some $x \in (a - \delta, a + \delta) \setminus \{a\}$ so that $|f(x) - L| \geq \varepsilon$. We use this to define a sequence.

In particular, fix this value of ε and then set $\delta = \frac{1}{n}$ for any $n \in \mathbb{N}$. Hence there is some $x_n \in (a - \delta, a + \delta) \setminus \{a\}$ so that $|f(x_n) - L| \geq \varepsilon$. This sequence (x_n) converges to a, and each $x_n \neq a$ for all n, but $f(x_n)$ does not converge to L. Hence the contrapositive holds. ■

Combining these two lemmas we have a nice result that explains the relationship between limits and convergence of sequences.

Corollary 3.1.8 Sequences and Limits. *The following two statements are logically equivalent*

- $\lim_{x\to a} f(x) = L$
- *for all sequences* (x_n) *with* $x_n \neq a$ *for every* $n \in \mathbb{N}$, *if* $x_n \to a$ *then* $f(x_n) \to L$.

3.1.2 Properties of Limits of Functions

All that work that we just did to show that limits of sequences and limits of functions are *essentially* the same is about to pay off! Recall we saw in a previous section that sequences satisfy the arithmetic that we expect 2.3.3.9. Using this previous work, we can establish the anaolgous statements for limits of functions. You are likely familiar with these already from differential calculus class.

Theorem 3.1.9 Arithmetic of limits. *Let* f, g *be functions defined on* $\mathbb{R}$ *except possibly at* $x = a$. *And let*

$$\lim_{x\to a} f(x) = F \qquad \text{and} \qquad \lim_{x\to a} g(x) = G$$

then

$$\lim_{x\to a}(f(x) \pm g(x)) = F \pm G$$
$$\lim_{x\to a}(f(x) \cdot g(x)) = F \cdot G$$
$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{F}{G} \qquad \text{provided } G \neq 0$$

Proof. The proofs of these statements are quite standard — see the treatment here[5] .

Alternatively, you can prove these statements using the analogous Arithmetic of limits 2.3.3.9 that you (hopefully!) proved when we were studying sequences. See Exercise 3.1.3.9! ■

We can also prove the sandwich- (or squeeze- or pinch-) theorem[6].

Theorem 3.1.10 The sandwich theorem. *Let* f, g, h *be functions on* $\mathbb{R}$ *satisfying*

$$f(x) \leq g(x) \leq h(x)$$

everywhere except perhaps $x = a$. *If* $f(x) \to L$ *and* $h(x) \to L$ *as* $x \to a$, *then we*

[5] https://personal.math.ubc.ca/~CLP/CLP1/clp_1_dc/sec_proof_arith_lim.html

[6] We note that there is also the less-kosher (and somewhat unrelated!) "ham-sandwich theorem". The interested reader should forgive the pun and search engine their way to the result.

also have

$$\lim_{x\to a} g(x) = L.$$

Proof. Let $\varepsilon > 0$. Now

- since $f(x) \to L$ there exists $\delta_f > 0$ so that when $0 < |x-a| < \delta_f$, $|f(x) - L| < \varepsilon$, and
- similarly since $h(x) \to L$ there exists $\delta_h > 0$ so that when $0 < |x-a| < \delta_h$, $|h(x) - L| < \varepsilon$.

Thus we know that when $0 < |x-a| < \delta = \min\{\delta_f, \delta_h\}$

$$L - \varepsilon < f(x) \leq g(x) \leq h(x) < L + \varepsilon$$

and so

$$-\varepsilon < g(x) - L < +\varepsilon$$

and so $|g(x) - L| < \varepsilon$. Hence $g(x) \to L$ as required. ■

3.1.3 Exercises

1. A simple limit. Use the ε-δ definition of a limit to prove

$$\lim_{x\to 5}(4x + 2) = 22$$

2. A slightly less simple limit. Use the ε-δ definition of a limit to prove

$$\lim_{x\to -2} x^3 = -8$$

As you can tell from doing the previous couple exercises, proving limits of "simple" functions directly from the ε-δ definition is tedious, at best. It is a good thing that there is a simpler way!

3. Limits of polynomials. In this exercise, we will deduce rigorously that all polynomials can be computed by just "plugging in" the limit point. In particular, this implies that polynomial functions are **continuous**, a concept introduced in the next section.

- If $f(x) = c$ is a constant function, prove that $\lim_{x\to a} f(x) = c$.
- If $f(x) = x$, prove that $\lim_{x\to a} f(x) = a$.
- For any polynomial $P(x)$ with real coefficients and any $a \in \mathbb{R}$, show that

$$\lim_{x\to a} P(x) = P(a).$$

4. Limits two ways. For each of the following, prove the desired limits in two different ways: (1) directly using the ε-δ definition; and (2) using properties of limits that we've shown like arithmetic of limits, the "niceness"

of polynomials, etc.

- $\lim_{x\to 1} \frac{1}{x} = 1.$
- $\lim_{x\to 0} \frac{x^3 + 2x^2 + 1}{x^2 - 1} = -1.$

Hint. For the first limit, you can insist that δ is small enough so that you stay away from the "bad" point $x = 0$. Similarly, in the second limit you can choose δ to avoid the bad points at $x = \pm 1$.

5. More limits. Prove the following limits *without* using compositions of continuous functions (we haven't technically learned what a **continuous** function is or how it behaves nicely with limits). Instead, use tools that we have developed so far.

- $\lim_{x\to -1} |2x - 1| = 3.$
- $\lim_{x\to 0} \sqrt{1 + x^2} = 1.$

6. Prove limit is not 3. Suppose someone claims that

$$\lim_{x\to 2} f(x) = 3.$$

What would you have to show to prove that this is *not* the case?

7. Quantifiers are important. Why is it important that the definition of a limit uses *for all* $\varepsilon > 0$ What would happen instead if it just required that the logical statement held *for some* $\varepsilon > 0$?

8. The limit does not exist![7]**.** Each of the following limits do not exist, at least according to our definition. Prove this is the case, and state in words what the problem is.

- $\lim_{x\to 1} \sqrt{x^2 - 1}$
- $\lim_{x\to 0^+} \log(x)$
- $\lim_{x\to 0} \frac{|x|}{x}$

9. Arithmetic of limits. Prove Theorem 3.1.9 using Exercise 2.3.3.9.

Takeaway: Arithmetic of limits of functions and arithmetic of limits of sequences are "basically" the same!

10. Infinite limits, revisited. Recall in Exercise 2.3.3.13, we saw what it

[7] A shout out to the movie "Mean Girls" in which the lead character wins a mathematics competition by correctly asserting that the limit does not exist.

meant for $x_n \to \infty$ for a sequnce $(x_n)_n$. Similarly, we can write

$$\lim_{x\to a} f(x) = +\infty \qquad (\text{resp. } = -\infty)$$

to mean that for all $M > 0$, there exists $\delta > 0$ such that whenever $0 < |x - a| < \delta$, there holds $f(x) \geq M$ (resp. $f(x) \leq -M$). In this case, we may use the words **diverges to infinity**, just as for sequences.

Or, instead of the image diverging to invinity, we could have the *domain* diverging to infinity. More precisely, we will write

$$\lim_{x\to\infty} f(x) = L \qquad (\text{resp. } x \to -\infty)$$

to mean that for all $\varepsilon > 0$, there exists $R > 0$ such that whenever $x \geq R$ (resp. $x \leq -R$), there holds $|f(x) - L| < \varepsilon$.

Or, just to upset you, we could mix both of the previous definitions and write

$$\lim_{x\to\infty} f(x) = +\infty$$

to mean that for all $M > 0$, there exists $R > 0$ such that whenever $x \geq R$, there holds $f(x) \geq M$. Of course, you could swap out $-\infty$ for $+\infty$ with proper modification, but we won't make you do that here.

Using these definitions, prove the following.

- Prove that $\lim_{x\to a} \dfrac{1}{(x-a)^2} = +\infty$.
- Show that $\lim_{x\to\infty} \dfrac{2x+5}{x+1} = 2$.
- Prove that $\lim_{x\to\infty} x^2 = +\infty$.

3.2 Continuity, Bolzano and the IVT

Consider the function

$$f : \mathbb{R} \to \mathbb{R} \qquad f(x) = x^2 - 2$$

From all our caluculusing we know that it has a minimum of -2 at $x = 0$ and then "smoothly" increases as x moves away from 0 in either direction. It doesn't jump around or have gaps; it is a nicely behaved function. Indeed, polynomials in general are nicely behaved — no jumps or gaps. They are **continuous**.

For example, our function f above, we know $f(1) = -1 < 0$ and $f(2) = 2 > 0$. Since the function does not jump or have gaps, being *continuous* we know that somewhere between these two points, it must touch the x-axis. This property is summarised by the Intermediate Value Theorem which is a main goal for this section.

Theorem 3.2.1 The intermediate value theorem. *Let $f : [a, b] \to \mathbb{R}$ be a continuous function with $f(a) \neq f(b)$. Then for all y between $f(a)$ and $f(b)$ there exists some $x \in [a, b]$ so that $f(x) = y$.*

This definitely does not hold for functions over the rationals. For example, if we take

$$g : \mathbb{Q} \to \mathbb{Q} \qquad g(x) = x^2 - 2$$

then $g(0) = -2$ and $g(2) = 2$ but there is not *rational* number $c \in [0, 2]$ so that $g(c) = 0$.

Since this fails for the rationals, but we know[1] that it works for the reals, this must somehow be a consequence of the completeness of the reals (or equivalently the existence of suprema). Indeed, our proof will rely on the existence of the supremum.

3.2.1 Continuity

Start with a definition.

Definition 3.2.2 Continuous. Let f be a real function defined in the neighbourhood of some point $c \in \mathbb{R}$. We say that f is **continuous** at c when

$$\lim_{x \to c} f(x) = f(c)$$

Equivalently, f is continuous at c when for all $\varepsilon > 0$ there exists $\delta > 0$ so that

$$0 < |x - c| < \delta \quad \Longrightarrow \quad |f(x) - f(c)| < \varepsilon.$$

That is, we can confine f to a given ε-ball of $f(c)$ by requiring that x lies inside a punctured δ-ball of c for some $\delta > 0$.

Finally, if f is continuous at every point in the interval $[a, b]$ then we say that is continuous on that interval. $\Diamond$

Notice that one can embellish this definition to handle **one-sided continuity**, where one approaches limits from the left or the right. This is important at the endpoints of intervals.

Definition 3.2.3 Left-continuous and right-continuous. Let $f : D \to \mathbb{R}$ be a function with domain $D \subseteq \mathbb{R}$. We say that f is **left-continuous** at the point $c \in D$ when for all $\varepsilon > 0$ there exists $\delta > 0$ so that

$$c - \delta < x < c \quad \Longrightarrow \quad |f(x) - f(c)| < \varepsilon.$$

Similarly, f is **right-continuous** at the point $c \in D$ when for all $\varepsilon > 0$ there exists $\delta > 0$ so that

$$c < x < c + \delta \quad \Longrightarrow \quad |f(x) - f(c)| < \varepsilon.$$

Note that when a point is both left- and right-continuous it is continuous. $\Diamond$

[1] Okay, we were *told* that it is true on the reals. We will prove it for ourselves soon.

Notice that these definitions really just say that "we can make $f(x)$ arbitrarily close to $f(c)$ by bringing x very close to c". This idea can be generalised by thinking not just of *points* x, c on the real line, but also points in more general spaces (e.g., higher dimensions). Also we can generalise the idea of *close*. We will return to these things later.

Example 3.2.4 Some continuous functions. Consider the function

$$f(x) : \mathbb{R} \to \mathbb{R} \qquad f(x) = x^2 - 2$$

It is continuous everywhere in its domain. So let $c \in \mathbb{R}$. We need to show that $\lim_{x\to c} f(x) = f(c)$; that is for any $\varepsilon > 0$ there exists $\delta > 0$ so that

$$0 < |x - c| < \delta \implies |(x^2 - 2) - (c^2 - 2)| < \varepsilon.$$

We start with what we want to show. Namely

$$\varepsilon > |f(x) - f(c)| = |(x^2 - 2) - (c^2 - 2)| = |x^2 - c^2| = |x + c| \cdot |x - c|$$

Of course, in the limit we'll have $x \approx c$ so we expect $|x + c| \approx 2|c|$. We can be more precise. Let's require that $|x - c| < 1$ (so we required $\delta < 1$). Then we have

$$|x + c| = |(x - c) + 2c| \leq |x - c| + 2|c| < 1 + 2|c|.$$

Thus we can write

$$|x + c| \cdot |x - c| < |x - c| \cdot (2|c| + 1) < \varepsilon$$

where we want to control the size of $|x-c|$ so that the rightmost inequality holds. So we need

$$|x - c| < \frac{1}{2|c| + 1}\varepsilon$$

That is, we also require $\delta < \frac{\varepsilon}{1+2|c|}$.

We can now put everything together.

Let $c \in \mathbb{R}$ and $\varepsilon > 0$. Then set

$$\delta = \min\left\{1, \frac{\varepsilon}{1 + 2|c|}\right\}$$

When $|x - c| < \delta$, we have

$$|x + c| < 2|c| + 1.$$

Now consider

$$\begin{aligned} |f(x) - f(c)| = |x^2 - c^2| &= |x + c| \cdot |x - c| \\ &< |x + c| \cdot \delta \\ &< (2|c| + 1) \cdot \frac{\varepsilon}{1 + 2|c|} = \varepsilon \end{aligned}$$

Thus f is continuous at $x = c$. Since c was an arbitrary real number, this tells us that f is continuous on $\mathbb{R}$. □

One can also prove that continuity and arithmetic play nicely together. Here is an example. We'll do more in homework.

Example 3.2.5 Sum of continuous functions is continuous. Let $f, g : D \to \mathbb{R}$ be functions on some domain D. If f, g are continuous at $a \in D$, then their sum $h(x) = f(x) + g(x)$ is also continuous at $x = a$.

Let $\varepsilon > 0$. Then since f, g are continuous at $x = a$ we know that

- there exists δ_f so that when $|x - a| < \delta_f$, $|f(x) - f(a)| < \varepsilon/2$, and
- there exists δ_g so that when $|x - a| < \delta_g$, $|g(x) - g(a)| < \varepsilon/2$.

Now let $h = f + g$, and set $\delta_h = \min\{\delta_f, \delta_g\}$. Then when $|x - a| < \delta_h$ we have

$$\begin{aligned} |h(x) - h(a)| &= |f(x) - f(a) + g(x) - g(a)| \\ &\leq |f(x) - f(a)| + |g(x) - g(a)| \\ &< \varepsilon \end{aligned}$$

and so h is continuous at $x = a$. □

Notice how important the triangle inequality is in these proofs. When we generalise the idea of "close", it had better still obey the triangle inequality[2].

Example 3.2.6 Reciprocal of a continuous function is continuous. Let $f : D \to \mathbb{R}$ be a continuous function so that $f(x) \neq 0$ for all $x \in D$. Then $\frac{1}{f(x)}$ is also continuous on D.

Answer. Let f be as stated and let $a \in \mathbb{D}$ and let $f(a) = b$. It suffices to show that $\lim_{x\to a} \frac{1}{f(x)} = \frac{1}{f(a)} = \frac{1}{b}$. This is given to us by Theorem 3.1.9.

We can also prove it from first principles. Since f is continuous at $x = a$ we know that there is some $\delta_1 > 0$ so that

$$|f(x) - f(a)| = |f(x) - b| < |b|/2 \qquad \forall x \in (a - \delta_1, a + \delta_1)$$

so that

$$|f(x)| > |b|/2 \qquad \forall x \in (a - \delta_1, a + \delta_1).$$

Let $\varepsilon > 0$. Since f is continuous at a, we know that there is some $\delta_2 > 0$ so that

$$|f(x) - f(a)| < \frac{b^2}{2}\varepsilon \qquad \forall x \in (a - \delta_2, a + \delta_2)$$

Now let $\varepsilon > 0$, then set $\delta = \min\{\delta_1, \delta_2\}$, so that

$$|f(x) - b| < \frac{b^2}{2}\varepsilon \qquad \text{and} \qquad |f(x)| > |b|/2.$$

Then

$$\left|\frac{1}{f(x)} - \frac{1}{b}\right| = \frac{|f(x) - b|}{|f(x)||b|} < \frac{2}{|b|^2}|f(x) - b| < \varepsilon.$$

Thus $\frac{1}{f}$ is continuous at $x = a$. □

[2]we will be doing a lot of generalising in Chapter 5 where we define a **metric**. Spoiler alert: one of the key properties that a metric must satisfy is a triangle inequality.

As an extension of this, one can show that the composition of continuous functions is also continuous.

Lemma 3.2.7 Composition of continuous functions. *Let f, g be functions so that*

- *f is continuous at c*
- *g is continuous at $d = f(c)$*

then $h = g \circ f$ is continuous at c.

Proof. We need to show that for any $\varepsilon > 0$ there exists some $\delta > 0$ so that

$$|h(x) - h(c)| = |g(f(x)) - g(f(c))| < \varepsilon \qquad \forall x \in (c - \delta, c + \delta).$$

Let $d = f(c)$. Since g is continuous at d, there exists some $\delta_g > 0$ so that

$$|g(t) - g(d)| < \varepsilon \qquad \forall t \in (d - \delta_g, d + \delta_g).$$

Then, since f is continuous at c, we know that there is some $\delta_f > 0$ so that

$$|f(x) - f(c)| < \delta_g \qquad \forall x \in (c - \delta_f, c + \delta_f).$$

So now, if $x \in (c - \delta_f, c + \delta_f)$, then we know that $f(x) \in (f(c) - \delta_g, f(c) + \delta_g) = (d - \delta_g, d + \delta_g)$. Hence $g(f(x))$ must lie inside $(g(f(c)) - \varepsilon, g(f(c)) + \varepsilon)$ as required. ■

One should be careful with the above result; it is the continuity of the functions that makes everything play together so nicely. As soon as functions are not continuous the limits and compositions can be quite badly behaved.

Lemma 3.2.8 Limits and compositions. *Let f, g be real functions so that*

- *f is defined in some punctured neighbourhood of c, and $\lim_{x \to c} f(x) = d$,*
- *g is defined in some punctured neighbourhood of d, and $\lim_{t \to d} g(t) = L$.*

If g is, in fact, continuous at d, then

$$\lim_{x \to c} g(f(x)) = \lim_{t \to d} g(t) = L.$$

If g is not continuous then $\lim_{x \to c} g(f(x))$ need not equal $\lim_{t \to d} g(t)$.

Proof. First, note that f may not be defined at c, nor g at d. We can, however, artificially recify this by defining auxiliary functions $\tilde{f}$ and $\tilde{g}$ that "fill in" these holes so-to-say. More precisely, define

$$\tilde{f}(x) = \begin{cases} f(x) & x \neq c \\ d & x = c \end{cases} \quad \text{and} \quad \tilde{g}(x) = \begin{cases} g(x) & x \neq d \\ L & x = d \end{cases}$$

Then $\tilde{f}$ is continuous in a neighborhood of c and $\tilde{g}$ is continuous in a neighborhood of d, so the previous result 3.2.7 applies, and therefore, $\tilde{g} \circ \tilde{f}$ is continuous at c and so

$$\lim_{x \to c} \tilde{g}(\tilde{f}(x)) = \tilde{g}(\tilde{f}(c)) = \tilde{g}(d) = L.$$

Recall, however, that when we write "$\lim_{x \to c} \tilde{g}(\tilde{f}(x))$", we don't care about what happens at the limiting point c, instead only as x *approaches* c. Therefore,

$$\lim_{x \to c} g(f(x)) = \lim_{x \to c} \tilde{g}(\tilde{f}(x)) = L.$$

When the continuity condition on g is dropped the statement is false. See Example 3.2.9 below; it shows that the result fails when g is not continuous. ■

Example 3.2.9 Compositions, limits, and discontinuity. The continuity condition in the previous lemma is required, without it the result is false. Consider the following functions.

$$f(x) = 1$$
$$g(t) = \begin{cases} 2 & t \neq 1 \\ 3 & t = 1 \end{cases}$$

Then we have $\lim_{x \to 0} f(x) = 1$ and $\lim_{t \to 1} g(t) = 2$, but

$$\lim_{x \to 0} g(f(x)) = 3 \neq \lim_{t \to 1} g(t) = 2.$$

Or we can give an alternate example in which we cannot even make sense of the expression $g(f(x))$ for values of x close to a. Consider, for instance, the function g defined only on the punctured interval $[-1, 0) \cup (0, 1]$ and

$$f(x) = \begin{cases} x \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Then, the expression $g(f(x))$ is not even defined at any point $x = \frac{1}{n\pi}$. In particular, we cannot even make sense of the composition as $x \to 0$, nevermind trying to prove the limit exists! □

3.2.2 Continuity and sequences

Recall that in the previous section, we saw that limits of functions is equivalent 3.1.8 to limits of sequences. In the setting in which the function is continuous, we can state this result in a slightly nicer form, for a few reasons. One reason is that we have the words "continuous" to mean precisely that the limit $\lim_{x \to a} f(x)$ exists **and equals** a particular constant — in particular, the function value $f(a)$. Another reason that things are a bit nicer in the case of dealing with continuous functions is that we can drop the requirement that the sequences

satisfy $x_n \neq a$, because if $x_n = a$, then the condition that $f(x_n) = f(a)$ holds trivially. To summarize precisely, we have:

Corollary 3.2.10 Sequences and continuity. *The following two statements are logically equivalent*

- *the function f is continuous at $x = a$*
- *for all sequences (x_n), if $x_n \to a$ then $f(x_n) \to f(a)$.*

Characterising continuous functions using sequences is often useful for proving that a given function is *not* continuous. Let's see a couple examples of this.

Example 3.2.11 Continuity of the sign function. Consider the **sign** function

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

This function is not continuous at $x = 0$.

We can show that this is not continuous by finding a sequence $x_n \to 0$ so that $f(x_n) \not\to f(0) = 0$. So take $x_n = 1/n$ for $n \in \mathbb{N}$. Then $f(x_n) = f(1/n) = 1$ for all $n \in \mathbb{N}$, so $f(x_n) \to 1 \neq f(0)$. By Corollary 3.2.10 the function is not continuous at $x = 0$. □

Example 3.2.12 A very discontinuous function. Let $R : \mathbb{R} \to \mathbb{R}$ be defined by

$$R(x) = \begin{cases} x & x \in \mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$$

This function is continuous at $x = 0$ but nowhere else.

We first show the function is continuous at $x = 0$. Let $\varepsilon > 0$ and set $\delta = \varepsilon$. Then when $0 < |x - 0| < \delta$ we have

$$|f(x) - f(0)| = |f(x)| \leq |x| < \delta = \varepsilon$$

Thus f is continuous at $x = 0$.

Now let $a \neq 0$.

- if $a \in \mathbb{Q}$, then let $x_n = a + \frac{\sqrt{2}}{n} \notin \mathbb{Q}$. Then $f(x_n) = 0$ for all n, and thus $f(x_n) \to 0$. However $f(a) = a \neq 0$.
- Now assume that $a \notin \mathbb{Q}$. Let x_n be a rational number in $(a+1/n, a+1/(n-1))$ (since the rationals are dense in the reals, we can always find such a number). Then $f(x_n) = x_n$ and so $f(x_n) \to a$. However $f(a) = 0 \neq a$

In either case $\lim f(x_n) \neq f(a)$ and so by Corollary 3.2.10 the function is not continuous at $x = a$. □

3.2.3 Bolzano's theorem

We sneak up on the IVT 3.2.1 by proving it for the special case of a function crossing the x-axis. This is Bolzano's Theorem which he proved it in 1817; the IVT follows quite directly from it. Some credit should also go to Stevin[3] who essentially proved the IVT for polynomials. Note that Bolzano didn't prove it in quite the same manner that we will, instead he used something more "bisection" flavoured; something like the method we used showing the relationship between suprema and nested-intervals.

Theorem 3.2.13 Bolzano's Theorem. *Let $f : [a, b] \to \mathbb{R}$ be a continuous function and suppose that $f(a)$ and $f(b)$ have opposite signs. Then there exists $c \in (a, b)$ so that $f(c) = 0$.*

Notice that this is almost the full IVT; indeed we get the whole IVT from this by adding a constant to the function. More on this soon.

Let's assume (without loss of generality) that $f(a) < 0$. Then form the set of all points where the function is negative:

$$L = \{x \in [a, b] \mid f(x) < 0\}$$

Since $a \in L$ it is non-empty, and is bounded above by b. So it has a supremum. We "just"[4] need to show that the supremum gives the desired result, namely the function at that point is zero. Continuity must also play a role in that.

Now to "just" prove this we'll need a little helpful lemma. It says that if a *continuous* function is positive at a point $x = c$ then it must stay positive in some small region around c. A similar statement holds for negative values. Also, this is definitely not true without continuity.

Lemma 3.2.14 *Let f be a continuous function. If $f(c) > 0$ then there exists some $\delta > 0$ so that $f(x) > 0$ for all $c - \delta < x < c + \delta$.*

A similar statement holds when $f(c)$ is negative.

Proof. Let f be continuous and assume that $f(c) = q > 0$. Since f is continuous we know that there is some $\delta > 0$ so that

$$c - \delta < x < c + \delta \quad \Longrightarrow \quad |f(x) - f(c)| < q.$$

But this implies that

$$-q < f(x) - f(c) < q$$

and so

$$f(c) - q < f(x) < f(c) + q$$

And since $q = f(c)$, this gives us $f(x) > 0$ as required. ■

[3] Simon Stevin was a Flemish mathematician (1548–1620). He is also credited with introducing Europe to our modern decimal notation. Actually he was credited (by Europeans) with inventing it until around the middle of the 20th century when someone realised that it had been introduced by medieval Islamic scholars before the year 1000.

[4] It is a dangerous word.

Proof of Bolzano's Theorem 3.2.13. Let f be as stated and assume that $f(a) < 0$ and $f(b) > 0$. If not, then multiply the function by -1.

Form the set of points at which the function is negative, namely

$$L = \{x \in [a, b] \mid f(x) < 0\}$$

Since $a \in L$ this is a non-empty set of reals that is bounded above by b. Consequently its supremum exists. Let $c = \sup L$. We will show that $f(c) = 0$.

Since f is continuous and $f(a) < 0$ and $f(b) > 0$, Lemma 3.2.14 tells us that $a < c < b$ — the zero must lie away from the endpoints.

Since $f(c) \in \mathbb{R}$ we know that exactly one of $f(c) < 0, f(c) > 0, f(c) = 0$ is true.

- Assume that $f(c) > 0$. Since f is continuous, Lemma 3.2.14 tells us that there is some $\delta > 0$ so that $f(x) > 0$ for all $x \in (c - \delta, c + \delta)$.

  However, this now tells us that $f(z) > 0$ for all $z \in (c - \delta, b]$. But this means $c - \delta$ is a smaller upper bound for the set L than c — contradicting the fact that $c = \sup L$.

- Now assume that $f(c) < 0$. Again we can apply Lemma 3.2.14. So there is some $\delta > 0$ so that $f(x) < 0$ for all $x \in (c - \delta, c + \delta)$. This means that the point $z = c + \frac{\delta}{2}$ satisfies $f(z) < 0$ and so $z \in L$. But $z > c$ and so c is not an upper bound, giving a contradiction.

We, therefore, must have $f(c) = 0$ as required. ■

We can also prove this result using nested intervals 2.2.2 .

Proof of Bolzano's Theorem 3.2.13 using nested intervals. Homework! ■

3.2.4 Proof of the intermediate value theorem

It is just a short step to extend Bolzano's theorem 3.2.13 to the IVT 3.2.1.

Proof of IVT 3.2.1. Let f be as stated and let y be some real between $f(a)$ and $f(b)$. Then define $g(x) = f(x) - y$. Since $f(a) \neq f(b)$ we know that $g(a) \neq g(b)$, and further, one must be positive and the other negative. Bolzano's theorem 3.2.13 then tells us there is some $c \in (a, b)$ so that $g(c) = 0$. From this we have $f(c) = y$ as required. ■

3.2.5 Exercises

1. **Removable, jump and infinite don't cover all possibilities.** You probably recall the words **removable**, **jump**, and **infinite** discontinuities. Match each of these words with one of the following mathematical sentences. For each one, explain why such a function cannot be continuous by our ε-δ definition. In other words, removable, jump, and infinite discontinuities that you worked with in first year calculus are truly places that fail the

rigorous definition of continuity that we're working with here.

- the limit $\lim_{x\to a} f(x)$ exists but does not equal $f(a)$.
- both $\lim_{x\to a^+} f(x)$ and $\lim_{x\to a^-} f(x)$ both exist but are not equal.
- at least one of $\lim_{x\to a^+} f(x)$ or $\lim_{x\to a^-} f(x)$ is infinite.

Give an example of a point of discontinuity that is *not* one of these three.

2. Not continuous at a particular point. Let

$$f(x) = \begin{cases} x^2 & x \neq 2 \\ 5 & x = 2 \end{cases}$$

Prove, using the definition of continuity, that f is not continuous at $x = 2$.

3. Use a sequence to prove discontinuous. Let

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Show that f is not continuous at $x = 0$.

Hint. It may be helpful to use Corollary 3.2.10.

4. Rational/irrational. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ -x & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Is f continuous at any point? Prove that your answer!

Hint. This is similar to Example 3.2.12.

5. A very standard IVT exercise. Show that the equation

$$x^3 + x - 1 = 0$$

has at least one solution on the interval $(0, 1)$.

6. IVT to find more than one solution. Use the intermediate value theorem to show that the equation

$$x^3 = 2x^2 + 3x - 3$$

has at least 3 solutions.

Hint. Show that the function changes sign multiple times.

7. **IVT Converse? It's a trap!** Suppose f is a function defined on some interval $[a, b]$ with the property that for every value between $f(a)$ and $f(b)$, there exists $c \in (a, b)$ such that $f(c)$ is that value. Is it true that f much be continuous? If so prove it, if not then provide a counter-example.

   Hint. Find a function that does work, and then put a discontinuity in there somewhere.

8. **Surprisingly matching weather.** Show that there is at least one other point on the earth that has *precisely* the same temperature as where you are currently (you may have to make some assumptions here).

   Hint. The earth has some places that are (famously) always hot and some other places that are (famously) always cold. They might help you.

9. **There must be a fixed point.** Suppose $f : [0, 1] \to [0, 1]$ is continuous. Show that f has a **fixed point**. That is, there is some point $c \in [0, 1]$ with $f(c) = c$.

   Hint. The IVT will help you, so consider the function $g(x) = f(x) - x$. Where is it positive? Where is it negative? What happens if it is zero?

10. **Invertible functions go up or go down but not both.** Prove that any continuous **invertible** function on $\mathbb{R}$ is either strictly increasing or strictly decreasing.

    Remember, to be invertible, it must be **bijective**, and the authors hope you remember what bijective means.

    Hint. If the function is *not* strictly monotone, then it either goes up then down, or vice versa. So you can find $a < b < c$ such that

    $$f(a) < f(b) \text{ and } f(c) < f(b)$$

    or the same with the inequalities reversed. Why is this a problem?

11. **An odd composition.** Suppose $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(f(x)) = -x$ for all $x \in \mathbb{R}$. Show that f is not continuous.

    Remember that function composition is associative.

    Hint. Let $g = f \circ f$, so that $g(x) = -x$. Notice that $g(g(x)) = x$, and so g is its own inverse. With some careful counting of compositions, you can tweak this argument to show that $f(x)$ is invertible. You can then use the previous problem to conclude a contradiction.

3.3 The extreme value theorem

The IVT 3.2.1 is, in some sense, a very accurate way to characterise what it means for a function to be continuous — if the function takes on this value over here and that value over there, then it takes on all the values in-between. One can, and should, ask about the converse of the statement. Are there functions that satisfy the intermediate value property without being continuous? Func-

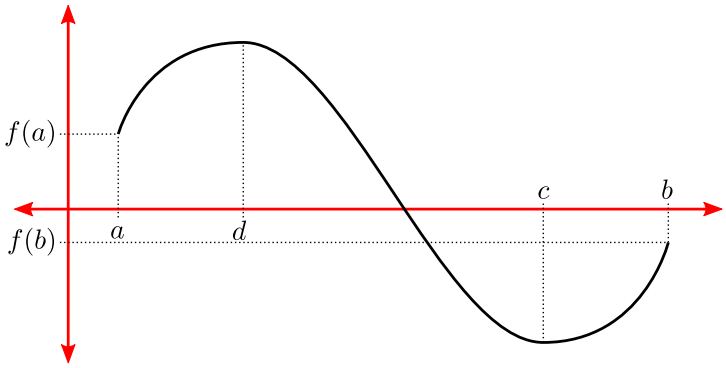
tions that satisfy the intermediate-value property are called **Darboux functions**.

Example 3.3.1 Not every Darboux function is continuous. Consider the function

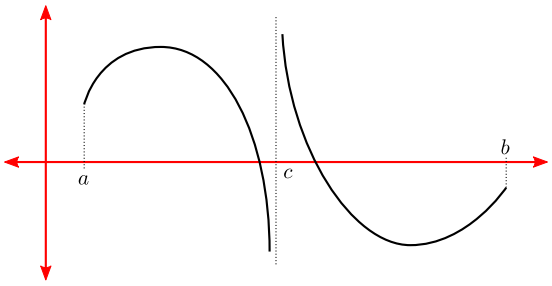
$$g : [0, \infty) \to [-1, 1] \qquad g(x) = \begin{cases} 0 & x = 0 \\ \sin(1/x) & x > 0 \end{cases}$$

This satisfies the intermediate-value property, so is Darboux, but is not continuous, as we saw in Exercise 3.2.5.3. □

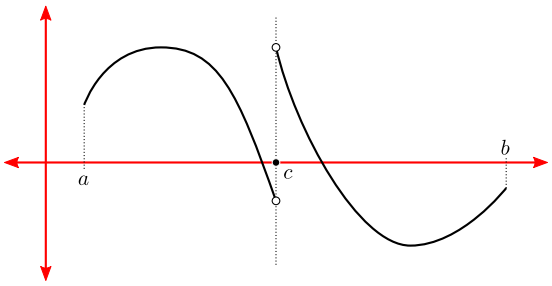
While Darboux functions are interesting, they are not something we are going to spend more time on (well, maybe on homework...). Consider the following image of a continuous function — you have likely seen something like it before; a sinusoidal squiggle starting from a increasing and then decreasing and then increasing and finally finishing at b. It takes its maximum value at some point $d \in [a, b]$ and its minimum at some $c \in [a, b]$, and thus is bounded $f(c) \leq f(x) \leq f(d)$.



Contrast this with a typical sketch of a discontinuous function. This function starts at $f(a)$ goes up a bit then down to $-\infty$ then "jumps" up to $+\infty$ then decreases for a while before finally arching up to $f(b)$. Since it "blows up" it is not bounded.



Of course, one can construct discontinuous functions that *do* have minimum and maximum values at points in their domain:



However, all continuous functions *must* take a minimum and maximum value in their domain and also be bounded. Actually, we had better be a bit more careful about this. Domains are not all created equal and we must be more precise in our statements.

Example 3.3.2 Be careful with the domain. Consider the functions

$$g : (0,1) \to (0,1) \qquad g(x) = x$$
$$h : (0,1) \to (1,\infty) \qquad h(x) = \frac{1}{x}$$

Both functions are continuous on their domains. Let $\varepsilon > 0$, and consider $a \in (0,1)$ then

- Set $\delta_g = \varepsilon$. For any $a - \delta_g < x < a + \delta_g$, we have $|g(x) - g(a)| = |x - a| < \varepsilon$
- Set $\delta_h = \min\{\frac{a^2\varepsilon}{2}, \frac{a}{2}\}$. For any x with $|x - a| < \delta_h$, we have $x > \frac{a}{2}$

$$|h(x) - h(a)| = \frac{|a - x|}{xa} < \frac{2|a - x|}{a^2} < \varepsilon$$

as required.

However, neither function takes a minimum nor a maximum on its domain. □

The critical difference here is that the domains are **open**; they do not include the endpoints $0, 1$. See Definition 2.2.1. The absence of those end points allows one to construct continuous functions that do strange things (like no maximum or minimum). Continuous functions on **closed** intervals behave better.

Before getting to that however, we would like to define what the words **bounded above** and **bounded below** mean for a function. Recall first from Chapter 2 that a function f being bounded 2.2.5 is equivalent to saying that there there exists some $M > 0$ so that $|f(x)| \leq M$ for all x in its domain.

Definition 3.3.3 Let $D \subseteq \mathbb{R}$ and let $f : D \to \mathbb{R}$. We say that f is **bounded above** if there exists some $M \in \mathbb{R}$ so that $f(x) \leq M$ for all $x \in D$. Similarly, we say that f is **bounded below** if there exists some $m \in \mathbb{R}$ so that $f(x) \geq m$ for all $x \in D$. ◊

The following lemma tells us that a continuous function on a closed interval must be bounded. Note that it does not tell us that the function has a maximum or minimum. Nor does it tell us what the bound is, just that it exists.

Lemma 3.3.4 Boundedness of continuous functions. *Let $g : [a, b] \to \mathbb{R}$ be a function on a closed interval of the real line. If g is continuous then it is bounded.*

A nested interval proof. Assume, to the contrary that g is not bounded above; a similar argument works when g is not bounded below. Let $a_1 = a, b_1 = b$ and let $I_1 = [a_1, b_1]$. Set $m = \frac{a+b}{2}$, then it must be the case that g is not bounded on either $[a, m]$ or $[m, b]$ (or potentially both). Set $I_2 = [a_2, b_2]$ be the subinterval on which g is unbounded (if there is a choice, then either will do). We can repeat this construction to get $I_2, I_3, \ldots$ satisfying

- $I_{n+1} \subseteq I_n$
- $|I_n| = (b - a)/2^{n-1}$, and
- g is unbounded on each I_n

The nested interval property then tells us that there is a unique $x \in \cap I_n$. Since g is unbounded above on each I_n, it follows that there exists some $x_n \in I_n$ so that $g(x_n) > n$ (otherwise the function would actually be bounded above). This sequence must converge to x. Since g is continuous, Corollary 3.2.10 implies that $g(x_n) \to g(x)$. However this gives a contradiction since we know that $g(x_n) > n$, the sequence $(g(x_n))$ diverges. Hence g must be bounded. ■

A proof using the supremum. Homework! ■

A sequencey proof. Assume, to the contrary that g is not bounded above; a similar argument works when g is not bounded below. Then this means that for any $n \in \mathbb{N}$ we can always find some $x \in [a, b]$ so that $g(x) > n$. This defines a sequence (x_n) contained in $[a, b]$. By

By Bolzano-Weierstrass 2.3.10 this sequence has a convergent subsequence (x_{n_k}). Notice that we must have $n_k \geq k$. Let the limit of this subsequence be α. And since $[a, b]$ is closed we know that $\alpha \in [a, b]$. Since g is continuous, Corollary 3.2.10 implies that

$$\lim_{x \to \infty} g(x_{n_k}) = g(\alpha)$$

However, $g(x_{n_k}) > n_k \geq k$ and so we must have that $g(x_{n_k})$ diverges. This gives a contradiction and so g must be bounded above. A nearly identical argument shows that g must also be bounded below. ■

The Extreme Value Theorem is really just a small extension of this making the result more precise. Not only is a continuous function on a closed interval bounded, it also takes a maximum and minimum.

Theorem 3.3.5 The extreme value theorem. *Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then there exists $c, d \in [a, b]$ so that*

$$f(c) \leq f(x) \leq f(d) \qquad \forall x \in [a, b]$$

and thus $f([a,b]) = [f(c), f(d)]$.

Proof. Let $B = \{f(x) \mid x \in [a,b]\}$ be the image of $[a,b]$ under f. It is trivially non-empty and by Lemma 3.3.4 we know that is is bounded. Hence its supremum $M = \sup B$ exists.

Now, either there is some $d \in [a,b]$ so that $f(d) = M$ or not. If there is, we are done. So assume that there is no such d. We will show that this gives a contradiction.

Let $g : [a,b] \to \mathbb{R}$ be defined by $g(x) = M - f(x)$. By assumption there is no $d \in [a,b]$ so that $g(d) = 0$. Additionally, it is a continuous function on $[a,b]$. Therefore, we saw in Example 3.2.6 that the reciprocal function

$$h(x) = \frac{1}{g(x)} = \frac{1}{M - f(x)}$$

is *also* a continuous function. Since it is continuous, it is also bounded.

But recall that $M = \sup B$, and so for all $\varepsilon > 0$ there is some $x \in [a,b]$ so that $f(x) > M - \varepsilon$. But this means that

$$h(x) = \frac{1}{M - f(x)} > \frac{1}{\varepsilon}$$

But this means that we can make h as large as we want — for example, setting $\varepsilon = \frac{1}{n}$ gives $h(x) > n$.

This contradicts Lemma 3.3.4. Thus we must have that $g(d) = 0$ for some $d \in [a,b]$ and thus there is some $d \in [a,b]$ so that $f(d) = M$ as required.

The proof for the minimum follows similarly. ■

Remark 3.3.6 Image of a closed interval under a continuous function. One very nice consequence of the EVT 3.3.5 is that it tells us that the image of a closed interval under a continuous function is also a closed interval.

One can cook up a simple example to show that continuity is required for this to be true. Consider the function

$$g : \mathbb{R} \to \mathbb{R} \qquad g(x) = \begin{cases} \frac{1}{2} & x = 0, 1 \\ x & x \neq 0, 1 \end{cases}$$

Then $g([0,1]) = (0,1)$.

Notice that the above remark does *not* have an anaologue for open sets. That is to say, it is not always true that the image of an open set is open; Consider $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$. Then the open interval $(-1,1)$ is mapped to $[0,1)$ which is neither open nor closed.

Also notice that we can extend the EVT from a closed interval to any closed bounded set of reals.

Corollary 3.3.7 An EVT for closed bounded sets. *Let K be a closed bounded subset of $\mathbb{R}$ and let $f : K \to \mathbb{R}$ be a continuous function. Then there*

exists $c, d \in K$ so that

$$f(c) \leq f(x) \leq f(d) \qquad \forall x \in K$$

Proof. Naturally, the proof relies on the properteis of closed and bounded sets that we discussed in Section 2.5. The proof of this theorem was thus given as an exercise in that section — see Exercise 2.5.7. ■

Exercises

1. **Does it have a min or max?** For each of the following functions, determine whether it achieves a maximum and/or minimum value over the given domain. Justify your answers.

   - $f(x) = x^2$ on $[0, 2]$
   - $f(x) = x^2$ on $(0, 2)$
   - $f(x) = \dfrac{1}{x}$ on $[1, 5]$
   - $f(x) = \dfrac{1}{x}$ on $(0, \infty)$

2. **Bounded function on an open set.** Suppose that f is a continuous function on a finite open interval (a, b). Suppose that

   $$\sup_{x \in (a,b)} |f(x)| < \infty.$$

   Is it necessarily true that

   $$\lim_{x \to a^+} f(x)$$

   exists? Either prove it or provide a counter-example.

   Hint. Think about an example that oscillates a lot.

3. **Stay away from zero.** Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous **positive** function, meaning that $f(x) > 0$ for every $x \in \mathbb{R}$. Show that, for any bounded set $K \subseteq \mathbb{R}$, that there exists some $\alpha > 0$ such that $f(x) \geq \alpha$ for all $x \in K$.

   Is the same thing true if K is not assumed to be bounded?

4. **Always hotter (by a lot).** Suppose at any moment in time, the temperature in Furnace Creek (in Death Valley) is always greater than that of Vancouver. Show that, over the year 2024, there is some positive temperature $c > 0$ such that the temperature in Furnace Creek is greater than the temperature in Vancouver plus c degrees.

5. **Justifying the "obvious".** Suppose that $f : (a, b) \to \mathbb{R}$ is continuous and

such that

$$\lim_{x\to a^+} f(x) = \lim_{x\to b^-} f(x) = -\infty.$$

Show that there is some $c \in (a,b)$ with the property that $f(x) \leq f(c)$ for all $x \in (a,b)$.

Hint. Split the interval into what happens close the end points $x = a, b$ and some closed interval in the middle.

Continuous functions play very nicely with closed bounded sets. Indeed, sets that are closed and bounded have nice properties, and will eventually lead us to **compactness** by the end of term. But for now we want to see some properties of intervals are preserved under continuous functions, and some that are not.

6. **You can map open to closed continuously.** Show there is a continuous function on the open interval $(0,1)$ such that $f((0,1)) = [0,1]$.

   Hint. Think of functions that oscillate a lot.

7. **You can map bounded to unbounded continuously.** Show that there is a continuous function $f : (0,1) \to \mathbb{R}$ where $f((0,1))$ is not bounded

8. **You can map closed to open (with some infinity).** Let K be a closed subset of $\mathbb{R}$. Show that there is a function $f : K \to \mathbb{R}$ so that $f(K)$ is not closed.Recall that $\mathbb{R}$ is both open *and* closed.There are many choices that will work here, but all require K to be unbounded. For example

   - $f : \mathbb{R} \to (0,1]$ defined by $f(x) = \frac{1}{1+x^2}$
   - $f : \mathbb{R} \to (-1,1)$ defined by $f(x) = \arctan(x)$
   - $f : \mathbb{R} \to (-1,1)$ defined by $f(x) = \frac{x\cdot|x|}{1+x^2}$

So now we have seen the continuous functions need not preserved openness, closeness, or boundedness. A continuous function does not preserve either the boundedness or openness of a bounded-open set. *BUT* a continuous function must map a closed and bounded set to a closed and bounded set — that is precisely Corollary 3.3.7.

9. **The importance of closed and bounded together.** Complete Exercise 2.5.7. That problem asks you to show that if K is a closed and bounded set, and $f : K \to \mathbb{R}$ is continuous, then the image $f(K)$ is also closed and bounded.

The following is really just an IVT exercise, but it fits in nicely with the last few problems.

10. **Continuous bijection from open to closed.** Show there does *not* exist a continuous bijection $f : (0,1) \to [0,1]$.

    Hint. Suppose that the image of f contains the endpoints 0 and 1. Then apply the Intermediate Value Theorem to conclude that $f([a,b]) = [0,1]$ for some $a, b \in (0,1)$. Conclude that it is impossible for such a function to be injective.

11. Uniformly continuous is easy if you're closed and bounded. Complete Exercise 2.5.8, which says that a continuous function f on a closed and bounded set K is actually **uniformly continuous** on K.

Chapter 4

Derivatives

4.1 Derivatives of functions

Let us start with the definition you should know well from your previous calculusing.

Definition 4.1.1 The derivative. Let f be a function defined in the neighbourhood of a point c. We say that f is **differentiable** at c when

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}=\lim_{h\to 0}\frac{f(c+h)-f(c)}{h}=f'(c)$$

exists. We call $f'(c)$ the **derivative** of f at c.

It is also common to use *Leibniz* notation for the derivative at a point c:

$$f'(c)=\frac{\mathrm{d}}{\mathrm{d}x}\bigg|_{x=c}f(x)=\frac{\mathrm{d}f}{\mathrm{d}x}\bigg|_{x=c}.$$

If f is differentiable at every point in an interval (a,b) then we say that it is **differentiable** on the interval (a,b). ♢

Example 4.1.2 Some simple derivatives. Consider the functions defined on the whole real line

$$\begin{aligned} f(x)&=c && \text{constant}\\ g(x)&=x^2\\ h(x)&=|x| \end{aligned}$$

We can then compute their derivatives at a generic point $a\in\mathbb{R}$.

$$\begin{aligned} f'(a)&=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}=\lim_{h\to 0}0=0\\ g'(a)&=\lim_{h\to 0}\frac{g(a+h)-g(a)}{h}\\ &=\lim_{h\to 0}=\lim_{h\to 0}\frac{2ah+h^2}{h} \end{aligned}$$

$$= \lim_{h\to 0} 2a + h = 2a$$

The derivative of h is more difficult since we have to simplify $|x|$ and that depends on the sign of x.

$$\lim_{x\to a} \frac{h(x) - h(a)}{x - a} = \lim_{x\to a} \frac{|x| - |a|}{x - a}$$

Now if $a > 0$ then we only care about what happens when x is very close to a and so we can further assume, as we compute the limit, that x is positive too. Similarly, if a is negative, then we can assume, when we compute the limit, that x is also negative. Hence

$$\frac{|x| - |a|}{x - a} = \frac{x - a}{x - a} = 1 \qquad \text{when } x, a > 0$$
$$\frac{|x| - |a|}{x - a} = \frac{-x + a}{x - a} = -1 \qquad \text{when } x, a < 0$$

And so

$$h'(a) = \begin{cases} 1 & a > 0 \\ -1 & a < 0 \end{cases}$$

However, this reasoning breaks down when $a = 0$; we cannot make this simplifying assumption about the sign of x. Indeed, we will get different limits if we approach from the left (ie $x < a$) or from the right (ie $x > a$).

$$\lim_{x\to 0^+} \frac{|x| - |a|}{x - a} = \lim_{x\to 0^+} \frac{x - a}{x - a} = 1$$
$$\lim_{x\to 0^-} \frac{|x| - |a|}{x - a} = \lim_{x\to 0^-} \frac{-x + a}{x - a} = -1$$

and thus the limit

$$h'(0) = \lim_{x\to 0} \frac{|x| - |a|}{x - a} \text{ does not exist.}$$

□

One of the first results we should prove is the relationship between **continuity** and **differentiability**.

Lemma 4.1.3 Differentiable implies continuous. *If a function is differentiable at c then it is continuous at c. The converse is false.*

Proof. Homework! ■

Definition 4.1.4 Let $f : D \to \mathbb{R}$ be a function. The **derivative function** denoted f' is the function

$$f'(x) = \lim_{h\to 0} \frac{f(x + h) - f(x)}{h}$$

The domain of f' is then the subset of points inside D at which this limit exists.

We will also occasionally use the *Leibniz* notation for the derivative function:

$$f'(x) = \frac{\mathrm{d}}{\mathrm{d}x} f(x) = \frac{\mathrm{d}f}{\mathrm{d}x}.$$

◊

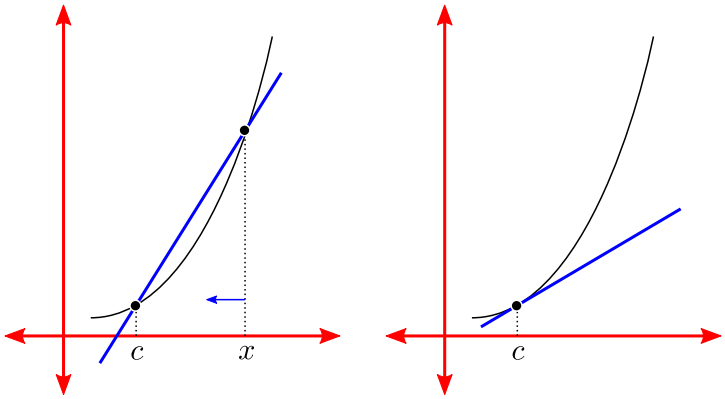
We can extend Lemma 4.1.3 to derivative-as-function. At the same time, it is important to note that while differentiability of a function implies *that* function is continuous, the derivative itself need not be continuous.

Lemma 4.1.5 Derivatives and continuity. *Let f be a function that is differentiable on a set D. Then f is continuous on D.*

At the same time, the derivative function f' need not be continuous on D.

Proof. The first statement follows directly from Lemma 4.1.3. The second statement can be shown by constructing a suitable example; see Example 4.1.12. We do not prove it directly here, but instead wait until we have stated the chain and product rules. ■

Recall that this definition stems from the idea of taking the graph of the function f and looking at the line joining two nearby points. By *squeezing* those points closer and closer we take the **secant** line joining those points and turn it into the **tangent** line.



Definition 4.1.6 Tangent line. Let $f : [a, b] \to \mathbb{R}$ be a function differentiable at $c \in (a, b)$. The **tangent line** to f at c is the linear function

$$T(x) = f(c) + f'(c)(x - c)$$

◊

The tangent line to a function at a given point is the *best* linear approximation in the following specific sense.

Lemma 4.1.7 Tangent line approximation. *Let f be a function differentiable at c, and let T be the tangent line to f at c. Then*

$$\lim_{x \to c} \frac{f(x) - T(x)}{x - c} = 0$$

Further, T is the unique linear function with this property.

Proof. Let f, T be as stated. Then

$$\begin{aligned}\frac{f(x)-T(x)}{x-c} &= \frac{f(x)-(f(c)+f'(c)(x-c))}{x-c}\\ &= \frac{f(x)-f(c)}{x-c} - f'(c)\end{aligned}$$

and so in the limit as $x \to c$ we have

$$\lim_{x\to c}\frac{f(x)-T(x)}{x-c} = \lim_{x\to c}\left(\frac{f(x)-f(c)}{x-c} - f'(c)\right) = 0$$

as required.

Uniqueness = a good homework problem! ■

This next lemma tells us that when a function is differentiable at $x = c$, it "looks something like" its tangent line. To be slightly more precise

$$f(x) = f(c) + (x-c)\cdot\varphi(x)$$

Now if that function $\varphi(x)$ were constant then we'd have a linear function. But in general we don't have that; what we do have is that as $x \to c$, the function $\varphi(x)$ has to "look like" $f'(c)$. That is $\lim_{x\to c}\varphi(x) = f'(c)$. So very roughly speaking, the lemma tells us that as we get really close to $x = c$, the function must look like its tangent line. Of course, one can make this even more precise, but this lemma will be quite useful; indeed it will help us prove the Chain Rule.

Theorem 4.1.8 *Let g be defined on some neighbourhood of a point c. If g is differentiable at c, then there is a function $\varphi(x)$ (which depends on g and c) so that*

$$g(x) - g(c) = \varphi(x)(x-c) \qquad \forall x \in (a,b)$$

and φ is continuous at c with

$$\lim_{x\to c}\varphi(x) = \varphi(c) = g'(c).$$

The converse also holds. Namely if one can find such a function φ, then g is differentiable at c with $g'(c) = \varphi(c)$.

Proof. Let

$$\varphi(x) = \begin{cases}\frac{g(x)-g(c)}{x-c} & x \neq c\\ g'(c) & x = c\end{cases}$$

so that $g(x) = g(c) + \varphi(x)(x-c)$. Then since g is differentiable at $x = c$ we know that

$$\lim_{x\to c}\varphi(x) = \lim_{x\to c}\frac{g(x)-g(c)}{x-c} = g'(c) = \varphi(c)$$

showing that φ is continuous at c.

Conversely, assume such a function φ exists. Then

$$\varphi(x) = \frac{g(x) - g(c)}{x - c}$$

and so if $\lim_{x \to c} \varphi(x)$ exists, then since

$$\varphi(c) = \lim_{x \to c} \varphi(x) = \lim_{x \to c} \frac{g(x) - g(c)}{x - c} = g'(c).$$

Thus g is differentiable at c. ■

Example 4.1.9 Let $g(x) = x^3 + 2x + 8$ and $c = 2$. Then

$$g(x) - g(2) = (x^3 + 2x + 8) - 20 = x^3 + 2x - 12 = (x - 2)(x^2 + 2x + 6)$$

and so we have $\varphi(x) = x^2 + 2x + 6$. Then (using some calculusing)

$$g'(x) = 3x^2 + 2 \qquad g'(2) = 14$$

and

$$\lim_{x \to 2} \varphi(x) = 4 + 4 + 6 = 14$$

as required. □

This result makes proving the chain rule quite[1] simple. Typically we just assume that the functions in the chain rule have domain and codomain the whole real line, which is a bit sloppy. Here we will state the chain rule quite carefully to make sure the domains and co-domains of functions match up correctly so that the composition is well-defined.

Theorem 4.1.10 The chain rule. *Let f, g be functions so that*

- *f is defined on a neighbourhood of c and differentiable at c*
- *g is defined on a neighbourhood of $f(c)$ and is differentiable at $f(c)$*

then the composition $h(x) = g(f(x))$ is well-defined and differentiable at $x = c$. The derivative of h is

$$h'(c) = (g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

Proof. Let f, g, h be as stated. Theorem 4.1.8 tells us that there are functions φ, ψ so that

$$f(x) = f(c) + \varphi(x)(x - c)$$
$$g(t) = g(d) + \psi(t)(t - d)$$

for d in some neighbourhood around $f(c)$ and where φ, ψ are continuous with $\varphi(c) = f'(c)$ and $\psi(d) = g'(d)$.

[1] Well, "relatively" simple

So we substitute $t \mapsto f(x), d \mapsto f(c)$ in this equation

$$g(f(x)) = g(f(c)) + \psi(f(x))(f(x) - f(c))$$

but recall that our definition of φ above tells us that we can rewrite $f(x) - f(c)$:

$$g(f(x)) = g(f(c)) + \psi(f(x)) \cdot \varphi(x)(x - c).$$

Then

$$\begin{aligned} h(x) - h(c) &= g(f(x)) - g(f(c)) \\ &= \psi(f(x)) \cdot \varphi(x)(x - c) \end{aligned}$$

so that

$$\begin{aligned} \lim_{x \to c} \frac{h(x) - h(c)}{x - c} &= \lim_{x \to c} \psi(f(x))\varphi(x) \\ &= \lim_{x \to c} \psi(f(x)) \cdot \lim_{x \to c} \varphi(x) \\ &= \psi(f(c)) \cdot \varphi(c) && \text{since } \psi, \varphi \text{ cts} \\ &= g'(f(c)) \cdot f'(c) \end{aligned}$$

where we have used the continuity of φ, ψ and Lemma 3.2.7. ■

With similar work we can show how derivatives and arithmetic interact.

Theorem 4.1.11 Arithmetic of derivatives. *Let f, g be functions differentiable at c, and let $K \in \mathbb{R}$ be a constant.*

- *If $f(x) = K$ for all x then $f'(c) = 0$*
- *If $f(x) = x$ then $f'(c) = 1$*
- $(f \pm g)'(c) = f'(c) \pm g'(c)$
- $(f \cdot g)'(c) = f(c)g'(c) + f'(c)g(c)$
- $(f/g)'(c) = \dfrac{f'(c)g(c) - g'(c)f(c)}{g(c)^2}$

Proof of product rule. Let f, g be differentiable at c. Then Theorem 4.1.8 tells us that we can write

$$\begin{aligned} f(x) &= f(c) + \varphi(x)(x - c) & \lim_{x \to c} \varphi(x) &= f'(c) \\ g(x) &= g(c) + \psi(x)(x - c) & \lim_{x \to c} \psi(x) &= g'(c) \end{aligned}$$

Then a little massaging of equations gives us

$$f(x)g(x) - f(c)g(c) = (x - c)(f(c)\psi(x) + g(c)\varphi(x)) + \varphi(x)\psi(x)(x - c)^2.$$

So now we can compute the derivative of the product of f, g:

$$\begin{aligned}(f \cdot g)'(c) &= \lim_{x \to c} \frac{f(x)g(x) - f(c)g(c)}{x - c} \\ &= \lim_{x \to c} (f(c)\psi(x) + g(c)\varphi(x)) + \varphi(x)\psi(x)(x - c) \\ &= f(c)g'(c) + g(c)f'(c) + 0\end{aligned}$$

which gives us the product rule. ■

Example 4.1.12 The derivative need not be continuous. Consider the function

$$g(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Let $x \neq 0$ then we can apply the chain and product rules to show that

$$g'(x) = 2x \sin(1/x) - \cos(1/x).$$

When $x = 0$ then we have to be a little more careful:

$$\begin{aligned}g'(0) &= \lim_{x \to 0} \frac{g(x) - g(0)}{x} \\ &= \lim_{x \to 0} \frac{x^2 \sin(1/x) - 0}{x} \\ &= \lim_{x \to 0} x \sin(1/x) \\ &= 0\end{aligned}$$

where we have made use of the sandwich theorem 3.1.10 to evaluate the limit via the bound

$$0 \leq |x \sin(1/x)| \leq |x| \qquad \forall x \neq 0.$$

So the derivative of this function is then

$$g'(x) = \begin{cases} 2x \sin(1/x) - \cos(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

This function is not continuous at 0 because the limit $\lim_{x \to 0} g'(x)$ does not exist; the term $\cos(1/x)$ does not converge as $x \to 0$. □

This is a very valuable example because it highlights that in general the derivative might not be as nicely behaved as the examples one encounters in initial calculus courses. Here is another nasty example that shows that the derivative can exist at an isolated point.

Example 4.1.13 Continuous and differentiable at a single point. Let

$f, g : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} - \mathbb{Q} \end{cases}$$

$$g(x) = xf(x) = \begin{cases} x^2 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} - \mathbb{Q} \end{cases}$$

The function f is continuous at $x = 0$ and nowhere else, while the function g is differentiable at 0 and nowhere else.

Notice that $|f(x)| \leq |x|$ and $|g(x)| \leq x^2$. Now let $\varepsilon > 0$ then set $\delta = \varepsilon$. Then for any $x \in (-\delta, \delta)$ we have

$$|f(x) - f(0)| = |f(x)| \leq |x| \leq \delta = \varepsilon$$

Hence $f(x) \to f(0)$ as $x \to 0$, and thus f is continuous at 0. Similarly

$$g'(0) = \lim_{x \to 0} \frac{g(x) - g(0)}{x} = \lim_{x \to 0} f(x) = 0$$

and so g is differentiable at 0.

Without loss of generality let $c > 0$ and let $\varepsilon = \frac{c}{2}$ and pick any $\delta > 0$.

- If $c \in \mathbb{Q}$, then Result 2.1.5 ensures that there is always some irrational number $z \in (c, c + \delta)$. Then
$$|f(z) - f(c)| = |0 - c| = |c| > \varepsilon$$
- Now assume that c is irrational and positive. Then density of the rationals 2.1.4, ensures that we can find a rational z that satisfies $c < z < c + \delta$. Then
$$|f(z) - f(c)| = z > c > \varepsilon$$
Similarly, if c is irrational and negative, then we can find a rational $z \in (c - \delta, c)$ so that
$$|f(z) - f(c)| = |z| > |c| > \varepsilon.$$

In either case the limit $\lim_{x \to c} f(x)$ does not exist, and so f is not continuous at $x = c$. The proof for c negative is nearly identical.

To show that the derivative of g does not exist, one can use the contrapositive of Lemma 4.1.3. It suffices to show that g is not continuous at c. The argument is essentially identical to that of f above. Again, let us assume that $c > 0$ (the proof for negative c is very similar), and then let $\varepsilon = \frac{c^2}{2}$ and pick any $\delta > 0$.

- If $c \in \mathbb{Q}$ then you can find an irrational number $z \in (c, c + \delta)$ and then
$$|g(z) - g(c)| = |c^2| > \varepsilon$$
- Similarly, if c is irrational, then you can find a rational $z \in (c, c + \delta)$ and then
$$|g(z) - g(c)| = z^2 > c^2 > \varepsilon$$

In either case, the limit $\lim_{x \to c} g(x)$ does not exist. Since g is not continuous at c, it is not differentiable there. □

Remark 4.1.14 Derivative of common functions. For the exercises in this section you may freely use the derivatives of common functions (like $\sin(x), \exp(x)$) as long as the question does not specifically ask you to not do so.

You should keep in mind that this is kind of "cheating". Well, not serious cheating, but we haven't actually *proved* the derivatives of those functions. In fact, many of them require some heavy machinery to even *define*. So we don't expect you to compute the derivative of sine from first principles just for a quick exercise.

Exercises

1. **You need to start somewhere.** If $f(x) = x$, prove using the definition of the derivative that $f'(x) = 1$.
2. **Multiplication can smooth out a sharp point.** Let $f(x) = x|x|$. Using the definition of the derivative, show that f is differentiable at $x = 0$.
3. **The simplest power rule.** Let $f(x) = x^n$ for a natural number $n \in \mathbb{N}$. Prove that $f'(x) = nx^{n-1}$ two different ways:
   - using the product rule and induction on n, and
   - directly from the definition of the derivative using the fact that $(x + h)^n = x^n + nx^{n-1}h + h^2P_n(x, h)$ where $P_n(x, h)$ is a polynomial in x and h.
4. **Negative integer powers.** If $f(x) = x^n$ for some *negative* integer n, prove using the quotient rule that $f'(x) = nx^{n-1}$ as well.
5. **Rational powers.** If $f(x) = x^{\frac{m}{n}}$ for $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, prove the expected equation
$$f'(x) = \frac{m}{n}x^{\frac{m}{n}-1}.$$
Hint. Consider $f(x)^n = x^m$ and differentiate both sides (using chain rule on the left-hand side).

Note that proving the well-known formula $f'(x) = px^{p-1}$ for *non* rational powers $f(x) = x^p$ is actually quite subtle! It can be done, for instance, once the derivative of $\log(x)$ is determined. Part of the underlying issue here is the subtlety of when the function x^p is actually defined for irrational values p .

6. **Two ways.** Compute the following limits two ways: Directly using algebra and your knowledge of limits; and using the definition of the derivative.
   - $\lim_{h\to 0} \dfrac{(3+h)^2 - 9}{h}$
   - $\lim_{h\to 0} \dfrac{\sqrt{4+h} - 2}{h}$

- $\lim_{h\to 0} \dfrac{\dfrac{1}{2+h} - \dfrac{1}{2}}{h}$

7. **Small steps towards product rule.** Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable, and $a \in \mathbb{R}$ a constant. Prove that

$$\frac{\mathrm{d}}{\mathrm{d}x}(af(x)) = af'(x)$$

two different ways: directly from the definition of the derivative; and using the product rule.

8. **A pretty classic example.** Define the function $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} x^a \sin(x^{-c}) & \text{if } x \neq 0, \\ 0 & \text{if } x \leq 0, \end{cases}$$

where $a > 0$ and $c > 0$ are constants. Prove the following statements.

- f is continuous if and only if $a > 0$.
- $f'(0)$ exists if and only if $a > 1$.
- f' is bounded if and only if $a \geq 1 + c$.
- f' is continuous if and only if $a > 1 + c$.

4.2 Increasing, decreasing and the derivative

The derivative helps us to understand how a function changes as we change its input. In first year calculus, we learn (though likely don't prove) that when the derivative is positive the function increases, and when the derivative is negative the function decreases. Let us actually prove this here.

Lemma 4.2.1 Positive derivative, negative derivative. *Let*

- $f : D \to \mathbb{R}$
- $c \in D$ *and there is* $w > 0$ *so that* $(c - w, c + w) \subseteq D$
- f *is differentiable at* c*, with* $f'(c) > 0$

Then there is some $\delta > 0$ *so that*

- *if* $x \in (c, c + \delta)$ *then* $f(x) > f(c)$*, and*
- *if* $x \in (c - \delta, c)$ *then* $f(x) < f(c)$*.*

A similar result holds for when $f'(c) < 0$*.*

Proof. Again, we use Theorem 4.1.8 to write

$$f(x) - f(c) = \varphi(x)(x - c)$$

Now since $\varphi(c) = f'(c) > 0$ and φ is continuous, we know by Lemma 3.2.14 that there is some $\delta > 0$ so that

$$\varphi(x) > 0 \qquad \forall x \in (c - \delta, c + \delta)$$

- If $c < x < c + \delta$ then $(x - c) > 0$ and so $f(x) - f(c) = \varphi(x)(x - c) > 0$
- Similarly when $c - \delta < x < c$, $(x - c) < 0$ and so $f(x) - f(c) < 0$

giving us the result.

The proof $f'(c) < 0$ is essentially the same. ■

The above lemma helps us towards an understanding of maxima and minima. Recall the following definition:

Definition 4.2.2 Local maxima and minima. Let $f : D \to \mathbb{R}$ be a function. We say that $c \in D$ is a **local maximum** if there exists $\delta > 0$ so that

$$f(x) \leq f(c) \qquad \forall x \in (c - \delta, c + \delta) \cap D.$$

Similarly, we say that c is a **local minimum** when there is some $\delta > 0$ so that

$$f(c) \leq f(x) \qquad \forall x \in (c - \delta, c + \delta) \cap D.$$

Note that we look at x-values in $D \cap (c-\delta, c+\delta)$ so that endpoints of intervals can still be local maxima and minima. ◊

From the previous lemma and the above definition we can prove Fermat's Theorem[1]. This is not his big famous one, but it is still very useful. It allows us to characterise the extrema of functions — where they are maximised and where they are minimised.

Theorem 4.2.3 Fermat's Theorem. *Let f be a continuous function on a closed interval $[a, b]$. If f takes its maximum or minimum value at $x = c$, then*

- *f is not differentiable at c, or*
- *$f'(c) = 0$, or*
- *$c = a$ or $c = b$*

That is, a maximum or minimum must occur either at an end-point, at a point of non-differentiability, or at a point where the derivative is zero.

[1]The interested reader should look at this article which describes how Fermat understood this question, which he wrote about in 1636, about 30 years before Newton developed Calculus and about 50 years before Leibniz published his work.

Proof. Either f is differentiable at c or not. If it is not, we are done. If it is differentiable, then either it is an end-point or not. If it is an end-point then we are done. So now assume that we have $c \in (a, b)$ and $f'(c)$ exists.

Assume that c is a local maximum; the proof for a local minimum is very similar. Since c is a local maximum, there is some $\delta > 0$ so that

$$f(x) - f(c) \leq 0 \qquad \forall x \in (c - \delta, c + \delta)$$

Hence the one-sided limits satisfy

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c} \leq 0$$
$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} \geq 0$$

Since the derivative exists (by assumption), we must have that the limit is 0. ■

Alternative proof. As per the proof above, assume that the first and third possibilities do not hold, and assume c is a local maximum and $f'(c)$ exists. Assume, to the contrary that $f'(c) > 0$. Then by Lemma 4.2.1 you can find a point z just to the right of c so that $f(z) > f(c)$ — giving a contradiction. Instead assume that $f'(c) < 0$, then you can find a point w just to the left of c so that $f(w) > f(c)$ — again giving a contradiction. Thus we must have $f'(c) = 0$. ■

Note that one should be careful of the flow of logic here. Just because a point has derivative zero, it does not mean that the point is a local maximum or minimum. Consider the polynomial x^3; at $x = 0$ its derivative is zero, but that point is neither a maximum nor minimum.

We are approaching the main result of this section, namely the Mean Value Theorem[2]. This result is crucial for many applications of calculus because it tells us how to transform local information about the derivative into global information about the function, and vice-versa. A great many other results rely on it.

The usual approach is to start with Rolle's Theorem[3].

Theorem 4.2.4 Rolle's theorem. *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$ then there exists $c \in (a, b)$ so that $f'(c) = 0$.*

[2]The MVT dates back as far is Bashkara (1114-1185), Parameshvara (1380-1460) who performed many astronomical calculations and computations of inverse-sine; their work shows connections to the MVT but as an useful approximation.

[3]Due to Michel Rolle in 1691. He actually only proved it for polynomials; Cauchy proved the general form we state here in 1823. Rolle also invented Gaussian Elimination, well in Europe. Gaussian Elimination first appears in a Chinese text which dates back to around 150BC. In Europe, it appears first in work of Newton. As often happens in mathematics, it is now named after someone else. The interested reader can learn much of its history in this excellent article including why we now attribute it to Gauss.

Proof. By the Extreme Value Theorem 3.3.5, the continuity of f implies that there exists $c \in [a,b]$ so that f takes it minimum value at $x = c$. Similarly there is $d \in [a,b]$ so that f takes its maximum value at $x = d$:

$$f(c) \leq f(x) \leq f(d) \qquad \forall x \in [a,b].$$

Now if $c \in (a,b)$ then c is a local minimum and so by Fermat's theorem 4.2.3 we know that $f'(c) = 0$. Similarly, if $d \in (a,b)$ we know that $f'(d) = 0$. In either case we are done because we have found a point at which the derivative is zero.

So now suppose that c, d are either a or b. That is, the maxima and minima occur at the ends of the interval. Now since, by assumption, we know that $f(a) = f(b)$, it follows that a, b are both maxima and minima for f on $[a,b]$. But this then requires that

$$f(x) \leq f(a) = f(b) \qquad \text{and} \qquad f(x) \geq f(a) = f(b) \qquad \forall x \in [a,b]$$

and thus we must have that $f(x) = f(a)$ for all $x \in [a,b]$. Thus the derivative $f'(x) = 0$ for all $x \in (a,b)$, and so we can take any point $z \in (a,b)$ and we have $f'(z) = 0$ as required. ■

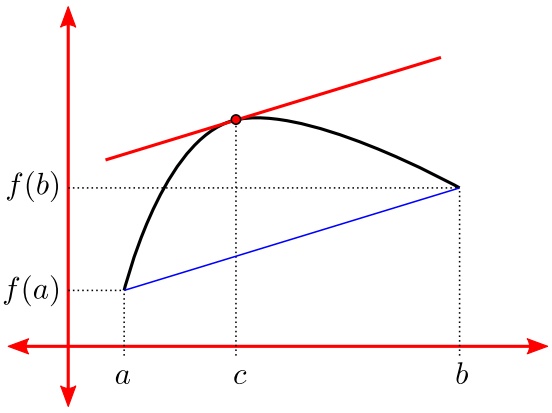
From here it is a quick jump to the Mean Value Theorem, often abbreviated "MVT"[4]. The proof takes the function and modifies it by subtracting off a linear function to obtain a function to which we can apply Rolle's Theorem 4.2.4. Translating back to the original function gives the result.

Theorem 4.2.5 The Mean Value Theorem. *Let $[a,b]$ be a closed interval and let f be continuous on $[a,b]$ and differentiable on (a,b). Then there is some $c \in (a,b)$ so that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently

$$f(b) - f(a) = f'(c)(b - a).$$



That is, there is a point $c \in (a,b)$ so that the derivative $f'(c)$ is equal to the

[4] the acronym MVT could also stand for "Most Valuable Theorem", which incidentelly, also describes the mean value theorem pretty well

slope of the line connecting the points $(a, f(a))$ *and* $(b, f(b))$.

Proof. Take f as stated, and use that to define a new function g on $[a, b]$:

$$g(x) = f(x) - (x - a)\frac{f(b) - f(a)}{(b - a)}.$$

so that

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Notice that since f is continuous on $[a, b]$, so is g. And since f is differentiable on (a, b), so is g. Finally notice that

$$\begin{aligned} g(a) &= f(a) + 0 = f(a) \\ g(b) &= f(b) - (b - a)\frac{f(b) - f(a)}{b - a} = f(a) \end{aligned}$$

Thus we can apply Rolle's theorem 4.2.4 and so find $c \in [a, b]$ so that $g'(c) = 0$. However, this implies that

$$\begin{aligned} f'(x) &= g'(x) + \frac{f(b) - f(a)}{b - a} && \text{and so} \\ f'(c) &= 0 + \frac{f(b) - f(a)}{b - a} \end{aligned}$$

as required. ■

The above form is sometimes known as Lagrange's MVT. One can generalise it in different ways; typically one encounters Cauchy's version:

Theorem 4.2.6 Cauchy's MVT or the Generalised MVT. *Let* $[a, b]$ *be a closed interval and let* f, g *be functions that are continuous on* $[a, b]$ *and differentiable on* (a, b). *Then there exists a point* $c \in (a, b)$ *so that*

$$f'(c)\big(g(b) - g(a)\big) = g'(c)\big(f(b) - f(a)\big)$$

Notice that by setting $g(x) = x$ *one recovers the usual MVT 4.2.5.*

Proof. Let

$$h(x) = f(x)(g(b) - g(a)) - g(x)(f(b) - f(a))$$

Notice that h is continuous on $[a, b]$ and differentiable on (a, b). Further, it satisfies

$$\begin{aligned} h(a) &= f(a)(g(b) - g(a)) - g(a)(f(b) - f(a)) = f(a)g(b) - g(a)f(b) \\ h(b) &= f(b)(g(b) - g(a)) - g(b)(f(b) - f(a)) = f(a)g(b) - g(a)f(b) = h(a) \end{aligned}$$

So we can apply Rolle's theorem 4.2.4 to get $c \in (a, b)$ with

$$h'(c) = 0 = f'(c)(g(b) - g(a)) - g'(c)(f(b) - f(a))$$

as required. ■

There is an even more general form expressed in terms of determinants, due to Peano. We won't use this in this course, but we state it here because some of you may find it interesting.

Theorem 4.2.7 Peano's MVT. *Let $[a,b]$ be a closed interval and let f,g,h be functions that are continuous on $[a,b]$ and differentiable on (a,b). Then there exists $c \in (a,b)$ so that the determinant*

$$\begin{vmatrix} f'(c) & g'(c) & h'(c) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix} = 0$$

Proof. Let

$$M(z) = \begin{vmatrix} f(z) & g(z) & h(z) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix}$$

then, a little linear algebra shows that $M(a) = M(b) = 0$ (since the matrix will have two identical rows). Applying Rolle's theorem 4.2.4 then gives $c \in (a,b)$ so that $M'(c) = 0$. That $M'(c)$ is precisely the determinant given. ■

Note that one recovers Cauchy's form of the MVT by setting $h(x) = 1$, and the usual MVT by setting $h(x) = 1, g(x) = x$.

Exercises

1. **MVT is good for inequalities.** Prove the following inequalities:
   - $|\cos(x) - \cos(y)| \leq |x - y|$ for $x, y \in \mathbb{R}$
   - $\frac{1}{2\sqrt{n+1}} < \sqrt{n+1} - \sqrt{n} < \frac{1}{2\sqrt{n}}$ for $n > 0$
2. **Zero derivatives and zeros.** True or False: A differentiable function that never has a flat slope, i.e., derivative equal to zero, has at most one root. Justify your answer.
3. **MVT fights speeding.** Police have two radar controls at a highway, say at kilometer 11 and at kilometer 20. Speed limit is 70 km/h. They measure a truck going through the first control, at 11:11am, at 65 km/h, and going through the second control at 11.17am, at 67 km/h. They issue a ticket. Why? (assuming it is for speeding and not, say, a broken light)
4. **Bounds on f' give bounds on f.** Suppose a function f is continuous on $[-2,7]$, differentiable on $(-2,7)$, $f(-2) = 3$, $f'(x) \leq 2$ for all $x \in (-2,7)$. Apply MVT to find largest possible value of $f(7)$.
5. **Prove there is a solution.** If

$$C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0,$$

prove that the equation

$$C_0 + C_1 x + \ldots C_{n-1} x^{n-1} + C_n x^n = 0$$

has at least one real solution x on the interval $[0, 1]$.

Hint. Find a function f with $f(0) = 0$ and $f(1) = 0$ and make judicious use of Rolle.

6. **One fixed point.** Suppose $f : [0, 1] \to [0, 1]$ is differentiable with $f'(x) < 1$ for all $x \in [0, 1]$. Prove that there is *exactly* one value $c \in [0, 1]$ with $f(c) = c$.

Hint. You've seen in Exercise 3.2.5.9 that there is at least one such point.

7. **A good application of the MVT.** A common theme in problems that apply the Mean Value Theorem is that you have some limit that you know exists, say

$$\lim_{x \to a} f'(x),$$

and then an application of the mean value theorem *for each individual* x value on the interval $[x, a]$ gives a new point c_x in between x and a. Now the limit

$$\lim_{x \to a} f'(c_x)$$

is not *quite* the same thing as the limit $\lim_{x \to a} f'(x)$ because the point c_x (between x and a) depends on the choice of x along the limit in possible some strange way. The whole point of this exercise is to show that this doesn't actually matter, i.e., that

$$\lim_{x \to a} f'(x) = \lim_{x \to a} f'(c_x).$$

In fact, it doesn't matter if the limiting x-value a is finite or infinite, or if the limiting $f(x)$-value L is finite or infinite either.

(a) Assume $g : \mathbb{R} \to \mathbb{R}$ is such that the limit $\lim_{x \to a^-} g(x)$ exists and equals L (here a, L are real numbers). If $c_x \in (x, a)$ for each x, prove that

$$\lim_{x \to a^-} g(c_x) = L.$$

(b) Assume $g : \mathbb{R} \to \mathbb{R}$ is such that the limit $\lim_{x \to \infty} g(x)$ exists and equals L (here L is a real number). If $c_x \in (x, \infty)$ for each x, prove that

$$\lim_{x \to \infty} g(c_x) = L.$$

(c) Assume $g : \mathbb{R} \to \mathbb{R}$ is such that the limit $\lim_{x \to a^-} g(x) = \infty$ (here a is a real number). If $c_x \in (x, a)$ for each x, prove that

$$\lim_{x \to a^-} g(c_x) = \infty.$$

(d) Assume $g : \mathbb{R} \to \mathbb{R}$ is such that the limit $\lim_{x\to\infty} g(x) = \infty$. If $c_x \in (x, \infty)$ for each x, prove that

$$\lim_{x\to\infty} g(c_x) = \infty.$$

8. It's gotta flatten out. Suppose that

$$\lim_{x\to\infty} f(x) \quad \text{and} \quad \lim_{x\to\infty} f'(x)$$

both exist (as real numbers). Prove that

$$\lim_{x\to\infty} f'(x) = 0.$$

Hint. Apply MVT to $f(x+1) - f(x)$. Also, note that you will have to apply the result of the previous exercise 4.2.7 in your proof.

9. Prove L'Hôptial's rule. Let f and g be continuous on $[a, b]$ and differentiable on (a, b) with $g'(x) \neq 0$ for all $x \in (a, b)$. Suppose $f(a) = g(a) = 0$ and that

$$\lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L.$$

Prove that

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

Hint. Use Cauchy's Mean Value Theorem and use the existence of the limit of the limit of the quotients of derivatives. Exercise 4.2.7 will also be helpful for some technical steps.

10. Darboux's theorem — an IVT for f'. Let $f : [a, b] \to \mathbb{R}$ be continuous and differentiable on the open interval (a, b). Suppose that $f'(a) < \lambda < f'(b)$. Then there exists some $c \in (a, b)$ such that $f'(c) = \lambda$.

Note that this does not imply that f' is a continuous function; not all functions that satisfy the conclusion of the Intermediate Value Theorem are continuous.

11. f' can be discontinuous, but only in certain ways. If f is a differentiable function, then f' has no removable or jump discontinuities.

Hint. Recall the definition of jump and removable discontinuity and use the previous exercise.

12. An excellent exam question. Suppose that $f : (-1, 1) \to \mathbb{R}$ is such that $f'(x)$ exists for all $x \neq 0$, and

$$\lim_{x\to 0} f'(x) = 3.$$

Show that f' is also differentiable at 0 and that $f'(0) = 3$ as well.

Hint. Think definition of derivative and MVT.

Also note that the constant 3 here is meaningless; any constant would do!

4.3 Consequences of the MVT

Let us now turn to some consequences of the MVT 4.2.5.

Corollary 4.3.1 *Let $[a,b]$ be a closed interval and let g be continuous on $[a,b]$ and differentiable on (a,b). Further, suppose that $g'(x) > 0$ for all $x \in (a,b)$. Then g is strictly increasing on $[a,b]$. That is, for all $x, y \in [a,b]$ with $x < y$ we have $g(x) < g(y)$.*

A similar result holds when $g'(x) < 0$.

Finally, if $g'(x) = 0$ on all of (a,b) then $g(x)$ is constant on $[a,b]$.

Proof. Let g be as stated, and let $x, y \in [a,b]$ with $x < y$. Notice that g satisfies the MVT 4.2.5 on the interval $[x,y]$. Thus there is $c \in (x,y)$ so that

$$g'(c)(y-x) = g(y) - g(x).$$

Since we know that $g'(c) > 0$ and, by assumption $(y-x) > 0$, it follows that $g(y) - g(x) > 0$ as required.

The proof for $g'(x) < 0$ is nearly identical.

Let g be as stated with $g'(x) = 0$ on $[a,b]$. Now assume that there are $x, y \in [a,b]$ with $x \neq y$ and $g(x) \neq g(y)$. By the MVT 4.2.5 there is some $c \in (x,y)$ so that $g'(c) = \frac{g(y)-g(x)}{y-x} \neq 0$. This contradicts our assumption, so there are no such x, y and g is constant. ■

Example 4.3.2 Two useful approximations. Using the fact that sine and cosine are bounded, we can use the MVT 4.2.5 to prove some simple bounds.

Let $f(x) = \sin(x)$. Now since $f'(x) = \cos(x)$, the MVT gives us

$$\sin(b) - \sin(a) = \cos(c)(b-a)$$

for some $c \in (a,b)$ Setting $a = 0$ reduces this to

$$\sin(b) = b\cos(c)$$

Now, if we restrict $b \leq \pi/2$, then we know that $c \in (0, \pi/2)$, and so $0 < \cos(c) < 1$, and hence we have

$$\sin(b) < b \qquad b \in (0, \pi/2).$$

With a little more work one can get similar useful bounds on cosine. Set $g(x) = 1 - x^2/2 - \cos(x)$. Then

$$\frac{g(b) - g(a)}{b-a} = g'(c) = \sin(c) - c$$

for some $c \in (a,b)$. Now, if we set $a = 0$, we get

$$\frac{1 - b^2/2 - \cos(b)}{b} = g'(c) = \sin(c) - c$$

and, by our work above, we know that $\sin(c) < c$, so the RHS is negative. Thus

$$\cos(b) > 1 - \frac{b^2}{2}$$

when $0 < b \leq \pi/2$.

One can actually keep iterating this procedure to get tighter and tighter bounds. But at that point, one is better off turning to Taylor approximations. □

Before transitioning topics slightly, we are *finally*[1] in a position where we can state and prove the **Inverse Function Theorem**. You likely will recognise this result from your first-year differential calculus class, but — as you will discover for yourself in your homework — the proof is quite subtle.

Theorem 4.3.3 Inverse Function Theorem. *Suppose $f : (a, b) \to \mathbb{R}$ is a differentiable function with $f'(x) > 0$ for every $x \in (a, b)$. Then f is invertible, and the inverse function — call it g — is also differentiable, and its derivative is calculated by*

$$g'(y) = \frac{1}{f'(g(y))}.$$

Proof. Homework!! ■

Now, I don't want to do Taylor series in their full glory. We just don't have time. However, I do want to give a taste of what one can do and prove.

Theorem 4.3.4 Taylor's theorem for tangent lines. *Let $[a, b]$ be a closed interval, let f be twice differentiable on (a, b) and let $c \in [a, b]$. The tangent line to f at c is*

$$T(x) = f(c) + (x - c)f'(c)$$

Then for any $x \in (a, b)$ with $x \neq c$, there exists q strictly between x and c so that

$$f(x) = T(x) + \frac{(x - c)^2}{2} \cdot f''(q)$$

Proof. Let f, x, c be as given. For any t between x and c define

$$F(t) = f(t) + f'(t)(x - t)$$

Notice that $F(x) = f(x)$ and $F(c) = f(c) + f'(c)(x - c) = T(x)$, and hence

$$F(x) - F(c) = f(x) - T(x).$$

Then set

$$G(t) = (x - t)^2$$

so that

$$G(x) - G(c) = 0 - (x - c)^2$$

[1] after a lot of work!

Now, the Cauchy MVT 4.2.6 tells us that there is some q between x and c so that

$$F(x) - F(c) = \frac{F'(q)}{G'(q)}(G(x) - G(c))$$

That is

$$f(x) - T(x) = -\frac{F'(q)}{G'(q)} \cdot (x-c)^2$$

The derivatives of F, G at $t = q$ are

$$F'(q) = f'(q) + f''(q)(x-q) - f'(q) = f''(q)(x-q)$$

and

$$G'(q) = -2(x-q)$$

So putting everything together gives us

$$f(x) - T(x) = \frac{1}{2}f''(q)(x-c)^2$$

as required. ■

Consequently, if we know that the second derivative is bounded on $[a, b]$ by M, then we know that

$$|f(x) - T(x)| \leq \frac{M}{2}|x-c|^2$$

and so we can control (or estimate) the error in approximating the function by its tangent line.

We can use this result to understand Newton's method, which we will do in the very next chapter! But first, exercise your brain a bit with the following problems.

Exercises

1. **Too good (bounds) to be true.** Suppose that $f : \mathbb{R} \to \mathbb{R}$ is such that

   $$|f(x) - f(y)| \leq (x-y)^2$$

   for all $x, y \in \mathbb{R}$. Prove that f is constant.

2. **Strictly monotone is bijective.** Let $f : (a, b) \to \mathbb{R}$ be a differentiable function so that $f'(x) > 0$ for all $x \in (a, b)$. Further let

   $$A = \lim_{x \to a^+} f(x) \quad \text{and} \quad B = \lim_{x \to b^-} f(x),$$

   where these can be infinite. Then show that by restricting the codomain, the function

   $$f : (a, b) \to (A, B)$$

   is a bijection.

Hint. The IVT is your friend.

3. **Positive derivative at infinity.** Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that

$$\lim_{x\to\infty} f'(x) > 0.$$

Prove that

$$\lim_{x\to\infty} f(x) = \infty.$$

Hint. Argue that you can find some R large enough so that $f'(x) \geq c > 0$ for all $x \geq R$. Then apply MVT on a very long interval after $x = R$.

4. **A small jiggle.** Suppose that $g : \mathbb{R} \to \mathbb{R}$ has a bounded derivative, i.e., that g' is a bounded function 2.2.4. Show that the function

$$f(x) = x + \varepsilon g(x)$$

is a bijection $\mathbb{R} \to \mathbb{R}$ if ε is small enough (depending on M).

Hint. The previous two exercises may be helpful.

5. **Convexity.** Suppose $f : [a, b] \to \mathbb{R}$ is continuous, with $a < b$, and $f''(x) \geq 0$ for all $x \in (a, b)$. Show that f is **convex**[2].

A function f is convex on $[a, b]$ whenever $a \leq x < y \leq b$ and $s \in [0, 1]$ we have

$$f(xs + y(1 - s)) \leq sf(x) + (1 - s)f(y).$$

Notice that this means that the function lies below the line-segment from $(x, f(x))$ to $(y, f(y))$.

Note that this does not require $f'(x)$ to be bounded above or below for $x \in (a, b)$. For example, let $[a, b] = [-1, 1]$ and $f(x) = -\sqrt{1 - x^2}$.

Hint. Let $z = xs + y(1 - s)$ and apply MVT to find points z_1, z_2 with $(f(z) - f(x))/(z - x) = f'(z_1)$ and $(f(y) - f(z))/(y - z) = f'(z_2)$ and use the fact that $f'(z_1) \leq f'(z_2)$ (why?).

4.4 Return to Newton

Let us return to Newton's method. Way back at the beginning of this text we looked at using Newton's method to solve $f(x) = x^2 - 2 = 0$ numerically.

In particular, one starts with an initial estimate x and then compute a (hopefully) better estimate y, via

$$y = x - \frac{f(x)}{f'(x)} = x - \frac{x^2 - 2}{2x}$$

We would like to understand under what conditions y is closer to $\sqrt{2}$ than x.

[2]Convexity of a function is important to many fields of mathematics, e.g., differential equations

More generally, let us take a twice differentiable function f, and let r be its zero. That is $f(r) = 0$. Using Taylor's Theorem 4.3.4 we know that

$$f(r) = 0 = f(x) + (r - x)f'(x) + \frac{1}{2}f''(q)(r - x)^2$$

for some q between x, r. Now, provided $f'(x) \neq 0$, we can rearrange this to give

$$r - x + \frac{f(x)}{f'(x)} = -\frac{f''(q)}{2f'(x)}(r - x)^2$$

But recall that given x, Newton's method gives y by exactly the LHS of this equation. Thus we have

$$r - y = -\frac{f''(q)}{2f'(x)}(r - x)^2$$

So in order to know if $|r - y| < |r - x|$, we need to understand the RHS here.

We don't actually know $f''(q)$ exactly, so this looks problematic. However but in many situations we can bound it. Note that the bound doesn't have to be especially good, just true. In the example, $f(x) = x^2 - 2$, we know that

$$f'(x) = 2x \qquad f''(x) = 2$$

So if our iterates are confined to $[1, 2]$ (which seems reasonable since $1 \leq \sqrt{2} \leq 2$), then we know that

$$2 \leq |f'(x)| \leq 4 \qquad f''(q) = 2$$

and so

$$\left|\frac{f''(q)}{2f'(x)}\right| \leq \frac{2}{4} = \frac{1}{2}$$

Thus

$$|r - y| \leq \frac{1}{2}|r - x|^2$$

So, if $|r - x| < 1$, then this will converge really quickly.

In particular,

- if we start with (say) $x = 2$, then $|r - x| < 1$, and so $|r - y| < \frac{1}{2}$.
- On the next iteration (setting x to the value of y from the previous iteration), since $|r - x| < \frac{1}{2}$, we know that $|r - y| < \frac{1}{8}$.
- Doing this again, $|c - x| < \frac{1}{8} = 2^{-3}$ and so $|r - y| < 2^{-7}$.

More generally, if we currently have $|r - x| \approx 2^{-k}$ then the next iterate y will give $|r - y| \approx 2^{-2k-1}$. Quadratic convergence!

We can make this same argument more generally. Namely, provided one can find a lower bound for $|f'(x)|$ and an upper bound for $f''(x)$, then provided we start close enough to the root, then the iterates will converge quadratically.

Of course, one can use other methods to make sure you are pretty close. For example, one can start in some big interval around the root and then close in by repeated bisection — much as we did back at the beginning of Section 2.2.

More generally, Newton's method can fail in various ways. For example take the function $g(x) = x^2 + 1$. Since g has no real zeros, we know that it shouldn't converge. Indeed, if one starts with $x = 0.1$, then you'll observe the iterates bounce around very chaotically. If you re-read the above analysis, you'll see that used the fact that $f(r) = 0$, and so we cannot repeat the analysis with our function g.

If we instead study the next iterate y as a function of the current iterate x we get

$$y = x - \frac{g(x)}{g'(x)} = x - \frac{x^2+1}{2x} = \frac{x^2-1}{2x}$$

The dynamics of the iterates are determined by the shape of the graph of this function. One can see, for example, that

- if $x = 0$ then y is not defined
- if $x = \pm\frac{1}{\sqrt{3}}$ then $y = \mp\frac{1}{\sqrt{3}}$ — and so the iterates cycle back and forth

One can construct other pathological examples. Take $h(x) = x^3 - 2x + 2$, which has a real zero at $x \approx -1.76929\ldots$. It we start Newton's method at $x = 0$, then the iterates alternate $0, 1, 0, 1, 0, 1, \ldots$. If you start at $x = 0.1$ then the iterates converge to this same 2-cycle. While if you start at $x = 0.2$, they bounce around a little before converging to the real-zero of $-1.76929\ldots$.

Exercises

1. **How many iterations?** How many iterations of Newton's method ($f(x) = x^2 - 2$ with initial guess $x = 1$) would you perform if you absolutely required an approximation of $\sqrt{2}$ that is accurate within 10^{-4}? What about 10^{-10}? 10^{-100}? Justify why your answer is correct.

   Note that we're not asking for a *sharp* answer in the sense that the number of iterations may not be the optimal one. Rather, we're asking for you to analyze the work done in the section to determine with certainty how many iterations is needed for you to be sure your answer is sufficiently precise.

2. **Cube root with quadratic convergence.** Show that Newton's method with the function $f(x) = x^3 - 2$ results in quadratic convergence when used to numerically approximate $\sqrt[3]{2}$, starting from an initial guess $x = 2$.

3. **Higher-order zero slows convergence.** For the sake of example, let's pretend that you're trying to use Newton's method to approximate the root of the function $f(x) = x^2$ starting from an initial guess of $x = 1$ (why is this a foolish endeavor?). Show that the convergence $x_n \to 0$ is *not* quadratic. The source of the issue here is that f has a **double root** at $x = 0$. Explain

mathematically what is going on here.

4. **A recipe for root-p.** Prove that Newton's method with the function $f(x) = x^2 - p$ will result in quadratic convergence when used to numerically approximate $\sqrt{p}$, for *any* positive choice of $p > 0$, as long as the initial guess is close enough.

   Hint. Proceed as in the notes to see that the convergence is quadratic as long as the successive error is decreasing.

Newton's method can also work on functions on the complex plain, $\mathbb{C}$. The "basins of attraction", meaning the regions in which convergence to a root occurs, can form very interesting sets in $\mathbb{C}$. This is somewhat beyond the scope of this course, but we encourage you to spend a few minutes reading through the Wikipedia page for *Newton Fractals*.

4.5 Getting closer

Recall the general statement of Taylor's Theorem from your earlier calculus courses:

Theorem 4.5.1 Taylor's theorem. *Let $[a, b]$ be a closed interval, let f be $(n+1)$-times differentiable on (a, b) and let $c \in [a, b]$. The nth Taylor polynomial of f about $x = c$ is*

$$T_n(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(c)(x - c)^k$$

Then for any $x \in (a, b)$ with $x \neq c$, there exists q strictly between x and c so that

$$f(x) - T_n(x) = \frac{(x - c)^{n+1}}{(n + 1)!} f^{(n+1)}(q)$$

Proof. The proof of this statement is quite similar to the proof of the linear case 4.3.4 we did above and the interested reader can find it here[1], for example. ■

In general there are two ways that we can improve the quality of the Taylor polynomial approximation.

- Bring x closer to c. If $(z - c) = \frac{1}{2}(x - c)$, then

$$\frac{|f(z) - T_n(z)|}{|f(x) - T_n(x)|} = \frac{1}{2^{n+1}} \frac{f^{(n+1)}(q_2)}{f^{(n+1)}(q_1)}$$

  So assuming that the derivative term does not change rapidly, the error reduces by a factor of about $1/2^{n+1}$.

[1] wikipedia.org/wiki/Taylor's_theorem#Proof_for_Taylor's_theorem_in_one_real_variable

- Increase n:
$$\frac{|f(x) - T_{n+1}(x)|}{|f(x) - T_n(x)|} = \frac{(x-c)}{(n+2)} \frac{f^{(n+2)}(q_2)}{f^{(n+1)}(q_1)}$$
so if the derivatives are approximately the same size, and $(x-c)$ isn't too large, then the error reduces by a factor of around n.

Now notice that we are measuring the distance between these two functions at a particular fixed point x. In many situations we would like a measure of closeness that is broader.

There are other (easier) ways that we can capture *closeness*. Recall the Maclaurin series for e^x:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots = \sum_{k \geq 0} \frac{1}{k!} x^k$$

and then the first few Maclaurin polynomials:

$$\begin{aligned} T_1(x) &= 1 + x \\ T_2(x) &= 1 + x + \frac{x^2}{2} \\ T_3(x) &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \\ T_4(x) &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} \end{aligned}$$

Notice that (unsurprisingly) each one gets "closer" to the Maclaurin series in that more and more terms of the expansion agree. The expansion of $T_n(x)$ agrees with the first $n+1$ terms of e^x (indeed it is constructed that way). We can write this slightly more formally as

$$e^x - T_n(x) = O(x^{n+1})$$

where we are making use of Landau's **big-O** notation.

Definition 4.5.2 Big-O notation. Let f, g be functions. We write

$$f(x) = O(g(x)) \qquad \text{as } x \to c$$

when there exists $\delta, M > 0$ so that for all $x \in (c - \delta, c + \delta)$ we know that

$$|f(x)| \leq M|g(x)|$$

$\Diamond$

This big-O notation is frequently used to describe the behaviour of a function as it approaches its limit. For example

$$2x + 6x^2 = O(x) \quad x \to 0$$

the function on the left "looks like" the function on the right up to some multiplicative factor. We can summarise Taylor's theorem using this notation:

$$f(x) - T_n(x) = O((x-c)^{n+1}) \quad x \to c$$

or, equivalently,

$$f(x) = T_n(x) + O((x-c)^{n+1}) \quad x \to c$$

meaning that $f(x)$ looks like $T_n(x)$ plus some function that is bounded by a multiple of $(x-c)^{n+1}$ as $x \to c$. The big-O term measures the closeness of the functions as $x \to c$.

There are other ways that we could describe "closeness"[2] of two function. For example, we might want to approximate our function $f(x)$ over a range of x-values, not just as $x \approx c$. So we could use an integral to measure the distance between two functions over the whole interval:

$$\|f - g\| = \int_a^b |f(x) - g(x)| \, \mathrm{d}x$$

One can then try to find approximations to f that minimise this measure of distance.

Of course, we have seen other measures of closeness in other contexts. For points in the plane (x, y) we measure the **Euclidean distance** as

$$d\left((x_1, y_1), (x_2, y_2)\right) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

and more general points in $\mathbb{R}$:

$$d\left(\vec{x}, \vec{z}\right) = \sqrt{\sum (x_k - z_k)^2}$$

Once we have a notion of distance we can recycle many of the notions we have been busy discussing in this course so far. We can talk about whether a sequence of objects (not numbers) converges to a limit. For example, our Taylor polynomials above *converge* to the Taylor series:

$$f(x) - T_n(x) = O(x^{n+1})$$

Under certain conditions (derivatives nicely behaved) we can assert that for fixed x, the RHS goes to zero as $n \to \infty$ and so $T_n(x) \to f(x)$.

This can be extremely useful. Here is some mathematical object that I want to understand x^*, the root of a polynomial or the solution of a DE, or the eigenvalue of a matrix, or …. I can't compute it exactly, but here is a guess x_0. I can then define some algorithm that takes that guess and produces x_1. It is better in the sense that

$$\|x^* - x_1\| < \|x^* - x_0\|$$

[2] in the next chapter, we will discuss extensively a generalised notion of measuring the distance between objects, called **metrics**

that is, x_1 is closer to x^* than x_0. This gives us a sequence of objects

$$x_0, x_1, x_2, \ldots,$$

that *converge* to the object we want x^* .

Now, one could prove such results for specific notions of distance for each particular problem, like those we have outlined above. It is far better to prove *general* theorems for *general* distances. We can then specialise those to specific problems.

Exercises

1. **Taylor polynomials to compute limits.** Taylor's Theorem can be used to quickly calculate annoying looking limits. For example, if f is $n+1$-times differentiable with bounded $(n+1)$'st derivative, first prove that

$$\lim_{x \to a} \frac{f(x) - P_n(x)}{(x-a)^n} = 0.$$

Here P_n denotes the order-n Taylor polynomial for f at $x = a$. Using this convergence, compute

- $$\lim_{x \to 0} \frac{\cos(x) - 1 + \frac{x^2}{2}}{11x^4}$$
- $$\lim_{x \to 0} \frac{e^{5x} - 1 - 5x - \frac{25x^2}{2}}{x^3}$$
- $$\lim_{x \to 0} \frac{\sin(x) - x + \frac{x^3}{6}}{12x^4}$$
- $$\lim_{x \to 0} \frac{e^{x^3}\log(x^5) + \cos(x^{95}) - \sin(x^5) - 1}{x^8}$$

In your work, you may use the results of Exercise 4.5.9 which says that the Taylor Polynomial of a product function can be found by multiplying the Taylor polynomials, and similarly for composition.

2. **Big-O and zero derivative.** Suppose that $f(x) - f(c) = ((x-c)^\alpha)$ as $x \to c$ for some $\alpha > 1$. Prove that $f'(c) = 0$.
3. **Big-O and not differentiable.** Suppose that $f(x) - f(c) \neq O((x-c)^\alpha)$ as $x \to c$ for *any* $\alpha \geq 1$. Prove that f is *not* differentiable at $x = c$.
4. **Big-O and maybe differentiable?** Suppose that $f(x) - f(c) = O(x-c)$ as $x \to c$. Show that f may or may not be differentiable at $x = c$, and if it is differentiable that $f'(c)$ could be any real number (depending on the

choice of function).

5. **Big-O does not play nicely with derivatives.** If $f = O(x^2)$ as $x \to 0$, then we have seen (in a previous exercise) that $f'(0)$ exists and is equal to 0. Show that it may *not* be true that $f'(x) = O(x)$ by considering the following function:

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

6. **Not everything can be Taylored.** Define the function

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

   - Use a proof by induction to show that for each $n \in \mathbb{N} \cup \{0\}$, there exists a polynomial $P_n(x)$ such that

$$f^{(n)}(x) = P_n(1/x)e^{-1/x^2}$$

     whenever $x \neq 0$.

   - Show that f is infinitely differentiable at $x = 0$ and that for all $n \in \mathbb{N}$, $f^{(n)}(0) = 0$.

   In particular, the Taylor series of f about $x = 0$ is identically zero, even though $f(x) \neq 0$ for $x \neq 0$. In particular, the Taylor series of a smooth function does not necessarily equal the function, except at the base point.

7. **Best approximation.** Suppose that f is $(n+1)$-times differentiable with $(n+1)$'st bounded derivative. Suppose also that $f(x) = Q_n(x) + O(x^{n+1})$ as $x \to 0$, where Q_n is some polynomial of order n. Prove that Q_n must be the Taylor polynomial of f centered at $x = 0$.

   Takeaway: Taylor polynomials are the *best* polynomial approximations to a function at a point.

   Hint. Both P_j and Q_j (we're calling Q_j to be the truncation of Q_n to degree j) approximate f at least to rate x^j.

8. **Big-O arithmetic.** Suppose that $f(x) = O(x^p)$ and $g(x) = O(x^q)$ for $p, q \in \mathbb{N}$.

   - Prove that $f + g = O(x^{\min\{p,q\}})$.

   - Prove that $fg = O(x^{p+q})$.

   - For some given $r > 0$, prove that $f(x)^r = O(x^{rp})$.

9. **Shortcuts.** A common shortcut introduced in first year calculus is that Taylor Polynomials can be multiplied together and composed. Here, you

will prove these statements rigorously. We will write P_n and Q_n for the degree n Taylor polynomials (centered at $x = 0$) for "nice" functions f, g respectively (let's assume "nice" means infinitely differentiable with bounded derivatives of all orders, for simplicity).

(a) Prove that
$$f(x)g(x) = P_n(x)Q_n(x) + O(x^{n+1}).$$
Conclude, using the previous exercise, that the degree n Taylor polynomial for the product fg is precisely the product of $P_n(x)$ and $Q_n(x)$, truncated to degree n.

(b) A similar result holds for compositions. If $g(0) = 0$, then show that
$$f(g(x)) = P_n(Q_n(x)) + O(x^{x+1}).$$
In particular, Exercise 4.5.7 implies that the degree n Taylor polynomial of the composition $f \circ g$ is the composition of P_n and Q_n, truncated to degree n. As a particular case of this, if you want to find the Taylor polynomial for, say $\sin(x^2)$, you can just substitute x^2 for x in the Taylor polynomial for $\sin(x)$.

(c) Using the above, compute the degree-16 Taylor polynomial (centered at $x = 0$) for the function $f(x) = (1 - \cos(x^2))e^{x^3}$.

Hint.

(a) You may use the formula
$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}(fg)(0) = \sum_{j=0}^{n} \binom{n}{j} f^{(j)}(0)g^{(n-j)}(0).$$
You should try to convince yourself why this equation is true!

(b) Since $g(0) = 0$, you can make a substitution $X = g(x)$ into the equation $f(X) = P_n(X) + O(X^{n+1})$.

Chapter 5

Metric spaces

As we discussed at the end of the last section, we need to understand how two mathematical objects can be "close" or "far". For real numbers, we typically characterise this using the absolute value function

$$\text{distance}(x, y) = |x - y|$$

For points in $\mathbb{R}^n$ we generalise this with a little Pythagoras:

$$\text{distance}(\vec{x}, \vec{y}) = \sqrt{\sum (x_k - y_k)^2}$$

giving us the Euclidean distance. However, another perfectly reasonable notion of distances between points in $\mathbb{R}^n$ is the **taxi-cab** metric

$$\text{distance}(\vec{x}, \vec{y}) = \sum |x_k - y_k|.$$

More generally we want to understand a set of mathematical objects equipped with some notion of distance between those objects. We call those more general measures of distance **metrics**, and a set with a metric is called a **metric space**. The benefit of working with metric spaces is that many of our ideas of convergence carry across quite directly to this general setting.

5.1 Metrics and metric spaces

When we try to generalise the notion of distance, we need to think about what a "distance" needs to do.

- a distance is a non-negative real number: $\rho(x, y) \geq 0$
- the distance between an object and itself should be zero: $\rho(x, x) = 0$
- if objects are different, their distance cannot be zero: $x \neq y \implies \rho(x, y) \neq 0$

- order doesn't matter: $\rho(x, y) = \rho(y, x)$

We can summarise these as

- $\rho(x, y) \geq 0$
- $\rho(x, y) = 0$ if and only if $x = y$
- $\rho(x, y) = \rho(y, x)$

But we are not quite there yet.

Implicit in the idea of distance is a notion of "minimality". If you are told that the distance between two points x, y is $\rho(x, y) = q$, then you cannot find some path between those points that is smaller. In particular, you can't sneak from x to some other point z and then from z to y and somehow end up finding a short-cut. That is,

$$\sim (\exists z \text{ so that } \rho(x, y) > \rho(x, z) + \rho(z, y))$$

which is precisely the triangle inequality

$$\forall z, \rho(x, y) \leq \rho(x, z) + \rho(z, y)$$

And you should recall that a very large number of our proofs relied on the fact that the absolute value satisfies the triangle inequality.

Definition 5.1.1 Metric. Let A be a set and let $a, b, c \in A$. A **metric** on A is a function $\rho : A \times A \to \mathbb{R}$ that satisfies

- positive definite: $\rho(a, b) \geq 0$ and further, $\rho(a, b) = 0$ iff $a = b$
- symmetry: $\rho(a, b) = \rho(b, a)$
- triangle inequality: $\rho(a, b) \leq \rho(a, c) + \rho(c, b)$

$\Diamond$

Example 5.1.2 A very boring metric. Let A be any set, then we define the **discrete metric** to be

$$\rho(a, b) = \begin{cases} 0 & b = a \\ 1 & \text{otherwise} \end{cases}$$

Of course, we should prove that is a metric.

Answer.

Proof. We have to show that ρ satisfies the three properties of a metric. So let $a, b, c \in A$

- By definition, $0 \leq \rho(a, b)$. Further, by definition, $\rho(a, b) = 0 \iff a = b$. So ρ is positive-definite.

- Also by definition $\rho(a,b) = \rho(b,a)$, so ρ is symmetric
- Finally, consider $\rho(a,b)$. Either $a = b$ or $a \neq b$
  - when $a = b$, we know that $\rho(a,b) = 0$, and since $0 \leq \rho(a,c), \rho(b,c)$, we have $\rho(a,b) \leq \rho(a,c) + \rho(b,c)$
  - When $a \neq b$, we know that $\rho(a,b) = 1$. Now since $a \neq b$, we cannot have $c = a$ and $c = b$ (transitivity of equality), and so we must have $c \neq a$ or $c \neq b$. In either case, $\rho(a,c) + \rho(b,c) \geq 1$, and thus $\rho(a,b) \leq \rho(a,c) + \rho(b,c)$.

  In either case the triangle inequality holds.

Thus ρ is a metric. ■

□

Example 5.1.3 A more interesting metric. Let $\rho : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$\rho(\vec{x}, \vec{y}) = |x_1 - y_1| + |x_2 - y_2|$$

This function is the **taxi-cab** metric. So-called as it describes the distance a car drives on a grid of city-blocks — it must drive east or west, and then north or south. Again, we prove it is a metric.

Answer.

Proof. We have to show that ρ satisfies the three properties of a metric. So let $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^2$

- Consider the distance between $\vec{a}, \vec{b}$:

  $$\rho(\vec{a}, \vec{b}) = |a_1 - b_1| + |a_2 - b_2|$$

  Now, since the absolute value function is non-negative, we immediately get that $\rho \geq 0$. Further, since $|x| = 0$ if and only if $x = 0$, we know that $\rho(\vec{a}, \vec{b}) = 0$ if and only if $a_1 = b_1$ and $a_2 = b_2$. That is $\vec{a} = \vec{b}$. Thus ρ is positive-definite.
- Since $|x - y| = |y - x|$, it follows that $\rho(\vec{a}, \vec{b}) = \rho(\vec{b}, \vec{a})$ and so is symmetric.
- Consider $\rho(\vec{a}, \vec{b})$:

  $$\begin{aligned} \rho(\vec{a}, \vec{b}) &= |a_1 - b_1| + |a_2 - b_2| \\ &= |a_1 - c_1 + c_1 - b_1| + |a_2 - c_2 + c_2 - b_2| \\ &\leq |a_1 - c_1| + |c_1 - b_1| + |a_2 - c_2| + |c_2 - b_2| \\ &= \rho(\vec{a}, \vec{c}) + \rho(\vec{c}, \vec{b}) \end{aligned}$$

  Hence ρ obeys the triangle inequality.

So the taxi-cab metric is actually a metric. ■

□

Example 5.1.4 Metric on continuous functions. Let $C[a,b]$ be the set of continuous functions on the closed interval $[a,b]$. For two functions in $C[a,b]$ define

$$\rho(f,g) = \sup\{|f(x) - g(x)| \mid a \leq x \leq b\}$$

Prove that ρ is a metric.

Answer.

Proof. First, note that since f and g are both continuous, so is $|f(x) - g(x)|$. In particular, the quantity

$$\rho(f,g) = \sup\{|f(x) - g(x)| \mid a \leq x \leq b\}$$

is *finite* by the Extreme Value Theorem 3.3.5. This shows that ρ is at least *well-defined*. That is good, but now we have to show the properties of a metric.

- Clearly if $f = g$ then $\rho(f,g) = 0$. Similarly, if $\rho(f,g) = 0$ then we must have that $|f(x) - g(x)| = 0$ everywhere in $[a,b]$. Thus $f = g$ everywhere on $[a,b]$.
- By the definition of ρ, it is symmetric in its arguments.
- We make use of the triangle inequality to prove the triangle inequality. Namely

$$\begin{aligned}
\rho(f,g) &= \sup_{x\in[a,b]} |f(x) - g(x)| \\
&= \sup_{x\in[a,b]} |f(x) - h(x) + h(x) - g(x)| \\
&\leq \sup_{x\in[a,b]} (|f(x) - h(x)| + |h(x) - g(x)|) \\
&\leq \sup_{x\in[a,b]} |f(x) - h(x)| + \sup_{x\in[a,b]} |h(x) - g(x)| \\
&= \rho(f,h) + \rho(h,g)
\end{aligned}$$

  Where we have used the fact that for any two functions f, g $\sup(f+g) \leq \sup(f) + \sup(g)$.

■

□

Example 5.1.5 Another metric on continuous functions. Again let $C[a,b]$ be the set of continuous functions on $[a,b]$. We define ρ by

$$\rho(f,g) = \int_a^b |f(x) - g(x)| \, \mathrm{d}x$$

Show that ρ is a metric.

Answer. We'll make use of the following "facts" from integral calculus without

proof.

- If $f(x) \leq g(x)$ on $[a,b]$ then $\int_a^b f(x)\,\mathrm{d}x \leq \int_a^b g(x)\,\mathrm{d}x$
- $\displaystyle\int_a^b f\,\mathrm{d}x = \int_a^c f\,\mathrm{d}x + \int_c^b f\,\mathrm{d}x$
- $\displaystyle\int_a^b (f+g)\,\mathrm{d}x = \int_a^b f\,\mathrm{d}x + \int_a^b g\,\mathrm{d}x$

Proof. Let f, g, h be continuous functions on $[a,b]$. Then since $|f(x)-g(x)| \geq 0$ we know that

$$\rho(f,g) = \int_a^b |f(x) - g(x)|\,\mathrm{d}x \geq 0.$$

Now, if $f = g$ then $f(x) - g(x) = 0$ on all of $[a,b]$ and so $\rho(f,f) = 0$. Now assume that $f \neq g$, and thus there is some $c \in [a,b]$ so that $|f(c) - g(c)| > 0$. Since f, g are continuous, there is some $\delta > 0$ so that $|f(x) - g(x)| > 0$ for all $x \in (c-\delta, c+\delta)$. And hence

$$\rho(f,g) = \underbrace{\int_a^{c-\delta} |f-g|\,\mathrm{d}x}_{\geq 0} + \underbrace{\int_{c-\delta}^{c+\delta} |f-g|\,\mathrm{d}x}_{>0} + \underbrace{\int_{c+\delta}^{b} |f-g|\,\mathrm{d}x}_{\geq 0} > 0$$

Hence $\rho(f,g) > 0$. Thus $\rho(f,g) = 0$ if and only if $f = g$.

The symmetry of ρ is immediate from its definition.

Then to show that ρ satisfies the triangle inequality, consider:

$$\begin{aligned}
\rho(f,g) &= \int_a^b |f-g|\,\mathrm{d}x \\
&= \int_a^b |f-h+h-g|\,\mathrm{d}x \\
&\leq \int_a^b |f-h| + |h-g|\,\mathrm{d}x \\
&= \int_a^b |f-h|\,\mathrm{d}x + \int_a^b |h-g|\,\mathrm{d}x \\
&\leq \rho(f,h) + \rho(h,g)
\end{aligned}$$

Thus ρ is a metric on the space of continuous functions. ■

□

Now that we have defined metrics and looked at some examples, we can define **metric spaces**.

Definition 5.1.6 Metric space. A **metric space** $M = (X, \rho)$ is a set X together with a metric ρ on that set. ◊

Metric spaces were introduced by Fréchet in 1906, but the term is due Haus-

dorff.

Example 5.1.7 Examples of metric spaces. From our work in the above we have some good examples of metric spaces.

- The real numbers with the usual distance $\rho(x, y) = |x - y|$
- Any non-empty set A with the discrete metric 5.1.2
- The set of points in $\mathbb{R}^2$ with the taxi-cab metric 5.1.3
- The set $C[a, b]$ of continuous functions on $[a, b]$ with $\rho(f, g) = \max\{|f(x) - g(x)| \mid a \leq x \leq b\}$

□

Once we have a metric space (X, ρ) we can generate lots more, just by restricting to some subset of X.

Definition 5.1.8 Metric subspaces. Let (X, ρ) be a metric space. Now let $Y \subseteq X$, and let ρ' be the restriction of ρ to Y. That is, $\rho'(a, b) = \rho(a, b)$ for any $a, b \in Y$. Then (Y, ρ') is a **metric subspace**.

Note that this notation is often abused, and we will simply write this as (Y, ρ), dropping the prime on ρ. ◊

Exercises

1. **You'll never forget the properties of a metric space after this.** For each of the following sets and corresponding function, decide whether or not it is a metric space. Justify your answers.

   (a) $X = \mathbb{R}, \quad d(x, y) = |x - y|^3$

   (b) $X = \mathbb{R}, \quad d(x, y) = \sqrt{|x - y|}$

   (c) $X = \mathbb{R}, \quad d(x, y) = |x - y| \cdot |x + y|$

   (d) $X = \mathbb{R}, \quad d(x, y) = |x^2 - y^2|$

   (e) $X = \mathbb{Z}, \quad d(x, y) = |x - y|$

   (f) $X = \mathbb{R}, \quad d(x, y) = \frac{|x-y|}{1+|x-y|}$.

   (g) $X = \mathbb{R}_{\geq 0}, \quad d(x, y) = \max\{x, y\}$

   (h) $X = \mathbb{R}, \quad d(x, y) = \min\{1, |x - y|\}$

   (i) $X = \mathbb{R}, \quad d(x, y) = |x - 2y|$

   (j) $X = \mathbb{R}, \quad d(x, y) = \sin(|x - y|)$

Hint. In Item 5.1.1.f, when considering the triangle inequality, think about the function

$$f(t) = \frac{t}{1+t}$$

and whether or not it satisfies $f(a+b) \leq f(a) + f(b)$.

2. **Are we out of the woods?**[1]**.** Suppose you're on a hike through the woods along a fixed path, and you want to define a function ρ which takes two points x, y along the path and returns the amount of energy you've exerted getting from x to y. Is this ρ a metric? Explain.

3. **A strange metric.** Consider the set of all finite length strings composed of the letters $a, b, c, \ldots, y, z$. Given strings A and B, define $d(A, B)$ to be the minimum number of individual changes (either additions of, removal of, or substituting of individual letters) to change A to B. Is d a metric on this set? Why or why not?

4. **A strange metric on positive reals.** Let $\mathbb{R}^+$ be the set of real *positive* numbers $x > 0$. Consider the following strange function $d : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}$ given by

$$d(x, y) = |x - y| + \left|\frac{1}{x} - \frac{1}{y}\right|$$

which looks like the sum of the distance between the two numbers, and the distance between their reciprocals.

   - Show that $(\mathbb{R}^+, d)$ is a metric space.
   - Consider some convergent sequence of positive numbers with respect to the standard Euclidean metric. Does this sequence also converge[2] in the metric space?

5. **Draw the "circles".** Consider the following three metrics on the set $\mathbb{R}^2$:

   - Euclidean: $d_1((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
   - Taxicab: $d_2((x_1, y_1), (x_2, y_2)) = |x_2 - x_1| + |y_2 - y_1|$
   - Chebyshev: $d_3((x_1, y_1), (x_2, y_2)) = \max\{|x_2 - x_1|, |y_2 - y_1|\}$

   Verify that these are, in fact, metrics. Also for each of them, draw the set

$$S = \{(x, y) \in \mathbb{R}^2 : d_i((x, y), (0, 0)) = 1\}$$

6. **Isometric (indistinguishable) metrics.** Given two metric spaces (X, d_1) and (Y, d_2), we say that they are **isometric** if there exists a distance-preserving bijection between them. In other words, there is some bijection

[1] one of the authors is a Swiftie

[2] We have not told you what it means for a sequence to converge in a general metric space yet. Check out Definition 5.3.1

$\varphi : X \to Y$ with the property that

$$d_1(x, y) = d_2(\varphi(x), \varphi(y))$$

for all choices of $x, y \in X$. Isometric metric spaces are **indistinguishable** given only the properties of the given metrics. For this question, let $(\mathbb{R}^2, d_2)$ and $(\mathbb{R}^2, d_3)$ be as in the previous question.

- Let $f : (\mathbb{R}^2, d_2) \to (\mathbb{R}^2, d_3)$ be given by $f(x, y) = (x + y, x - y)$. Show that f is a bijection.
- Show that f preserves distances.

Thus, we have shown (or you will show) that $(\mathbb{R}^2, d_2)$ and $(\mathbb{R}^2, d_3)$ are isometric metric spaces!

The following exercise might come up later when we discuss compactness.

7. Extended real numbers. Let $X = \mathbb{R} \cup \{\pm\infty\}$. Note the symbols $+\infty$ and $-\infty$ at this point are just that — *symbols* — don't think of them as anything but two new elements of a set. Define a function $\varphi : X \to \mathbb{R}$ by

$$\varphi(x) = \begin{cases} \arctan(x) & x \in \mathbb{R} \\ \pi/2 & x = +\infty \\ -\pi/2 & x = -\infty. \end{cases}$$

Then define $d : X \times X \to \mathbb{R}$ by

$$d(x, y) = |\varphi(x) - \varphi(y)|.$$

(a) Prove that (X, d) is a metric space.

(b) Define an ordering $<$ on X by saying that $x < y$ if $\varphi(x) < \varphi(y)$. Show that this is, in fact, an ordering of X with $+\infty$ being an upper bound and $-\infty$ a lower bound. Also show that it preserves the natural ordering on $\mathbb{R}$ that we defined in chapter 2, meaning that if $x, y \in \mathbb{R}$ then $x < y$ in this ordering if and only if $x < y$ as real numbers.

(c) Given the ordering $<$ on X as defined in (b), show that *every* subset of X has a least upper bound and greatest lower bound (not just every bounded set).

5.2 Open and closed sets

5.2.1 Open and closed sets

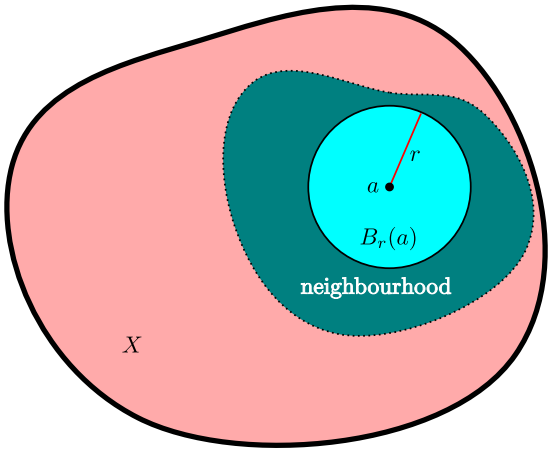
Metric spaces allow us to abstract many of the concepts we have learned about real numbers earlier in this course, to more general objects. For example, we can

describe open and closed sets in our metric space. We generalize the definition of open and closed sets of $\mathbb{R}$ that we saw in Section 2.2. In particular, recall that we called a set $A \subseteq \mathbb{R}$ **open** if for each point $x \in A$, there is some $r > 0$ such that the small **neighbourhood** $(x - r, x + r)$ of x should be fully contained in A. This idea naturally generalises to an arbitrary metric space.

Definition 5.2.1 Open ball. Let (X, ρ) be a metric space and let $a \in X$ and $r > 0$. The **open ball about** a **of radius** r is the set

$$B_r(a) = \{x \in X \mid \rho(x, a) < r\}$$

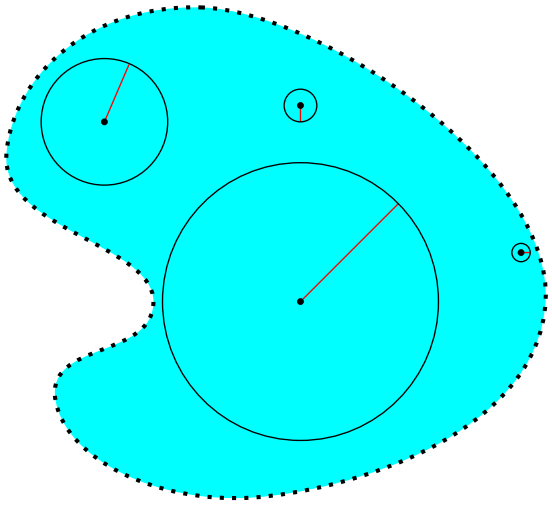
Further, the set $A \subseteq X$ is a **neighbourhood** of a if there is some $\varepsilon > 0$ so that $B_\varepsilon(a) \subseteq A$. That is, A contains some open ball about a. ◊



From the definition of open balls, we can define more general **open** sets.

Definition 5.2.2 Open sets. Let (X, ρ) be a metric space and let $A \subseteq X$. We say that A is **open** when for every $a \in A$ there is $\varepsilon > 0$ so that $B_\varepsilon(a) \subseteq A$.

Hence a set A is open when it is a neighbourhood of every $a \in A$. ◊



Example 5.2.3 This is consistent with what we did in Section 2.2. We already used the word "open" when discussing the real numbers. Indeed, recall in Definition 2.2.3 that we defined a set $A \subseteq \mathbb{R}$ to be open if for every $x \in A$ there is some $r > 0$ such that $(x - r, x + r) \subseteq A$. This is consistent with Definition 5.2.2 because the metric we are implicitly using in $\mathbb{R}$ is the Eucliden metric $d(x, y) = |x - y|$, and in this case $B_r(x) = (x - r, x + r)$. So all the results about open sets in Section 2.2 continue to be true with this definition of **open** sets.

In particular, open intervals like $(0, 1)$ are open, closed intervals like $[0, 1]$ and half-open-half-closed intervals like $[0, 1)$ are all not open $\mathbb{R}$ (with the Euclidean metric).

That Y is not open, does not imply that Y is **closed**. Some more work is required before we can define a **closed** set. □

We should prove that the **open balls** that we have defined are compatible with our definition of **open**.

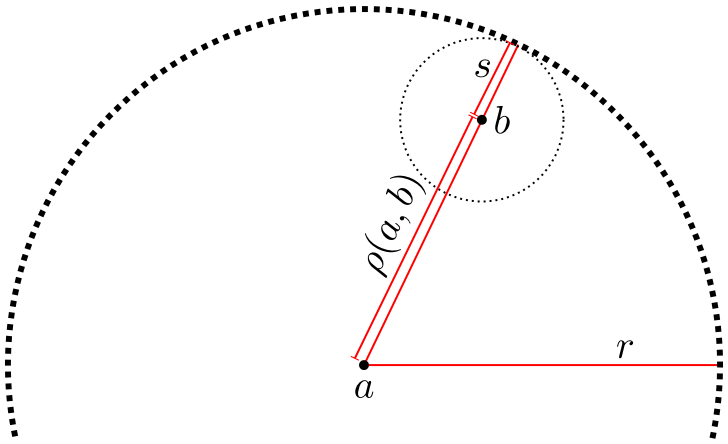
Lemma 5.2.4 Open balls are open. *Let (X, ρ) be a metric space, let $a \in X$ and let $r > 0$. The open ball $B_r(a)$ is an open set.*

Proof. Fix a, r and set $Y = B_r(a)$. To show that Y is open we must show that any $b \in Y$ can be contained in a ball that is itself contained in Y.

Now since $b \in Y$, we know that $\rho(a, b) < r$ and thus $s = r - \rho(a, b) > 0$. We will show that the ball $B_s(b)$ is contained in Y. Let $c \in B_s(b)$, so that $\rho(c, b) < s$. Now consider $\rho(a, c)$. By the triangle inequality

$$\rho(a, c) \leq \rho(a, b) + \rho(b, c) < \rho(a, b) + s = \rho(a, b) + (r - \rho(a, b)) = r$$

Thus $c \in Y$. Hence $B_s(b) \subseteq B_r(a)$, and so $B_r(a)$ is an open set as required.



■

So far, we've defined open sets (for a general metric space) in a synonymous manner to the one we used to define open sets in $\mathbb{R}$ in Section 2.2. Recall there, we decided to make the definition that a set $A \subseteq \mathbb{R}$ is called "closed"

if its complement[1] is open. For example, the interval $A = [0,1] \subseteq \mathbb{R}$ is the complement (in the reals) of the open set

$$(-\infty, 0) \cup (1, \infty).$$

Since that all seemed to work out OK for us, we will do the same thing in the general case as well. After all, the general notion of a metric space is kind of a natural generalization of the real numbers.

Definition 5.2.5 Closed sets. Let (X, ρ) be a metric space, and let $A \subseteq X$. We say that A is **closed** in X when its complement $A^c = X - A$ is open. ◊

Thus $(0,1)$ is open in the reals, while $[0,1]$ is closed. The set $(0,1]$ is neither open nor closed. The following result tells us that there are always two sets in our metric space that are both open and closed.

Warning 5.2.6 Closed is not "not open". The examples just above show us an important point. Just because a given set is not open, it does not follow that is closed. Indeed, the set $(0,1]$ as a subset of the real line is neither open nor closed.

Recall that we showed in Exercise 2.2.3.8 that the only subsets of $\mathbb{R}$ (with the Euclidean metric) that is both open *and* closed is the emptyset $\emptyset$ and the entire real set itself. A similar result is true in an arbitrary metric space (X, ρ), though in general there may be other sets which are both open and closed (see Exercise 5.2.4.9, for example).

Result 5.2.7 *Let (X, ρ) be a metric space. The set X is both open and closed. Consequently, the empty set is also both open and closed.*

Proof. Consider $x \in X$. Since the open ball $B_1(x)$ is defined to be a subset of X, it follows that every point in X is in an open ball that is contained in X. Thus X is open. By definition, this implies that $X^c = \varnothing$ is closed.

Now consider the empty set. Since it contains no points, it is vacuously true that every point in $\varnothing$ is in a ball that is contained in $\varnothing$. Hence the empty set is open, and, by definition, $\varnothing^c = X$ is closed. ■

Example 5.2.8 The underlying metric space matters. Consider the set $A = \{0\}$ inside the reals with the usual, i.e., Euclidean, metric $d(x,y) = |x - y|$. The complement of this set is $(-\infty, 0) \cup (0, \infty)$ is open. Thus A is closed.

Now change the underlying metric to the discrete metric:

$$\rho(x,y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

Let $\varepsilon = \frac{1}{2}$, then the open ball about 0 is

$$B_\varepsilon(0) = \left\{ x \in \mathbb{R} \mid \rho(x,0) < \frac{1}{2} \right\} = \{0\}$$

[1] Don't confuse the "complement" with the "compliment".

since every non-zero point is distance 1 from 0. Thus A is open in the reals with the discrete metric.

We can actually go further and show that $A = \{0\}$ is also closed in the reals with the discrete metric. Consider any element q of $A^c = (-\infty, 0) \cup (0, \infty)$. By the reasoning above, there is always a ball of radius $1/2$ about q that is contained in A^c. Thus A^c is open. But this then implies that $(A^c)^c = A$ is closed in the reals with the discrete metric.

Indeed, under the discrete metric, every set is both open and closed. □

By definition of **closed** we know how openness and closedness interact with taking the complement. We can, and should, understand how the openness and closedness of sets interact with intersections and unions. Finite unions and intersections preserve openness and closedness, but more care is required for infinite unions and intersections. One can quickly cook up examples such as

$$\bigcup_{n\in\mathbb{N}} [0, 2 - 1/n] = [0, 2)$$

and

$$\bigcap_{n\in\mathbb{N}} (-1, 1/n) = (-1, 0]$$

which demonstrate this.

Theorem 5.2.9 Unions and intersections of open and closed sets. *Let $\mathcal{A} = \{A_i \mid i \in I\}$ be a collection of open sets. Then*

- *The union $\bigcup_{i\in I} A_i$ is open*
- *If $\mathcal{A}$ is finite, then the intersection $\bigcap_{i\in I} A_i$ is open*

Then let $\mathcal{C} = \{C_i \mid i \in I\}$ be a collection of closed sets. Then

- *The intersection $\bigcap_{i\in I} C_i$ is closed*
- *If $\mathcal{C}$ is finite, then the union $\bigcup_{i\in I} C_i$ is closed*

Proof. The proofs are nearly identical to the analogous claims in the standard Euclidean case that we discussed in Section 2.2. The only difference is you replace $(x - r, x + r)$ in the Euclidean case with $N_r(x)$ in the general case. The details are left to the reader. ■

5.2.2 Interior, closure and boundary

Defining closed sets as complements of open sets feels more than a little indirect. We'll work, slowly, towards a somewhat more direct definition, but we'll get there via a bit of a digression[2]. Consider the following (related) intervals $[0, 1], (0, 1), [0, 1)$.

[2]Ironic, given that we are trying to be more direct.

- $[0,1]$ is closed, but $(0,1)$ is its largest open subset
- $(0,1)$ is open, but $[0,1]$ is its smallest closed superset
- $[0,1)$ is neither open nor closed, but $(0,1)$ is its largest open subset, and $[0,1]$ is its smallest closed superset

In each case the points $0, 1$ somehow lie right at the edge between being "inside" and "outside" the set. Let us make all these ideas more precise.

Definition 5.2.10 Interior, closure and boundary. Let (X, ρ) be a metric space and let $A \subseteq X$. Then

- the **interior** of A is the set

$$\operatorname{int}(A) = \bigcup \{B \mid B \text{ is open and } B \subseteq A\}.$$

  and is also sometimes denoted by A° or $\langle A \rangle$.

- the **closure** of A is the set

$$\operatorname{cl}(A) = \bigcap \{C \mid C \text{ is closed and } C \supseteq A\}$$

  and is also sometimes denoted $\overline{A}$.

- the **boundary** of A is the set

$$\delta(A) = \operatorname{cl}(A) - \operatorname{int}(A)$$

$\Diamond$

We should confirm that these definitions match up with our intuitive ideas of what it means to be in the **interior** of an object, or at the **boundary**. First of all, the interior should be *inside* the set, while the closure should *contain* all of the set.

Returning to the example of the interval $A = [0,1)$ in the reals with the usual metric, then

$$\operatorname{int}(A) = (0,1) \qquad \operatorname{cl}(A) = [0,1] \qquad \delta(A) = \{0,1\}$$

Notice that

- For any point in $\operatorname{int}(A)$ we can always find some open ball around it that still lies within A
- For any point in $\operatorname{cl}(A)$, every ball around contains some element of A
- For any point in $\delta(A)$, every ball contains an element of A and an element outside A

Also notice that for $A^c = (-\infty, 0) \cup [1, \infty)$ we have

$$\begin{aligned} \operatorname{int}(A^c) &= (-\infty, 0) \cup (1, \infty) = (\operatorname{cl}(A))^c \\ \operatorname{cl}(A^c) &= (-\infty, 0] \cup [1, \infty) = (\operatorname{int}(A))^c \\ \delta(A^c) &= \{0, 1\} = \delta(A) \end{aligned}$$

Suggesting a close relationship between interiors, closures, boundaries and complements.

Lemma 5.2.11 Open interior and closed closure. *Let (X, ρ) be a metric space and let $A \subseteq X$. Then*

$$\operatorname{int}(A) \subseteq A \subseteq \operatorname{cl}(A)$$

and $\operatorname{int}(A)$ *is open, while* $\operatorname{cl}(A)$ *and* $\delta(A)$ *are closed.*

Proof. Let $a \in \operatorname{int}(A)$. By definition, $a \in B$ where B is some open subset of A. Thus $a \in A$. Let $x \in A$ and let C be any closed superset of A. Then $x \in C$. Thus $x \in \operatorname{cl}(A)$.

Since $\operatorname{int}(A)$ is the union of open sets, and $\operatorname{cl}(A)$ is the intersection of closed sets, by Theorem 5.2.9, the interior is open and the closure is closed. We can write the boundary as $\operatorname{cl}(A) \cap (X - \operatorname{int}(A))$ and so it is the intersection of two closed sets, and thus also closed by Theorem 5.2.9. ■

Now, if something is the **interior** of the set, then there should be some finite distance between it and the "outside" of the set. That is, we should be able to put a small ball around any point that lies in the interior.

Theorem 5.2.12 Properties of the interior. *Let (X, ρ) be a metric space and let $A \subseteq X$. Then*

- $a \in \operatorname{int}(A)$ *if and only if there is $\varepsilon > 0$ so that $B_\varepsilon(a) \subseteq A$*
- *A is open if and only if $A = \operatorname{int}(A)$*
- $\operatorname{int}(\operatorname{int}(A)) = \operatorname{int}(A)$

Proof. We prove each statement in turn.

- ◦ Let $a \in \operatorname{int}(A)$. By Lemma 5.2.11 $\operatorname{int}(A)$ is open and so there is some $B_\varepsilon(a) \subseteq \operatorname{int}(A) \subseteq A$.
  ◦ Now assume that there is some $B_\varepsilon(a) \subseteq A$. Since that ball is an open subset of A it is, by definition, also a subset of the interior. Thus $a \in \operatorname{int}(A)$.
- ◦ Assume that $A = \operatorname{int}(A)$. By Lemma 5.2.11 $\operatorname{int}(A)$ is open, and thus A is open.
  ◦ Now assume that A is open. By Lemma 5.2.11 $\operatorname{int}(A) \subseteq A$, so it suffices to show that $A \subseteq \operatorname{int}(A)$. Let $a \in A$, then since A is open, we know that we can find some open ball $B_\varepsilon(a)$ about a so that $B_\varepsilon(a) \subseteq$

A. Then since this ball is open and a subset of A, by definition, $B_\varepsilon(a) \subseteq \operatorname{int}(A)$, and thus $a \in \operatorname{int}(A)$. Thus $A \subseteq \operatorname{int}(A)$ as required.

- Let $B = \operatorname{int}(A)$, so it is sufficient to prove that $\operatorname{int}(B) = B$. Now, by Lemma 5.2.11, we know that B is open (being an interior), and so by the proof above, we know that $B = \operatorname{int}(B)$ as required.

■

This theorem gives us a good criterion to show that a set is *not* open. We just[3] need to find a point $a \in A$ that does not lie in the interior. To do that, we can show, for example, that any ball about that point contains elements outside of A. See Example 5.2.21.

If an object lies in the **closure** of a set, then it should always be "close" to the set. That is, no matter how small a ball we draw around the point, that ball should always intersect the set.

Theorem 5.2.13 Properties of the closure. *Let (X, ρ) be a metric space and let $A \subseteq X$. Then*

- *$a \in \operatorname{cl}(A)$ if and only if for all $\varepsilon > 0$, $B_\varepsilon(a) \cap A \neq \varnothing$*
- *A is closed if and only if $A = \operatorname{cl}(A)$*
- $\operatorname{cl}(\operatorname{cl}(A)) = \operatorname{cl}(A)$

Proof. We prove each statement in turn.

- We prove both implications by their contrapositives.
  - Assume that $x \notin \operatorname{cl}(A)$, and so $x \in X - \operatorname{cl}(A)$ which is an open set. Thus there is some $\varepsilon > 0$ so that $B_\varepsilon(x) \subseteq X - \operatorname{cl}(A)$. Hence $B_\varepsilon(x)$ is disjoint from A .
  - Now assume that there is some $\varepsilon > 0$ so that $B_\varepsilon(x) \cap A = \varnothing$ But then $X - B_\varepsilon(x)$ is closed and a superset of A. Since $\operatorname{cl}(A)$ is the intersection of all closed supersets of A, it follows that $x \notin \operatorname{cl}(A)$.
- We prove each implication directly.
  - Assume that $A = \operatorname{cl}(A)$. By Lemma 5.2.11 $\operatorname{cl}(A)$ is closed, and thus A is closed.
  - Now assume that A is closed. By Lemma 5.2.11 we know that $A \subseteq \operatorname{cl}(A)$, so it suffices to show that $\operatorname{cl}(A) \subseteq A$. Let $x \in \operatorname{cl}(A)$, then by definition x is in every closed superset of A. Since A is a closed superset of itself, it follows that $x \in A$. So we have shown that $A = \operatorname{cl}(A)$.
- Finally, write $B = \operatorname{cl}(A)$, so by Lemma 5.2.11 B is a closed set. Then by the result above, we immediately have that $B = \operatorname{cl}(B)$ as required.

[3] Of course, this might not actually be all that easy.

■

Just as Theorem 5.2.12 gave us a way to show that a set is not open, this theorem gives us a good criterion to show that a set is *not* closed. We need to find a point $a \notin A$ that lies in the closure of A. To show that it is in the closure, we can show that any ball about the point always contains a point of A. See Example 5.2.21.

Interiors and closures play very nicely (as one might expect) with set complements. The following result formalises this and shows how one can write the closure in terms of the interior and vice-versa.

Lemma 5.2.14 Interior and closure complements. *Let (X, ρ) be a metric space and let $A \subseteq X$. Then*

$$X - \operatorname{cl}(A) = \operatorname{int}(X - A) \qquad \textit{and} \qquad X - \operatorname{int}(A) = \operatorname{cl}(X - A)$$

and, equivalently,

$$\operatorname{cl}(A) = X - \operatorname{int}(X - A) \qquad \textit{and} \qquad \operatorname{int}(A) = X - \operatorname{cl}(X - A).$$

Proof. Let $x \in X - \operatorname{cl}(A)$. Since $x \notin \operatorname{cl}(A)$, by Theorem 5.2.13, there exists some $\varepsilon > 0$ so that $B_\varepsilon(x) \cap A = \varnothing$. Thus $B_\varepsilon(x) \subseteq X - A$. Thus x is in some open subset of $X - A$ and thus in $\operatorname{int}(X - A)$.

Conversely, let $x \in \operatorname{int}(X - A)$. By Theorem 5.2.13 this means that there is some $B_\varepsilon(x) \subseteq X - A$. However, this means that $(X - B_\varepsilon(x))$ is a closed superset of A that does not contain x. Since $\operatorname{cl}(A)$ is the intersection of all of its closed supersets, it follows that $x \notin \operatorname{cl}(A)$. ■

Finally, as its name suggests, the boundary must be those points that are both nearly *inside* the set and at the same time nearly *outside*. Indeed, if we draw a small ball around any boundary point, that ball should always contain some point inside the set, and some point outside the set.

Theorem 5.2.15 Properties of the boundary. *Let (X, ρ) be a metric space and let $A \subseteq X$. Then*

$$\delta(A) = \operatorname{cl}(A) \cap \operatorname{cl}(X - A) = \delta(X - A)$$

Additionally, let $x \in \delta(A)$, then for all $\varepsilon > 0$, $B_\varepsilon(x) \cap A \neq \varnothing$ and $B_\varepsilon(x) \cap (X - A) \neq \varnothing$.

Proof. The equality follows quite directly from Lemma 5.2.14.

$$\begin{aligned}
\delta(A) &= \operatorname{cl}(A) - \operatorname{int}(A) \\
&= \operatorname{cl}(A) \cap (X - \operatorname{int}(A)) \\
&= \operatorname{cl}(A) \cap \operatorname{cl}(X - A) \\
&= (X - \operatorname{int}(X - A)) \cap \operatorname{cl}(X - A) \\
&= \operatorname{cl}(X - A) - \operatorname{int}(X - A) = \delta(X - A).
\end{aligned}$$

Now let $x \in \delta(A)$. Hence, by the above, $x \in \operatorname{cl}(A)$ and $x \in \operatorname{cl}(X - A)$. Thus

by Theorem 5.2.13, for any $\varepsilon > 0$, the open ball $B_\varepsilon(x)$ intersects both A and $X - A$. ■

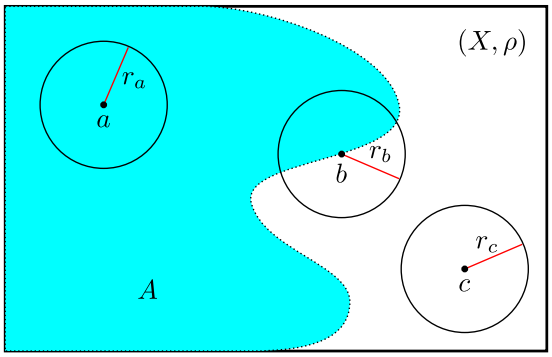


Figure 5.2.16 A subset A of a metric space (X, ρ). The point $a \in A$ is in the interior of A since there is some open ball $B_{r_a}(a)$ around a that is a subset of A. The point c lies outside of A and its closure. More precisely it lies in the interior of the complement of A since there is an open ball $B_{r_c}(c)$ contained in the complement of A. The point b lies on the boundary of A since for every r_b, the ball $B_{r_b}(b)$ will contain points both inside and outside A.

Example 5.2.17 $\mathbb{Q}$ is not so nice (part 1). By this point, we're all familiar and comfortable with the set of rational numbers $\mathbb{Q}$. But let's spend a moment exploring this set in a little more detail, in the setting of metric spaces. We will consider this set with its "usual" metric $d(x, y) = |x - y|$. The question we'd like to answer with this metric space is the following:

Question: Is the set $A = [-\sqrt{2}, \sqrt{2}] \cap \mathbb{Q}$, an open or closed (or neither) subset of $(\mathbb{Q}, d)$?

If you prefer, you can define A as $A = \{q \in \mathbb{Q} \mid q^2 \leq 2\}$.

Take a moment to think about it before looking at the solution.The authors apologise for the trick question. The answer is that A is *both* open and closed! It is open because A is really just the open ball in $\mathbb{Q}$ about 0 or radius $\sqrt{2}$:

$$A = B_{\sqrt{2}}(0),$$

We know that the open ball is open by Lemma 5.2.4. At the same time it is closed because its complement

$$\mathbb{Q} \setminus A = (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$$

can be written as the union of open balls:

$$\mathbb{Q} \setminus A = \left(\bigcup_{n \geq 1} B_{n-\sqrt{2}}(-n) \right) \bigcup \left(\bigcup_{n \geq 1} B_{n-\sqrt{2}}(n) \right)$$

Since the complement is the union of open sets it is open, and hence the original set is closed.

In fact, whenever one has a sub-metric space (Y, d) of a larger space (X, d) — in this case, $\mathbb{Q} \subset \mathbb{R}$ — then the open (resp. closed) sets of Y are precisely the

intersections of open (resp. closed) sets in X with the subset Y. In this case

$$A = [-\sqrt{2}, \sqrt{2}] \cap \mathbb{Q} = (-\sqrt{2}, \sqrt{2}) \cap \mathbb{Q}$$

can be written as the intersection of $\mathbb{Q}$ with an open set in $\mathbb{R}$, that is $(-\sqrt{2}, \sqrt{2})$, and it can *also* be written as the intersection of $\mathbb{Q}$ with a closed set in $\mathbb{R}$, namely $[-\sqrt{2}, \sqrt{2}]$. The reason why this works is that the points that we would consider to be the boundary points of A are not contained in the metric space $(\mathbb{Q}, d)$!

More on this a bit later in Example 5.3.8. □

5.2.3 Back to closed sets

Let's get back to closed sets. We have been working with closed sets as complements of open sets. However, Theorem 5.2.13 can lead us to an alternative. In particular, we know that

$$a \in \mathrm{cl}\,(A) \quad \iff \quad \forall \varepsilon > 0, B_\varepsilon(a) \cap A \neq \varnothing$$

So, around any point in the closure, we can always find a point of A. We should really bit a little more nuanced about this. Since $a \in B_\varepsilon(a)$, we should perhaps consider what happens if we want there to be some *other* point of the set A nearby. That is

$$\forall \varepsilon > 0, \exists x \in B_\varepsilon(a) \cap A \text{ s.t. } x \neq a.$$

For example, if we consider the set $A = (0, 1) \cup \{2\}$, then both $a = 1$ and $a = 2$ lie in the closure of A but in different ways. If we take any $B_\varepsilon(1)$ then it contains many other points in A, however $B_{1/2}(2) \cap A = \{2\}$. This shows how the point 2 is *isolated* from the rest of the set, while any open ball about 1 contains an infinite number of points of A. This leads to the following definition.

Definition 5.2.18 Accumulation points and isolated points. Let (X, ρ) be a metric space and let $A \subseteq X$.

- We say that $x \in X$ is an **accumulation point** (or **limit point**) of A when for every $\varepsilon > 0$, the ball $B_\varepsilon(x)$ contains some $a \in A$ with $a \neq x$.
- By contrast, we say that $a \in A$ is an **isolated point** of A when there exists some $\varepsilon > 0$ so that $B_\varepsilon(a) \cap A = \{a\}$.

That is, every open ball about an accumulation point contains an infinite number of points of A, and there exists an open ball about an isolated point that contains no other elements of A.

Note that an **accumulation point** of a set need not be in that set, while an isolated point must be. For example, the set $A = (0, 1) \cup \{2\}$ has one isolated point ($x = 2$, which is in A) and many accumulation points (the whole interval $[0, 1]$ are accumulation points) with not all of them being contained in A. ◊

Theorem 5.2.19 Accumulation points, isolated points and closure. *Let*

(X, ρ) be a metric space and let $A \subseteq X$. Then

$$\mathrm{cl}\,(A) = \{x \in X \mid x \text{ is an accumulation point of } A.\} \cup A.$$

We can actually go further and write the closure as the disjoint union of all the accumulation points of A and all the isolated points of A.

Proof. We prove each inclusion in turn.

Let $x \in \mathrm{cl}\,(A)$. If $x \in A$ then we are done. So now assume that $x \notin A$. However, since $x \in \mathrm{cl}\,(A)$, by Theorem 5.2.13, for any $\varepsilon > 0$, we know that $B_\varepsilon(x) \cap A \neq 0$. Since $x \notin A$, we know that this ball contains some point of A that is not x. Hence x is an accumulation point of A.

Now let $x \in RHS$. If $x \in A$, then, by Lemma 5.2.11 we know that $A \subseteq \mathrm{cl}\,(A)$, so $x \in \mathrm{cl}\,(A)$. So now assume that $x \notin A$, and so it must be an accumulation point of A. Thus any ball around x contains a point in A and so by Theorem 5.2.13 we know $x \in \mathrm{cl}\,(A)$.

To complete the proof it suffices to show that any point in A is either an accumulation point or an isolated point. So let $a \in A$. If a is an accumulation point we are done. On the other hand, if it is not an accumulation point then there exists some $\varepsilon > 0$ so that $B_\varepsilon(a) \cap A$ only contains a and no other points of A. Hence a is an isolated point. ■

Corollary 5.2.20 Closure and accumulation points. *Let (X, ρ) be a metric space and let $A \subseteq X$. Then, A is closed if and only if it contains all its accumulation points.*

Proof. Assume A contains all its accumulation points. Then, by the result we have just proved, we know that $A = \mathrm{cl}\,(A)$ and so, by Theorem 5.2.13, we know A is closed.

Now assume that A is closed. Hence we know that A is equal to its closure. So by the result we have just proved, A must contain all its accumulation points. ■

These results show us that we can define a closed set in terms of accumulation points rather than as the complement of an open set. It is also a very useful way to show that a set is not closed, we just need to find an accumulation point that is outside the set.

Example 5.2.21 Neither closed nor open. Let $A = (0, 1]$. This set is neither closed nor open. We can prove the former by finding an accumulation point outside the set, and we can prove the latter by finding a point in the set that is not an interior point.

Consider the point $x = 0$ which lies outside A. However, for any $\varepsilon > 0$, the point $a = \varepsilon/2$ lies inside the ball $B_\varepsilon(x)$. Thus $x = 0$ is an accumulation point of A but lies outside A. So by Theorem 5.2.19 is not closed.

Now consider the point $a = 1 \in A$. Let $\varepsilon > 0$. The point $b = 1 + \varepsilon/2 > 1$ lies outside A but lies inside the ball $B_\varepsilon(a)$. Hence for any $\varepsilon > 0$ $B_\varepsilon(a) \not\subseteq A$. Hence by Theorem 5.2.12 $a \notin \mathrm{int}\,(A)$, and so A is not open. □

After that little digression, lets get back to limits.

5.2.4 Exercises

1. **No limit points.** Give an example of a set in $\mathbb{R}$ that has no limit points.
2. **Limit points of a pedagogical example.** Let $A = \left\{ \frac{1}{n} \;\middle|\; n \in \mathbb{N} \right\}$. What are the limit points of A?
3. **A contrived construction.** Construct a bounded set of real numbers with exactly three limit points.

   Hint. Maybe think how you might do it with exactly two limit points and then generalise.
4. **Did you pay attention?** True or False: Every limit point of a set $A \subseteq \mathbb{R}$ must belong to A.
5. **Finite sets are all isolated points.** Show that a finite set has no limit points.
6. **Limit points of limit points.** Suppose $A \subseteq \mathbb{R}$ and A' denotes the set of limit points of A. Prove or disprove: $(A')' = A'$.
7. **Closures and unions.** Let $A_1, A_2, A_3, \ldots$ be subsets of a metric space.

   (a) If $B_n = \bigcup_{i=1}^{n} A_i$, prove that $\operatorname{cl}(B_n) = \bigcup_{i=1}^{n} \operatorname{cl}(A_i)$ for $n = 1, 2, 3, \ldots$

   (b) If $B = \bigcup_{i=1}^{\infty} A_i$, prove that $\operatorname{cl}(B) \supseteq \bigcup_{i=1}^{\infty} \operatorname{cl}(A_i)$.

   (c) Show, by an example, that the inclusion in (b) can be proper.
8. **Limit points in $\mathbb{R}^2$.** Is every point of every open set $E \subseteq \mathbb{R}^2$ a limit point of E? Answer the same question for closed sets in $\mathbb{R}^2$. Note: We are considering $\mathbb{R}^2$ equipped with the "standard", i.e., Euclidean, metric $\rho(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$.
9. **Both open and closed.** Consider an arbitrary set X equipped with the discrete metric d. Show that *every* subset $A \subseteq X$ is both open *and* closed.
10. **More properties of interiors.** Let $\operatorname{int}(E)$ denote the set of all interior points of a set E. Above we proved that $\operatorname{int}(E)$ is always open.

    (a) If $G \subseteq E$ and G is open, prove that $G \subseteq \operatorname{int}(E)$.

    (b) Do E and $\operatorname{cl}(E)$ always have the same interiors?

    (c) Do E and $\operatorname{int}(E)$ always have the same closures?
11. **Parsing the words can be hard.** Show that the complement of the interior of a set is the closure of its complement.
12. **Limit points and open sets (this is useful later!).** Show that if x is a limit point of A, then every open set containing x contains infinitely many points of A.

Hint. Find a sequence of points in A approaching x by induction.

13. **Limit points is closed.** Prove that the set of limit points of a set $A \subseteq \mathbb{R}$ is closed.

    Hint. One method would be to use the triangle inequality to show that A' contains all its limits points.

14. **Density Equivalences.** Let E be a subset of a metric space X. Prove that the following statements are equivalent:

    (a) E is dense in X.

    (b) The only closed set which contains E is X.

    (c) The only open set disjoint from E is the empty set.

    (d) E intersects every non-empty open set.

    (e) E intersects every ball in X.

    Hint. You can prove all are equivalent by showing that each one implies the next and the last implies the first.

5.3 Limits in metric spaces

Our metric spaces are now equipped with a generalisation of distance from $|x-y|$ to $\rho(x, y)$, which, in turn, allowed us to define more general open and closed sets. Using all of this we can (re)define limits and, later, continuity in this more general setting.

Recall our definition of the limit of a sequence 2.3.1 of real numbers. We can map this to our more general metric spaces setting quite directly. Essentially all that is required is to change

$$|a_n - L| < \varepsilon$$

with the more general

$$a_n \in B_\varepsilon(L).$$

Notice that both of these statements just state that $\rho(a_n, L) < \varepsilon$.

Definition 5.3.1 Sequences and limits in metric spaces. Let (X, ρ) be a metric space. Let $(a_1, a_2, a_3, \dots) = (a_n)$ be a sequence of elements of X. We say that the sequence **converges** to the **limit** $L \in X$ when for any $\varepsilon > 0$ there exists an integer N so that if $n > N$ then $a_n \in B_\varepsilon(L)$. In this case we write $a_n \to L$.

Note that $a_n \in B_\varepsilon(L)$ is equivalent to $\rho(a_n, L) < \varepsilon$. ◊

So, all our standard examples of the convergence of sequences of real numbers carry over to this new definition. But we can also make some new ones.

Example 5.3.2 A sequence in $\mathbb{R}^2$. Consider a sequence (a_n) of points in $\mathbb{R}^2$, i.e., the plane, defined by

$$a_n = (\cos(n)/n, \sin(n)/n^2)$$

Intuitively, this sequence points spiral around and converge to the origin $(0,0)$. Of course, to make this rigorous, we need to know which metric we are using. Let's try two.

- Let $\rho((x,y),(z,w)) = \sqrt{(x-z)^2+(y-w)^2}$ be the usual Euclidean metric. Then notice that

$$\begin{aligned}\rho(a_n,(0,0)) &= \sqrt{\cos^2(n)/n^2 + \sin^2(n)/n^4} \\ &\leq \sqrt{\cos(n)^2/n^2 + \sin^2(n)/n^2} = \frac{1}{n}\end{aligned}$$

  Thus given any $\varepsilon > 0$, pick $N = \lceil 1/\varepsilon \rceil$. Now if $n > N$, we know that

$$\rho(a_n,(0,0)) \leq \frac{1}{n} < \frac{1}{N} \leq \varepsilon.$$

  So $a_n \in B_\varepsilon(0,0)$ and hence the sequence converges to $(0,0)$.

- Instead consider the taxi-cab metric $\rho((x,y),(z,w)) = |x-z| + |y-w|$. Then notice that

$$\rho(a_n,(0,0)) = |\cos(n)/n| + |\sin(n)/n^2| \leq \frac{2}{n}.$$

  So given any $\varepsilon > 0$, set $N = \lceil \frac{2}{\varepsilon} \rceil$. Then when $n > N$, we know that

$$\rho(a_n,(0,0)) \leq \frac{2}{n} < \frac{2}{N} \leq \varepsilon$$

  Hence, again, $a_n \to (0,0)$.

□

Example 5.3.3 Convergence depends on the metric. Consider the sequence of real numbers $(a_n) = (1/n)$. If we take the usual metric space on the reals (ie $\rho(a,b) = |a-b|$) then we know that this converges to 0. In particular, given any $\varepsilon > 0$ we pick $N = \lceil 1/\varepsilon \rceil$ and then when $n > N$ we know that

$$|a_n - 0| = \frac{1}{n} < \frac{1}{N} \leq \varepsilon.$$

However, if we change to the discrete metric 5.1.2 then since $a_n \neq 0$, we know that

$$\rho(a_n, 0) = 1 \quad \text{for all } n$$

And hence for $\varepsilon = 1/2$, $a_n \notin B_\varepsilon(0)$ for any n.

Indeed, convergence under the discrete metric is quite boring (though it does make useful counter-examples). To be more precise a sequence (a_n) converges to L under the discrete metric if and only if there is N so that when $k \geq N$, $a_k = L$. That is, the sequence converges if and only if its terms are eventually identical to the limit. □

Of course, mucking around with real numbers isn't such a generalisation. Instead consider convergence in the set of functions!

Example 5.3.4 Convergence of a sequence of functions. Recall Example 5.1.4 and consider the metric space defined on the set $C[0,1]$ of continuous functions on the closed interval $[0,1]$ with the metric

$$\rho(f,g) = \max\{|f(x) - g(x)| \mid a \leq x \leq b\}$$

Now, define a sequence (f_n) of *functions* in $C[0,1]$ defined by

$$f_n(x) = x^n(1-x)^n$$

and let $g(x) = 0$ be the constant zero function. Then $f_n \to g$.

To see this, consider $\rho(f_n, g)$:

$$\rho(f_n,g) = \max\{|f(x) - g(x)| \mid x \in [0,1]\} = \max\{|f(x)| \mid x \in [0,1]\}.$$

Now, with a little calculus 4.2.3 one can prove that on $[0,1]$, f_n takes its maximum at $x = 1/2$ and so $|f_n(x)| \leq 4^{-n}$. So given any $\varepsilon > 0$, we can pick $N = \lceil -\log_4 \varepsilon \rceil$. Then if $n > N$ we have

$$\rho(f_n, g) = 2^{-2n} < 4^{-N} < \varepsilon$$

as required.

We will revisit examples like this more in Section 5.6. □

Notice that the definition of limits looks very similar to the definition of accumulation point 5.2.18.

Lemma 5.3.5 Limits and accumulation points. *Let (X,ρ) be a metric space and let $A \subseteq X$. Then let (a_n) be a convergent sequence of elements of A with limit L. Then $L \in \mathrm{cl}(A)$ and so $L \in A$ or L is an accumulation point of A.*

Further, let $L \in \mathrm{cl}(A)$, then there is some sequence in A that converges to L.

Proof. Let $\varepsilon > 0$. Since $a_n \to L$, we know that for k sufficiently large, $a_k \in B_\varepsilon(L)$, and so $B_\varepsilon(L) \cap A \neq \varnothing$. Thus L is in the closure of A. By Theorem 5.2.19 we know that L is an accumulation point of A or lies in A.

To prove the reverse implication let $L \in \mathrm{cl}(A)$. If $L \in A$ then the constant sequence $a_n = L$ trivially converges to L. So now assume that $L \notin A$, and so by Theorem 5.2.19, we know that L is an accumulation point of A. So now let $n \in \mathbb{N}$ and set $\varepsilon = 1/n$; since L is an accumulation point, the intersection $B_\varepsilon(L) \cap A$ is non-empty. Let $a_n \in A$ be some point in that intersection. In this

way we construct a sequence (a_n) in A that converges to L. ■

This result leads us very quickly to another property of closed sets - this is essentially a restatement of Theorem 5.2.19.

Theorem 5.3.6 Limits and closed sets. *Let (X, ρ) be a metric space, let $A \subseteq X$, and let $\mathcal{A}$ be the set of sequences in A that are convergent. Then A is closed if and only if all $(a_n) \in \mathcal{A}$ converge to limits in A.*

Proof. Assume A is closed, and let (a_n) be a sequence in A that converges. By Lemma 5.3.5, the limit of the sequence, L, lies in the closure of A. And since A is closed, $L \in A$.

Now assume that all convergent sequences converge to points in A. Now let $x \in \mathrm{cl}(A)$, by Lemma 5.3.5 there is a sequence in A that converges to x, and so, by assumption, $x \in A$. Thus $A = \mathrm{cl}(A)$ and so A is closed. ■

Notice that this theorem gives us a very good way to show that a set is *not* closed. Namely, find a sequence that converges to a point outside of the set. We can construct a very similar example to Example 5.2.21.

Example 5.3.7 Example 5.2.21 (via a sequence). Again, take the set $A = (0, 1]$. We can show that A is not closed by finding a sequence in A that converges to $0 \notin A$. This is completely analogous to how we proved that A is not closed by showing that 0 is an accumulation point.

So let $a_n = 1/n \in A$. Then since $a_n \to 0$ and $0 \notin A$, it follows that A cannot be closed since there is a sequence in A that converges to a point outside A. □

Example 5.3.8 $\mathbb{Q}$ is not so nice (part 2). Let's go back to Example 5.2.17, that is, the space $\mathbb{Q}$ with its usual (Euclidean) metric $d(x, y) = |x - y|$. We saw in Example 5.2.17 that the set

$$A = [-\sqrt{2}, \sqrt{2}] \cap \mathbb{Q}$$

is both open and closed. Now that we have the understanding that closed sets are precisely those which contain all their limit points, we know that this must apply to this set A. In particular, the point of this example is to drive home the fact that $\pm\sqrt{2}$ are **not** limits points of A, for the reason that these points are not in the space!

Takeaway: In order for a point x to be a limit point of a set A, the point x has to be in the metric space! □

Exercises

1. **Constant sequences.** Show that a constant sequence in any metric space converges.

2. **Meaning of "non-convergence".** Write down — in logical quantifiers — what it means for a sequence $(x_n)_n$ in a metric space (X, d) to *not* converge.[1]

3. **Non-convergence and infinitely many points.** Suppose a sequence $(x_n)_n$ does not converge to a point x (in a metric space (X, d)). Prove that there is some $\varepsilon > 0$ and a subsequence $(x_{n_k})_k$ for which

$$d(x_{n_k}, x) \geq \varepsilon$$

for all k.

4. **A strange metric on $\mathbb{N}$.** Consider the set of natural numbers $\mathbb{N}$. Prove that

$$d : \mathbb{N}^2 \to \mathbb{R} \qquad d(n, k) = \left| \frac{1}{n} - \frac{1}{k} \right|$$

is a metric.

Then determine (with proof) what the convergent sequences are in this space.

After reading through the next section (about completeness), revisit this problem. Is $(\mathbb{N}, d)$ complete?

Hint. Assume the sequence (a_n) converges to L. Show that we must have $L \in \mathbb{N}$. Then show that for sufficiently large n we must have $a_n > L - 1$, and then also $a_n < L + 1$.

5. **Limits are unique.** Prove that if a sequence (x_n) in a metric space to converge, then the limit is unique. In other words, if $x_n \to x$ and $x_n \to y$, then $x = y$.

Hint. The triangle inequality will help you.

6. **Double sequences and convergence.** Let (x_n) and (y_n) be sequences in a metric space (X, d) such that $x_n \to x$ and $y_n \to y$. Prove that $d(x_n, y_n) \to d(x, y)$.

Hint. The triangle inequality is your friend.

7. **Convergence of subsequences.** Suppose (x_n) converges to x in a metric space (X, d). Prove that any subsequence of (x_n) also converges to x.

8. **Limit points can be the original set.** Give an example of a set whose set of limit points is equal to the set itself. Prove that your answer is correct.

9. **Convergent sequences are bounded.** Prove that a convergent sequence (a_n) in a metric space is necessarily bounded. That is, there is some c and $M > 0$ so that

$$\forall n, \quad d(x_n, c) \leq M$$

Hint. Compare to Lemma 2.3.4.

10. **Discrete metric has boring converging subsequences.** Prove that convergence in a discrete metric space implies the sequence is eventually

[1] This will be a useful condition in later work.

constant.

11. **Different metrics change convergence.** Give an example of a sequence that converges in one metric but not in another on the same underlying set.

12. **A strange, but equivalent, subsequence condition.** Fix some metric space (X, d) and a point $y \in X$. Let (x_n) be a sequence in X such that every subsequence of (x_n) has a further subsequence that converges to y. Show that the original sequence converges to y.

    Hint. Write down what it means for the sequence to not converge to L.

13. **Does it still work without the fixed point y?** Let (x_n) be a sequence in a metric space (X, d) such that every subsequence of (x_n) has a further subsequence that converges (but we don't know a priori what each subsequence converges to). Is it true that the original sequence must converge? Prove your answer is correct.

5.4 Completeness in metric spaces

The notion of **completeness** for metric spaces is the generalisation of the completeness of the reals 2.3.23. Much of what we do here mirror quite directly the work in this section 2.3.2.

Definition 5.4.1 Cauchy sequences in metric spaces. Let (X, ρ) be a metric space. The sequence $(a_n)_n$ in X is a **Cauchy sequence** when

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ so that } \forall k, \ell > N, \rho(a_k, a_\ell) < \varepsilon.$$

That is, the terms of the sequence eventually become arbitrarily close together. ◊

Just as was the case for convergent sequences of reals 2.3.17, any convergent sequence is Cauchy.

Lemma 5.4.2 Any convergent sequence is Cauchy. *Let (X, ρ) be a metric space. If the sequence $(a_n)_n$ converges in X then it is Cauchy.*

Proof. The proof is essentially the same as that of Lemma 2.3.17. ■

It is easy to see that not all Cauchy sequences converge.

Example 5.4.3 For convergence to be true, the limit has to be in the set. Consider the metric space given by the set $(0, 1]$ with the usual metric on the reals. Then the sequence $a_n = 1/n$ is Cauchy but not convergent *in the metric space*.

First, notice that the sequence converges to 0 which is outside the space. You can make this into a straightforward proof by contradiction. For example, assume, to the contrary that $a_n \to q \in (0, 1]$. However, we can then set $\varepsilon = q/2$. However, now, no matter what N we choose, there will always be n so that $1/n < \varepsilon$ (via Archimedes 2.1.2). That is, $a_n \notin B_\varepsilon(q)$. Hence $a_n \not\to q$.

At the same time, we can readily prove that the sequence is Cauchy. Let $\varepsilon > 0$, then pick $N = \lceil 2/\varepsilon \rceil$. Then when $k, \ell > N$, we know that $k, \ell > 3/\varepsilon$ and so $0 < 1/k, 1/\ell < \varepsilon/3$. Thus, by the triangle inequality, $|1/k - 1/\ell| < \frac{2}{3}\varepsilon < \varepsilon$ as required. □

Of course, the above example is a little contrived. We took a perfectly reasonable metric space $[0, 1]$ and a sequence that converged to the boundary point 0, and then removed 0 to make our example. We can build more sophisticated examples when they are needed, and we will.

However, this also points out an important result, if we have a closed set then we know, by Theorem 5.3.6, that all convergent sequences converge to points in the set. So we can cook up lots of examples like the one above. Take any sequence that converges to a point on the boundary. By Lemma 5.4.2 it is Cauchy. Now tweak the set to remove that boundary point, and bingo! We have a Cauchy sequence that does not converge. Notice that this also suggests that closed sets are important for completeness, and they are.

Theorem 5.4.4 Closed subsets of complete metric spaces are complete metric subspaces. *Let (X, ρ) be a complete metric space and let Y be a subset of X. Then the metric subspace (Y, ρ) is complete if and only if Y is closed.*

Proof. Let us start by assuming that Y is a closed subset of X, and form the metric subspace (Y, ρ) (where we restrict ρ to just points in Y). Now let $(a_n)_n$ be a Cauchy sequence in Y. Since $Y \subseteq X$, we know that $(a_n)_n$ is also a sequence in X, and thus, it converges to a limit $\alpha \in X$. Since Y is closed, Theorem 5.3.6 tells us that $\alpha \in Y$. Thus $a_n \to \alpha \in Y$. Thus Y is complete.

Now assume that Y is complete. We can prove that Y is closed, via Theorem 5.2.19, by showing that it contains all its accumulation points. So let α be an accumulation point of Y. Then for every $\varepsilon > 0$, the ball $B_\varepsilon(\alpha) \cap Y$ is non-empty. Now let $\varepsilon = 1/n$, then we define a_n to be any element of $B_\varepsilon(\alpha) \cap Y$. Hence $a_n \to \alpha$ and so by Lemma 5.4.2 is Cauchy in X, and so Cauchy in Y. Since Y is complete, it follows that $a_n \to \alpha \in Y$. Thus Y is closed. ■

From this we immediately get that any closed interval of the reals $[a, b]$ forms a complete metric space. Notice that this is also a consequence of the Bolzano-Weierstrass Theorem 2.3.10.

Corollary 5.4.5 *Let $[a, b]$ be a closed interval of reals equipped with the usual metric. Then $[a, b]$ is complete.*

Proof. This was actually covered in Exercise 2.5.5, but now we can use our more general tools and knowledge of metric spaces to prove it.

This follows directly from Theorem 5.4.4.

To see that it follows from Theorem 2.3.10, let $(x_n)_n$ be a Cauchy sequence in $[a, b]$. By Bolzano-Weierstrass Theorem 2.3.10 (x_n) has a subsequence that converges to $\alpha \in [a, b]$. Then by Lemma 2.3.19 we know that the original sequence must also converge to α. Hence $[a, b]$ is complete. ■

We'll move on to continuity next, and then come back to consider complete-

ness of spaces of functions.

Exercises

1. **Discrete metric is complete.** Show that any set X with the discrete metric
$$d(x,y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$
forms a complete metric space.

   Hint. Make ε small enough and everything follows.

2. **A complete metric on $\mathbb{Q}$.** The set $\mathbb{Q}$ is not complete with respect to its usual (Euclidean) metric. Is there a metric on $\mathbb{Q}$ which *does* make it complete? Justify your answer.

3. **An unusual metric on $\mathbb{N}$, revisited.** Recall Exercise 5.1.4 in which you proved that the function $d : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}$ given by
$$d(x,y) = |x-y| + \left|\frac{1}{x} - \frac{1}{y}\right|,$$
is a metric on the set $X = \mathbb{R}^+$ (real positive numbers), so that (X,d) is a metric space. Show that this space is actually complete.

   Hint. A Cauchy sequence with respect to d is also Cauchy with respect to the Euclidean metric, and so it has a nonnegative limit. Then you can break into cases depending on whether the limit is positive or zero. Show the limit can't actually be zero. Then recall some results about limits of reciprocals to finish things off.

4. **If a group of you stick together and some of you are going to Cleveland, then you're all going to Cleveland[1].** Show that in a metric space, if a Cauchy sequence has a convergent subsequence and then the full sequence converges.

   Hint. Very similar to Lemma 2.3.19.

5. **Union of complete sets.** True or false? The union of two complete subsets of a metric space is complete. Either prove it or provide a counterexample.

   Hint. The previous exercise.

6. **Completeness and isolated points.** Let (X,d) be a complete metric space and $c \in X$. Prove that $(X \setminus \{c\}, d)$ is complete if and only if c is isolated.

[1] attributing this quote to Professor Brian Marcus (I don't think he will mind). The group of people is a sequence, you stick together, i.e., Cauchy, and some of you are going to Cleveland, i.e., subsequence converging

7. THE COMPLETION. Let (X, d) be a metric space. In this problem, you will construct its completion using equivalence classes of Cauchy sequences. This actually gives an alternate (and perhaps more "traditional"?) approach to constructing the real numbers from the rationals.

(a) Let
$$\mathcal{C} = \{(x_n) \in X^{\mathbb{N}} : (x_n) \text{ is a Cauchy sequence in } X\}.$$
Recall what it means for a sequence to be Cauchy in a metric space, and show that $\mathcal{C}$ is nonempty.

(b) Define a relation $\sim$ on $\mathcal{C}$ by
$$(x_n) \sim (y_n) \quad \text{if} \quad \lim_{n\to\infty} d(x_n, y_n) = 0.$$
Prove that $\sim$ is an equivalence relation. Henceforth, we will write $\widehat{X} = \mathcal{C}/\sim$, and denote the equivalence class of a sequence (x_n) by $[(x_n)]$.

(c) Define a distance function $\widehat{d}$ on $\widehat{X}$ by
$$\widehat{d}\,([(x_n)], [(y_n)]) = \lim_{n\to\infty} d(x_n, y_n).$$
Show that $\widehat{d}$ is a well-defined metric on $\widehat{X}$. (You have to show that the limit exists, is independent of the choice of representatives, and satisfies the properties of a metric).

(d) Define a map $\varphi : X \to \widehat{X}$ by $\varphi(x) = [(x, x, x, \dots)]$. Show that φ is an **embedding**, that is, it is an isometry onto its image. In particular, you only need to show that it preserves distances $d(x, y) = \widehat{d}(\varphi(x), \varphi(y))$ (make sure you understand why this is sufficient). [2]

(e) Show $\varphi(X)$ is dense in $\widehat{X}$.

(f) Prove that $(\widehat{X}, \widehat{d})$ is a complete metric space by following the outline described below.

(i) Let $[(x_n^{(m)})]_{m=1}^{\infty}$ be a Cauchy sequence in $\widehat{X}$ (here m is the index denoting which sequence it is). Define a new sequence in X by setting
$$z_n := x_n^{(n)}.$$
In other words, you're taking the sequence composed of the diagonal. Show that (z_n) is a Cauchy sequence in X, in other words that it is a member of $\mathcal{C}$. *Hint:* It may be helpful to notice that
$$d(z_n, z_m) = d(x_n^{(n)}, x_m^{(m)}) \le d(x_n^{(n)}, x_n^{(m)}) + d(x_n^{(m)}, x_m^{(m)})$$

(ii) Show that $[(z_n)]$ is the limit of $[(x_n^{(m)})]_{m=1}^{\infty}$.
Hint: Similar trick as before; you can write

$$d(x_n^{(m)}, z_n) \leq d(x_n^{(m)}, x_n^{(n)}) + d(x_n^{(n)}, z_n)$$

If you've made it to the end of this problem, CONGRATULATIONS. You've constructed the **completion** of an arbitrary metric space X. For some context, if you take $X = \mathbb{Q}$, you've just shown an alternative method of constructing the real numbers $\mathbb{R}$, albeit with some more heavy machinery than we had at the beginning of term.

5.5 Continuity in metric spaces

Just as we generalised limits, lets continue on and generalise continuity. First, recall our definition of the limit of a real function 3.1.1. To move to metric spaces, we replace $|x - a| < \delta$ by $x \in B_\delta(a)$ and $|f(x) - L| < \varepsilon$ by $f(x) \in B_\varepsilon(L)$.

Definition 5.5.1 Limits of functions. Let (X, ρ) and (Y, τ) be two metric spaces and let $f : X \to Y$ be a function. The **limit** of f as $x \to a \in X$ is $L \in Y$ when

- for every $\varepsilon > 0$, there exists $\delta > 0$ so that if $x \in B_\delta(a)$ then $f(x) \in B_\varepsilon(L)$

In that case we write $\lim_{x \to a} f(x) = L$, or $f(x) \to L$ as $x \to a$. ♢

From this generalisation of limits of functions we immediately get the generalisation of continuity 3.2.2.

Definition 5.5.2 Continuous functions. Let (X, ρ) and (Y, τ) be two metric spaces and let $f : X \to Y$ be a function, and let $a \in X$. We say that f is **continuous at** a when $f(x) \to f(a)$ as $x \to a$.

Equivalently, f is continuous at a when for any $\varepsilon > 0$ there is $\delta > 0$ so that when $x \in B_\delta(a)$ we have $f(x) \in B_\varepsilon(f(a))$.

Finally, if f is continuous at every $a \in A$, then we say that f is **continuous** on A. ♢

We can use this to generalise, for example, Corollary 3.2.10 to show that limits of sequences and continuous functions continue[1] to play nicely together in this more general metric space setting.

Theorem 5.5.3 Sequences, limits and continuity. *Let (X, ρ) and (Y, τ) be two metric spaces and let $f : X \to Y$ be a function. f is continuous at $L \in X$ if and only if whenever a sequence (a_n) in X converges to L, then $f(a_n) \to f(L)$.*

[2]Injectivity follows from distance preservation (why?), and any function is automatically surjective onto its image, so this **distance preservation** implies that φ is a bijection onto its image (hence why we call it an *isometry*, which we previously defined to mean a bijection that is distance preserving).

[1]Ha!

Proof. Let f be continuous at L, and let (a_n) be a sequence in X that converges to L. Let $\varepsilon > 0$. Since f is continuous at L, we know that there is $\delta > 0$ so that when $a_n \in B_\delta(L)$ then $f(a_n) \in B_\varepsilon(f(L))$. Since $a_n \to L$, we know that there is N so that when $n > N$ we have $a_n \in B_\delta(L)$. And thus when $n > N$ we know that $f(a_n) \in B_\varepsilon(f(L))$. Hence $f(a_n) \to f(L)$ as required.

We prove the contrapositive of the converse. Assume that f is not continuous at L, so there is some $\varepsilon > 0$ so that for every $\delta > 0$ there is some $x \in B_\delta(L)$ with $f(x) \notin B_\varepsilon(f(L))$. By setting $\delta = 1/n$, we define a sequence of balls each of which contains some a_n so that $f(a_n) \notin B_\varepsilon(f(L))$. Thus the sequence $a_n \to L$ but $f(a_n) \not\to f(L)$. ■

Continuity also plays nicely with open and closed sets. Though, perhaps this is not immediately obvious. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Then $f((-1, 1)) = [0, 1)$. So the *image* of an open set need not be open. However, the *preimage* of (say) $(4, 9)$ is the open set $(-3, -2) \cup (2, 3)$. More generally, preimages of continuous functions preserve openness and closedness.

Theorem 5.5.4 Characterising continuity. *Let (X, ρ) and (Y, τ) be metric spaces and let $f : X \to Y$. Then the following are equivalent.*

1. *f is continuous on X*
2. *If D is* any *open subset of Y then $f^{-1}(D)$ is an open subset of X*
3. *If C is* any *closed subset of Y then $f^{-1}(C)$ is an closed subset of X*

Proof. First, recall (say, from a previous course on proof and logic) that if $f : X \to Y$ and $D \subseteq Y$ then

$$f^{-1}(Y - D) = X - f^{-1}(D).$$

From this we can see that (2) and (3) are equivalent. Assume (2) and then let $D = Y - C$ is open, then we know that

$$f^{-1}(C) = f^{-1}(Y - D) = X - f^{-1}(D)$$

By (2), we know that $f^{-1}(D)$ is open, and so $X - f^{-1}(D)$ is closed as required. One can similarly show that (3) implies (2).

So now assume (1), that f is continuous. Let D be an open subset of Y and let $a \in f^{-1}(D)$. Since $f(a) = b \in D$ and D is open, there is some $\varepsilon > 0$ so that $B_\varepsilon(b) \subseteq D$. Since f is continuous, there is some $\delta > 0$ so that $x \in B_\delta(a)$ implies that $f(x) \in B_\varepsilon(b)$. Thus $B_\delta(a) \subseteq f^{-1}(B_\varepsilon(b)) \subseteq f^{-1}(D)$. Thus around any $x \in f^{-1}(D)$ there is always an open ball contained in $f^{-1}(D)$. Thus $f^{-1}(D)$ is open.

Now assume (2). Let $\varepsilon > 0$ and $a \in X$. The set $B_\varepsilon(f(a))$ is open, and so by (2) we know that its preimage is open. Thus there is some $\delta > 0$ so that $B_\delta(a) \subseteq f^{-1}(B_\varepsilon(f(a)))$. Thus when $x \in B_\delta(a)$ we know that $f(x) \in B_\varepsilon(f(a))$. Hence f is continuous at a. ■

For general metric spaces we don't have arithmetic, so we are not going to prove that the sum of continuous functions is continuous etc etc. However, we can prove that the composition of continuous functions is continuous.

Theorem 5.5.5 Continuity and compositions. *Let $(X,\rho),(Y,\tau)$ and (Z,σ) be metric spaces and let $f: X \to Y, g: Y \to Z$ be continuous. Then $g \circ f$ is continuous.*

A proof using open sets. Assume that f, g are continuous and let C be any open subset of Z. Then, since g is continuous, Theorem 5.5.4 tells us that $B = g^{-1}(C)$ is an open subset of Y. Similarly, since f is continuous, $f^{-1}(B) = f^{-1}(g^{-1}(C))$ is an open subset of X. Thus the preimage of an open set under the composition is open and so by Theorem 5.5.4, the composition is continuous. ■

A proof using sequences. Assume that f, g are continuous and let (a_n) be a convergent sequence in X with limit $\alpha \in X$. Since f is continuous, we know by Theorem 5.5.3 that $b_n = f(a_n)$ is a sequence in Y which converges to $f(\alpha) = \beta \in Y$. Similarly, since g is continuous, we know that $g(b_n) = g(f(a_n))$ is a sequence in Z that converges to $g(\beta) = g(f(\alpha)) \in Z$. Hence, by Theorem 5.5.3, the composition is continuous. ■

We can impose stronger conditions than continuity. Recall that in our definition of continuity 5.5.2, for each point a, no matter which ε, there is a δ so that $x \in B_\delta(a)$ implies that $f(x) \in B_\varepsilon(f(a))$. A subtle point here is that δ can depend on both ε and the point a. A stronger notion of continuity requires that δ is independent of the point a, and must hold for everywhere in X. This is **uniform continuity**.

Definition 5.5.6 Uniform continuity. Let (X,ρ) and (Y,τ) be two metric spaces and let $f: X \to Y$ be a function. We say that f is **uniformly continuous** when for all $\varepsilon > 0$ there is some $\delta > 0$ so that for any $a, b \in X$ if $\rho(a,b) < \delta$ then $\tau(f(a), f(b)) < \varepsilon$. ♢

Notice that in the above definition, after ε is set, the chosen δ has to work for all a, b. For the usual notion of continuity we choose a first and then set ε and δ.

Example 5.5.7 One reason why uniform continuity is important. Uniform continuity plays very nicely with *compact* subsets of metric spaces (we will discuss compactness later in Chapter 6). In fact, the *Heine-Cantor Theorem* asserts that if $f: K \to \mathbb{R}$ is a continuous function and $K \subseteq X$ is compact, then in fact f is *uniformly continuous* on K. This result is extremely useful - in particular, it gives an easy[2] proof of the fact that the definite integral of a continuous function $\int_a^b f(x)\,\mathrm{d}x$ is well-defined. □

One can go even further and require that δ is a simple function of ε. This gives **Lipschitz continuity**.

[2] this is a dangerous word in mathematics

Definition 5.5.8 Lipschitz continuity. Let (X, ρ) and (Y, τ) be two metric spaces and let $f : X \to Y$ be a function. We say that f is **Lipschitz continuous** when there exists a constant K so that for any $a, b \in X$

$$\tau(f(a), f(b)) \leq K \cdot \rho(a, b)$$

The constant K is called the **Lipschitz constant**. When the Lipschitz constant satisfies $K < 1$, we call f a **contraction** (see Section 5.7) ♢

Notice that Lipschitz continuity "effectively" puts a bound on how quickly a function can change. Small change in the input x cannot incur a large change in the output. More precisely, the size of the change in the output is bounded by a constant times the change in the input. This smells a little like a bound on the derivative of the function — but of course, we don't actually require that the function be differentiable.

Example 5.5.9 One reason why Lipschitz continuity is important. Lipschitz continuity shows up in all sorts of areas of mathematics. One such example is as follows: Suppose $(f_n)_n$ is a sequence of real-valued functions on a closed finite interval $[a, b]$, and that each f_n is Lipschitz continuous with constant K, and that this contstant K is the same for each $n \in \mathbb{N}$. Then the *Arzela-Ascoli Theorem*[3] implies that there is some subsequence $(f_{n_k})_k$ which **converges uniformly**[4] on $[a, b]$. Again, this is beyond the scope of this course, but we want these definitions to feel somewhat motivated. □

We will also discuss contraction mappings (which are just functions from a space X to itself which is also a contraction, in the sense of Definition 5.5.8) later in this chapter as another reason why Lipschitz continuity is an important concept.

Of course, before we move on, we ought to formally justify all the (non)implications between all these notions of continuity. Let's do that now.

Lemma 5.5.10 Lipschitz implies uniform implies ordinary. *Let (X, ρ) and (Y, τ) be two metric spaces and let $f : X \to Y$ be a function.*

- *If f is Lipschitz continuous, then it is uniformly continuous*
- *If f is uniformly continuous, then it is continuous*

The converses of these statements are false.

Proof. Let X, Y, f be as given, and assume that f is Lipschitz continuous. Hence there exists some K so that for any $a, b \in X$

$$\tau(f(a), f(b)) \leq K \cdot \rho(a, b)$$

So now let $\varepsilon > 0$ and set $\delta = \varepsilon / K$. Then if $\rho(a, b) < \delta$, we must have that

$$\tau(f(a), f(b)) \leq K \cdot \rho(a, b) < K \cdot \delta = \varepsilon$$

[3] this is beyond the scope of this course

[4] this term will be defined in the next section

Hence f is uniformly continuous.

Now let f be uniformly continuous and let $a \in X, \varepsilon > 0$. Then since f is uniformly continuous, we know that for any $x \in X$

$$\rho(x,a) < \delta \implies \tau(f(x), f(a)) < \varepsilon$$

and hence f is continuous.

Now consider the function $f : [0,1] \to [0,1]$ with $f(x) = \sqrt{x}$. We can prove that this is uniformly continuous but not Lipschitz continuous.

- Let $\varepsilon > 0$ and then set $\delta = \varepsilon^2$. Now if $|a-b| < \delta$ we have

$$\begin{aligned} |\sqrt{a} - \sqrt{b}|^2 &= |\sqrt{a} - \sqrt{b}||\sqrt{a} - \sqrt{b}| \\ &\leq |\sqrt{a} - \sqrt{b}||\sqrt{a} + \sqrt{b}| \\ &= |a-b| \leq \delta = \varepsilon^2 \end{aligned}$$

  Using the fact that the square-root is increasing, we have that $|\sqrt{a} - \sqrt{b}| < \varepsilon$ as required.

- Now assume that f is Lipschitz, and so there is some K so that

$$|\sqrt{a} - \sqrt{b}| < K|a-b|$$

  for all $a, b \in [0,1]$. Now pick $b = 0$, then we have that $\sqrt{a} < Ka$. Massaging this around gives

$$\frac{1}{\sqrt{a}} < K$$

  But this is not true for all a. As $a \to 0$ the left-hand-side can be made arbitrarily large. Indeed, pick any $0 < a < \frac{1}{K^2}$. Thus f is not Lipschitz.

Notice that this (roughly speaking) occurs because the derivative of $\sqrt{x}$ becomes unbounded as $x \to 0$.

Let $g : (0,1) \to (1,\infty)$ be defined by $g(x) = 1/x$. Then g is continuous but not uniformly continuous.

- Let $a \in (0,1)$ and let $\varepsilon > 0$. Let $\delta = \min\{a/2, a^2\varepsilon/2\}$. Now assume that $|x-a| < \delta$, so that

$$a/2 < x < 3a/2$$

  Then

$$\left|\frac{1}{x} - \frac{1}{a}\right| = \frac{|x-a|}{xa} \leq \frac{\delta}{a^2/2} < \varepsilon$$

  Hence $f(x) \to \frac{1}{a}$ as $x \to a$. Since a was arbitrary, it follows that f is continuous on $(0,1)$.

- Now we show that f is not uniformly continuous. This means that we must show that there is some $\varepsilon > 0$ so that for any $\delta > 0$, we can find $|x - a| < \delta$ and $|1/x - 1/a| > \varepsilon$. So pick $\varepsilon = 1$, and then for any δ, set $x = \min\{1/2, \delta\}$, and then pick $y = x/2$. Then we have

$$|x - y| = x/2 < \delta/2 < \delta$$

and at the same time

$$\frac{1}{x} - \frac{1}{y} = \frac{1}{x} - \frac{2}{x} = \frac{1}{x} \geq 2 > \varepsilon.$$

Thus f is not uniformly continuous.

Notice that this (roughly speaking) occurs because the value of $1/x$ becomes unbounded as $x \to 0$. ■

Exercises

1. **Is it continuous?** Suppose f is a real function defined on $\mathbb{R}$ which satisfies

$$\lim_{h \to 0} [f(x + h) - f(x - h)] = 0$$

for every $x \in \mathbb{R}$. Does this imply that f is continuous? Justify your answer either with a proof or a counter-example.

2. **Distance function is continuous.** Given any metric space (X, d) and a fixed $y \in X$, show that the function $f(x) = d(x, y)$ is continuous.

3. **Continuity and closures.** Let f be a continuous function from a metric space (X, d_X) to another metric space (Y, d_Y).

- Prove that
$$f(\mathrm{cl}\,(E)) \subseteq \mathrm{cl}\,(f(E))$$
for every set $E \subset X$. Note that $\mathrm{cl}\,(E)$ is the closure of E with respect to d_X, and $\mathrm{cl}\,(f(E))$ is the closure of $f(E)$ with respect to d_Y.

- Show, by an example, that $f(\mathrm{cl}\,(E))$ can be a proper subset of $\mathrm{cl}\,(f(E))$.

Hint. Take some $y \in f(\mathrm{cl}\,(E))$ and some $x \in \mathrm{cl}\,(E)$ with $f(x) = y$, then break into cases of whether or not x is a limit point of E.

There are many possible examples, but if you're having difficulty coming up with one, consider a function on $\mathbb{R}$ that does not attain its supremum.

4. **Continuity and density.** Let f and g be continuous functions from a metric space (X, d_X) to another metric space (Y, d_Y), and let E be a dense subset of X.

- Prove that $f(E)$ is dense in $f(X)$. (See the previous exercise)
- If $g(p) = f(p)$ for all $p \in E$, prove that $g(p) = f(p)$ for all $p \in X$.
- If $X = Y = \mathbb{R}$ and it is known that $f(x) = 0$ for all rational numbers x (we are still assuming f is continuous here), show that f is the zero function.

This demonstrates that a continuous function is determined by its values on a dense subset of its domain.

5. **Zero set is closed.** Let f be a continuous real function on a metric space X. Let $Z(f)$ be the set of all $p \in X$ at which $f(p) = 0$ (the **zero set** of f). Prove that $Z(f)$ is closed.

Hint. An exercise above will help you.

6. **Identity continuous?** Consider the identity map $\mathrm{id} : \mathbb{R} \to \mathbb{R}$. Let d_E and d be the Euclidean and discrete metrics on $\mathbb{R}$ respectively. Show that the identity map one way, $\mathrm{id} : (\mathbb{R}, d) \to (\mathbb{R}, d_E)$ is continuous but the identity map back, $\mathrm{id} : (\mathbb{R}, d_E) \to (\mathbb{R}, d)$ is not!

Of course, these are not really identity maps, since they are not mapping a metric space back to itself — the metric changes.

Hint. Consider what $B_r(x)$ is in each metric for small r, and use the ε-δ definition of continuity.

7. **Popcorn**[5]**.** Every rational number can be written in the form $x = m/n$, where $n > 0$ and $m, n \in \mathbb{Z}$ have no common divisors (lowest form). When $x = 0$, we take $n = 1$. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & x \text{ is irrational} \\ \frac{1}{n} & x = m/n \text{ in lowest form.} \end{cases}$$

- Prove that f is discontinuous at every rational point.
- Prove that f is continuous at every irrational point.

Hint. Use the fact that you can find an irrational number arbitrarily close to any real number.

Given any $\varepsilon > 0$, the set of rational numbers $x = m/n$ which are "reasonably close" to a given irrational number, with $n \leq 1/\varepsilon$, is finite.

5.6 Metric space of continuous bounded functions

Let us spend a little time on metric spaces that are not (immediately) based on subsets of reals. A very important example is the metric space of bounded functions, and the metric subspace of continuous bounded functions. Indeed one

[5]The function in this example is often called the "popcorn function"

of the drivers of the original work generalising results about real numbers was to understand how one might sensibly understand convergence within the space of functions. This will take a little work and so we've made it into a separate section. The reader should take a moment to revise Example 5.1.4 which is related.

Definition 5.6.1 Bounded functions. Let X be a set, then we define $B(X)$ to be the set of functions $f : X \to \mathbb{R}$ that are bounded. That is, the set

$$\{f(x) \mid x \in X\} \subseteq \mathbb{R}$$

is bounded.

We then define the **supremum metric** between functions in $B(X)$ by

$$\rho(f, g) = \sup\{|f(x) - g(x)| \mid x \in X\} = \|f - g\|$$

When there is little chance of ambiguity we will take $B(X)$ to be the metric space of bounded functions equipped with this metric. ♢

In many contexts people consider the **supremum norm** being $\|f\|_\infty = \sup_{x\in X} |f(x)|$. We are skipping over the development of **normed vector spaces** and the interested reader should search-engine their way to more on that topic.

We will prove two main results in this section

- For any X, the space of bounded functions $B(X)$ is complete
- For any X, the space of continuous bounded functions $C(X)$ is also complete

To sneak up on our first result, namely that $B(X)$ is complete, by breaking it into 3 parts

- Show that for any fixed x, the sequence $(f_n(x))_n$ converges, and so we can use that limit to define $f(x)$ for each $x \in X$.
- Show that the function $f(x)$ is bounded and so lies in $B(X)$.
- Show that $f_n \to f$ in $B(X)$ (essentially uniform convergence, the same ε for all $x \in X$)

We make the first two steps into lemmas and then combine them with the third to prove the full result.

Lemma 5.6.2 *Let X be a set and let $B(X)$ be the metric space of bounded functions with the supremum metric. Let $(f_n)_n$ be a Cauchy sequence of functions in $B(X)$. Then for any $x \in X$, the sequence $(f_n(x))_n$ is a Cauchy sequence of reals and so converges.*

*That is, the sequence of functions converges **pointwise**.*

Proof. Let $(f_n)_n$ be a Cauchy sequence in $B(X)$ and fix $a \in X$. We can then

define

$$a_n = f_n(a) \in \mathbb{R}$$

which gives us a sequence of reals. We show that this is a Cauchy sequence of reals and so converges.

Let $\varepsilon > 0$. Since (f_n) is Cauchy in $B(X)$, there is some N so that for any $k, \ell > N$ we know that

$$\|f_k - f_\ell\| = \sup\{|f_k(x) - f_\ell(x)| \mid x \in X\} < \varepsilon$$

However, for any fixed $a \in X$, $|f_k(a) - f_\ell(a)| \leq \|f_k - f_\ell\|$, and so we know that

$$|f_k(a) - f_\ell(a)| = |a_k - a_\ell| \leq \|f_k - f_\ell| < \varepsilon.$$

Hence $(a_n)_n$ is Cauchy. Then since the reals are complete, we know that $a_n \to \alpha \in \mathbb{R}$. ■

So we know that the sequence of functions converge point-by-point. We now use that to the limit function. We can show that function is bounded and so in $B(X)$.

Lemma 5.6.3 *Let $(f_n)_n$ be a Cauchy sequence in $B(X)$. Define a new function $f : X \to \mathbb{R}$ by*

$$f(x) = \lim_{n\to\infty} f_n(x) \qquad \text{for each } x \in X.$$

Then f is bounded and so $f \in B(X)$.

Proof. Let $(f_n)_n$ be as stated and let $\varepsilon = 1$. Thus we know that there is some N so that for all $k, \ell > N$,

$$\|f_k - f_\ell\| < \varepsilon$$

Additionally, since the $f_n \in B(X)$, we know that for a given ℓ there is some $M \geq 0$ so that

$$|f_\ell(x)| \leq M \qquad \text{for all } x \in X.$$

Of course, here M depends on ℓ (but not x), so we can't just immediately infer that f is bounded.

However, we can use the Cauchy property to get us there; for any given $x \in X$ we have

$$\begin{aligned} |f_k(x)| &= |f_k(x) - f_\ell(x) + f_\ell(x)| \\ &\leq |f_k(x) - f_\ell(x)| + |f_\ell(x)| \\ &< 1 + M \end{aligned}$$

So Cauchy and the bound on f_ℓ allow us to bound $f_k(x)$ by a quantity that is independent k, x. Letting $k \to \infty$ we then have

$$|f(x)| = |\lim_{k\to\infty} f_k(x)| \leq M + 1$$

Hence f is bounded and so is an element of $B(X)$.

Notice that the strict inequality becomes less-than-or-equal-to. This is precisely that we are including a limit point which might lie right on the boundary. For example, $1 - 1/k < 1$ but $1 - 1/k \to 1$. ■

We can now prove that $B(X)$ is complete, by showing that $\|f_n - f\| \to 0$.

Theorem 5.6.4 Space of bounded functions is complete. *Let X be a set and let $(B(X), \rho)$ be the metric space of bounded functions defined above. It is complete.*

Proof. We have to show that any Cauchy sequence of functions $(f_n)_n$ in $B(X)$ converges in $B(X)$. So let $(f_n)_n$ be a Cauchy sequence in $B(X)$.

By the previous two lemmas, we know that for any given $a \in X$, that $f_n(a) \to f(a)$ for some function $f \in B(X)$. That is, there is a point-wise limit of this sequence of functions. Indeed, we define f as this point-wise limit.

So we have to show that $f_k \to f$ in this supremum norm. That is

$$\|f_k - f\| \to 0 \qquad \text{as} k \to \infty$$

Or, equivalently, for any $\varepsilon > 0$, there is some N so that when $k > N$

$$\sup_{x \in X} |f_k(x) - f(x)| < \varepsilon$$

We already know that this is true for each x, but now we must show that it works for an ε that is independent of x. This is **uniform convergence**. Our proof is very similar to that used in the previous lemma.

Let $\varepsilon > 0$, then since the sequence of functions is Cauchy (by assumption), there is N so that when $k, \ell > N$, we have

$$\|f_k - f_\ell\| = \sup |f_k(x) - f_\ell(x)| < \varepsilon/2.$$

Notice that the N does not depend on x, but must hold for all $x \in X$.

Thus, for any given $x \in X$ we also know that $|f_k(x) - f_\ell(x)| < \varepsilon/2$. Then

$$\begin{aligned} |f_k(x) - f(x)| &= |f_k(x) - \lim_\ell f_\ell(x)| \\ &= \lim_\ell |f_k(x) - f_\ell(x)| \leq \varepsilon/2 \end{aligned}$$

(Again notice that the strict inequality becomes non-strict when we take this limit.) Since this result does not depend on x, we can take the supremum over x to show that

$$\|f_k - f\| \leq \varepsilon/2 < \varepsilon$$

Thus $f_k \to f$ as required. ■

Thankfully we do not have to prove everything from scratch to show that the completeness of the set, $C_b(X)$, of continuous bounded functions on X. We can take advantage of Theorem 5.4.4; it suffices to show that $C_b(X)$ is a closed subset of $B(X)$. To show closure, it suffices, by Theorem 5.3.6, to show that the

limit of any sequence in $C_b(X)$ is also continuous.

Theorem 5.6.5 The space of continuous bounded functions is complete. *Let X be a metric space, then the set $C_b(X)$, of continuous bounded functions on X with the supremum metric, is complete.*

Proof. By Theorem 5.4.4 this result will follow if we can show that the set of continuous functions within $B(X)$ forms a closed set. We do that via Theorem 5.3.6; if we can show that any convergent sequence of continuous functions, $(f_n)_n$ has a continuous limit, f, then we are done. We do this by manipulating

$$\begin{aligned} |f(x) - f(a)| &= |f(x) - f_k(x) + f_k(x) - f_k(a) + f_k(a) - f(a)| \\ &\leq |f(x) - f_k(x)| + |f_k(x) - f_k(a)| + |f_k(a) - f(a)| \end{aligned}$$

and bounding each piece in turn, which tells us that f is continuous at a.

Let $(f_n)_n$ be a convergent sequence of continuous functions in $B(X)$ and let $a \in X$ and $\varepsilon > 0$. Since $f_n \to f$, we know that there is some N so that for all $k > N$, $\|f_k - f\| < \varepsilon/3$. From this we know that

$$|f_k(a) - f(a)| < \varepsilon/3 \qquad \text{and} \qquad |f_k(x) - f(x)| < \varepsilon/3$$

for any $x \in X$. At the same time, f_k is continuous, so there is some $\delta > 0$ so that when $x \in B_\delta(a)$ we know that $|f_k(x) - f_k(a)| < \varepsilon/3$.

Putting these together, given any $\varepsilon > 0$, there is some $\delta > 0$ so that when $x \in B_\delta(a)$, we know that

$$\begin{aligned} |f(x) - f(a)| &\leq |f(x) - f_k(x)| \qquad + |f_k(x) - f_k(a)| + |f_k(a) - f(a)| \\ &< \varepsilon/3 \qquad\qquad + \varepsilon/3 + \varepsilon/3 \\ &= \varepsilon \end{aligned}$$

Thus f is continuous at $a \in X$. Since a was arbitrary, we know that f is continuous on all of X.

Hence the limit of the sequence of bounded continuous functions is also a bounded continuous function. Thus the set $C_b(X)$ is closed, and so is complete. ■

Example 5.6.6 Two sequences of bounded continuous functions. Consider again the two sequences of continuous functions, $(f_n)_n, (g_n)_n$ on the unit interval, defined by interval:

$$f_n = x^n \qquad \text{and} \qquad g_n = x^n(1-x)^n$$

It is not hard to show that these are continuous and bounded. Indeed, they map the unit interval to itself.

At first glance one might think that by the previous theorem, these sequences of functions both have continuous limits. However this is not the case. As we

saw earlier, the first sequence converges point-wise to the limit

$$f_n(x) \to f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}$$

which is not continuous at $x = 1$. While the second converges pointwise to the constant zero function.

$$g_n(x) \to g(x) = 0$$

which is continuous. How do we reconcile this with the theorem above?

Simply put, the sequence $(f_n)_n$ is not Cauchy (with respect to the supremum metric), while $(g_n)_n$ is.

To show that $(f_n)_n$ is not Cauchy we must show that there is some $\varepsilon > 0$ so that for all $N \in \mathbb{N}$ there exist $k, \ell > N$ with $\rho(f_k, f_\ell) > \varepsilon$. That is, we must be able to choose x, k, ℓ so that

$$|f_k - f_\ell| = x^\ell(1 - x^{k-\ell}) > \varepsilon$$

So take $\varepsilon = \frac{1}{2}$. Now, fix ℓ at some value and notice that

- we can take $x > (\frac{3}{4})^{1/\ell}$, so that the term $\frac{3}{4} < x^\ell < 1$.
- at the same time, for any $x < 1$, we can choose sufficiently large k so that $0 < x^{k-\ell} < \frac{1}{4}$. This means we have $\frac{3}{4} < 1 - x^{k-\ell} < 1$

Hence there are x, k, ℓ so that

$$x^\ell \cdot (1 - x^{k-\ell}) > (3/4)^2 > 1/2 = \varepsilon$$

And thus $\rho(f_k, f_\ell) > \varepsilon$ and so the sequence is non-Cauchy.

Now turning to the other sequence, we know that it converges (in the supremum norm) to $g(x) = 0$. So we can argue that since it converges to a limit in the space, it must also be Cauchy. Equivalent, we can simply use the ideas behind the proof that convergence implies Cauchy convergence. Namely, since $g_n \to g$ we know that there N so that for any $k > N$, we have $\rho(g_k, g) < \varepsilon/2$. Hence for any $k, \ell > N$

$$\rho(g_k, g_\ell) \leq \rho(g_k, g) + \rho(g_\ell, g) < \varepsilon.$$

Thus the sequence $(g_n)_n$ is Cauchy. □

Exercises

1. **Continuity preserved by pointwise limits?** Let (f_n) be a sequence of continuous functions $f_n : \mathbb{R} \to \mathbb{R}$. Suppose the pointwise limit $f(x) = \lim_{n\to\infty} f_n(x)$ exists for every $x \in \mathbb{R}$. Is f necessarily continuous? Either prove it or provide a counter-example.

2. Pointwise vs uniform. For each $n \in \mathbb{N}$, define a continuous function f_n on $[0,1]$ by

$$f_n(x) = \frac{x^2}{x^2 + (1-nx)^2}$$

- Show that $f_n \to 0$ pointwise.
- Does $\|f_n\| \to 0$? Show that your answer is correct.

Hint. Where does f_n achieve its maximum?

3. Pointwise convergence can have "bad" limits. For each $n \in \mathbb{N}$, define

$$f_n(x) = \lim_{m\to\infty} \cos(n\pi x)^{2m}.$$

It is a fact that f_n converges pointwise to some function. Determine that function.

Hint. Break into cases of whether x is an integer or not.

4. Converges uniformly equivalent condition. Define

$$f_n(x) = \frac{nx}{1+n^2x^2}$$

- Show that f_n converges pointwise on all of $\mathbb{R}$ and calculate the pointwise limit.
- Show that f_n converges *uniformly* on $A \subseteq \mathbb{R}$ if 0 is *not* a limit point of A.
- (harder) Show that the converse to the previous part is also true. That is, if the f_n converge uniformly on A, then 0 is not a limit point of A.

Reminder: The condition that 0 is a limit point does have any bearing on whether or not it is contained in the set A itself (meaning you'll likely have to consider the case $x = 0$ separately).

Hint.

- You may have to treat $x = 0$ separately.
- If $0 \notin A'$, this means that there is some $\delta > 0$ such that any $x \in A\backslash\{0\}$ satisfies $|x| \geq \delta$.
- By contradiction: If $0 \in A'$, this means you can find a sequence $x_k \in A \setminus \{0\}$ with $|x_k| \to 0$. Now with this sequence, you need to find a way to bound each element x_k from both above and below (in terms of some n_k) in order to use an appropriate inequality in both the numerator and denominator.

5. Uniform convergence is convenient. Suppose $f_n \in \mathcal{C}(X)$ for each $n \in \mathbb{N}$, and that $f_n \to f$ uniformly for some $f \in \mathcal{C}(X)$. Prove that

$$\lim_{n\to\infty} f_n(x_n) = f(x)$$

for every sequence of points $x_n \in X$ with $x_n \to x$.

Hint. Add and subtract $f(x_n)$ and use the triangle inequality.

6. Uniform convergence implies integrals converge. Suppose that $f_n \in \mathcal{C}[a,b]$ for $n \in \mathbb{N}$ and that $f_n \to f$ uniformly as $n \to \infty$. Prove that

$$\int_a^b f_n(x)\,\mathrm{d}x \to \int_a^b f(x)\,\mathrm{d}x$$

as $n \to \infty$. In other words, uniform convergence implies that integrals also converge. Compare this result with the following exercise.

Hint. You may have to recall some basic facts about integrals from first year calculus.

7. Integrals converging does not imply uniform convergence. For each $n \in \mathbb{N}$, define a continuous function f_n on $[0,1]$ by

$$f_n(x) = nx(1-x^2)^n.$$

(a) Show that $f_n \to 0$ pointwise on $[0,1]$.

(b) Calculate

$$\int_0^1 f_n(x)\,\mathrm{d}x$$

for each $n \in \mathbb{N}$. What happens to this quantity as $n \to \infty$? Compare the result to the previous problem. Is there any inconsistency?

Hint. You may use (without proof) that exponential functions grow faster than polynomials.

8. Converges in one metric, but not the other. Let $C([0,1])$ be the set of all continuous real-valued functions on $[0,1]$. Consider two different metrics on this space:

$$d_1(f,g) = \sup_{t\in[0,1]} |f(t)-g(t)| \quad \text{and} \quad d_2(f,g) = \int_0^1 |f(t)-g(t)|\,\mathrm{d}t.$$

- We have already seen in class that d_1 is really a metric. Prove that d_2 is as well.
- Show that $f_n(x) = x^n$ converges (to some function) with respect to d_2, but *not* with respect to d_1.

Hint. The tricky part here is to show that $d_2(f,g) > 0$ if $f \neq g$. For

this, consider a point $x_0 \in [0,1]$ such that $|f(x_0) - g(x_0)| > 0$ and use the continuity to obtain a lower bound for the integral.

For the last task, uniform convergence helps.

9. Back to polynomials. Let $a, b \in \mathbb{R}$, $a < b$. Let $\mathcal{C}([a,b])$ denote the collection of all continuous, real-valued functions defined on the closed interval $[a,b]$ with the supremum metric d.

- Show that the set $\mathcal{P}$ of all polynomials in the metric space $(\mathcal{C}[a,b], d)$ is not an open set in $\mathcal{C}[a,b]$.
- Show that $\mathcal{P}$ is not a closed subset of the metric space $\mathcal{C}[a,b]$.

Remark: There is a very subtle result[1] (well beyond the scope of this course) which states that $\mathcal{P}$ is actually a *dense* subset of $(\mathcal{C}[a,b], d)$.

Hint. For any given $\varepsilon > 0$, consider the function $g(x) = f(x) + \varepsilon/2 \sin(x)$. Consider something like $\sin(x)$ and apply Taylor's Theorem.

5.7 Contraction mappings

Think back to the start of the course and our discussion of Newton iteration 1.1 and Picard iteration 1.2. One of the main goals of this course is to understand, quite generally, when and why these methods work. The language and machinery of metric-spaces gives us the required tools.

Recall that

- Newton's method to solve $f(x) = 0$. Start with initial guess x_0, and then from a current estimate x_n construct a better estimate $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.
- Picard iteration to solve $y' = f(x,y), y(0) = y_0$. Start with initial guess $\varphi_0(x) = y_0$, and then from current estimate $\varphi_n(x)$ construct a better estimate $\varphi_{n+1} = y_0 + \int_0^x f(t, \varphi_n(t))\,\mathrm{d}t$.

In both cases we expect (and one can prove) that these sequences of objects converge to the correct limit according to some sensible metric.

More generally, given a metric space (X, ρ)

- Start from a point x_0
- Given a point x_n construct a new point $x_{n+1} = T(x_n)$

where $T : X \to X$ maps the metric space to itself. Under what circumstances does a unique limit $x_n \to x^*$ exist? One approach is to show that the mapping T always[1] moves objects closer to each other. That is

$$\rho(T(x), T(y)) < \rho(x,y) \text{ for any } x, y \in X.$$

[1] called the *Weierstrass* (or, more generally, the *Stone-Weierstrass*) Theorem

[1] Or, rather, understand under what circumstances we can guarantee that T moves objects closer together.

Then we can imagine that by repeatedly applying T, our initial object x_0 will be drawn closer and closer to the limiting object x^*

This is the right idea — but we need a little more care (and pedantry). We need **contraction mappings**.

Definition 5.7.1 Contraction mapping. Let (X, ρ) be a metric space and let $T : X \to X$. If there is a constant $K \in [0, 1)$ so that

$$\rho(T(x), T(y)) \leq K\rho(x, y)$$

for any $x, y \in X$, then we say that T is a **contraction mapping**.

Note: The constant K must work for all $x, y \in X$ simultaneously in order for T to be a contraction. ♢

Remark 5.7.2 Notice that **contraction mappings** are a special subset of Lipschitz continuous functions 5.5.8. Also note that we require a fixed constant $K < 1$. The condition $\rho(T(x), T(y)) \leq \rho(x, y)$ is not sufficient.

Also notice that in the context of functions on $[a, b] \subseteq \mathbb{R}$ there is something a little "mean value theorem" about contraction mappings. In particular, the MVT tells us that there is some $c \in [a, b]$ so that

$$T(a) - T(b) = T'(c) \cdot (a - b)$$

so if we can bound $|T'(c)| \leq K < 1$ then we know T is a contraction mapping.

Here is a very simple example which illustrates why this works.

Example 5.7.3 Simple contraction mapping. Let $X = \mathbb{R}$ and define $T(x) = \frac{x}{2}$. Then since

$$|T(x) - T(y)| = \left|\frac{x - y}{2}\right| = \frac{1}{2}|x - y|$$

the function T is a contraction mapping with Lipschitz constant $1/2$.

Now, consider what happens when we repeatedly apply T to a pair of points x, y:

$$\begin{aligned} T(x) - T(y) &= \frac{x - y}{2} \\ T^2(x) - T^2(y) &= \frac{T(x) - T(y)}{2} = \frac{x - y}{4} \\ T^k(x) - T^k(y) &= \frac{x - y}{2^k} \end{aligned}$$

Since x, y are fixed, as we increase $k \to \infty$, the distance between the iterates of x, y go to zero.

Indeed, from any starting x we have that $T^k(x) \to 0$, and that $T(0) = 0$.

One can push this example a little further, if we take any linear function $f(x) = mx + b$, then the behaviour of iterates will be governed by the gradient

m. In particular, if $|m| < 1$, then the mapping contracts, while if $|m| > 1$, the mapping expands. Since we can approximate a single variable function about a given point by its tangent line, this suggests that the derivative at that point will play a crucial role (and also matches up with our expectation that the MVT is lurking here somewhere). □

This example shows how contraction mappings work. If $K < 1$ then after n iterations, we know that the distance between points will be reduced by at least a factor of K^n. Of course, some care is needed around the edges and we'll give a few examples to illustrate that.

Theorem 5.7.4 Contraction mapping theorem. *Let (X, rho) be a non-empty, complete metric space, and let $T : X \to X$ be a contraction mapping with Lipschitz constant $K < 1$. Then T has a unique fixed point, i.e., there is unique $x^* \in X$ so that $T(x^*) = x^*$.*

Further, let $x_0 \in X$ be an initial point and define the iterates $x_n = T(x_{n-1})$. Then we can bound the distance between the nth iterate and the fixed point x^* by

$$\rho(x_n, x^*) \leq \frac{K^n}{1-K}\rho(x_0, x_1).$$

Proof. We show

- that any fixed point must be unique
- that the sequence $(x_n)_n$ is Cauchy
- and since the space is complete, we know that the sequence converges in X
- we must show that the limit is a fixed point
- and along the way we establish the bounds stated.

Assume, to the contrary that we have two fixed points $z, q \in X$. Since T is a contraction mapping, we must have that $\rho(T(z), T(q)) < K\rho(z, q)$. However, by assumption, $T(z) = z, T(q) = q$, and thus we have

$$\rho(T(z), T(q)) = \rho(z, q) < K\rho(z, q)$$

Since $\rho(z, q) \geq 0$ we must have that $\rho(z, q) = 0$ and thus $z = q$.

Now let x_0 be some arbitrary point in X and let the sequence of iterates under T be $(x_n)_n$ as stated. In order to show that the sequence is Cauchy, we need to bound $\rho(x_k, x_\ell)$. In order to do that, we first bound $\rho(x_k, x_{k-1})$ and then leverage the triangle inequality. So first notice that

$$\begin{aligned}
\rho(x_1, x_2) &< K\rho(x_0, x_1) \\
\rho(x_2, x_3) &< K\rho(x_1, x_2) \\
\rho(x_3, x_4) &< K\rho(x_2, x_3)
\end{aligned}$$

and so on. Thus, for any k

$$\rho(x_k, x_{k+1}) < K^k \rho(x_0, x_1).$$

Now consider $\rho(x_k, x_\ell)$ with $k > \ell$. The triangle inequality gives us

$$\begin{aligned}\rho(x_k, x_\ell) &\leq \rho(x_k, x_{k-1}) + \rho(x_{k-1}, x_\ell) \\ &\leq \rho(x_k, x_{k-1}) + \rho(x_{k-1}, x_{k-2}) + \rho(x_{k-2}, x_\ell)\end{aligned}$$

keep applying the triangle inequality

$$\begin{aligned}&\leq \rho(x_k, x_{k-1}) + \rho(x_{k-1}, x_{k-2}) + \cdots + \rho(x_{\ell+1}, x_\ell) \\ &\leq \left(K^{k-1} + K^{k-2} + \cdots K^\ell\right) \rho(x_0, x_1) \\ &\leq K^\ell \left(1 + K + K^2 + \cdots\right) \rho(x_0, x_1) \\ &\leq \frac{K^\ell}{1-K} \rho(x_0, x_1)\end{aligned}$$

So, for fixed x_0, we can write $D = \rho(x_0, x_1)/(1-K)$ and then for any $k > \ell$ we have that

$$\rho(x_k, x_\ell) \leq DK^\ell.$$

Now, for any $\varepsilon > 0$ we can set $N = \log(\varepsilon/D)/\log(K)$, and then for any $k > \ell > N$, we have

$$\rho(x_k, x_\ell) \leq DK^\ell < DK^N = \varepsilon$$

and thus the sequence $(x_n)_n$ is Cauchy.

Now since $(x_n)_n$ is Cauchy and the space X is complete (by assumption), we know that x_n converges to some limit x^*. We still need to show that this limit is a fixed point of T. Thankfully we have already done all the hard work required for this; since T is Lipschitz continuous, Lemma 5.5.10, tells us that it is also continuous. And thus, by Theorem 5.5.3 we have

$$T(x^*) = T(\lim_n x_n) = \lim_n T(x_n) = \lim x_{n+1} = x^*.$$

■

Example 5.7.5 Iterated cosines. Let $X = [-1, 1]$ and define $T(x) = \cos x$. Since X is a closed subset of the reals it is complete (for example, by Bolzano-Weierstrass Theorem 2.3.10). To show that T is a contraction, we can use the MVT; for $x, y \in [-1, 1]$, we know that there is some $c \in [-1, 1]$ so that

$$|T(x) - T(y)| \leq |\sin(c)||x - y|.$$

Now, since $\sin(x)$ is increasing on $[-1, 1]$ (you can easily verify this by looking at its derivative) we can bound $|\sin(x)| \leq |\sin(\pm 1)| < 1$. Thus T is a contraction, with Lipschitz constant $K \leq \sin(1) < 0.842$.

Hence we can apply the contraction mapping theorem and from any starting point $x_0 \in [-1, 1]$ the sequence x_n converges to a unique fixed point, x^*. Further

the sequence should converge as

$$|x_n - x^*| \leq \frac{K^n}{1-K}|x_0 - x_1|$$

So, for example, if we start with $x_0 = 1$, then we get

$$\begin{aligned} x_1 &= 0.540302305868 & x_2 &= 0.857553215846 \\ x_3 &= 0.6542897905 & x_4 &= 0.793480358743 \\ x_5 &= 0.701368773623 & x_6 &= 0.7639596829 \end{aligned}$$

and so on, converging to $x^* = 0.7390851332151$. Notice that since $K^{13} \approx 0.1$, we should expect to take about 13 iterations to "fix" another digit.

However, recall that our estimate of the Lipschitz constant was somehow the worst possible, and we should expect that as get closer and closer to the limit, that the speed of contraction should be governed by the derivative close to the fixed point — namely the size of $\sin(x^*) \approx 0.674$. Since $0.674^6 \approx 0.1$, we actually see that it takes about 6 iterates to fix another digit. □

Example 5.7.6 Iterated sines. Continuing the previous example, again take $X = [-1, 1]$ but now set $T(x) = \sin(x)$. This function has a fixed point at $x = 0$ since $\sin(0) = 0$, however it is not a contraction mapping. This is because $T'(0) = \cos(0) = 1$, and indeed, taking a point x close to zero and using a Taylor series:

$$|\sin(0) - \sin(x)| = |\sin(x)| = |x + \frac{x^3}{6} + \dots|$$

So we know that we can make $|\sin(0) - \sin(x)|/|0 - x|$ arbitrarily close to 1, and thus the Lipschitz constant cannot be smaller than 1.

Despite this, the iterates do converge, just incredibly slowly. Using the inequality $0 < \sin\theta < \theta$, one can show that for $0 < x < 1$, the iterates form a monotonically decreasing sequence bounded below by 0, and thus converge. Similarly, if one starts at a negative iterate. However, since $T'(x) \to 1$ as $x \to 0$, the convergence slows as the iterates approach 0.

Let us be non-rigorous for a moment. We know, from Taylor series, that for small x,

$$\sin(x) \approx x - \frac{x^3}{6}$$

So if $x_n = 0.1$, then $x_{n+1} = \sin(x_n) \approx x_n - \frac{x_n^3}{6} = 0.1 - \frac{0.001}{6}$. So one iteration reduces the iterate at 0.1 by a tiny $0.001/6$ — we barely move at all!

Indeed, if we start at $x_0 = 1$ then it takes nearly 300 iterations to reach $x = 0.1$, and then about 30000 iterations to reach $x = 0.01$. □

Example 5.7.7 A non-example. Let $X = [1, \infty)$ and define $T : X \to X$ by $T(x) = x + \frac{1}{x}$. Then

$$(x + 1/x) - (y + 1/y) = (x - y) + (1/x - 1/y) = (x - y)(1 - 1/xy)$$

so $|T(x) - T(y)| < |x - y|$ since $x, y > 1$. However, there is no *fixed* constant $K < 1$ so that $|T(x) - T(y)| < K|x - y|$; you can take x, y bigger and bigger, to make $|T(x) - T(y)|$ closer and closer to $|x - y|$.

Indeed, since $T(x) > x$ for all $x \in X$, there is no $x^* \in X$ so that $T(x^*) = x^*$. Thus it is not too unreasonable to expect that $T^k(x) \to \infty$. With a little work one can convince oneself that

$$T^k(1) \approx \sqrt{2k}.$$

□

Let us revisit Newton's method again.

Example 5.7.8 Returning to Newton. Let us again return to Newton's method and use the contraction mapping theorem 5.7.4 to understand when it will converge.

We take the whole real line as the underlying metric space and then let $f(x) = x^2 - 2$, giving us the mapping

$$T(x) = x - \frac{f(x)}{f'(x)} = x - \frac{x^2 - 2}{2x} = \frac{x}{2} + \frac{1}{x}$$

Since we know that $1 < \sqrt{2} < 2$, it makes sense to focus on the interval $[1, 2]$, and show that T is a contraction mapping on that closed interval. Then Theorem 5.7.4 ensures that the Newton iterates will converge. The fact that $[1, 2]$ is closed ensures that it is *complete* which we must have.

The derivative is

$$T'(x) = \frac{1}{2} - \frac{1}{x^2} = \frac{x^2 - 2}{2x^2}.$$

and so $T(x)$ takes its minimum at $x = \sqrt{2}$. Since $T(1) = \frac{3}{2}, T(\sqrt{(2)}) = \sqrt{(2)}$ and $T(2) = \frac{3}{2}$, we know that $T([1, 2]) \subseteq [1, 2]$.

Further, the derivative is an increasing function of x and so we know that

$$T'(1) = -\frac{1}{2} \leq T'(x) \leq \frac{1}{4} = T'(2)$$

Hence for any $a, b \in [1, 2]$, the MVT ensures that there is some c so that

$$T(a) - T(b) = T'(c)(a - b)$$

and so

$$|T(a) - T(b)| \leq \frac{1}{2}|a - b|$$

That is, T is a contraction mapping on $[1, 2]$ with Lipschitz constant $\frac{1}{2}$. This agrees with our previous analysis 4.5. □

Example 5.7.9 Newton non-example. What goes wrong if we apply New-

ton's method to $g(x) = x^2 + 1$?. We have

$$T(x) = \frac{x}{2} - \frac{1}{2x} = \frac{x^2 - 1}{2x}$$

Notice that if $x > 0$ then $T(x) < x$. So if we take an interval $[a, b]$ on the positive real line, then $T([a, b]) \not\subseteq [a, b]$. For example $T([1, 2]) = [0, \frac{3}{4}]$. Similarly, for any interval $[a, b]$ in the negative real line $T([a, b]) \not\subseteq [a, b]$. Notice that T does not always make intervals shorter; $T([1/2, 1]) = [-3/4, 0]$.

So while $T(x)$ does contract (some) intervals, it does not map any interval into itself. Hence we cannot apply the contraction mapping theorem. □

See this page[2] for information on how contraction mappings are use to prove Picard's theorem[3] — which guarantees the existence and uniqueness of the solution of an ordinary differential equation.

Exercises

1. **Completeness is important.** Let (X, d) be a metric space that is not complete. Give an example of a contraction mapping $f : X \to X$ which does not have a fixed point.

   Be sure to verify that f is a contraction and explain why the fixed point theorem does not apply.

2. **A way to calculate square roots.** Define $f(x) = \frac{1}{2}\left(x + \frac{c}{x}\right)$ for $c > 0$ and $x > 0$.

   (a) Show that $\sqrt{c}$ is a fixed point of f.

   (b) Show that you can choose positive $a < \sqrt{c} < b$ such that the image of f, when restricted to the closed interval $[a, b]$, is contained in $[a, b]$ and that f is a contraction there.

   (c) Show that $x_{n+1} = f(x_n)$ converges to $\sqrt{c}$ whenever the initial "guess" x_0 is contained in $[a, b]$.

   (d) Manually compute a few digits of $\sqrt{2}$ using this method, without a calculator.

   This iterative process is the classical iteration used in the "Babylonian method" for computing square roots, and dates back to the 17th century BCE.

   Hint. For (b), use the continuity of f and f'.

3. **Maps and fixed points.** Pretend that you have printed out a map of Canada and you lay it on the ground in front of you. Is there necessarily

[2] www.jirka.org/ra/html/sec_metpicard.html
[3] www.jirka.org/ra/html/sec_picard.html

a point on your map that is *exactly* on top of its respective point on the actual planet? What if you crumple the piece of paper first? What if you rip the piece of paper first?

The authors are writing this in Canada, and are assuming you are reading this in Canada. If you are reading this in a different country, then please replace "Canada" with the country in which you are reading. If you are reading this somewhere that is not a country, then we'd love to hear where you are and why you're doing math problems where you are (also, please pretend for the sake of this problem that you are reading it in Canada).

4. **Read the theorem carefully!** Construct a function $f : \mathbb{R} \to \mathbb{R}$ that is differentiable, and that $|f'(x)| < 1$ for every $x \in \mathbb{R}$, but which has *no* fixed point. Why does this not contradict the Contraction Mapping Theorem 5.7.4?

Hint. Consider a function that is almost $f(x) = x$, but differs by a small amount.

5. **Some trickery involved.** Suppose $f : X \to X$ is a continuous function on a complete metric space X such that for some $n \in \mathbb{N}$, the composition $h = \underbrace{f \circ f \circ \cdots \circ f}_{n}$ is a contraction mapping. Does it follow that f has a unique fixed point? Either prove it or provide a counterexample.

Hint. Once you've found a fixed point of the composition, try applying f to both sides.

Chapter 6

Compactness

Some preliminary notes on sequential compactness and compactness.

6.1 Towards compactness

Let's now think back to what helped us get everything working for the reals: limits 3.1, continuity 3.2, IVT 3.2.1, EVT 3.3.5, MVT 4.2.5. We started by defining digit strings and using those to show that the reals have a supremum — every non-empty set with an upper bound has a best upper bound. The supremum allowed us to fill in the ``gaps'', like $\sqrt{2}$, in the rationals.

We then showed that the supremum was equivalent to two other properties

- the nested interval property 2.2.2 (plus the Archimedean property 2.1.2), and
- Bolzano-Weierstrass 2.3.10 (which helped us build completeness of Cauchy sequences 2.3.23)

If we now try to generalise all this to our new world of metric spaces, it makes sense that we might try to generalise one of these three properties:

- supremum — best upper bound
- nested intervals — intersection of nested intervals contains a unique element
- Bolzano-Weierstrass — every bounded sequence has a convergent subsequence.

Let's start with the supremum, it is easy to define **bounded** in a general metric space:

Definition 6.1.1 Bounded. Let (X, ρ) be a metric space, and let $A \subseteq X$. The diameter of A is defined by

$$d(A) = \sup \{\rho(x, y) \colon x, y \in A\}$$

If $d(A) < \infty$ then we say that A is **bounded**. ◊

However, the idea of an "upper bound" no longer makes sense, away from the 1d world of $\mathbb{R}$. Consequently we cannot really define a *best* upper bound.

Then, given how nice the nested interval property was to work with in many of our proofs, one might try to generalise that. And indeed, one can, however it turns out (after much work by very clever people) to be not quite the right generalisation to drive things.

This leaves us (somewhat intentionally for this discussion) with Bolzano-Weierstrass:

Definition 6.1.2 Bolzano-Weierstrass property. Let (X, ρ) be a metric space and let $A \subseteq X$. We say that the set A has the Bolzano-Weierstrass property, when every sequence in A has a convergent subsequence with limit in A. ◊

It turns out that the B-W property is somehow just ``weak enough'' to make the set A "nice", in the sense that it behaves nicely with respect to things we might ask of it involving sequences, continuity, and so on. If this does not feel immediately obvious, you are not alone. Determining that this is "The Right Thing" was, arguably, a major achievement of mathematics in the late 19th century.

As an aside, I recommend looking at the article A pedagogical history of compactness[1].

So — we have defined this desirable property. I want all sequences in my set A to have convergent subsequences. What sets might do this? That is, can we determine what properties might imply B-W. Let's build up to that by first considering some sets inside $(\mathbb{R}, |.|)$. Perhaps the simplest starting point is a set that consists of a single element.

6.1.1 Finite sets

Let $A = \{0\}$. Then, trivially, any sequence in A is the constant and so is convergent. This example is rather too simple, so we should build things up a bit more.

Now let $A = \{y_1, y_2, \ldots, y_N\}$ be any finite subset of $\mathbb{R}$ and let (a_n) be any sequence in A. Form the sets

$$I_k = \{n : a_n = y_k\}$$

for $k = 1, 2, \ldots, N$. It is impossible for all the I_k to be finite otherwise the sequence would only have finite length. Hence one of the I_k must be infinite. The indices in that set give us a constant subsequence. Hence our set A has a convergent subsequence.

This shows us that all finite sets have the B-W property. One of the very strange things about the B-W property, and compactness more generally, is that

[1] arxiv.org/abs/1006.4131

tells us that our set "behaves" a lot like a finite set. We'll see this more as we proceed.

Time to embrace infinity.

6.1.2 Infinite sets

From our previous work we know that open sets will fail to have the B-W property. The open-interval $A = (0, 1) \subset \mathbb{R}$ contains the sequence $a_n = 1/n$. Since $a_n \to 0$, we know that all subsequences also converge to $0 \notin A$. In this way, there is a sequence in $A = (0, 1)$ for which no subsequence converges inside A.

More generally, if a set A has an accumulation point $L \notin A$, then we can construct 5.3.5 a sequence (a_n) in A so that $a_n \to L$.

To be more precise, since L is an accumulation point, we know that for any $\varepsilon > 0$ the ball $B_\varepsilon(L)$ contains some point of A. So by setting $\varepsilon = 1/n$ we can use that element to define the sequence term a_n. Then $\rho(a_n, L) < \varepsilon$, and so $a_n \to L$. All subsequences of (a_n) also converge to $L \notin A$ and so A fails to have the B-W property.

So we need our set A to contain all its accumulation points, and so, be closed. Our first encounter with the B-W property illustrates exactly this. The B-W Theorem 2.3.10 tells us that a sequence in $[a, b]$ has a subsequence that converges to some point in $[a, b]$.

But closed is not enough.

6.1.3 Closed is not enough

We know that the rationals have "bad" properties with respect to limits. So, consider the metric space $(\mathbb{Q}, |.|)$ — ie rationals with the usual euclidean metric. And now consider the set

$$A = \mathbb{Q} \cap [0, 2]$$

Then this set is closed because its complement

$$\{q \in \mathbb{Q} \colon q < 0 \vee q > 2\}$$

is open.

To see that, take any $p \in B$. If $p > 2$, set $\varepsilon = (p - 2)/2$, while if $p < 0$ set $\varepsilon = -p/2$. Then the open ball

$$B_\varepsilon(p) = \{x \in \mathbb{Q} \colon |x - p| < \varepsilon\}$$

is contained in B. Hence B is open and so A is closed.

So now we have a perfectly good metric space containing a perfectly good closed set. Now consider any sequence of rational numbers that converge to $\sqrt{2}$. If you want a specific example, you can do a little Newton's method (or Horner's method) to get

$$a_1 = 1 \qquad\qquad a_n = \frac{x}{2} + \frac{1}{x}.$$

This is sequence in A, however it converges[2] to $\sqrt{2} \notin A$, and so all its subsequences converge to $\sqrt{2} \notin A$.

This is precisely because $\mathbb{Q}$ is not complete — it is full of holes that our sequences can sneak their way into.

6.1.4 So closed and complete? Alas, no.

Nope[3]. Consider $\mathbb{Z}$ inside $(\mathbb{R}, |.|)$. It is closed and complete, but the sequence $a_n = n$ diverges and all its subsequences also diverge.

This sequence has the property that the next point is always "far" from all the previous points. This also means that it has *no Cauchy subsequences*. This is slightly stronger than "no convergent subsequences" since every convergent sequence is Cauchy, but the converse is false.

We need some sort of boundedness condition, so we cannot construct a sequence that ``escapes'' off towards infinity like this.

So, one might want something like: we require that there exists some $M > 0$ and $a \in A$ so that $A \subseteq B_M(a)$. That is, the set is completely contained in a ball of (large) finite radius. That would certainly fix things in $\mathbb{R}$ with the usual euclidean metric.

Notice, also, that if we can cover a set, say $[0, 2]$, with a single ball of radius 1.1:

$$[0, 2] \subseteq B_{1.1}(1)$$

then we can "cover" it with two balls of radius 0.55:

$$[0, 2] \subseteq B_{0.55}(1/2) \cup B_{0.55}(3/2)$$

and three balls of radius 0.34:

$$[0, 2] \subseteq B_{0.34}(1/3) \cup B_{0.34}(1) \cup B_{0.34}(4/3)$$

This makes sense. We are trying to cover a "volume", and as we reduce the volume of the covering balls, we need more of them — but always a finite number. Indeed, we expect some sort of linear[4] relationship between the number of balls and the inverse-radius.

However, metrics can do strange things. We know the discrete metric is a metric and it brings all objects back to a distance of at most 1. Look more carefully at the reals with the discrete metric. Then $\mathbb{Z} \subseteq B_{1.1}(0)$. Even though $\mathbb{Z}$ is bounded (by the discrete metric) it contains a sequence $a_n = n$ which does not converge. Additionally, this sequence, $a_n = n$ has the property that the next term in the sequence is distance at least 0.5 from all previous terms. Hence it has no Cauchy subsequences, and no convergent subsequences. It fails to have the B-W property.

[2]Think of our space as being a metric-subspace of $\mathbb{R}$ with the usual metric.

[3]Sigh. Mathematics can be fickle.

[4]In $\mathbb{R}^2$ we'd expect a quadratic relation, and in $\mathbb{R}^3$ we need a cubic relation. The interested reader should search-engine their way to "Hausdorff dimension".

This is also reflected in the fact that if we reduce the radius of the ball from 1.1 to 0.55 then a ball can only ``cover'' a single point. Hence to cover $\mathbb{Z}$ we need a ball at each integer:

$$\mathbb{Z} \subseteq \cup_{n\in\mathbb{Z}} B_{1/2}(n)$$

So by reducing the radius we have jumped from 1 ball to an infinite number. This "feels" unnatural and it is because of the weird distortions induced by the discrete metric.

Now, this might feel like a pathological example, however, we can deform any metric so that the maximum distance in our space is 1. That is, we can bring objects that are nearly "at infinity" back to distance less than one. For example, given (X, ρ) we can define a new metric, τ by

$$\tau(x, y) = \frac{\rho(x, y)}{1 + \rho(x, y)}$$

Notice that $0 \leq \tau < 1$. It is not too hard to check that τ is symmetric and positive-definite. The triangle inequality takes a little more work:

$$\begin{aligned}
\tau(x, y) = \frac{\rho(x, y)}{1 + \rho(x, y)} &= 1 - \frac{1}{1 + \rho(x, y)} \\
&\leq 1 - \frac{1}{1 + \rho(x, z) + \rho(z, y)} = \frac{\rho(x, z) + \rho(z, y)}{1 + \rho(x, z) + \rho(z, y)} \\
&= \frac{\rho(x, z)}{1 + \rho(x, z) + \rho(z, y)} + \frac{\rho(z, y)}{1 + \rho(x, z) + \rho(z, y)} \\
&\leq \frac{\rho(x, z)}{1 + \rho(x, z)} + + \frac{\rho(z, y)}{1 + \rho(x, z)} = \tau(x, z) + \tau(z, y).
\end{aligned}$$

This deformed metric will brings points way off nearly at infinity and bring them back to distance around 1, and so our whole space can be placed inside a ball of radius 1. So, while the discrete metric is a somewhat extreme example, it is far from unique in having a finite maximum distance.

By deforming things in this way we are not simply "scaling" the space, rather we are compressing regions by different amounts. Nearby "volumes" don't change much, but "volumes" a long way away are compressed down into tiny regions. However, if we move away from the origin towards those distant regions, they expand again.

We need a notion of boundedness that resists these sorts of distortions. To get there, let's think about what is "going wrong" for $\mathbb{Z} \subseteq \mathbb{R}$ with either the euclidean or discrete metrics. Now I don't just want to think about "escaping off to infinity", since the discrete metric (and lots of others) brings infinity back to a finite distance. We need another way to think about "what goes wrong". We know that $\mathbb{R}$ is complete, so in order for a sequence to have no convergent subsequence, it cannot have a *Cauchy subsequence*.

So let's try to construct a "bad" sequence in a metric space (X, ρ) that has no Cauchy subsequences. So lets fix some $\varepsilon > 0$ and start at a point x_1. Now when we pick the next point, we want it to be "far" from the previous points --- so pick x_2 so that $\rho(x_1, x_2) > \varepsilon$. Now we pick x_3 so that it is "far" from both x_1, x_2:

$$\rho(x_1, x_3), \rho(x_2, x_3) > \varepsilon$$

That is, we need $x_3 \in X - \cup_{k=1}^{2} B_\varepsilon(x_k)$. Similarly pick x_4 to be far from all the previous points. Continue this all the way to x_n. Now we pick x_{n+1} so that

$$x_{n+1} \in X - \cup_{k=1}^{n} B_\varepsilon(x_k)$$

So, as long as we can keep doing this we ensure that we *can* construct a sequence that will have no Cauchy subsequence because we have $\rho(x_k, x_\ell) > \varepsilon$ for all k, ℓ.

Now we can definitely do this for $\mathbb{Z}$ inside $\mathbb{R}$ with either the euclidean or discrete metric — simply pick balls of radius $1/2$ and put one at each point.

However we cannot do this for $[0, 1]$ inside $\mathbb{R}$ with the euclidean metric. No matter what $\varepsilon > 0$ you pick, we will cover $[0, 1]$ in a finite number of balls of radius ε and so be *forced* to pick the next term in our sequence inside one of the balls we have already placed.

This suggests a link between the B-W property and the idea that your set can be "covered" by a finite number of ε balls no matter what ε you choose. Consequently we have the following more "refined" definition of boundedness

Definition 6.1.3 Totally bounded. Let (X, ρ) be a metric space, and let $A \subseteq X$. We say that A is **totally bounded** when for any $\varepsilon > 0$, there is a finite set of points $\{a_k : k = 1, \ldots, n\}$ so that

$$A \subseteq \cup_{k=1}^{n} B_\varepsilon(a_k)$$

That is, for any $\varepsilon > 0$ we can always "cover" the set A by a finite number of ε -balls. $\Diamond$

Notice that the definition of totally-bounded somehow "feels" volumes more carefully than bounded. It is more resistant to the possible distortions induced by a particular metric.

Now, we can prove that totally bounded is equivalent to the existence of a Cauchy subsequence (notice some similarities here between how we proved B-W using the NIP).

Theorem 6.1.4 Totally bounded and Cauchy subsequences. *Let (X, ρ) be a metric space. Then X is totally bounded if an only if all sequences in X have Cauchy subsequences.*

Proof. Assume that all sequences have Cauchy subsequences and let $\varepsilon > 0$. We construct as sequence much as we did above. Pick some point $x_1 \in X$, then pick x_2 so that $\rho(x_1, x_2) > \varepsilon$. Continue this picking x_n so that $\rho(x_k, x_n) > \varepsilon$ for all $k = 1, 2, \ldots, n-1$. Now, if we can continue this to infinity, then we have constructed a sequence with no Cauchy subsequences -- a contradiction. Hence

this must terminate after a finite number of steps --- meaning that there is no "uncovered" point. So X is totally bounded.

Now assume that X is totally bounded and let (a_n) be any sequence in X. We need to show that there is some Cauchy subsequence of (a_n). Since X is totally bounded, there is a finite set of points $\{x_{1,1}, x_{1,2}, x_{1,3}, \dots x_{1,n_1}\}$ so that

$$X \subseteq \cup_{k=1}^{n_1} B_1(x_{1,k})$$

Now, one of these balls must contain an infinite number of terms of the sequence — call that infinite subsequence $(a_{1,n})$.

Again, since X is totally bounded, there is a finite set of points $\{x_{2,1}, x_{2,2}, x_{2,3}, \dots x_{2,n_2}\}$ so that

$$X \subseteq \cup_{k=1}^{n_2} B_{1/2}(x_{2,k})$$

Again, since this is a finite collection of balls we know that one of these must contain an infinite number of points from the sequence $(a_{1,n})$ — call that infinite subsequence $a_{2,n}$.

We can keep going in this way — covering X with balls of radius $1/3$, in order to find the subsequence $a_{3,n}$, radius $1/4$ to find the subsequence $a_{4,n}$ and so on. Notice that all the points in the sequence $a_{1,n}$ sit inside a ball of radius 1, those in $a_{2,n}$ sit inside a ball of radius $1/2$, and those in $a_{k,n}$ sit inside a ball of radius $1/k$.

Now consider the sequence the *diagonal* sequence $c_n = a_{n,n}$ — this is a subsequence of the original sequence b_n. Let $\varepsilon > 0$, and pick N sufficiently large so that $1/N \leq \varepsilon/2$. Now, for any $k, \ell > N$, the terms c_k, c_ℓ are both terms in the sequence $a_{k,N}$ and so are inside a ball of radius $1/N \leq \varepsilon/2$. Thus $\rho(c_k, c_\ell) < \varepsilon$. Hence the sequence c_n is Cauchy. ■

Oof!

But now we are ready — we need closed, complete and totally bounded to ensure a set has the B-W property.

6.2 Sequential compactness

Metric spaces with the B-W property are called "sequentially compact". Why not just "compact"? — we'll get to that soon.

Definition 6.2.1 Sequentially compact. We say that a set A in a metric space (X, ρ) is sequentially compact when every sequence in A has a convergent subsequence that converges to some point in A. ◊

And here is our first big result on compactness.

Theorem 6.2.2 Characterising sequential compactness. *Let (X, ρ) be a metric space. Then X is sequentially compact if and only if it is both complete and totally bounded.*

Proof. We prove each implication in turn.

- ($\Rightarrow$): Assume that X is sequentially compact and let (x_n) be a Cauchy sequence in X. Since X is sequentially compact we know that (x_n) has a convergent subsequence. However, any Cauchy sequence with a convergent subsequence is itself convergent. So (x_n) converges. Thus X is complete.

  Similarly, let y_n be any sequence in X. Then since X is sequentially compact, we know that y_n has a convergent subsequence. That subsequence is Cauchy. Hence any sequence has a Cauchy subsequence, and thus X is totally bounded (by our work above).

- ($\Leftarrow$): Now assume that X is totally bounded and complete. Let (x_n) be any sequence in X. Then since X is totally bounded, this has a Cauchy subsequence. Since X is complete, we know that this Cauchy subsequence converges. Thus (x_n) has a convergent subsequence and so X is sequentially complete.

■

Following quickly from this we can show that sequential compactness plays nicely with closures.

Lemma 6.2.3 *A closed subset of a sequentially compact metric space is sequentially compact.*

Proof. Let $A \subseteq X$ and let (a_n) be a sequence in A. Since X is sequentially compact, we know that (a_n) has a convergent subsequence. Since A is closed, we know that the limit of that subsequence lies in A. Thus A is sequentially compact. ■

We also get the following generalisation of the extreme value theorem:

Lemma 6.2.4 Generalised EVT. *The image of a sequentially compact set under a continuous function is sequentially compact.*

Proof. Let $f : X \to Y$ be a continuous function from (X, ρ) to (Y, τ). Let $A \subseteq X$ be a sequentially compact set in X and let $B = f(A)$ be its image in Y.

Not let (b_n) be a sequence in B. We need to show that b_n has a convergent subsequence in B. Now, since $b_n \in f(A)$ there is some $a_n \in A$ so that $b_n = f(a_n)$. Since A is sequentially compact, we know that (a_n) has a convergent subsequence $a_{n_k} \to L \in A$. Then since f is continuous, we know that the subsequence $b_{n_k} = f(a_{n_k}) \to f(L) \in f(A) = B$. Hence $B = f(A)$ is sequentially compact. ■

We also have the following Big Theorem.

Theorem 6.2.5 Heine-Borel Theorem. *Let $A \subseteq \mathbb{R}^n$ with the usual euclidean metric. Then A is sequentially compact if and only if it is closed and bounded.*

Proof.

- ($\Rightarrow$): If A is sequentially compact then from the above we know it is closed and totally bounded. If A is totally bounded then set $\varepsilon = 1$ and

then there is a finite set of point $\{a_1, \ldots, a_N\}$ so that $A \subseteq \cup B_1(a_k)$. Let $M = \max\{\rho(a_k, a_\ell)\} + \varepsilon$ and then $A \subseteq B_M(a_1)$. Hence A is bounded.

- ($\Leftarrow$): Now assume that A is closed and bounded. Hence there is some $M > 0$ so that A is a closed subset of $[-M, M]^n$. So it suffices to show that $[-M, M]^n$ is sequentially compact.

  Now let (x_k) be a sequence in $\mathbb{R}^n$. Then the first-ordinate of that sequence $x_{k,1}$ is a sequence in $[-M, M]$ and so, by Bolzano-Weierstrass, we know it has a convergent subsequence $x_{k_\ell,1} = y_\ell$.

  Consider the second-ordinate of this sequence, $y_{\ell,2}$. This is a sequence in $[-M, M]$, and so by B-W, we know it has a convergent subsequence $y_{\ell_j,2} = z_j$. This has the property that its first two ordinates converge in $[-M, M]$ as $j \to \infty$.

  Continuing in this way constructs a subsequence of the original sequence for which all its n components converge. Thus our original sequence has a subsequence that converges. So $[-M, M]^n$ is sequentially compact.

■

This tells us that closed and bounded is enough in the sorts of spaces we typically think of, namely $\mathbb{R}, \mathbb{R}^2$ and $\mathbb{R}^3$ with the usual euclidean metric and subspaces of them. However, we now know plenty of other weirder spaces where this definitely fails; for example $\mathbb{Z}$ inside $\mathbb{R}$ with the discrete metric is closed and bounded, but definitely not sequentially compact.

6.3 What about just "compact"?

Compactness is weird. It *almost* says that our set is finite. Indeed, it is arguably as close as we can get to having the properties of a finite set without it actually being finite. This article[1] is a nice introduction.

Let us start with a definition, which generalises what we did to define totally bounded. A set was totally bounded if for any δ we could "cover" it by some finite collection of δ-balls. Instead, let us "cover" a set by any collection of open sets — not just balls of a given radius.

Definition 6.3.1 Open cover. Let (X, ρ) be a metric space and let $\mathcal{O} = \{O_i : i \in I\}$ be a set of open sets in X. We say that $\mathcal{O}$ is an **open cover** of X when $X = \cup_{i \in I} O_i$. If $J \subseteq I$ then $\{O_j : j \in J\}$ is a subcover of X. If J is a finite set, then $\{O_j : j \in J\}$ is a **finite subcover**. ◊

So far so good. Now the weird bit.

Definition 6.3.2 Compact. Let (X, ρ) be a metric space. We say that it is compact when every open cover of X has a finite subcover. ◊

[1] blogs.scientificamerican.com/roots-of-unity/what-does-compactness-really-mean/

This definition is far from intuitive, and it is definitely a piece of mathematics for which the famous quote of von~Neumann is true:

> In mathematics you don't understand things. You just get used to them.

From the name, there is clearly a link between compactness and sequential compactness. Indeed in metric spaces they are equivalent. In more general settings they are not equivalent — but the counter-examples are, to say the least, more than a little involved. For example

- The set of functions $f : \mathcal{P}(\mathbb{N}) \to [0,1]$ is compact but not sequentially compact. See this article[2].
- The long-line is sequentially compact but not compact. This example is highly non-trivial and you can find more information on wikipedia[3].

Notice that any finite set is compact. The reals are not compact since though we can cover the reals by

$$\mathbb{R} = \cup_{n\in\mathbb{Z}}(-n, n)$$

this covering has no finite subcover. Similarly the set $(0,1)$ is not compact since we can cover the set with

$$(0,1) = \cup_{n\in\mathbb{N}}(1/n, 1)$$

but this has no finite subcover. In homework you proved that $[0,1]$ is compact, but that proof is far from immediate.

Our main result in this section is that compactness and sequential compactness are equivalent. We have some work to do to get us there. To warm up, we prove the following (which should feel similar to the nested-interval property):

Lemma 6.3.3 *Let X be a compact metric space and let $A_1, A_2, \ldots$ be a sequence of non-empty closed sets in X so that*

$$A_1 \supset A_2 \supset A_3 \supset \ldots$$

Then $\cap A_k \neq \emptyset$.

Proof. We prove this by contradiction. Assume that X is compact but that $\cap A_k = \emptyset$. Then the open sets, $B_k = X - A_k$, form an open-cover of X and also satisfy

$$B_1 \subset B_2 \subset B_3 \subset \ldots$$

Since the B's form an open cover, it has a finite subcover, say $B_{n_1}, B_{n_2}, \ldots, B_{n_N}$. This means that $B_{n_N} = X$ (since the B's are nested) and so $A_{n_N} = \emptyset$. However this gives a contradiction, since all the A's are non-empty. ■

[2] asgarli.wordpress.com/2014/09/24/compact-space-that-is-not-sequentially-compact/
[3] en.wikipedia.org/wiki/Long_line_(topology)

We can now show that compact implies sequentially compact. Given any sequence, (x_n) we can form the set of "tail-closures":

$$A_n = \mathrm{cl}\left(\{x_k \colon k \geq n\}\right)$$

which is a set of closed sets that satisfy the lemma above, and hence their intersection is non-empty. A point in that intersection will help us construct a convergent subsequence.

Lemma 6.3.4 Compact implies sequentially compact. *A compact metric space is sequentially compact.*

Proof. Assume that X is a compact metric space. Let (x_n) be a sequence in X. For each n define the tail of the sequence and their closures:

$$\begin{aligned} T_n &= \{x_k \colon k \geq n\} \\ A_n &= \mathrm{cl}\left(T_n\right). \end{aligned}$$

The A's satisfy

$$A_1 \supset A_2 \supset A_3 \supset \ldots$$

and so we can apply the previous result to show that their intersection is non-empty.

Now let $\alpha \in \cap A_n$. Hence α lies in the closure of every tail. We will construct a sequence x_{n_k} so that for any k, $\rho(x_{n_k}, a) < \frac{1}{k}$.

Since $a \in A_1 = \mathrm{cl}\left(T_1\right)$ there is some point, $x_{n_1} \in T_1$ so that $\rho(a, x_1) < 1$. Since $a \in \mathrm{cl}\left(T_{n_1+1}\right)$ there is some $x_{n_2} \in T_{n_1+1}$ so that $\rho(x_{n_2}, a) < 1/2$. We can continue this to construct a subsequence x_{n_k} that satisfies $\rho(x_{n_k}, a) < 1/k$. Hence this subsequence converges to a.

Thus X is sequentially compact. ■

Oof! The converse is also not so easy. A useful stepping stone is the Lebesgue Covering Lemma. It (effectively) tells us that for any open cover, there is some minimum distance δ so that each point is at least distance δ inside a part of that cover.

Lemma 6.3.5 Lebesgue covering lemma. *Let X be sequentially compact and let $\mathcal{O} = \{O_i \colon i \in I\}$ is an open cover of X. Then there exists some $\delta > 0$ so that for every $x \in X$ there is some $O_i \in \mathcal{O}$ so that $B_\delta(x) \subseteq O_i$.*

Proof. Assume, to the contrary, that no such δ exists. Then for every n there is a point x_n so that $B_{1/n}(x_n)$ is not contained in any part of the open cover.

By sequential compactness, we know that this sequence has a convergent subsequence, $x_{n_k} \to \alpha$. Since α is in some O_ℓ and O_ℓ is open, we know that there is some $\varepsilon > 0$ so that $B_\varepsilon(\alpha) \subseteq O_\ell$.

But now, since $x_{n_k} \to \alpha$, we know that we can pick a term, n_k, sufficiently far down that subsequence so that $\rho(x_{n_k}, \alpha) < \varepsilon/2$ and that $1/n_k < \varepsilon/2$. But then $B_{1/n_k}(x_{n_k}) \subset B_\varepsilon(\alpha) \subset O_\ell$ — which contradicts the choice of the x_n. ■

This lemma does most of the hard work needed to show the converse.

Lemma 6.3.6 Sequentially compact implies compact. *A sequentially compact metric space is compact.*

Proof. Suppose that X is sequentially compact and let $\mathcal{O} = \{O_i : i \in I\}$ is an open cover of X. By the previous lemma we know there is $\delta > 0$ so that for every $x \in X$, the ball $B_\delta(x) \in O_i$ for some $O_i \in \mathcal{O}$.

Now since X is sequentially compact, we know that it is totally bounded. Hence there is a finite set $\{x_1, x_2, \ldots, x_n\}$ so that $X \subseteq \cup_{k=1}^{n} B_\delta(x_k)$. However, we know that each $B_{\delta(x_k)} \subset O_k$ for some $O_k \in \mathcal{O}$. Then these O_k form a finite subcover of X. Thus X is compact. ■

And now we have

Theorem 6.3.7 Compact is sequentially compact. *Let X be a metric space. Then X is compact if and only if it is sequentially compact.*

Appendix A

Constructing the real numbers

Some preliminary notes on building the real numbers via intervals of rational numbers.

A.1 Decadic fractions, intervals and chains

From the discussion above, we are going to make heavy use of rational numbers with denominators that are powers of 10. We'll refer to then as "decadic" numbers; unfortunately there is not a good common name or notation for this set since it does not have good closure properties.

Definition A.1.1 Decadic numbers. Let $n \in \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then we define $\mathbb{D}_n$ to be the set of rational numbers of the form

$$\mathbb{D}_n = \left\{ \frac{a}{10^n} \;\middle|\; a \in \mathbb{Z} \right\} \qquad \text{and} \qquad \mathbb{D}_* = \bigcup_{n \in \mathbb{N}_0} \mathbb{D}_n.$$

We will refer to $\mathbb{D}_*$ as the set of decadic rationals. ◊

We note that a similar set using rational numbers of the form $a/2^n$ are referred to as dyadic rationals — these are much more common in mathematics.

The following lemma will be useful in much of what follows. It is a variant of the well-ordering principle which will help us prove existence of narrow sequences of intervals.

Lemma A.1.2 *Let $A \subseteq \mathbb{D}_n$ be a non-empty set. If A is bounded above then* $\max(A)$ *exists, and if A is bounded below then* $\min(A)$ *exists.*

Proof. You should do this in the exercise below. ■

Exercise A.1.3 Prove a lemma! Prove Lemma A.1.2

Hint. Multiply by a big power of ten and then everything is integer and you can use the well-ordering principle.

Solution.

Proof. Let $B = \{10^n a \mid a \in A\}$ so that $B \subseteq \mathbb{Z}$. If A is bounded above, then

so is B, and by the well-ordering principle the maximum of B exists. Dividing by 10^n then gives the maximum of A. The minimum is obtained similarly. ■

To help our definition of sequences of nested intervals, we define the following set of closed intervals. Normally when we write $[a, b]$ we think of the subset of reals:

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\} \subset \mathbb{R}.$$

We don't yet have the reals, so we will define our intervals to be subsets of $\mathbb{D}_*$.

Definition A.1.4 Intervals. Let $\mathbb{I}_n$ be the set of closed intervals of $\mathbb{D}_*$ whose endpoints are taken from $\mathbb{D}_n$. That is

$$\mathbb{I}_n = \{[a, b] \subset \mathbb{D}_* \mid a \leq b \in \mathbb{D}_n\}. \qquad \text{and} \qquad \mathbb{I}_* = \bigcup_{n \in \mathbb{N}_0} \mathbb{I}_n.$$

Note that $\mathbb{I}_n \subseteq \mathbb{I}_{n+1}$ for all $n \in \mathbb{N}_0$. Then for any $X = [a, b] \in \mathbb{I}_*$ we denote

- its width by $|X| = b - a$, and,
- its endpoints by $\max(X) = \overline{X} = b$ and $\min(X) = \underline{X} = a$.

◊

We can now define our main mathematical object, "chains" — sequences of nested intervals.

Definition A.1.5 Chain. Consider a sequence $(A_n)_{n \in \mathbb{N}_0} = (A_0, A_1, A_2, \dots)$ of intervals in $\mathbb{I}_*$ so that

- $A_n \in \mathbb{I}_n$, and
- they are nested $A_{n+1} \subseteq A_n$, and
- their widths become arbitrarily small, i.e., for any $\varepsilon > 0$ there is some N so that for any $k \geq N$ we have $|A_k| < \varepsilon$.

We call such a sequence a chain. Let $\mathcal{C}$ be the set of all chains. ◊

So for example

$$A = ([1, 2], [14/10, 15/10], [141/100, 142/100], [1414/1000, 1415/1000], \dots) \qquad \text{and}$$
$$B = ([-1, 2], [8/10, 11/10], [90/100, 93/100], [914/1000, 917/1000], \dots)$$

are both chains. The first was generated from the decimal expansion of $\sqrt{2}$

$$A_n = \left[\left\lfloor 10^n \sqrt{2} \right\rfloor / 10^n, \left\lceil 10^n \sqrt{2} \right\rceil / 10^n \right]$$

The intervals are as tight as possible; the widths are $|A_n| = 10^{-n}$. The second was generated from the decimal expansion of Catalan's constant[1]

$$G = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots \approx 0.91596559\dots$$

[1] It is thought to be transcendental, but we cannot even show that it is irrational!

Again, we take the decimal expansion but now deliberately widen each interval a little.

$$B_n = \left[\lfloor 10^n G - 1 \rfloor / 10^n, \lceil 10^n G + 1 \rceil / 10^n\right]$$

Consequently (and very deliberately) the intervals are not tight; their widths are $|B_n| = 3 \times 10^{-n}$.

Exercise A.1.6 Prove something is a chain. Redo Exercise 1.5.5, but now make sure your answer is a chain.

The definition of a chain does not require that the width of each interval to be as narrow as possible. Soon we'll consider chains with that extra condition. (Spoiler: They will be called refinements … but more on that later). Before we get there, we'll explore some properties of these chains and prove some results about them. First notice that the upper and lower ends of intervals within the chains are weakly decreasing and increasing (respectively).

Lemma A.1.7 *For any $A \in \mathcal{C}$, the sequence of $(\underline{A_n})$ is weakly increasing, while $(\overline{A_n})$ is weakly decreasing.*

Proof. This follows immediately from the containment property of intervals within the chain. ■

From this lemma we can (sensibly) define what it means for a chain, $(A_0, A_1, A_2, \dots)$, to be positive: $\underline{A_n} > 0$ for all n. That is — all left endpoints of the intervals of the chain are strictly larger than 0. We can similarly define what it means for a chain to be negative: $\overline{A_n} < 0$ for all n.

However, these definitions are a little dissatisfying. Consider, for example, the chain $(A_0, A_1, \dots)$ defined by

$$A_n = \left[\frac{1}{2} - \frac{1}{n+1}, \frac{1}{2} + \frac{1}{n+1}\right]$$

The first few terms of the sequence are then

$$[-1/2, 3/2] \supseteq [0, 1] \supseteq [1/6, 5/6] \supseteq [1/4, 3/4] \supseteq [3/10, 7/10] \dots$$

Since the first two intervals contain 0, this sequence is not positive (in the sense described above). However, zero lies to the left of every subsequent interval. So even though it is not positive, eventually its intervals become positive.

So if we want our chains to represent real numbers (as we'll get to later), this seems to not be the "correct" definition of what a **positive** chain should be. But that is OK! Remember, at the end of the day, definitions in mathematics is nothing but assigning some mathematical meaning to words, and nothing more. A definition can never be *good* or *bad*, just more (or less) *useful*. I hope that you can convince yourself that a more *useful* definition of a chain being positive is if its "tail-end", i.e., for n arbitrarily large, is positive. Consequently, we define eventually-positive and eventually-negative.

Definition A.1.8 Let $A \in \mathcal{C}$. We say that

- the chain A is eventually-positive when there exists n so that $\underline{A_n} > 0$, and
- the chain A is eventually-negative when there exists n so that $\overline{A_n} < 0$.

Note that due to Lemma A.1.7 we can equivalently state that

- the chain A is eventually-positive when there exists N so that $\underline{A_n} > 0$ for all $n \geq N$, and
- the chain A is eventually-negative when there exists N so that $\overline{A_n} < 0$ for all $n \geq N$.

$\Diamond$

Exercise A.1.9 Not every chain is eventually positive, or eventually negative. Construct three chains: one that is eventually positive, one that is eventually negative, and one that is neither.

Solution. There are lots of possibilities, but the following three chains work.

$$\left([0,1], \left[\frac{9}{10}, 1\right], \left[\frac{99}{100}, 1\right], \left[\frac{999}{1000}, 1\right], \dots\right)$$
$$\left([-1,0], \left[-1, \frac{-9}{10}\right], \left[-1, \frac{-99}{100}\right], \left[-1, \frac{-999}{1000}\right], \dots\right)$$
$$\left([-1,1], \left[\frac{-1}{10}, \frac{1}{10}\right], \left[\frac{-1}{100}, \frac{1}{100}\right], \left[\frac{-1}{1000}, \frac{1}{1000}\right], \dots\right)$$

From this definition we get a trichotomy of chains. You should view this as a step towards showing that the reals are trichotomous — namely for any $x \in \mathbb{R}$, we know that exactly one of $x < 0, x = 0, x > 0$ is true.

Lemma A.1.10 Trichotomy 1. *Let $A \in \mathcal{C}$. Then exactly one of the following is true:*

- *A is eventually-positive,*
- *A is eventually-negative, or*
- *for all n, $0 \in A_n$.*

Proof. Let $A \in \mathcal{C}$. If $0 \in A_n$ for all n then we are done. Otherwise, there exists some N so that $0 \notin A_N$. Hence either $0 < \underline{A_N}$ or $0 > \overline{A_N}$. In the first case, it follows that A is eventually-positive, while in the second A is eventually-negative.

Assume that A is eventually-positive and, to the contrary, that A is also eventually-negative at the same time. Hence there are n, k so that $\underline{A_n} > 0$ and $\overline{A_k} < 0$. Now let $\ell \geq \max\{n, k\}$, then we have

$$\overline{A_\ell} \leq \overline{A_k} < 0 < \underline{A_n} \leq \underline{A_\ell}$$

However, this gives $\overline{A_\ell} < 0 < \underline{A_\ell}$ giving a contradiction. ■

Eventually-positive, eventually negative and this trichotomy result, extend naturally to the definition of an order on chains:

Definition A.1.11 Let $A, B \in \mathcal{C}$. Then we write

- $A < B$ when there exists n so that $\overline{A_n} < \underline{B_n}$,
- $A > B$ when there exists n so that $\underline{A_n} > \overline{B_n}$,
- $A \equiv B$ when for all n, $A_n \cap B_n \neq \emptyset$.

Further, we write $A \leq B$ when $A < B$ or $A \equiv B$, and similarly write $A \geq B$ when $A > B$ or $A \equiv B$.Lemma A.1.7

- $A < B$ when there exists N so that $\overline{A_n} < \underline{B_n}$ for all $n \geq N$, and
- $A > B$ when there exists N so that $\underline{A_n} > \overline{B_n}$ for all $n \geq N$, and

◊

The relation $<$ for chains is a condition that only needs to hold "eventually", i.e., at some point along the chain, and not necessarily for all intervals, which is a good thing to keep in mind as we proceed.

Exercise A.1.12 The condition only needs to hold eventually. Write down an example of two chains A and B such that $A < B$, but such that it is *not* true that $\overline{A_n} < \underline{B_n}$ for every $n \in \mathbb{N}$.

It is important to note that the relation $<$ for chains is a condition that only needs to hold "eventually", i.e., at some point along the chain, and not necessarily for all intervals. This is a good thing to keep in mind as we proceed.

Solution. There are many possibilities here. A simple example is

$$A = \left([0,2], \left[0, \frac{2}{10}\right], \left[0, \frac{2}{100}\right], \left[0, \frac{2}{1000}\right], \dots \right)$$

and

$$B = \left([1,2], \left[1, \frac{11}{10}\right], \left[1, \frac{101}{100}\right], \left[1, \frac{1001}{1000}\right], \dots \right)$$

Notice that $\overline{A_0} = 2 \not< \underline{B_0} = 1$ but we do have $\overline{A_n} < \underline{B_n}$ for all $n \geq 1$.

The notation "$A \equiv B$" is strongly suggestive that this is an equivalence relation on chains — and it is! We'll prove it very shortly. Also notice that setting $B = ([0,0],[0,0],\dots)$ recovers the definition of eventually-positive and eventually-negative.

The above leads quickly to another trichotomy result. This is almost what we need. It is a little weaker since it gives equivalence rather than equality.

Lemma A.1.13 Trichotomy 2. *Let $A, B \in \mathcal{C}$. Then exactly one of the following is true:*

$$A < B \qquad or \qquad A \equiv B \qquad or \qquad A > B.$$

Consequently if $A \leq B$ and $A \geq B$ then $A \equiv B$.

Proof. See the exercise below. ■

Exercise A.1.14 Prove a lemma! Prove Lemma A.1.13.

Hint. This should be very similar to the proof of Lemma A.1.10.

The relation $<$ on chains is very similar to the relation $<$ on reals. It is a strict partial order, meaning it is:

- irreflexive — $\forall A, A \not< A$
- asymmetric — if $A < B$ then we have $B \not< A$, and
- transitive — if $A < B$ and $B < C$, then $A < C$.

Note that the relation $\leq$ on chains is not a partial order because it is not anti-symmetric since $A \leq B$ and $B \leq A$ implies $A \equiv B$ but not $A = B$.

Theorem A.1.15 *The relation $<$ on chains is a strict partial order. That is, it is irreflexive, asymmetric and transitive.*

Exercise A.1.16 Prove it! Prove Theorem A.1.15.

Hint. Lemma A.1.13 is useful for showing that $<$ is irreflexive and asymmetric, while transitivity follows from the definition of $<$.

Solution. An excellent homework problem!

We've been calling $\equiv$ "equivalence" without proving it, so it is about time that we did. The proof that it is reflexive and symmetric is relatively straight forward, but a little care is required for transitivity.

But before we leap into proving that we do have an equivalence relation you should make sure that you recall some definitions.

Exercise A.1.17 Recall some definitions. You should recall that an equivalence relation on a set A must be reflexive, symmetric and transitive. Please define the terms **reflexive**, **symmetric** and **transitive**.

Solution. A relation R on a set A is

- reflexive — for all $a \in A$, we have aRa
- symmetric — if aRb then bRa
- transitive — if aRb and bRc then aRc

Theorem A.1.18 *The relation $\equiv$ on chains is an equivalence relation.*

Proof. We start by showing it is reflexive and symmetric:

- Since $A_n \cap A_n = A_n$, reflexive follows immediately.

- Assume that $A \equiv B$ so that for every n, $A_n \cap B_n \neq \emptyset$. Hence $B_n \cap A_n \neq \emptyset$ and thus $B \equiv A$.

To prove transitivity we need to use the fact that we know that the widths of the intervals are going to zero. Assume that $A \equiv B$ and $B \equiv C$, but, to the contrary, that $A \not\equiv C$. Thus for all n we know that

$$A_n \cap B_n \neq \emptyset \qquad \text{and} \qquad B_n \cap C_n \neq \emptyset$$

but that there exists some N so that $A_N \cap C_N = \emptyset$.

WLOG, assume that $A_N < C_N$ (in the sense that $\overline{A_N} < \underline{C_N}$). Now let $k \geq N$, then since the intervals are nested, we know that for any $z \in A_k, w \in C_k$ we must have $z \leq \overline{A_N}, w \geq \underline{C_N}$ and so

$$|w - z| = w - z \geq \underline{C_N} - \overline{A_N} = \delta$$

which is some fixed positive rational. At the same time, by assumption $A_k \cap B_k \neq \emptyset$ and $B_k \cap C_k \neq \emptyset$, and so there exist $x, y \in B_k$ so that

$$x \in A_k \qquad \text{and} \qquad y \in C_k$$

and thus $|B_k| \geq \delta$. This contradicts the fact that the width of the intervals goes to zero. Thus, we must have that $A_n \cap C_n \neq \emptyset$ for all n. ■

Thankfully our order $<$ on chains is compatible with our equivalence relation $\equiv$ on chains; it preserves order. So, if we know that $A < B$, and also that $A \equiv X$ then we also know that $X < B$. Before we prove this result, which is given as Theorem A.1.21, we first give a technical lemma about equivalent chains which we'll need in that proof. It tells us that if two chains are equivalent, then the endpoints of their intervals must become arbitrarily close together.

Lemma A.1.19 *Let $A, B \in \mathcal{C}$ so that $A \equiv B$. Then*

$$A_n \cup B_n = \left[\min\left\{\underline{A_n}, \underline{B_n}\right\}, \max\left\{\overline{A_n}, \overline{B_n}\right\}\right]$$

and

$$|A_n \cup B_n| \to 0$$

in the sense that for any $\varepsilon > 0$ there is N so that if $n > N$ then $|A_n \cup B_n| < \varepsilon$.

Exercise A.1.20 Prove this technical lemma. Prove Lemma A.1.19.

Solution.

Proof. Let A, B be as given. For any n, we define $X_n = A_n \cup B_n$. Since $A \equiv B$ we know that $A_n \cap B_n \neq \emptyset$ and so we can write $X_n = [l, r]$ where $l = \min\left\{\underline{A_n}, \underline{B_n}\right\}$ and $r = \max\left\{\overline{A_n}, \overline{B_n}\right\}$. Notice that we must have

$$|X_n| \leq |A_n| + |B_n|$$

Now let $\varepsilon > 0$, then since $A, B \in \mathcal{C}$ there is N so that if $n > N$ then $|A_n|, |B_n| < \varepsilon/3$. Hence $|X_n| < \varepsilon$ as required. ■

Theorem A.1.21 Equivalence preserves order. *Let $A, B, X, Y \in \mathcal{C}$ so that $A \equiv X$ and $B \equiv Y$. Then*

$$A < B \qquad \Longleftrightarrow \qquad X < Y$$

Proof. Homework!!! ■

Another piece of notation that is related to the equivalence $\equiv$, is that of containment of chains.

Definition A.1.22 Let $A, B \in \mathcal{C}$. We write $A \subseteq B$ when for all n we have $A_n \subseteq B_n$. ◊

We can easily show that containment implies equivalence, and that the converse is false.

Lemma A.1.23 *Let $A, B \in \mathcal{C}$. If $A \subseteq B$ then $A \equiv B$.*

Proof. Since $A \subseteq B$ we know that $A_n \subseteq B_n$ for every n, and thus $A_n \cap B_n = A_n \neq \emptyset$. So $A \equiv B$ as required. ■

Exercise A.1.24 Equivalence does not imply containment. Prove that the converse of Lemma A.1.23 is false.

Hint. Find two chains that intersect so that neither contains the other.

Solution. Consider chains defined by $A_n = [-1/10^n, 2/10^n]$ and $B_n = [-2/10^n, 1/10^n]$. Then for every n, $A_n \cap B_n = [-1/10^n, 1/10^n] \neq \emptyset$ and $A_n \not\subseteq B_n$ and $B_n \not\subseteq A_n$. Hence $A \equiv B$ but $A \not\subseteq B$ and $B \not\subseteq A$.

A.2 Addition and subtraction of chains

Interval arithmetic is a widely in computations, especially when rigorous bounds are required. We leverage ideas from that area of mathematics to define arithmetic on chains.

Definition A.2.1 Addition of intervals. For any n, let $A, B \in \mathbb{I}_n$ then

$$A + B = [\underline{A} + \underline{B}, \overline{A} + \overline{B}] = \{x + y \mid x \in A, y \in B\}$$
$$A - B = [\underline{A} - \overline{B}, \overline{A} - \underline{B}] = \{x - y \mid x \in A, y \in B\}$$

These are equivalent to the Minkowski sums and differences of the intervals.

We also define

$$-B = [-\overline{B}, -\underline{B}] = \{-y \mid y \in B\}$$

so that $A - B = A + (-B)$. ◊

From addition and subtraction of intervals we then define addition and subtraction of chains:

Definition A.2.2 Addition of chains. Let $A, B \in \mathcal{C}$ then we can add and subtract them term-by-term:

$$A \oplus B = (A_0 + B_0, A_1 + B_1, A_2 + B_2, A_3 + B_3, \dots)$$

$$A \ominus B = (A_0 - B_0, A_1 - B_1, A_2 - B_2, A_3 - B_3, \dots)$$

We similarly define

$$-B = (-B_0, -B_1, -B_2, -B_3, \dots)$$

so that $A \ominus B = A \oplus (-B)$. ◊

Note: We are deliberately using the somewhat strange notation of $\oplus$ and $\ominus$ (and later, $\oplus$ for multiplication), because we want to save the more common symbols $+, -, \cdot$ to mean addition, subtraction, and multiplaction respectively of *refined* chains (we will call these **refinements**). More on that later. Just be aware of the intentional use of this strange notation for now, and bear with us.

Exercise A.2.3 Consider the two chains coming from the decimal approximates of π and e respectively that we found earlier:

$$A = \left([3,4], \left[\frac{31}{10}, \frac{32}{10}\right], \left[\frac{314}{100}, \frac{315}{100}\right], \left[\frac{3141}{1000}, \frac{3142}{1000}\right], \dots\right)$$
$$B = \left([2,3], \left[\frac{27}{10}, \frac{28}{10}\right], \left[\frac{271}{100}, \frac{272}{100}\right], \left[\frac{2718}{1000}, \frac{2719}{1000}\right], \dots\right),$$

and verify that $A \oplus B$ seems to be converging to the value of $e + \pi \approx 5.859874482\dots$

The chains A and B in the previous exercise are the *best* approximates the irrational numbers π and e respectively; each interval A_n (resp. B_n) has length precisely 10^{-n} and contains π (resp. e). Of course, we could write down many different chains that also approximate these numbers, such as the following examples:

$$A = \left([2,5], \left[\frac{30}{10}, \frac{35}{10}\right], \left[\frac{311}{100}, \frac{317}{100}\right], \left[\frac{3137}{1000}, \frac{3148}{1000}\right], \dots\right)$$
$$B = \left([1,3], \left[\frac{26}{10}, \frac{29}{10}\right], \left[\frac{271}{100}, \frac{273}{100}\right], \left[\frac{2716}{1000}, \frac{2719}{1000}\right], \dots\right),$$

This is why we need to work with equivalences in chains; we don't want to spend any time worrying about choosing a slightly different chain. That being said, we will see later there is a canonical, i.e., natural, choice of chain in each equivalence class. We will eventually call this natural choice **refinements**.

Of course, just because we have defined things this way, it does not mean that they do what we want. In particular, we need to prove that $A \oplus B, A \ominus B$ are actually chains.

Lemma A.2.4 *The set of chains is closed under addition and subtraction as defined above.*

Exercise A.2.5 Prove it! Prove Lemma A.2.4.

Hint. Given chains A, B, you need to show that $A \oplus B$ is also a chain. That is, you must show:

- $(A \oplus B)_n \in \mathbb{I}_n$,
- $|(A \oplus B)_n| \to 0$, and
- the intervals $(A \oplus B)_n$ are nested.

Solution.

Proof. Let A, B be chains and then set $X = A \oplus B$ and $Y = A \ominus B$.

- Since $A_n, B_n \in \mathbb{I}_n$, it follows that $X_n, Y_n \in \mathbb{I}_n$.
- Then notice that
$$0 \leq |X_n|, |Y_n| \leq |A_n| + |B_n|$$
and since the widths of A_n, B_n go to zero, it follows that the widths of X_n, Y_n also go to zero.
- Finally, we show that the terms of X, Y form nested sequences of intervals. Fix n and let $k > n$. Let $z \in A_k + B_k$, which implies that there are $x \in A_k, y \in B_k$ so that $x + y = z$. Since $A_k \subseteq A_n$ and $B_k \subseteq B_n$ we know $x \in A_n, y \in B_n$ and thus $z = x + y \in A_n + B_n = X_n$. Thus $X_k \subseteq X_n$. A similar argument shows that $Y_k \subseteq Y_n$.

Thus $X, Y \in \mathcal{C}$ and so the set of chains is closed under addition and subtraction as required. ■

Now we need to do a little technical checking to confirm that addition of chains "behaves" the way we want it to.

Lemma A.2.6 *Addition of chains is commutative and associative.*

Proof. For commutativity, note that for any n

$$\begin{aligned} A_n + B_n &= \{a + b \mid a \in A_n, b \in B_n\} \\ &= \{b + a \mid a \in A_n, b \in B_n\} = B_n + A_n. \end{aligned}$$

Similarly for associativity, for any n

$$\begin{aligned} A_n + (B_n + C_n) &= \{a + x \mid a \in A_n, x \in (B_n + C_n)\} \\ &= \{a + (b + c) \mid a \in A_n, b \in B_n, c \in C_n\} \\ &= \{(a + b) + c \mid a \in A_n, b \in B_n, c \in C_n\} \\ &= \{y + c \mid y \in (A_n + B_n), c \in C_n\} = (A_n + B_n) + C_n. \end{aligned}$$

■

Helpfully addition and subtraction are compatible with our notion of equivalence of chains:

Lemma A.2.7 Addition, subtraction and equivalence. *Let* $A, B, X, Y \in \mathcal{C}$

so that $A \equiv X$ and $B \equiv Y$. Then

$$A \oplus B \equiv X \oplus Y \qquad \textit{and} \qquad A \ominus B \equiv X \ominus Y.$$

Proof. Let A, B, X, Y be as stated and let $n \in \mathbb{N}_0$ be fixed. Then

$$A_n \cap X_n \neq \emptyset \qquad \text{and} \qquad B_n \cap Y_n \neq \emptyset$$

Hence there is $a \in A_n \cap X_n$ and $b \in B_n \cap Y_n$. Thus $a + b \in A_n + B_n$ and at the same time $a + b \in X_n + Y_n$. This means that $(A_n + B_n) \cap (X_n + Y_n) \neq \emptyset$. Since n was arbitrary, it follows that $A \oplus B \equiv X \oplus Y$. The argument for subtraction is almost identical. ■

We have an equivalent of zero:

Definition A.2.8 Define the zero-chain to be

$$\mathbf{0} = ([0,0],[0,0],[0,0],\dots)$$

◊

Now a few exercises to show that the zero-chain is an additive identity, and that it (mostly) behaves as we would like it to do so.

Exercise A.2.9 Adding zero does nothing. Let $X \in \mathcal{C}$, then show that

$$X \oplus \mathbf{0} = X$$

Solution. This follows pretty immediately from the definitions

Exercise A.2.10 Subtracting yourself gives something equivalent to zero. Let $X \in \mathcal{C}$ then show that

$$\mathbf{0} \equiv X \oplus (-X).$$

Solution. Again, this follows quite directly from the definition

Exercise A.2.11 Subtracting yourself need not give exactly zero. Let $X \in \mathcal{C}$ then show that $X \oplus (-X)$ need not be *equal to* zero.

Solution. Let X be a chain with $X_n = [0,1]$ for all $n = 0, 1, 2, \dots$. Then $-X_n = [-1, 0]$ and

$$(X \oplus (-X))_n = [0,1] + [-1,0] = [-1,1] \neq [0,0].$$

We can also show that addition and order are compatible.

Lemma A.2.12 *Let $A, B, X \in \mathcal{C}$. If $A < B$ then $A \oplus X < B \oplus X$.*

Proof. Let $A, B, X \in \mathcal{C}$ so that $A < B$. Then for any n

$$\underline{(B_n + X_n)} - \overline{(A_n + X_n)} = \underline{B_n} - \overline{A_n} + \overline{X_n} - \underline{X_n} = \underline{B_n} - \overline{A_n} + |X_n|$$

It suffices to show that we can find n so that this is positive. Since $A < B$ we know that there is N so that $\overline{A_N} < \underline{B_N}$. Set $\varepsilon = \underline{B_N} - \overline{A_N}$. Since $X \in \mathcal{C}$ there

is K so that $|X_K| < \varepsilon$. Then for any $n > \max\{N, K\}$ we have

$$\underline{(B_n + X_n)} - \overline{(A_n + X_n)} > 0$$

as required. ■

A.3 Multiplication of chains

A.3.1 Definitions and properties

Multiplication of intervals and chains is more complicated due to possible sign changes.

Definition A.3.1 Multiplication of intervals. For any n, let $A, B \in \mathbb{I}_n$ then

$$\begin{aligned} A \cdot B = [p_\ell, p_r] &= \{xy \mid x \in A, y \in B\} \in \mathbb{I}_{2n} \\ p_\ell &= \min\left\{\underline{AB}, \underline{A}\overline{B}, \overline{A}\underline{B}, \overline{AB}\right\} \\ p_r &= \max\left\{\underline{AB}, \underline{A}\overline{B}, \overline{A}\underline{B}, \overline{AB}\right\} \end{aligned}$$

These is equivalent to the Minkowski product the intervals. ◊

Notice that the product of two intervals in $\mathbb{I}_n$ gives an interval in II_{2n}. We can work around this using

$$X = \left[\frac{a}{10^{2n}}, \frac{b}{10^{2n}}\right] \in \mathbb{I}_{2n} \qquad \Longleftrightarrow \qquad X = \left[\frac{10a}{10^{2n+1}}, \frac{10b}{10^{2n+1}}\right] \in \mathbb{I}_{2n+1}$$

Definition A.3.2 Multiplication of chains. Let $A, B \in \mathcal{C}$, then we define their product as

$$P = A \odot B = (P_0, P_1, P_2, \dots)$$

where $P_{2n} = A_n \cdot B_n$ and $P_{2n+1} = A_n \cdot B_n$. ◊

Before checking any properties of $\odot$, we want you to try your hand at multiplying chains by hand.

Exercise A.3.3 Multiply by hand. Compute the product $A \odot B$ where

$$\begin{aligned} A &= ([-1, 2], [5/10, 15/10], [1, 1], [1, 1], \dots) \\ B &= ([-2, 1], [-1, 5/10], [0, 0], [0, 0], \dots) \end{aligned}$$

The product is

$$A \odot B = \left([-4, 2], [-4, 2], \left[\frac{-15}{10}, \frac{75}{100}\right], \left[\frac{-15}{10}, \frac{75}{100}\right], [0, 0], \dots\right)$$

We need this to work like multiplication should:

Lemma A.3.4 *The set of chains is closed under multiplication as defined above. Further, multiplication is commutative.*

Proof. Let $A, B \in \mathcal{C}$ and set $P = A \odot B$. We need to show that $P \in \mathcal{C}$.

- By construction, for each $k, P_k \in \mathbb{I}_k$. Further, we know that $P_{2n+1} = P_{2n}$. It suffices to show that $P_{2n+2} \subseteq P_{2n}$. So let $z \in P_{2n+2}$, then there are $a \in A_{n+1}, b \in B_{n+1}$ so that $z = ab$. However since $A, B \in \mathcal{C}$ we also know $a \in A_n, b \in B_n$ and thus $z \in P_{2n} = A_n \cdot B_n$. Hence the intervals of P are nested.

- It suffices to show that $|P_{2n}| \to 0$. Let $M = \max\left\{|\underline{A_0}|, |\overline{A_0}|, |\underline{B_0}|, |\overline{B_0}|\right\}$. Notice that $A_k, B_k \subseteq [-M, M]$ for any k.

  Now write write $P_{2n} = [xz, yw]$ where $x, y \in A_n$ and $z, w \in B_n$. Then (in a calculation reminiscent of the product rule):

$$\begin{aligned} |P_{2n}| = |xz - yw| &= |x(z-w) + w(x-y)| \\ &\leq |x||z-w| + |w||x-y| \\ &\leq M|B_n| + M|A_n| \end{aligned}$$

  So, for any $\varepsilon > 0$, there is N so that when $n \geq N$, $|A_n|, |B_n| < \varepsilon/2M$, and thus $|P_{2n}| < \varepsilon$ as required.

Hence $P \in \mathcal{C}$ and so chains are closed under multiplication.

For commutativity, note that for any n

$$\begin{aligned} A_n \cdot B_n &= \{a \cdot b \mid a \in A_n, b \in B_n\} \\ &= \{b \cdot a \mid a \in A_n, b \in B_n\} = B_n \cdot A_n. \end{aligned}$$

■

Associativity requires more care, and indeed, multiplication is not associative, but it is "nearly associative".

Lemma A.3.5 Nearly associative. *Let $A, B, C \in \mathcal{C}$ then*

$$(A \odot B) \odot C \equiv A \odot (B \odot C).$$

Proof. First note that multiplication of intervals is associative. This can be proved much as we proved the associativity of addition of intervals above. Let A, B, C be as given. Then

$$\begin{aligned} LHS &= ((A \odot B) \odot C)_{2n} = ((A \odot B) \odot C)_{2n+1} = A_{\lfloor n/2 \rfloor} B_{\lfloor n/2 \rfloor} C_n \\ RHS &= (A \odot (B \odot C))_{2n} = (A \odot (B \odot C))_{2n+1} = A_n B_{\lfloor n/2 \rfloor} C_{\lfloor n/2 \rfloor} \end{aligned}$$

Notice that since A, B, C are chains, we know that

$$A_n \subseteq A_{\lfloor n/2 \rfloor} \qquad \text{and} \qquad C_n \subseteq C_{\lfloor n/2 \rfloor}$$

and thus the set $A_n B_{\lfloor n/2 \rfloor} C_n \neq \emptyset$ is a subset of both the LHS and RHS. Thus $LHS \cap RHS \neq \emptyset$ and the result follows. ■

Exercise A.3.6 Equivalence not equality. Lemma A.3.5 involves equivalence between the two sides of the equation, not equality. Produce an example of three chains A, B, C to show that equality need not hold. That is

$$(A \odot B) \odot C \neq A \odot (B \odot C)$$

Hint. Look back at the proof of the equivalence $(A \odot B) \odot C \equiv A \odot (B \odot C)$ for an indication for what terms in the product may not be equal. This may help you both come up with an example and show that your example is correct.

Multiplication is compatible with equivalence of chains:

Lemma A.3.7 Multiplication and equivalence. *Let $A, B, X, Y \in \mathcal{C}$ so that $A \equiv X$ and $B \equiv Y$. Then*

$$A \odot B \equiv X \odot Y$$

Exercise A.3.8 Prove it! Prove Lemma A.3.7

Solution.

Proof. Let A, B, X, Y be as stated and let $n \in \mathbb{N}_0$ be fixed. Then

$$A_n \cap X_n \neq \emptyset \qquad \text{and} \qquad B_n \cap Y_n \neq \emptyset$$

Thus there is $a \in A_n \cap X_n$ and $b \in B_n \cap Y_n$. Thus $ab \in A_n \cdot B_n$ and at the same time $ab \in X_n \cdot Y_n$. Thus $A_n B_n \cap X_n Y_n \neq \emptyset$. Since n was arbitrary it follows that $A \odot B \equiv X \odot Y$. ■

Chains have an equivalent of one — when you multiply it you get back what you started with

Definition A.3.9 Define the one-chain to be

$$\mathbf{1} = ([1,1],[1,1],[1,1],\dots)$$

◊

It is nearly the multiplicative identity:

Lemma A.3.10 *Let $X \in \mathcal{C}$ then $X \odot \mathbf{1} \equiv X$.*

Proof. We have

$$X \odot \mathbf{1} = (X_0, X_0, X_1, X_1, X_2, \dots)$$

Since $X_n \subseteq X_{\lfloor n/2 \rfloor}$ the result follows. ■

Multiplication and addition are nearly distributive — they are called sub-distributive:

Lemma A.3.11 Sub-distributive. *Let $A, B, C \in \mathcal{C}$ then*

$$A \odot (B \oplus C) \subseteq (A \odot B) \oplus (A \odot C).$$

and hence both sides are equivalent.

Proof. Let A, B, C be as stated then

$$LHS_{2n} = LHS_{2n+1} = A_n \cdot (B_n + C_n)$$
$$RHS_{2n} = RHS_{2n+1} = (A_n \cdot B_n) + (A_n \cdot C_n)$$

If $x \in LHS$ then there are $a \in A_n, y \in (B_n + C_n)$ so that $x = ay$. Then since $y \in (B_n + C_n)$ there are $b \in B_n$ and $c \in C_n$ so that $y = b + c$. Hence $x = a(b + c) = ab + ac$. Since $ab \in A_n \cdot B_n$ and $ac \in A_n \cdot C_n$, it follows that $x \in RHS$. This is sufficient to show equivalence. ■

Unfortunately the reverse inclusion does not hold in general. The following exercise asks you to construct an example that shows that.

Exercise A.3.12 It really is SUB-distributive. Construct a simple example to show that in general

$$(A \odot B) \oplus (A \odot C) \not\subseteq A \odot (B \oplus C)$$

Hint. Choose a very simple interval for A, and then try choosing B, C so that $B \oplus C$ is perhaps the simplest chain you know, and so very easy to work with.

Solution. Consider chains A, B, C with their first intervals being $A_0 = [1, 2]$ and $B_0 = [-1, -1], C_0 = [1, 1]$. Then

$$LHS = [1, 2] \cdot [-1, -1] + [1, 2] \cdot [1, 1] = [-2, -1] + [1, 2] = [-1, 1].$$
$$RHS = [1, 2] \cdot ([-1, -1] + [1, 1]) = [1, 2] \cdot [0, 0] = [0, 0]$$

Multiplication and ordering of chains are compatible:

Lemma A.3.13 *Given two chains $A, B \in \mathcal{C}$, then each of the following holds:*

- *If $A > \mathbf{0}$ and $B > \mathbf{0}$, then $A \odot B > \mathbf{0}$.*
- *If $A < \mathbf{0}$ and $B > \mathbf{0}$, then $A \odot B < \mathbf{0}$.*
- *If $A < \mathbf{0}$ and $B < \mathbf{0}$, then $A \odot B > \mathbf{0}$.*

Proof. We prove the first case in which both chains are positive. The remaining cases are left as an exercise.

Since $X, Y > \mathbf{0}$ we know that there is some N so that for all $n \geq N, 0 < \underline{X_n}, \underline{Y_n}$. Thus we know for all $k \geq 2N$ we have $\underline{(X \odot Y)_k} > 0$ and so the result follows. ■

Exercise A.3.14 You finish it off. Complete the proof of Lemma A.3.13 by proving the remaining two cases.

A.3.2 Where we are up to

At this point we have shown that chains behave a lot like real numbers under addition and multiplication, but not quite the same. For chains X, Y and Z:

- Closed: $X \oplus Y, X \odot Y \in \mathcal{C}$
- Commutative: $X \oplus Y = Y \oplus X$ and $X \odot Y = Y \odot X$
- Associative (nearly): $X \oplus (Y \oplus Z) = (X \oplus Y) \oplus Z$ and $X \odot (Y \odot Z) \equiv (X \odot Y) \odot Z$.
- Distributive (nearly): $X \odot (Y \oplus Z) \equiv (X \odot Y) \oplus (X \odot Z)$
- Additive identity: $X \oplus \mathbf{0} = X$
- Additive inverse (nearly): $X \oplus (-X) \equiv \mathbf{0}$
- Multiplicative identity (nearly): $X \odot \mathbf{1} \equiv X$ (with $1 \neq 0$)
- Transitive: If $X < Y$ and $Y < Z$ then $X < Z$.
- Trichotomy (nearly): Exactly one of $X < Y, X \equiv Y, X > Y$ holds.
- Order and addition: If $X < Y$ then $X \oplus Z < Y \oplus Z$
- Order and multiplication: If $\mathbf{0} < X, Y$ then $\mathbf{0} < X \odot Y$

So, we have a list of things to fix up; they nearly work except we have equivalence instead of equality:

- associativity of multiplication,
- distribution of addition and multiplication,
- the additive inverse,
- the multiplicative identity, and
- trichotomy,

and a (thankfully shorter) list of things we are yet to look at:

- the multiplicative inverse, and
- the least upper bound property.

Oof!

Thankfully the items on the "they nearly work" list become fairly straight forward when we stop looking at all chains an instead focus on chains that are as narrow as possible. Roughly speaking, those narrowest chains give us a way to work with each equivalence class of chains via a unique representative. But there is still some hard work to do on the least upper bound and on multiplicative inverses.

A.4 Narrowing and refinements

In order to get the arithmetic properties of chains to align with those of real numbers, we need to concentrate on chains that are as narrow as possible. These narrowest chains give us a unique representative of each equivalence class or chains. These unique representatives will act just like real numbers.

So imagine, for the moment, that we have some chain $B \in \mathcal{C}$. We would like to find a chain that is equivalent to it, but narrower. One way to do this is to look for a chain A so that each of its intervals are contained within the intervals of B. Going back to our Catalan's constant example

$$\begin{aligned} A &= ([0,1],[9/10,10/10],[91/100,92/100],[915/1000,916/1000],\dots) \\ &\subseteq ([-1,2],[8/10,11/10],[90/100,93/100],[914/1000,917/1000],\dots) = B \end{aligned}$$

Since the intervals of A are contained in the intervals of B we have $A \equiv B$ and also $|A_n| \leq |B_n|$ at every n.

Given any chain we can ask for the "narrowest" chain that is equivalent to it by asking for a chain that is contained within all equivalent chains. Given a specific chain X, it is not difficult to do this explicitly.

Exercise A.4.1 Consider the following chain, obtained from the constant π:

$$A = \left([-1, 5], \left[\frac{29}{10}, \frac{33}{10}\right], \left[\frac{298}{100}, \frac{325}{100}\right], \left[\frac{3140}{1000}, \frac{3146}{1000}\right], \dots \right).$$

Write down a chain B such that $B \subseteq A$ and that the width of the first four intervals in B are as small as possible.

Solution. Knowing that the decimal expansion[1] of π is 3.14159265...

$$B = \left([3, 4], \left[\frac{31}{10}, \frac{32}{10}\right], \left[\frac{314}{100}, \frac{315}{100}\right], \left[\frac{3141}{1000}, \frac{3142}{1000}\right], \dots \right).$$

Or, instead of using the decimal expansion, we would be able to still know for certain that the first 3 intervals

$$B = \left([3, 4], \left[\frac{31}{10}, \frac{32}{10}\right], \left[\frac{314}{100}, \frac{315}{100}\right], \dots \right)$$

are as "tight" as possible, because the fourth interval in A, that is A_4 is

$$\left[\frac{3140}{1000}, \frac{3146}{1000}\right],$$

which satisfies

$$\overline{A_4} - \underline{A_4} = \frac{3146}{1000} - \frac{3140}{1000} = \frac{6}{1000} < 10^{-3}.$$

So, in particular, we know that we can specify the first 3 intervals in B to be as tight as possible.

Definition A.4.2 Let $A \in \mathcal{C}$. We say that A is a **refinement**[2] if for all $B \equiv A$ we have $A \subseteq B$. Let $\mathcal{R} \subseteq \mathcal{C}$ be the set of all refinements. Note that the zero-chain and one-chain are refinements. ◊

Exercise A.4.3 Zero and one are refined. Show that the zero-chain **0** and one-chain **1** are refinements (don't just take our word for it).

Of course, just because we have defined refinements, it does not mean they exist, nor does it mean they are unique. There is work to do!

Lemma A.4.4 Uniqueness of refinements. *Let $A \in \mathcal{C}$. There exists at most one $X \in \mathcal{R}$ so that $X \equiv A$. That is, if A has a refinement then that refinement is unique.*

Exercise A.4.5 Prove it! Prove Lemma A.4.4.

Hint. You need to show that given any chain X, if Y and Z are both refinements of X, then $Y = X$.

[1] using a decimal expansion is kind of "cheating" here, because decimal expansions is one representation of the real numbers, which is precisely what we are trying to construct! But oh well

[2] you can think of refinements as a type of "minimal" chain

Solution.

Proof. Let $X, Y \in \mathcal{R}$ so that $X \equiv A$ and $Y \equiv A$. Since both X, Y are minimal with respect to containment, it follows that

$$X \subseteq Y \qquad \text{and} \qquad Y \subseteq X.$$

Hence for all n we know that $X_n = Y_n$ and so $X = Y$. ■

Given a chain A, we will prove the existence of an equivalent refinement, $\hat{A}$ by construction.

Lemma A.4.6 Existence of refinements. *Given any $A \in \mathcal{C}$ and then define $Z = \hat{A}$ as*

$$Z = \hat{A} = ([\alpha_0, \beta_0], [\alpha_1, \beta_1], [\alpha_2, \beta_2], \dots)$$

where for each $n \in \mathbb{N}_0$

$$\alpha_n = \max\left\{\underline{B_n} \;\middle|\; B \in \mathcal{C}, B \equiv A\right\} \qquad \beta_n = \min\left\{\overline{B_n} \;\middle|\; B \in \mathcal{C}, B \equiv A\right\}$$

Then $Z \in \mathcal{R}$ and is a refinement equivalent to A.

Proof. We start by showing that $Z = \hat{A}$ is a chain.

- We first show that the endpoints of intervals are well-defined. Let $A \in \mathcal{C}$ and fix some $n \in \mathbb{N}_0$. Consider the set

  $$U_n = \left\{\overline{B_n} \;\middle|\; B \equiv A\right\} \subseteq \mathbb{D}_n$$

  Since $B \equiv A$, $B_n \cap A_n \neq \emptyset$, and we can bound $\underline{A_n} \leq \overline{B_n}$. Thus the set U_n is bounded below and $\beta_n = \min(U_n)$ exists. By similar reasoning, we know that α_n exists.

- Next we prove that $\alpha_n \leq \beta_n$, and so each pair actually defines an interval. Assume, to the contrary that $\alpha_n > \beta_n$. Since α_n, β_n are defined as the maximum and minimum of given sets, there must be elements in those sets that give the maximum and minimum. Hence there exist $X, Y \equiv A$ so that

  $$\alpha_n = \underline{X_n} \qquad \beta_n = \overline{Y_n}$$

  This implies that $\overline{Y_n} < \underline{X_n}$ and so $X_n \cap Y_n = \emptyset$. Hence $X \not\equiv Y$. However, this contradicts the transitivity of the equivalence relation. So, we must have $\alpha_n \leq \beta_n$.

- Finally, we show that the intervals are nested by proving that the sequences (α_n) and (β_n) are weakly increasing and decreasing respectively. By construction, for any n we know that there is some $B \equiv A$ so that $\alpha_n = \underline{B_n}$. Since $B \in \mathcal{C}$ we know that $\underline{B_n} \leq \underline{B_{n+1}}$. By construction we then know $\underline{B_{n+1}} \leq \alpha_{n+1}$. Hence $\alpha_n \leq \alpha_{n+1}$ making the sequence (α_n) weakly increasing. By similar reasoning we know that the sequence (β_n) is weakly decreasing.

Oof! Hence $Z = \hat{A}$ is a chain. It remains to show that Z is a refinement of A. So let B be any chain equivalent to A. Then for any n, by construction, we know that

$$\underline{B_n} \leq \alpha_n \leq \beta_n \leq \overline{B_n}$$

and so $[\alpha_n, \beta_n] \subseteq B_n$. This means that $Z = \hat{A} \subseteq B$. Since B was arbitrary, it follows that $Z = \hat{A}$ is a refinement of A. ■

Now that we know every chain has a unique refinement, we can start to show what they look like. The next lemma shows that the widths of intervals in a refinement are as narrow as possible. That is, the endpoints of an interval X_n are either equal (if the refinement represents a number in $\mathbb{D}_n$) or they differ by exactly 10^{-n}.

The following result formalizes this. Although it seems intuitively obvious[3], it is actually quite subtle to prove it.

Lemma A.4.7 *Let $A \in \mathcal{C}$ and let $X = \hat{A}$ be its refinement. Then for each $n \in \mathbb{N}_0$*

$$0 \leq |X_n| = \overline{X_n} - \underline{X_n} \leq 10^{-n}$$

Proof. Let A, X be as given, and then construct

$$S = \bigcap_{k \in \mathbb{N}_0} X_k.$$

and for each $n \in \mathbb{N}_0$ set $S_n = \mathbb{D}_n \cap S$. We claim that S_n contains at most one element.

Assume, to the contrary that S_n contains two or more elements. Then there is a minimum element $a \in \mathbb{D}_n$ and a distinct maximum element $b \in \mathbb{D}_n$; these satisfy $b - a \geq 10^{-n}$. Since both $a, b \in S_n$, we know $a, b \in S$, and so for all k, $a, b \in X_k$. But this means that $|X_k| \geq 10^{-n}$, which contradicts the fact that $|X_k| \to 0$. Thus the intersection contains at most one element.

Now we fix $n \in \mathbb{N}_0$. Then either $S_n = \emptyset$ or $S_n = \{q\}$ for some $q \in \mathbb{D}_n$.

- First let $S_n = \{q\}$ for some $q \in \mathbb{D}_n$. Since $q \in \mathbb{D}_k$ for all $k \geq n$, this implies that $S_k = S \cap \mathbb{D}_k = \{q\}$. Then let

  $$Z = ([\alpha_0, \beta_0], \ldots, [\alpha_{n-1}, \beta_{n-1}], [q, q], [q, q], [q, q], \ldots)$$

  Notice that $Z \in \mathcal{C}$, and for all ℓ, $Z_\ell \cap X_\ell \neq \emptyset$. So $Z \equiv X$ and by transitivity, $Z \equiv A$. Consequently we must have that $X \subseteq Z$, and so for all $k \geq n$, $\alpha_k = \beta_k = q$. Hence $|X_k| = 0$ for all $k \geq n$.

- Now assume that $S_n = \emptyset$. Since X is a chain, we know there exists $k > n$ so that $\beta_k - \alpha_k < 10^{-n}$. Further, since $S_n = \emptyset$, there must be $a \in \mathbb{Z}$ so that

  $$p = \frac{a}{10^n} < \alpha_k \leq \beta_k < \frac{a+1}{10^n} = q.$$

[3] this is a dangerous word in mathematics

That is, at some depth, k, we must have an interval that is narrower than the space between two points in $\mathbb{D}_n$. Now form the chain

$$Z = ([\alpha_0, \beta_0], \ldots, [\alpha_{n-1}, \beta_{n-1}], [p, q], [p, q], \ldots, [p, q], [\alpha_k, \beta_k], \ldots)$$

that is

$$Z_j = \begin{cases} [p, q] & n \leq j < k \\ X_j & \text{otherwise} \end{cases}.$$

Notice that $Z \equiv X$, and so, by the minimality of X we must have that $\alpha_n \geq p$ and $\beta_n \leq q$ and so

$$\beta_n - \alpha_n = |X_n| \leq 10^{-n}$$

In both cases we recover the required inequality. ■

This restriction on the width of refinements, in turn, restricts how the endpoints of intervals within a refinement can change with n. We know that they weakly increase and decrease, but the following lemma shows they cannot increase or decrease by very much. Roughly speaking, if one thinks of the corresponding decimal expansions, then this result says that only the last digit of the expansion changes.

Lemma A.4.8 *Let $X \in \mathcal{R}$. Then for all $n \in \mathbb{N}_0$*

$$\underline{X_n} \leq \underline{X_{n+1}} < \underline{X_n} + 10^{-n}$$

and

$$\overline{X_n} - 10^{-n} < \overline{X_{n+1}} \leq \overline{X_n}$$

Proof. Let $X \in \mathcal{R}$. Because of this, X is also a chain and so for all $n \in \mathbb{N}_0$ we have $X_{n+1} \subseteq X_n$. Hence $\underline{X_n} \leq \underline{X_{n+1}}$ as required. To prove the other inequality, assume, to the contrary, that there is $n \in \mathbb{N}_0$ so that

$$\underline{X_n} + 10^{-n} \leq \underline{X_{n+1}}$$

This implies that

$$\underline{X_n} < \underline{X_n} + 10^{-n} \leq \overline{X_{n+1}} \leq \overline{X_n}$$

when means that

$$|X_n| = \overline{X_n} - \underline{X_n} > 10^{-n}$$

which contradicts the previous lemma. This proves the result for left endpoints; the proof for right endpoints is similar. ■

So we can summarise with the following theorem.

Theorem A.4.9 *Let $A \in \mathcal{C}$ then there exists a unique $X \in \mathcal{R}$ so that for all $B \equiv A$, $X \subseteq B$. Further, for all $n \in \mathbb{N}_0$*

$$|X_n| \leq 10^{-n}, \quad 0 \leq \underline{X_{n+1}} - \underline{X_n} < 10^{-n} \quad \textit{and} \quad 0 \leq \overline{X_n} - \overline{X_{n+1}} < 10^{-n}$$

Notice that our results above about ordering and trichotomy of chains carry over to refinements. However, we can now make stronger statement which looks just like what we need for the reals. This then gives us a strict total order on refinements.

Corollary A.4.10 Trichotomy. *Let $X, Y \in \mathcal{R}$, then exactly one of the following is true*

$$X < Y \qquad or \qquad X = Y \qquad or \qquad X > Y.$$

Consequently, the relation $<$ is both transitive and trichotomous and so is a strict total order on $\mathcal{R}$.

Proof. Let X, Y be as given. Then since $X, Y \in \mathcal{C}$ we know that exactly one of

$$X < Y \qquad \text{or} \qquad X \equiv Y \qquad \text{or} \qquad X > Y.$$

So, now assume that $X \equiv Y$. Since both X, Y are refinements, we know that $X \subseteq Y$ and $Y \subseteq X$. Hence $X = Y$. Thus we have trichotomy. Transitivity was proved for $<$ on chains, and so holds for $\mathcal{R} \subseteq \mathcal{C}$. ■

A.5 Arithmetic on refinements

Consider two refinements $X, Y \in \mathcal{R}$. We can add, subtract and multiply them as chains, the resulting chains $X \oplus Y, X \ominus Y, X \odot Y$ need not be refinements. However, we know that there is a unique refinement for every chain, so we simply take the result and ask for the equivalent refinement[1].

Definition A.5.1 Addition, subtraction and multiplication of refinements. Let $X, Y \in \mathcal{R}$, then define

$$X + Y = \widehat{X \oplus Y} \qquad X - Y = \widehat{X \ominus Y} \qquad X \cdot Y = \widehat{X \odot Y}$$

where we use $\hat{A}$ to denote the refinement of any chain A. ◊

Now we can leverage the fact that equivalence is compatible with arithmetic and ordering of chains. This gives us the following theorem.

Theorem A.5.2 Arithmetic and ordering of Refinements. *Let $X, Y, Z \in \mathcal{R}$. Then*

- *Closure: $X + Y, X - Y, X \cdot Y \in \mathcal{R}$*
- *Commutative: $X + Y = Y + X$ and $X \cdot Y = Y \cdot X$*

[1]There is a good reason why we have been using the sort of clumsy symbols $\oplus, \ominus, \oplus$; We wanted to save the more traditional symbols $+, -, \cdot$ for the respective operations on refinements (instead of just ordinary chains), because refinements will be playing the role of real numbers for us

- *Associative:* $X + (Y + Z) = (X + Y) + Z$ *and* $X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$
- *Distributive:* $X \cdot (Y + Z) = (X \cdot Y) + (X \cdot Z)$
- *Additive identity:* $X + \mathbf{0} = X$
- *Additive inverse:* $X + (-X) = \mathbf{0}$
- *Multiplicative identity:* $X \cdot \mathbf{1} = X$
- *Trichotomy: exactly one of* $X < Y, X = Y, X > Y$ *holds*
- *Transitive:* $(X < Y) \wedge (Y < Z) \implies (X < Z)$
- *Order and addition:* $(X < Y) \implies (X + Z) < (Y + Z)$
- *Order and multiplication:* $(\mathbf{0} < X, Y) \implies (\mathbf{0} < X \cdot Y)$

Proof. Thankfully much of this holds for chains and so holds automatically for refinements. We have left to prove:

- associativity of addition,
- associativity of multiplication,
- distribution of addition and multiplication,
- the existence of additive inverse, and
- the existence of multiplicative identity.

Notice that for any X, Y, Z we know the follow (as chains)

$$\begin{aligned} X \oplus (Y \oplus Z) &= (X \oplus Y) \oplus Z \\ X \odot (Y \odot Z) &\equiv (X \odot Y) \odot Z \\ X \odot (Y \oplus Z) &\equiv (X \odot Y) \oplus (X \cdot Z) \\ X \oplus (-X) &\equiv \mathbf{0} \\ X \odot \mathbf{1} &\equiv X \end{aligned}$$

Since each chain has a unique equivalent refinement, the refinements of both sides of each equation are equal. Hence as refinements we have

$$\begin{aligned} X + (-X) = \overbrace{X \oplus (-X)} = \widehat{\mathbf{0}} = \mathbf{0} \\ X \cdot \mathbf{1} = \widehat{X \odot \mathbf{1}} = \widehat{X} = X \end{aligned}$$

We note that we don't have a good way to render a really wide angle "hat" over some of these expressions, so we have used a wide brace instead just for this proof.

To show "Associativity of addition", we have to make sure that when we take the refinement at each step of addition that we aren't messing something up.

$$\begin{aligned} X + (Y + Z) &= \overbrace{X \oplus (Y + Z)} \\ &= \overbrace{X \oplus \widehat{Y \oplus Z}} \\ &\subseteq X \oplus \widehat{Y \oplus Z} \\ &\subseteq X \oplus (Y \oplus Z), \end{aligned}$$

which implies $X + (Y + Z) \equiv X \oplus (Y \oplus Z)$. Similarly, we have that

$$(X + Y) + Z \subseteq (X \oplus Y) \oplus Z,$$

which implies $(X + Y) + Z \equiv (X \oplus Y) \oplus Z$. Finally, we know that

$$X \oplus (Y \oplus Z) = (X \oplus Y) \oplus Z$$

by the associativity of $\oplus$. So we must have that

$$(X + Y) + Z = X + (Y + Z)$$

since both sides are refinements.

Showing the "associativity of multiplication"

$$X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$$

is extremely similar to the above, so we think you should do it! Also we leave the details of "distribution of addition and multiplication"

$$X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$$

to the reader, because it is good for you. ■

Almost everything! Next up, multiplicative inverse.

A.6 Multiplicative inverse

We need to show that for any refinement $X \neq \mathbf{0}$ there exists a unique refinement Y so that $X \cdot Y = \mathbf{1}$. To do this we will make use of interval arithmetic. In particular, given a positive interval $[0, 0] < [a, b]$, one can define

$$[a, b]^{-1} = [1/b, 1/a] = \{1/x \mid x \in [a, b]\}$$

with a similar definition for negative intervals. Unfortunately this does not extend trivially to chains for two reasons:

- if $a = 0$ then $[a, b]^{-1}$ is not defined, and

- $[a,b] \in \mathbb{I}_n$ does not imply that $[a,b]^{-1} \in \mathbb{I}_n$.

Exercise A.6.1 An example to tell you to be cautious. Consider a chain equivalent to the Euler–Mascheroni constant:

$$\gamma = \lim_{n\to\infty} \left(\log n - \sum_{k=1}^{n} \right) = 0.5772156\ldots$$

This number shows up all over mathematics[1]. The corresponding refinement would be

$$X = ([0,1],[5/10,6/10],[57/100,58/100],[577/1000,578/1000],\ldots).$$

Show that $Y = (X_0^{-1}, X_1^{-1}, \ldots)$ is *not* a chain because both of the following undesirable things occur:

- the inverse of at least one interval X_n^{-1} is not defined; and
- at least one of the intervals X_n^{-1} is not be contained in $\mathbb{I}_n$.

Solution. The reciprocals of the first few intervals are

$$([1,\text{undefined}],[10/6,10/5],[100/58,100/57],\ldots)$$

which is not a chain, and fails both conditions.
We can, however, fix both of these issues.

- We know that $X > \mathbf{0}$. Hence there is N so that $\underline{X_N} > 0$. Set $u = \lceil 1/\underline{X_N} \rceil$, and then build the first few intervals using this constant. That is, set $Y_k = [0,u]$ for $k = 0,1,\ldots,N$.
- Now let $k > N$. Then $X_k = [a,b]$ with $a,b \in \mathbb{D}_k$. We build $Y_k \in \mathbb{I}_k$ so that if $x \in X_k$ then $1/x \in Y_k$:

$$Y_k = \left[\lfloor 10^k/b \rfloor \, 10^{-k}, \lceil 10^k/a \rceil \, 10^{-k} \right].$$

For our Euler-Mascheroni constant example above this reciprocal is

$$Y = ([0,2],[0,2],[172/100,176/100],[1730/1000,1734/1000],\ldots)$$

and thankfully each interval $Y_n \in \mathbb{I}_n$ and contains $1/\gamma = 1.73245471\ldots$. We also observe that the intervals are nested and narrowing, but they are not a refinement. There is some work to be done.

Theorem A.6.2 *Let $X \in \mathcal{R}$ so that $X > 0$. Then there exists positive $Z \in \mathcal{R}$ so that $X \cdot Z = \mathbf{1}$. A similar result holds for $X < \mathbf{0}$.*

[1]It is one of *those* numbers

Proof. Let X be positive as stated. We then construct $Y \in \mathcal{C}$ as described above. Since X is positive, we know that $\overline{X_n} > 0$ for all n. At the same time, there is $N \in \mathbb{N}_0$ so that $\underline{X_N} > 0$. Let $u = \left\lceil 1/\underline{X_N} \right\rceil \geq 1$. We set $Y_k = [0, u]$ for $k = 0, 1, \ldots, N$. Now consider $n > N$. We know that $X_n = [a_n, b_n]$ with $0 < a_n \leq b_n$ and $a_n, b_n \in \mathbb{D}_k$. Set

$$Y_n = \left[\lfloor 10^n/b_n \rfloor \, 10^{-n}, \lceil 10^n/a_n \rceil \, 10^{-n} \right].$$

We claim that the resulting Y is a chain with $Y > \mathbf{0}$.

- We first show that the intervals within Y are nested. Notice that $Y_0 = Y_1 = \cdots = Y_N$, and so are nested.

  Then to show $Y_{N+1} \subseteq Y_N$:

  $$0 = \underline{Y_N} \leq \underline{Y_{N+1}}.$$

  and since $a_N \leq a_{N+1}$

  $$\overline{Y_N} = u = \lceil 1/a_N \rceil \geq \lceil 1/a_{N+1} \rceil \geq \left\lceil 10^{N+1}/\underline{X_{N+1}} \right\rceil 10^{-N-1} = \overline{Y_{N+1}}$$

  Finally, for $n > N$, we know that $1/a_n \geq 1/a_{n+1}$ and so

  $$\begin{aligned} \overline{Y_n} &= \lceil 10^n/a_n \rceil \, 10^{-n} \\ &\geq \lceil 10^n/a_{n+1} \rceil \, 10^{-n} \\ &\geq \left\lceil 10^{n+1}/a_{n+1} \right\rceil 10^{-n-1} \\ &= \overline{Y_{n+1}} \end{aligned}$$

  Thus the $\overline{Y_n}$ are weakly decreasing and a similar argument shows that the $\underline{Y_n}$ are weakly increasing.

  Thus the intervals are nested.

- We next show that the width of the intervals tends to zero. Let $n > N$ then

  $$\begin{aligned} |Y_n| &= 10^{-n} \left[\lfloor 10^n/a_n \rfloor - \lceil 10^n/b_n \rceil \right] \\ &\leq 10^{-n} \left[10^n/a_n - 10^n/b_n \right] \\ &= \frac{1}{a_n} - \frac{1}{b_n} = \frac{b_n - a_n}{a_n b_n}. \end{aligned}$$

  Since $a_n, b_n \geq \underline{X_N} > 0$, we obtain from this that

  $$|Y_n| \leq \frac{b_n - a_n}{a_n b_n} \leq \frac{b_n - a_n}{\underline{X_N}^2}$$

  Since $\underline{X_n}$ is fixed and $b_n - a_n \leq 10^{-n}$, we know that $|Y_n| \to 0$ as required.

Next we show that $X \cdot Y \equiv \mathbf{1}$. Since chains are closed under multiplication, we know that $X \cdot Y$ is a chain. Consequently it suffices to show that $1 \in X_n \cdot Y_n$ for large n. When $n > N$ we have

$$\begin{aligned} X_n \cdot Y_n &= [a_n, b_n] \cdot [\underline{Y_n}, \overline{Y_n}] \\ &= [a_n 10^{-n} \lfloor 10^n / b_n \rfloor, b_n 10^{-n} \lceil 10^n / a_n \rceil] \end{aligned}$$

Examine the right endpoint:

$$\begin{aligned} \frac{b_n}{10^n} \left\lceil \frac{10^n}{a_n} \right\rceil &\geq \frac{b_n}{10^n} \frac{10^n}{a_n} \\ &= \frac{b_n}{a_n} \\ &\geq 1 \end{aligned}$$

Similarly, the left endpoint:

$$\begin{aligned} \frac{a_n}{10^n} \left\lfloor \frac{10^n}{b_n} \right\rfloor &\leq \frac{a_n}{10^n} \frac{10^n}{b_n} \\ &= \frac{a_n}{b_n} \\ &\leq 1 \end{aligned}$$

Thus $1 \in X_n \cdot Y_n$, and so $\mathbf{1} \equiv X \cdot Y$.

Since Y is a chain, we know that it has a unique refinement, $Z = \hat{Y}$. Since equivalence is preserved under multiplication, we know that multiplying as chains gives us $X \cdot Z \equiv \mathbf{1}$, and so multiplying as refinements gives the required $X \cdot Z = \mathbf{1}$ as required. ■

All that remains now is the least upper bound property.

A.7 Bounds, good bounds and the supremum

The least upper bound property separates $\mathbb{R}$ from $\mathbb{Q}$. The starting point, unsurprisingly, is a definition of bounds.

Definition A.7.1 [Upper and lower bounds] Let $\mathcal{A} \subseteq \mathcal{R}$ be a set of refinements. We say that $U \in \mathcal{R}$ is an upper bound for $\mathcal{A}$ when for every $A \in \mathcal{A}$ we have $A \leq U$. Similarly, we say that L is a lower bound for $\mathcal{A}$ when for every $A \in \mathcal{A}$, $L \leq A$. ◊

So, for example, we could construct set of refinements corresponding to the real numbers $\{\sqrt{2}, 3/2, \sqrt{3}, 9/5\}$:

$$\begin{aligned} \mathcal{A} = \Big\{ &([1,2], [14/10, 15/10], [141/100, 142/100], \dots), \\ &([1,2], [15/10, 15/10], [150/100, 150/100], \dots), \end{aligned}$$

$$([1,2],[17/10,18/10],[173/100,174/100],\dots),$$
$$([1,2],[18/10,18/10],[180/100,180/100],\dots)\Big\}$$

Then $\mathcal{A}$ is bounded above by, say

$$U = ([2,2],[20/10,20/10],[200/100,200/100],\dots)$$

corresponding to $2 \in \mathbb{R}$ and also bounded below

$$L = ([1,2],[12/10,13/10],[123/100,124/100],\dots)$$

corresponding to $\sqrt{5}-1 \in \mathbb{R}$. Neither of these bounds are particularly "tight" and can be improved. For example

$$V = ([1,2],[19/10,19/10],[190/100,190/100],\dots)$$

corresponding to $19/10$, is a "better" upper bound since $V < U$, and

$$K = ([1,2],[14/10,14/10],[140/100,140/100],\dots)$$

corresponding to $7/5$ is a "better" lower bound since $L < K$.

Once we realise that we can "improve" bounds on a set, it is not unreasonable to ask for the "best" bounds on a set. What do we mean by that? If U is the best upper bound on a set $\mathcal{A}$, then no other upper bound can be smaller than U. Another way of saying that is that if we take any $V \in \mathcal{R}$ so that $V < U$, then V is not an upper bound. That, in turn, means that there must be some $A \in \mathcal{A}$ so that $V < A$ — which stops V from being an upper bound.

What we are busy describing here as the "best" upper bound, is called the supremum. When reading the following definition, go back and compare it to the *Least Upper Bound* in the setting of rational numbers as discussed in Section 1.5 and Exercise 1.5.8. You'll notice the definitions are the same. The difference is that the "least upper bound property" does not hold for the rational number $\mathbb{Q}$, but we will see that it does hold for the set of refinements $\mathcal{R}$ equipped with the order $<$, which we will eventually consider to be the set of real numbers.

Definition A.7.2 The supremum and infimum. Let $\mathcal{A} \subseteq \mathcal{R}$. We say that U is the least upper bound, or supremum (and will use the notation $U = \sup A$), of $\mathcal{A}$ when

- for all $A \in \mathcal{A}$, $A \leq U$, and
- if $V < U$ then there is $A \in \mathcal{A}$ so that $V < A$.

Notice that the first of these means that U is an upper bound, while the second says that no smaller refinement is an upper bound.

We similarly say that L is the greatest lower bound, or infimum (and will use the notation $L = \inf A$), of $\mathcal{A}$ when

- for all $A \in \mathcal{A}$, $L \leq A$, and

- if $L < K$ then there is $A \in \mathcal{A}$ so that $A < K$.

$\Diamond$

It is certainly far from obvious that the supremum or infimum exist. Indeed, one can readily construct a set of rational numbers that has upper and lower bounds, but does not have an infimum or supremum.

$$\{q \in \mathbb{Q} \mid q^2 < 2\}$$

This set is bounded below by (say) -2 and above by $+2$. However, one can show (and it makes an excellent homework problem), that since $\sqrt{2} \notin \mathbb{Q}$, no matter what rational upper (or lower) bound you construct, you can always construct a better rational upper (or lower) bound. That is, there is no supremum nor infimum over the rationals.

Theorem A.7.3 Least upper bound property of the reals. *Let $\mathcal{A} \subseteq \mathcal{R}$ be a non-empty set with an upper bound $U \in \mathcal{R}$. Then $Z = \sup \mathcal{A}$ exists and $Z \in \mathcal{R}$.*

Proof. Let $n \in \mathbb{N}_0$ and form the set

$$L_n = \left\{ \underline{A_n} \;\middle|\; A \in \mathcal{A} \right\}.$$

Then since $A \leq U$, we know that L_n is bounded above by $\overline{U_n}$, and so we know that $\max L_n$ exists. Hence we can fix

$$\underline{X_n} = \max L_n \qquad \text{and} \qquad \overline{X_n} = \underline{X_n} + 10^{-n}.$$

We now need to show that $X \in \mathcal{C}$. By construction $|X_n| \to 0$, so it suffices to show that the intervals are nested. We show that

$$\underline{X_n} \leq \underline{X_{n+1}} < \overline{X_{n+1}} \leq \overline{X_n}$$

for each n. The middle inequality follows by our choice of $\overline{X_n}$ above, so we only need to show the first and third inequalities.

- Since $\underline{X_n} = \max L_n$, there exists $A \in \mathcal{A}$ so that $\underline{X_n} = \underline{A_n}$. Since $A \in \mathcal{R}$, we know that $\underline{A_n} \leq \underline{A_{n+1}} \leq \underline{X_{n+1}}$, and so

  $$\underline{X_n} = \underline{A_n} \leq \underline{A_{n+1}} \leq \underline{X_{n+1}}$$

  as required.

- By construction, we have $A, B \in \mathcal{A}$ so that $\underline{X_n} = \underline{A_n}$ and $\underline{X_{n+1}} = \underline{B_{n+1}}$. A previous lemma gives us the following inequality linking $\underline{B_n}$ and $\underline{B_{n+1}}$:

  $$\underline{B_{n+1}} - 10^{-n} < \underline{B_n}$$

  Since $\underline{X_n}$ is defined by the maximum left end-point, we must have $\underline{B_n} \leq \underline{A_n}$ and so

  $$\underline{B_{n+1}} - 10^{-n} < \underline{B_n} \leq \underline{A_n}$$

$$\underline{B_{n+1}} < \underline{A_n} + 10^{-n}$$

Since all these quantities lie in \(\DD_{n+1} \), we can rewrite the strict inequality as a weak inequality:

$$\underline{B_{n+1}} + 10^{-n-1} \leq \underline{A_n} + 10^{-n}$$

And so we obtain $\overline{X_{n+1}} \leq \overline{X_n}$ as required.

So we have shown that $X \in \mathcal{C}$.

Next X is an upper bound for $\mathcal{A}$ and that there is no better upper bound.

- Let $A \in \mathcal{A}$. By construction, we have $A_n \leq X_n$ for all $n \in \mathbb{N}_0$. Hence $A \leq X$. Thus X is an upper bound for $\mathcal{A}$.
- Now let $Y < X$. By definition, there is $n \in \mathbb{N}_0$ so that $\overline{Y_n} < \underline{X_n}$. By construction, there is $A \in \mathcal{A}$ so that $\underline{X_n} = \underline{A_n}$. Hence we must have $Y < A$. So Y is not an upper bound.

Thus X is the supremum of $\mathcal{A}$. Unfortunately, while X is a chain we don't actually know it is a refinement. However, we do know that X has a unique refinement. Let $Z \in \mathcal{R}$ so that $Z \equiv X$. Since the equivalence relation preserves ordering, $\sup \mathcal{A} = Z \in \mathcal{R}$ as required. ■

We have now shown that refinements together with addition, multiplication and an order do everything that real numbers do. This demonstrates that we can construct mathematical objects that behave just like real numbers. This is a non-trivial achievement and, as you have seen, it takes work! It shows that when we play with elements of $\mathbb{R}$, add them, subtract them, multiply them, find bounds and so forth, there really are underlying mathematical objects that we *could* use to represent them.

Now, in practice, we don't use intervals and refinements to represent real numbers. Instead, we typically use them at "arm's length", eg, representing root-two as $\sqrt{2}$, and just using its algebraic or arithmetic properties when needed: $(\sqrt{(2)})^2 = 2$. If really pressed, we write down the start of their decimal expansion $\sqrt{2} = 1.414\ldots$; this is pretty much writing down the first few intervals in the refinement that represents $\sqrt{2}$. Consequently, in the remaining chapters we won't make use of refinements. Think of the work we have done in this chapter as "proof of concept" and a thoroughly good warm-up for what is to come.

There will one more optional section in this chapter that is not quite ready. We show that you can embed $\mathbb{Q}$ inside $\mathcal{R}$. This is really a technical exercise that shows that $\mathbb{R}$ is a *natural* superset of $\mathbb{Q}$. When we find $\mathbb{Q}$ inside $\mathcal{R}$ we want to know that it really behaves as it should with respect to arithmetic and ordering. That we haven't somehow distorted it while building the reals. Note that we are used to playing this game when we represent rationals like $\frac{1}{3}$ both as its fraction and as its decimal expansion, and being able to do arithmetic on both, moving back and forth between these two representations. It gets pretty technical and

the less-than-interested reader can skip it and just take our word for it — it all works out okay.

Appendix B

Selected Solutions

2 · Consequences of the Supremum
2.1 · Archimedes

· Exercises

2.1.1. No such thing as infinitesimals.

If $x > 0$ were a smallest number, then the Archimedean Principle guarantees the existences of some $n \in \mathbb{N}$ with $0 < \frac{1}{n} < x$, contradicting the supposed minimality of x.

2.1.2. Trapped between two integers.

Fix $x \in \mathbb{R}$, so by Archimedes we are guaranteed the existence of some $N \in \mathbb{N}$ with $x < N$. This shows that the set

$$A = \{m \in \mathbb{N} : m \leq x\}$$

has a largest element by the well-ordering principle 1.4.2 and Corollary 1.4.3. This largest element is our n. Note that it must be true that $x < n+1$, because otherwise this would contradict that n is the largest element of A.

2.1.3. A useful thing for proving inequalities.

Suppose to the contrary that $y < x$, which means that $0 < x - y$. The Archimedean Principle guarantees the existence of some $n \in \mathbb{N}$ with $0 < \frac{1}{n} < x - y$. Then we have

$$y < y + \frac{1}{n} < x.$$

But now if we take $\varepsilon = \frac{1}{2n}$, we obtain the absurd inequality

$$y + \frac{1}{n} < x \leq y + \frac{1}{2n} \;\Rightarrow\; y < y - \frac{1}{2n} < y.$$

2.1.4. Irrationals are dense too.

Let $x, y \in \mathbb{R}$ with $x < y$. Then $\frac{x}{\sqrt{2}} < \frac{y}{\sqrt{2}}$ (since $\sqrt{2} > 0$) and so by the density of

the rational numbers, we can find some $r \in \mathbb{Q}$ with

$$\frac{x}{\sqrt{2}} < r < \frac{y}{\sqrt{2}}.$$

Then by multiplying both sides of the equation by $\sqrt{2}$, we have

$$x < r\sqrt{2} < y.$$

Moreover it is impossible for $r\sqrt{2}$ to be rational, as this would imply that $\frac{r\sqrt{2}}{r} = \sqrt{2}$ would also be rational (and we know it isn't).

2.2 · Nested intervals
2.2.3 · Exercises

2.2.3.1. Nested open intervals don't work.

Let $I_n = (0, 1/n)$. Notice that for any $n \in \mathbb{N}$ we have $I_{n+1} \subseteq I_n$ and also that $|I_n| \to 0$. We assert that $\bigcap I_n = \emptyset$

Assume, to the contrary that there is some $x \in \bigcap I_n$. This means that $x > 0$, since it must be an element of all the I_n. However, by the Archimedean property, there exists some $k \in \mathbb{N}$ so that $x > \frac{1}{k}$, and so $x \notin I_k$. Consequently $x \notin \bigcap I_n$, and so we have that the intersection is empty.

2.2.3.2. Neither open nor closed.

Let $r > 0$. Then the set $(1 - r, 1 + r)$ always contains the point $x = 1 + r/2$. Since $r > 1$, we know $r \notin A$ and so $(1 - r, 1 + r) \not\subseteq A$ for any positive r, and so A is not open.

Similarly, consider the set $(-r, r)$. It contains the point $r/2 > 0$. Hence $(-r, r) \not\subseteq \mathbb{R} - A$ for any positive r, and so $\mathbb{R} - A$ is not open, and so A is not closed. The result follows.

2.2.3.3. Alternative definition of bounded.

Assume that A is bounded, then there are $a, b \in \mathbb{R}$ so that $a \leq x \leq b$ for every $x \in A$. So then set $M = \max\{|a|, |b|\}$ and the result holds. In the converse direction, set $a = -R, b = +R$, and then the set is bounded by the interval $[-R, R]$.

2.2.3.4. Open, closed, union, intersection.

Let U_α be a set of open subsets of reals (indexed by $\alpha \in I$), and let $A = \bigcup U_\alpha$. To show A is open we need to show that you can make a small open interval about any point of A. So let $x \in A$. By definition, we have that $x \in U_\alpha$ for some α. Since U_α is open, there is some $r > 0$ so that $(x - r, x + r) \subseteq U_\alpha$. This means that $(x - r, x + r) \subseteq A$. This shows A is open.

Now let C_α be a set of closed subsets of reals, and let $B = \bigcap C_\alpha$. Then the complement of B is

$$\mathbb{R} - B = \mathbb{R} - \bigcap C_\alpha = \bigcup (\mathbb{R} - C_\alpha).$$

Since the complement is a union of open sets, we know it is open. Hence the

original set B must be closed.

2.2.3.5. Infinity does not play nicely.

For each statement we need to prove each side is included in the other.

(a) Let $x \in LHS$. Hence there is some n so that $1/n \leq x \leq 1 - 1/n$. This, in turn implies that $0 \leq x \leq 1$ and so $x \in RHS$.

Now let $x \in RHS$. Since $x > 0$, we know, by the Archimedean property 2.1.2 that there is $n \in \mathbb{N}$ so that $1/n < x$. Similarly, since $1 - x > 0$, there is $k \in \mathbb{N}$ so that $1 - x < 1/k$. Thus setting $m = \max\{n, k\}$ we have that $1/m < x < 1 - 1/m$, and so $x \in LHS$.

(b) Let $x \in RHS$. We know that for all $n \in \mathbb{N}$, $1/n > 0$. Hence for all n we have $-1/n < 0 \leq x \leq 1 < 1 + 1/n$. Thus $x \in LHS$.

We prove the contrapositive of the reverse inclusion. So assume that $x \notin RHS$. Hence either $x > 1$ or $x < 0$. If $x > 1$, then by the Archimedean property 2.1.2, there is some $n \in \mathbb{N}$ so that $1 < 1 + 1/n < x$. Similarly if $x < 0$ then there is some $k \in \mathbb{N}$ so that $x < -1/k < 0$. In either case $x \notin LHS$.

2.2.3.6. Finite unions and intersections are nice.

Let $U_n, n = 1, 2, \ldots, N$ be a finite set of open sets. Let $x \in A = \bigcap U_n$. Since the U_n are all open, there are r_n so that $(x - r_n, x + r_n) \subseteq U_n$. Now set $r = \min\{r_1, r_2, \ldots, r_n\}$, so that $(x - r, x + r) \subseteq U_k$ for every $k = 1, 2, \ldots, N$. Hence $A = \bigcap U$ is open as required.

The proof that the finite union of closed sets is closed follows by taking complements.

2.2.3.7. Open is open and closed is closed.

We start by showing (a, b) is open. So let $x \in (a, b)$ and then set $r = \min\{x - a, b - x\}$. Notice that $(x - r, x + r) \subseteq (a, b)$ and so (a, b) is open.

Now consider $[a, b]$. Notice that $\mathbb{R} - [a, b] = (-\infty, a) \cup (b, \infty)$. This is the union of two open sets and so, by an exercise above, is also open. This means that $[a, b]$ is closed.

Similarly, notice that $\mathbb{R} - \{a\} = (-\infty, a) \cup (a, \infty)$ and so is open. This means that $\{a\}$ is closed as required.

2.2.3.8. Both open and closed.

Let $A \subseteq \mathbb{R}$. First notice that $\mathbb{R}$ is open, and so $\emptyset$ is closed. Similarly, notice that $\emptyset$ is open (vacuously) since it contains no points, there is nothing to test against.

So now assume that A is not empty and that $B = \mathbb{R} - A$ is not empty. Since A is both open and closed, we know that B must also be both open and closed. And without loss of generality, we assume there are x, y so that $x < y$ and $x \in A$ and $y \in B$. Now form the set

$$S = \{t \in A \mid t < y\}$$

We know that $x \in S$ and so $S \neq \emptyset$. We also know that S is bounded above by

y. Hence its supremum $z = \sup S$ exists. We assert that $z \notin A$ and $z \notin B$ and so arrive at a contradiction.

Now, assume that $z \in A$. Since A is open, there is some $r > 0$ so that $(z - r, z + r) \subseteq A$. However this means that the point $t = z + r/2 > z$ lies in A. Hence z is not an upper bound for S giving a contradiction.

So now assume that $z \in B$ (and so is not in A). Then, since B is open, there is some $r > 0$ so that $(z - r, z + r) \subseteq B$. But then the point $t = z - r/2$ would be a better upper bound for S contradiction the assumption that z is the supremum.

Hence no such z can exist, meaning that A cannot be both open and closed.

2.3 · Cauchy and Completeness
2.3.3 · Exercises

2.3.3.1. Seems obvious, but you should check.

Let $(x_n)_n$ be a constant sequence, i.e., that there exists $x \in \mathbb{R}$ such that $x_n = x$ for every $n \in \mathbb{N}$. Then for any $\varepsilon > 0$, we can take $N = 1$. So whenever $n \geq N$, we have

$$|x_n - x| = |x - x| = 0 < \varepsilon.$$

Thus, $x_n \to x$.

2.3.3.2. Is it complete?

- $(0, 1) \cup [2, 4]$ is *not* complete because the sequence $x_n = 1/n$ is contained in the set, it converges (to 0) and is therefore Cauchy, but the limiting point is not in the set.
- Any finite set *is* complete because any Cauchy sequence contained in a finite set must actually be eventually constant, and thus converges to that constant (which is in the set).
- $\{1/n : n \in \mathbb{N}\} \cup \{0\}$ *is* complete. Any Cauchy sequence in this set is either (a) eventually constant, or (b) converging to 0. In either case, the limiting value is in the set, so it is complete.

2.3.3.3. Non-convergence.

This is what it means for a sequence (x_n) to converge:

$$\exists x \in \mathbb{R}, \forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, |x_n - x| < \varepsilon.$$

The negation of this is:

$$\forall x \in \mathbb{R}, \exists \varepsilon > 0, \forall N \in \mathbb{N}, \exists n \geq N, |x_n - x| \geq \varepsilon.$$

2.3.3.4. A simple construction.

Let $x_n = (-1)^n$. The overall sequence does not converge because for any $x \in \mathbb{R}$, we can take $\varepsilon = 1$, $N = 1$ and $n \in \mathbb{N}$ either even (if $x < 0$) or odd (if $x \geq 0$).

Then $|x - x_n| \geq 1$. So it must not be a converging sequence. On the other hand, the even numbered terms and odd number terms are both converging subsequences because they're constant.

2.3.3.5. 2 converging subsequences.

The boundedness implies there exists $R > 0$ large enough so that $|x_n| \leq R$ for all $n \in \mathbb{R}$. For each $N \in \mathbb{N}$, break up the interval $[-R, R]$ into N subintervals of equal length $2R/N$ (call these $I_1, \ldots I_N$). Then for some N large enough, there is some index $j \leq N - 2$ such that I_j and I_{j+2} each contain infinitely many terms, i.e., two distinct subsequences of the original sequence (x_n). The reason why is that otherwise, we would be able to say that for each such N, all but finitely many terms are at most $4R/N$ distance apart, but this implies the sequence is actually Cauchy and thus convergent. Then Bolzano-Weierstrass 2.3.10 can be applied to each of these subsequences to demonstrate the existence of a converging subsequence in each "bin".

2.3.3.6. Convergence and subsequences.

We prove each direction in turn.

($\Rightarrow$) Let $(x_{n_k})_k$ be a subsequence and let $\varepsilon > 0$. By the convergence of the original sequence, there exists $N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$. Then we may assume $K \in \mathbb{N}$ is large enough so that $n_k \geq N$ for all $k \geq K$. This shows the convergence of the subsequence.

($\Leftarrow$) Assume that $x_n \not\to x$, which is to say that there exists some $\varepsilon > 0$ such that for any $N \in \mathbb{N}$, there exists $n \geq N$ such that

$$|x_n - x| \geq \varepsilon.$$

Subsequently choose a sequence of points as follows. Let $N_1 = 1$ and $n_1 \geq N_1$ as above and take x_{n_1} to be the first member of the sequence. Now given n_k, let $N_{k+1} = n_k + 1$ and repeat the process with this new choice of N. The result is a subsequence $(x_{n_k})_k$ such that

$$|x_{n_k} - x| \geq \varepsilon$$

for every $k \in \mathbb{N}$. This subsequence cannot converge to x, a contradiction.

2.3.3.7. Unbounded sequences don't converge[1].

Recursively find a sequence of $(x_n)_n$ as follows. Take $x_{n_1} = x_1$. Given x_{n_k} choose $x_{n_{k+1}}$ by requiring that

$$|x_{n_{k+1}}| \geq |x_{n_k}| + 1.$$

This is always possible because we are assuming that the original sequence is not bounded.

For any $x \in \mathbb{R}$, we will show that $(x_n)_n$ cannot converge to this choice of x. Since this potential limiting value was arbitrary, we will have shown that the sequence is not convergent. To show this, note that

$$|x_{n_k} - x| \geq |x_{n_k}| - |x| \geq (k - 1) + x_1 - |x|.$$

The right-hand side is not smaller than say 1 as long as k is large enough (depending on x). Therefore, this sequence does not converge to x because this subsequence doesn't converge to x.

2.3.3.8. Convergence and Subsequences 2.

The "only if" direction is identical to the previous exercise, so we will only show the "if" direction and assume for a contradiction that $(x_n)_n$ does *not* converge. Break into two cases: the sequence is bounded, or not.

Case 1: Assume here that the sequence $(x_n)_n$ is bounded. Then by Exercise 2.3.3.5, there exists two distinct subsequences of x_n that converge to different values (call them $x_{n_k^1}$ and $x_{n_k^2}$). We can form a subsequence by taking

$$x_{n_k} = \begin{cases} x_{n_k^1} & k \text{ is odd} \\ x_{n_k^2} & k \text{ is even.} \end{cases}$$

This subsequence does not converge because the even terms are converging to one limit, and the odd terms another.

Case 2: Now assume, that the sequence $(x_n)_n$ is *unbounded*. This means that for any $R > 0$, there exists $n \in \mathbb{N}$ with $|x_n| \geq R$. In particular, we can find a subsequence x_{n_k} with $|x_{n_k}| \geq k$. So this subsequence is unbounded, and therefore must not be convergent by Exercise 2.3.3.7, a contradiction.

2.3.3.9. Arithmetic of limits.

- (Sum Rule) Let $\varepsilon > 0$ and take $N_1 \in \mathbb{N}$ large enough so that

  $$|x_n - x| < \frac{\varepsilon}{2}$$

  whenever $n \geq N_1$, and $N_2 \in \mathbb{N}$ large enough so that

  $$|y_n - y| < \frac{\varepsilon}{2}$$

  whenever $n \geq N_2$. Then with $n \geq \max\{N_1, N_2\}$, we have

  $$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

- (Product Rule) First, by the triangle inequality, we have

  $$\begin{aligned} |x_n y_n - xy| &\leq |x_n y_n - x_n y| + |x_n y - xy| \\ &= |x_n||y_n - y| + |x_n - x||y| \end{aligned}$$

  for any $n \in \mathbb{N}$. Since $x_n \to x$, this sequence is bounded. So we know there is some $R > 0$ such that $|x_n| \leq R$, regardless of the value of $n \in \mathbb{N}$. So for any $\varepsilon > 0$, we can take $N_1 \in \mathbb{N}$ large enough so that

  $$|x_n - x| < \frac{\varepsilon}{2|y|}$$

whenever $n \geq N_1$, and $N_2 \in \mathbb{N}$ large enough so that

$$|y_n - y| < \frac{\varepsilon}{2} R$$

whenever $n \geq N_2$. Taking $N = \max\{N_1, N_2\}$ proves the result.

- (Quotient Rule) First, take $N_1 \in \mathbb{N}$ large enough to guarantee that

$$|y_n - y| \leq |y|/2$$

for $n \geq N_1$. In particular, this implies that $|y_n| \geq |y|/2$ for all such n. We will restrict to n at least this large to guarantee that x_n/y_n is not causing a division by zero.

$$\begin{aligned}\left|\frac{x_n}{y_n} - \frac{x}{y}\right| &\leq \left|\frac{x_n}{y_n} - \frac{x_n}{y}\right| + \left|\frac{x_n}{y} - \frac{x}{y}\right| \\ &= \frac{|x_n|}{|y y_n|}|y_n - y| + \frac{1}{|y|}|x_n - x|.\end{aligned}$$

Since $x_n \to x$, this sequence is bounded. So we know there is some $R > 0$ such that $|x_n| \leq R$, regardless of the value of $n \in \mathbb{N}$. So for any $\varepsilon > 0$, we can take $N_2 \in \mathbb{N}$ large enough so that

$$|x_n - x| < \frac{\varepsilon}{2}|y|$$

whenever $n \geq N_2$, and $N_3 \in \mathbb{N}$ large enough so that

$$|y_n - y| < \frac{\varepsilon}{4} R|y|^2$$

whenever $n \geq N_3$. Taking $N = \max\{N_1, N_2, N_3\}$ proves the result.

2.3.3.10. Recursive sequence.

- We will show that $x_n < 2$ by induction. We know the base case is true because $\sqrt{2} < 2$. Now assume $x_n < 2$, so then we have

$$x_{n+1} = \sqrt{x_n + 2} < \sqrt{2 + 2} = 2$$

where we have used the fact that the square root function is increasing to preserve the inequality.

- We have

$$x_{n+1} - x_n = \sqrt{x_n + 2} - x_n = \frac{x_n + 2 - x_n^2}{\sqrt{2 + x_n} + x_n}.$$

The latter is a positive quantity as long as $0 < x_n < 2$. Thus, the sequence is increasing as long as it stays less than 2, which is always true.

- This is a converging sequence because it is increasing and bounded. So it has some positive limit x. But this limiting value must satisfy

$$x = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} \sqrt{x_n + 2} = \sqrt{\lim_{n\to\infty} x_n + 2} = \sqrt{x+2}.$$

(where we are making use of Corollary 3.2.10). The only positive value x that satisfies this is $x = 2$.

2.3.3.11. Now you do it.

We will show that $x_n < \frac{5}{2}$ by induction. We know the base case is true because $2 < \frac{5}{2}$. Now assume $x_n < \frac{5}{2}$, so then we have

$$x_{n+1} = \frac{x_n + 5}{3} < \frac{\frac{5}{2} + 5}{3} = \frac{5}{2}.$$

This concludes the induction. Now we show the sequence is increasing:

$$x_{n+1} - x_n = \frac{x_n + 5}{3} - x_n = \frac{-2x_n + 5}{3}$$

which is positive as long as $x_n < \frac{5}{2}$ (and this is always true). So the sequence is always increasing and bounded, thus convergent to some $x \in \mathbb{R}$. This limiting value must satisfy

$$x = \lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} \frac{x_n + 5}{3} = \frac{x+5}{3} \;\Rightarrow\; x = \frac{5}{2}.$$

2.3.3.12. Know the Cauchy condition precisely.

We know

$$|\sqrt{n+1} - \sqrt{n}| = \frac{1}{|\sqrt{n+1} + \sqrt{n}|} \leq \frac{1}{2\sqrt{n}}.$$

Thus for any $\varepsilon > 0$, we can take $N \in \mathbb{N}$ large enough so that $N \geq \frac{1}{\varepsilon^2}$. So for any $n \geq N$, we have

$$|\sqrt{n+1} - \sqrt{n}| \leq \frac{1}{2\sqrt{n}} \leq \frac{1}{2\sqrt{N}} \leq \frac{\varepsilon}{2} < \varepsilon.$$

This proves the desired limit. Yet $\sqrt{x}$ grows without bound as x gets large, so it is impossible for

$$|x_n - x_m| = |\sqrt{n} - \sqrt{m}|$$

to remain small (take m to be fixed and n very large for instance). So (x_n) is not Cauchy.

2.3.3.13. Infinite (improper) limits.

- Suppose, for contradiction, that $\lim_{n\to\infty} x_n = L \in \mathbb{R}$. Take $\varepsilon = 1$. Then for large n, $|x_n - L| < 1$, so $x_n < L + 1$. But letting $M = L + 2$ in the

definition of $x_n \to \infty$ gives large n with $x_n \geq M = L + 2$, a contradiction. Hence no finite limit exists.

- Given $M > 0$, choose $N = \lceil M^2 \rceil$. For $n \geq N$, $\sqrt{n} \geq \sqrt{N} \geq M$. Thus $\sqrt{n} \to \infty$.
- Let $M > 0$. Since $x_n \to \infty$, there exists N_1 such that for $n \geq N_1$, $x_n \geq M$. Also, by hypothesis there exists N_0 such that for $n \geq N_0$, $y_n \geq x_n$. With $N = \max\{N_0, N_1\}$, for $n \geq N$ we have $y_n \geq x_n \geq M$, so $y_n \to \infty$.

2.5 · An important aside to closed and bounded sets of reals

· Exercises

2.5.1. You really neeed both.

The sets $A = (0, 1)$ is not closed and $B = \mathbb{N}$ is not bounded. Let's see why these examples don't "work".

For A, we can take a sequence $x_n = 1/n$, which *does* converge in the real numbers, to $x = 0$ (and thus so do all subsequences of it, by Exercise 2.3.3.6), but this limiting value is not contained in A. Therefore it does not satisfy Theorem 2.5.1.

For B, we can take a sequence $x_n = n$, for which every subsequence (x_{n_k}) is an unbounded sequence of real numbers, and thus cannot converge in the real numbers by Exercise 2.3.3.7, nevermind in the set $\mathbb{N}$. Therefore it does not satisfy Theorem 2.5.1.

2.5.2. Unbounded function and a bad point.

First, that f is unbounded means that

$$\forall M > 0,\ \exists x_M \in K \ \text{ such that } \ |f(x_M)| \geq M.$$

Thus, we may apply the above condition with a sequence of values $M \to \infty$, say $M = n$. This gives a sequence of points $(x_n)_n$ with $|f(x_n)| \to \infty$.

Then since K is closed and bounded, there is some converging subsequence $(x_{n_k})_k$ with the limiting value — call is x — contained in K as well. Then for every $r > 0$, the set N_r contains infinitely many elements of the sequence $(x_{n_k})_k$, each of which satisfies $|f(x_{n_k})| \geq n_k$, and thus

$$\sup_{k \in \mathbb{N}} f(x_{n_k}) = \infty.$$

2.5.3. Closed sets contain their limits.

An excellent homework problem.

2.5.4. Limits contained implies closed.

An excellent homework problem.

2.5.5. Closed same as Complete (in $\mathbb{R}$).

The set A is complete if and only if every Cauchy sequence $(x_n) \subseteq A$ converges

to a point $x \in A$. But we know the sequence converges to *some* real number x (which, a priori, may or may not be contained in A). But the previous parts show that $x \in A$ for all such sequences iff A is closed.

2.5.6. Nested closed bounded sets — loosen the nested interval property.

Each A_n is non-empty, so we can find some $x_n \in A_n$. Write A to be the intersection

$$A = \bigcap_{n \in \mathbb{N}} A_n.$$

Since the $(A_n)_n$ are nested, each x_n is contained in this intersection A. Now, A is bounded because $A \subseteq A_1$, and the latter is a bounded set. Furthermore, A is closed because it is an intersection of closed sets.

Thus we have shown that A is closed and bounded. Since $(x_n)_n$ is a sequence contained in A, Theorem 2.5.1 guarantees some converging subsequence $(x_{n_k})_k$ to a point $x \in A$. Therefore, A is non-empty.

In the event that the diameters of the $(A_n)_n$ go to zero, then there can be at most one $x \in A$ because if $|x - y| < \varepsilon$ for every $\varepsilon > 0$, this implies that $x = y$.

2.5.7. Generalised Extreme Value Theorem.

Let $(y_n)_n$ be some sequence of points in the image $f(K)$. That is to say, there exists a sequence $(x_n)_n$ in K such that

$$f(x_n) = y_n \quad \text{for every} \quad n \in \mathbb{N}.$$

Since K is closed and bounded, we know by Theorem 2.5.1 that there is some subsequence $(x_{n_k})_k$ and a point $x \in K$ such that $x_{n_k} \to x$. Then since f is continuous, we obtain that

$$\lim_{k \to \infty} f(x_{x_k}) = f(x).$$

But since $y_{n_k} = f(x_{n_k})$, we have found a subsequence of the sequence (y_n) — namely, $(y_{n_k})_k$ — with a limiting point $y = f(x)$, which is in the image $f(K)$. Therefore, we can conclude by Theorem 2.5.1 that $f(K)$ is closed and bounded.

2.5.8. Uniformly continuous is easy if you're closed and bounded.

Suppose, for contradiction, that f is not uniformly continuous on K. Then there exists $\varepsilon > 0$ such that for every $\delta > 0$, there exist points $x, y \in K$ with

$$|x - y| < \delta \quad \text{but} \quad |f(x) - f(y)| \geq \varepsilon.$$

In particular, for each positive integer n, let $\delta = \frac{1}{n}$. Then there exist points $x_n, y_n \in K$ such that

$$|x_n - y_n| < \frac{1}{n}, \quad \text{and} \quad |f(x_n) - f(y_n)| \geq \varepsilon.$$

Since $(x_n) \subseteq K$, and K is compact, the sequence (x_n) has a convergent subse-

quence (x_{n_k}) such that $x_{n_k} \to x_0 \in K$. Now observe that

$$|y_{n_k} - x_{n_k}| < \frac{1}{n_k} \to 0,$$

so $y_{n_k} \to x_0$ as well. Since f is continuous, it follows that

$$f(x_{n_k}) \to f(x_0), \quad \text{and} \quad f(y_{n_k}) \to f(x_0),$$

2.6 · Series
2.6.8 · Exercises

2.6.8.1. Now you do it (the "hard" way).

Let

$$s_N = \sum_{n=0}^{N} \frac{1}{3^n} = 1 + \frac{1}{3} + \frac{1}{9} + \cdots + \frac{1}{3^N}$$

denote the N-th partial sum. Multiply this equation by 3 then subtract the above to obtain

$$2s_N = 3s_N - s_N = 3 - \frac{1}{3^N},$$

since all of the intermediate terms cancel. Therefore

$$2s_N = 3 - \frac{1}{3^N},$$

so

$$s_N = \frac{3}{2} - \frac{1}{2 \cdot 3^N}.$$

Taking $N \to \infty$, we have

$$\lim_{N\to\infty} s_N = \frac{3}{2}.$$

Since the sequence of partial sums converges, the series converges, and its value is

$$\sum_{n=0}^{\infty} \frac{1}{3^n} = \frac{3}{2}.$$

2.6.8.2. e is irrational.

a There are many ways to show convergence: ratio test; comparison to a series that we know converges (like $\sum 1/n^2$); or even "by hand" by showing that the partial sums are monotone increasing and bounded. Pick your poison.

b Let $s_n = \sum_{k=0}^{n} \frac{1}{k!}$. Then

$$e - s_n = \sum_{k=n+1}^{\infty} \frac{1}{k!}.$$

The $e - s_n > 0$ inequality is then clear because the RHS above is a sum of positive numbers. For the other inequality, we estimate term-by-term. For $k \geq n+1$,

$$k! = n!(n+1)(n+2)\cdots k \geq n!(n+1)^{k-n}.$$

Therefore,

$$e - s_n \leq \sum_{k=n+1}^{\infty} \frac{1}{n!(n+1)^{k-n}} = \frac{1}{n!} \sum_{j=1}^{\infty} \frac{1}{(n+1)^j}.$$

This is a geometric series, so

$$\sum_{j=1}^{\infty} \frac{1}{(n+1)^j} = \frac{\frac{1}{n+1}}{1 - \frac{1}{n+1}} = \frac{1}{n}.$$

Thus

$$0 < e - s_n < \frac{1}{n!\, n}.$$

c Suppose $e = \frac{p}{q}$ for integers p, q. Take $n = q$. Then

$$0 < e - s_q < \frac{1}{q!\, q}.$$

Multiply by $q!$ to obtain

$$0 < q!(e - s_q) < \frac{1}{q}.$$

Now note that $q!s_q$ is an integer, since

$$q!s_q = \sum_{k=0}^{q} \frac{q!}{k!},$$

and each term $\frac{q!}{k!}$ is an integer. Also $q!e = q!\frac{p}{q} = p \cdot (q-1)!$ is an integer. Hence

$$q!(e - s_q)$$

is an integer as the difference of two integers.

We have shown that there exists an integer strictly between 0 and 1, which is impossible (a consequence of the Archimedean principle).

2.6.8.3. A term-by-term product.

Since (b_n) is monotone and bounded, it converges to some limit b. A priori, we do not know anything about b (it could be positive, negative, or zero). But by taking ε to be the minimum of 1 and $|b|$ in the definition of convergence

(assuming $b \neq 0$, which requires a distinct argument) that for all $n \geq N$, b_n has a fixed sign and is bounded in magnitude by $|b| + 1$. Now apply the Cauchy convergence criterion: Fix $n \geq m \geq N$ so that

$$|s_n - s_m| = \left| \sum_{k=m}^{n} a_n b_n \right| \leq (|b| + 1) \left| \sum_{k=m}^{n} a_n \right|.$$

Because the $\sum a_n$ converges, it passes the Cauchy criterion, and we may increase N (if necessary) to ensure that

$$\left| \sum_{k=m}^{n} a_n \right| \leq \frac{\varepsilon}{|b| + 1}$$

If the monotonicity assumption is dropped, the result fails dramatically. For example, let $a_n = \frac{(-1)^n}{n}$ and $b_n = (-1)^n$.

2.6.8.4. Alternating series test.

As usual, let S_n denote the partial sum:

$$S_n = \sum_{k=1}^{n} (-1)^{k+1} a_k.$$

a First, we group the even partial sums in a useful way:

$$S_{2k} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2k-1} - a_{2k}).$$

And then since a_n is decreasing, we have

$$S_{2k+2} - S_{2k} = (a_{2k+1} - a_{2k+2}) \geq 0.$$

Therefore (S_{2k}) is increasing.

Similarly,

$$S_{2k+1} - S_{2k-1} = -(a_{2k} - a_{2k+1}) \leq 0,$$

which shows (S_{2k-1}) is decreasing.

Now note that

$$S_{2k} \leq S_{2k+1} \leq S_1,$$

and also $S_{2k} \geq S_2$. Thus both subsequences are monotone and bounded, hence both converge.

b Consider the difference

$$S_{2k-1} - S_{2k} = a_{2k}.$$

Taking the limit $k \to \infty$, we know that the limit of both terms on the LHS exist (as shown in (a)), and also the limit of the RHS is zero (this is one of the hypotheses). Therefore,

$$\lim_{k \to \infty} S_{2k} = \lim_{k \to \infty} S_{2k-1}.$$

Call the limiting value of both these limits L. To obtain the convergence of the original series S, we let $\varepsilon > 0$ and take N large enough so that

$$|S_{2k} - L| < \varepsilon$$

as well as

$$|S_{2k+1} - L| < \varepsilon$$

whenever $k \geq N_1$ (resp. N_2). Then if $N = \max\{2N_1, 2N_2 + 1\}$, we have

$$|S_n - L| < \varepsilon.$$

2.6.8.5. Some series.

a The series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^{3/2}}$$

converges *absolutely* by comparison to the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$$

b The series

$$\sum_{n=2}^{\infty} \frac{(-1)^{n}}{\ln(n)}$$

converges *conditionally* by the Alternating Series Test, and by comparison to the (divergent) series

$$\sum_{n=2}^{\infty} \frac{1}{\ln(n)} \geq \sum_{n=2}^{\infty} \frac{1}{n} = \infty$$

c The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n+1}$$

diverges because the terms do not converge to zero.

2.6.8.10. Convergence practice.

a The series converges by comparison with the convergent p-series $\sum \frac{1}{n^2}$.

b The series converges (absolutely), again by comparison to $\sum \frac{1}{n^2}$.

c The series converges (absolutely) by the ratio test:

$$\frac{|a_{n+1}|}{|a_n|} = \frac{3}{n+1} \to 0.$$

d The series diverges since it is a p-series with $p = \frac{3}{4} < 1$.

e $a_n = \frac{1}{n \ln n}$. The series diverges by a previous exercise (dyadic comparison).

f $a_n = (-1)^n \frac{n}{n^2+1}$. The series converges conditionally by the alternating series test since $\frac{n}{n^2+1} \searrow 0$.

g $a_n = (-1)^{n+1} \frac{1}{\sqrt{n}}$. The series converges conditionally by the alternating series test, but not absolutely since $\sum \frac{1}{\sqrt{n}}$ diverges.

h $a_n = \frac{3^n}{5^n+n}$. The series converges absolutely since

$$\frac{3^n}{5^n + n} \leq \left(\frac{3}{5}\right)^n,$$

and the geometric series converges.

2.6.8.11. An important trick.

Let's start with the simpler – or at least, more elementary – method. Applying the inequality

$$\sqrt{xy} \leq \frac{x+y}{2}$$

with $x = a_n$ and $y = \frac{1}{n^2}$, we obtain

$$\frac{\sqrt{a_n}}{n} \leq \frac{a_n}{2} + \frac{1}{2n^2}.$$

Summing over n and applying the comparison test (since both series on the RHS are convergent and all our terms are nonnegative) gives the result. As an aside, the hinted inequality follows by rearranging the more obvious

$$(\sqrt{x} - \sqrt{y})^2 \geq 0$$

This is a common trick in analysis – and is the crux of the Cauchy-Schwarz inequality (which you should still look up).

3 · Continuity
3.1 · Limits of functions
3.1.3 · Exercises

3.1.3.1. A simple limit.

Let $\varepsilon > 0$ and choose $\delta = \varepsilon/4$. Then whenever $0 < |x-5| < \delta$, we have

$$|(4x+2)-22| = |4x-20| = 4|x-5| < 4\delta = 4\frac{\varepsilon}{4} = \varepsilon.$$

3.1.3.2. A slightly less simple limit.

Let $\varepsilon > 0$ and let $\delta = \min\{1, \varepsilon/20\}$. Then for $|x+2| < \delta$, we have

$$|x^3 - (-8)| = |x^3+8| = |x+2||x^2-2x+4| < \delta|x^2-2x+4|.$$

Since we assume $\delta \leq 1$, we can say that $-3 \leq x \leq -1$ and so

$$|x^2-2x+4| = |(x-1)^2+3| \leq |x-1|^2+3 < 20.$$

Thus

$$|x^3-(-8)| < \delta|x^2-2x+4| < 20\delta = \varepsilon.$$

3.1.3.3. Limits of polynomials.

- For any $a \in \mathbb{R}$ and any $\varepsilon > 0$, we may take $\delta = 1$ so that whenever $0 < |x-a| < \delta$, we have $|f(x)-f(a)| = |c-c| = 0 < \varepsilon$.
- Let $a \in \mathbb{R}$ and let $\varepsilon > 0$ and take $\delta = \varepsilon$. Then $0 < |x-a| < \delta$ implies $|x-a| < \varepsilon$.
- Every polynomial is a finite sum of terms of the form $a_n x^n$. Thus any polynomial can be written as a sum of products of constant functions and $f(x) = x$. Then we can apply the arithmetic of limits to conclude the result.

3.1.3.4. Limits two ways.

In each case, the easy way is to use the niceness of polynomials and arithmetic of limits — just like back in first-year calculus. The ε-δ proofs are more interesting.

- Let $\varepsilon > 0$ and let $\delta = \min\{1/2, \varepsilon/2\}$. Then whenever $|x-1| < \delta$, we have
$$\left|\frac{1}{x} - 1\right| = \frac{1}{|x|}|x-1| < 2\delta \leq \varepsilon$$
where we have used the fact that $x \geq 1/2$ and so $1/|x| \leq 2$.
- Let $\varepsilon > 0$ and let $\delta = \min\{1/2, \varepsilon/8\}$. Then whenever $|x| < \delta$, we have
$$\begin{aligned}\left|\frac{x^3+2x^2+1}{x^2-1} - (-1)\right| &= \left|\frac{x^3+2x^2+1}{x^2-1} + \frac{x^2-1}{x^2-1}\right| \\ &= \left|\frac{x^3+3x^2}{x^2-1}\right| = |x|^2\left|\frac{x+3}{x^2-1}\right| \\ &< 8\delta \leq \varepsilon\end{aligned}$$
where we have used the fact that $|x| \leq 1/2$ and so $|x^2-1| \geq 1/2$ and $x+3 \leq 4$.

3.1.3.5. More limits.

- As long as we restrict to the interval $[-2, 0]$, then $|2x-1| = -(2x-1)$ and so we can just the "niceness" of polynomials to conclude the result (in the process, we assume that $\delta \leq 1$ to stay within the desired interval).
- Let $\varepsilon > 0$ and let $\delta = \sqrt{\varepsilon}$ (or even $\delta = \varepsilon$ would do if $\varepsilon \leq 1$). Then whenever $|x| < \delta$, we have

$$\begin{aligned}\left|\sqrt{1+x^2} - 1\right| &= \left|\frac{1+x^2-1}{\sqrt{1+x^2}+1}\right| \\ &\leq \frac{|x|^2}{2} \leq |x|^2 \\ &< \delta^2 = \varepsilon\end{aligned}$$

  as required.

3.1.3.6. Prove limit is not 3.

You would have to show the negation of the logical statement

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x-2| < \delta \Rightarrow |f(x) - 3| < \varepsilon.$$

A little care is required with the statement after the "such that"; implicitly this is "for all x so that $|x-2| < \delta$". Taking this into account we get

$$\exists \varepsilon > 0 \text{ such that } \forall \delta > 0, \exists x \text{ so that } (|x-2| < \delta) \wedge (|f(x) - 3| \geq \varepsilon).$$

That is for non-convergence, we don't need all x in that little δ-ball push the function away from 3, we only need one x-value in that ball that messes things up.

3.1.3.7. Quantifiers are important.

The importance of the $\forall \varepsilon$ is that we want the function to have to be forced *arbitrarily* close to the limiting value. If we had only $\exists \varepsilon$, then the function may stay a distance of say $\varepsilon/2$ away from the "limiting" value, but not actually get any closer.

3.1.3.8. The limit does not exist![2]**.**

- The problem here is the domain of the function: as $x \to 1$ from the left the function is not defined. In particular, for any $0 < 1 - x < \delta$, then as long as $\delta < 1$ (so that $x + 1 > 0$), we have

$$x^2 - 1 = -(x+1)(1-x) < 0.$$

  So $\sqrt{x^2-1}$ is not defined in this case.

- The problem here is that the "limit" is infinite. In particular, for any $M \geq 0$, we can take $\delta > 0$ small enough ($\delta = e^{-M}$) so that

$$\log(x) < \log(e^{-M}) = -M.$$

So in particular, it is impossible for

$$|\log(x) - L|$$

to be less than (say) 1 for all x as we take x closer and closer to zero, regardless of the real number L.

- The problem here is the left-hand and right-hand limits are not equal. In particular,

$$\frac{|x|}{x} = \begin{cases} 1 & x > 0 \\ -1 & x < 0. \end{cases}$$

So for any $\varepsilon > 0$, we take δ to be an arbitrary positive number in the definition of a one-sided limit to show that $\lim_{x\to 0^-} \frac{|x|}{x} = -1$ and $\lim_{x\to 0^+} \frac{|x|}{x} = 1$.

3.1.3.9. Arithmetic of limits.

We will only prove the quotient rule because it is the most difficult, and leave the sum and product rules as exercises to the reader.

Let f, g, a, F, G be as in the statement of the Theorem, i.e.,

$$\lim_{x\to a} f(x) = F, \quad \lim_{x\to a} g(x) = G, \quad G \neq 0.$$

Take any sequence of points $(x_n)_n$ with $x_n \to a$ and $x_n \neq a$ for all $n \in \mathbb{N}$. By the previous exercise, it suffices to show that

$$\lim_{n\to\infty} \frac{f(x_n)}{g(x_n)} = \frac{F}{G}.$$

But actually, this is known to be trye by Arithmetic of Limits (Sequences) 2.3.3.9 if it is known that $g(x_n) \neq 0$ for all $n \in \mathbb{N}$. Unfortunately, this condition may not actually be satisfied, but it will *eventually* be satisfied because we know that there is some $N \in \mathbb{N}$ such that for $n \geq N$ there holds

$$|g(x_n) - G| < \frac{G}{2} \text{ which implies } |g(x_n)| \geq \frac{|G|}{2} > 0.$$

This completes the proof.

3.1.3.10. Infinite limits, revisited.

- Let $M > 0$ and choose $\delta = 1/\sqrt{M}$. If $0 < |x-a| < \delta$, then $(x-a)^2 < 1/M$,

and therefore

$$\frac{1}{(x-a)^2} > M.$$

Thus the limit is $+\infty$.

- For $\varepsilon > 0$, note

$$\left|\frac{2x+5}{x+1} - 2\right| = \frac{3}{x+1}.$$

Choose $R = \max\{1, 3/\varepsilon - 1\}$. If $x \geq R$, then $x + 1 \geq R + 1 \geq 3/\varepsilon$, so

$$\left|\frac{2x+5}{x+1} - 2\right| \leq \frac{3}{x+1} \leq \varepsilon.$$

Hence the limit is 2.

- Given $M > 0$, take $R = \sqrt{M}$. If $x \geq R$, then $x^2 \geq M$. Therefore x^2 diverges to $+\infty$.

3.2 · Continuity, Bolzano and the IVT

3.2.5 · Exercises

3.2.5.1. Removable, jump and infinite don't cover all possibilities.

The three notions of a discontinuity given above are **removable**, **infinite**, and **jump** respectively. In fact, the given limits are the definition of these words. These three types however do not encapsulate all possible discontinuities of a function, as illustrated by the following example:

$$f(x) = \begin{cases} \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

In this example, neither $\lim_{x\to 0^-} f(x)$ nor $\lim_{x\to 0^+} f(x)$ exist (as either a real number or infinite), so it doesn't fall into any of the three categories above.

3.2.5.2. Not continuous at a particular point.

We need to demonstrate that there exists some $\varepsilon > 0$ such that for any $\delta > 0$, there exists some x with $0 < |x - 2| < \delta$ but with $|f(x) - f(2)| \geq \varepsilon$. So we can take $\varepsilon = 1/2$. Now notice that for any x satisfying

$$1.9 < x < 2.1$$

we have

$$3.6 < 3.61 = 1.9^2 < x^2 < 4.41 < 4.5.$$

So now let δ be any positive real, and pick any x that satisfies

$$x \in (1.9, 2.1) \cap (2 - \delta, 2 + \delta).$$

Notice that this covers the possibilities that δ is tiny and also that δ could be huge. Then

$$|f(x) - f(2)| = |f(x) - 4| \geq 1/2 = \varepsilon.$$

3.2.5.3. Use a sequence to prove discontinuous.

If f were continuous at $x = 0$, then for every sequence $x_n \to 0$, we would have $f(x_n) \to f(0) = 0$. But if we take $x_n = \frac{2}{(4n+1)\pi}$. Then $x_n \to 0$, but

$$f(x_n) = \sin\left(\frac{1}{x_n}\right) = \sin\left(\frac{(4n+1)\pi}{2}\right) = \pm 1$$

So $f(x_n)$ does not converge to 0. Therefore, f must not be continuous at $x = 0$.

3.2.5.4. Rational/irrational.

The only point of continuity is $x = 0$. To see it is continuous at $x = 0$, simply take $\delta = \varepsilon$ in the definition of continuity to see that

$$|f(x) - f(0)| = |x| < \varepsilon$$

whenever $|x| < \delta = \varepsilon$.

Now, to show that it is discontinuous at every other point, let $c \neq 0$ be fixed. Assume c is rational (a similar proof holds if c is irrational). Let $\varepsilon = |c|/2$. Then for any $\delta > 0$, we can find an irrational x (you proved this back in Exercise 2.1.4) such that

$$x \in (c - r, c + r)$$

where

$$r = \min\{|c|/2, \delta\}.$$

Then

$$|f(x) - f(c)| = |x - (-c)| \geq |c|/2 = \varepsilon.$$

3.2.5.5. A very standard IVT exercise.

To prove the existence of a zero we just need a point where the function is positive and one where it is negative. Then the result follows pretty immediately from the IVT. But, remember, we also need that the function is continuous, so we'd better explain that too.

Let $f(x) = x^3 + x - 1$. Since f is a polynomial, it is continuous on all $\mathbb{R}$ and so we can apply the IVT on any closed interval of the real line.

Now notice that $f(0) = -1$, $f(1) = 1$. Then since f is continuous, by the IVT, here must be some $c \in (0, 1)$ with $f(c) = 0$.

3.2.5.6. IVT to find more than one solution.

Proving the equation has at least 3 solutions is equivalent to proving that the function $f(x) = x^3 - 2x^2 - 3x + 3$ has at least 3 roots. To do this, we consider the function at four special points

$$f(-2) = -7,\ f(0) = 3,\ f(1) = -3,\ f(3) = 1$$

This implies that the function changes signs somewhere in between each pair of consecutive points. Since f is continuous, so it must be that f has a root $x_1 \in (-2, 0)$, $x_2 \in (0, 1)$, and $x_3 \in (1, 3)$. Of course, these roots must be distinct because the intervals $(-2, 0), (0, 1), (1, 3)$ are disjoint.

3.2.5.7. IVT Converse? It's a trap!

Let $a = -2$, $b = 2$, and

$$f(x) = \begin{cases} x^3 - x & x \neq 0 \\ 1 & x = 0. \end{cases}$$

Clearly, f is not continuous at $x = 0$, but every value in $[-6, 6]$ is attained in the image of f.

3.2.5.8. Surprisingly matching weather.

Consider a function that returns the temperature along the line of longitude that passes through your current location. The authors are writing from near Vancouver which is at about $123°W$. So let $f : [-\pi/2, \pi/2] \to (0, \infty)$ be the function that maps the latitude along this line to temperature (in Kelvin), with $\pm\pi/2$ be the south and north poles, and 0 being the equator. Let c be your current location (latitude). Since the temperature at the north and south poles are much colder than your current location, and the temperature at the equator is hotter, we know that

$$f(\pi/2), f(-\pi/2) \leq f(c) \leq f(0)$$

Now, assuming that $f(x)$ is a continuous function (not an unreasonable assumption), we know, by the IVT, that there must be some point $a \in (-\pi/2, 0)$ so that $f(a) = f(c)$.

3.2.5.9. There must be a fixed point.

First, if $f(0) = 0$ or $f(1) = 1$, then we are done. So now, $f(0) > 0$ and $f(1) < 1$. Then the function $g(x) = f(x) - x$ satisfies $g(0) > 0$ and $g(1) < 1$ and g is continuous (the sum two continuous functions is continuous). The IVT guarantees the existence of some $c \in (0, 1)$ with $g(0) = 0$, and thus $f(c) = c$.

3.2.5.10. Invertible functions go up or go down but not both.

Assume $a < b < c$ are as per the hint. Let $A = \max\{f(a), f(c)\}$, which is less than $f(b)$ and take some $B \in (A, f(b))$. The IVT guarantees the existence of points $x_1 \in (a, b)$ and $x_2 \in (b, c)$ with

$$f(x_1) = B = f(x_2).$$

But then f is not injective, and must therefore not bijective, and so is not invertible.

3.2.5.11. An odd composition.

We write $f^{(n)}(x)$ to be the n-fold composition of f with itself.

Since $f^{(2)}(x) = -x$ we know that $f^{(4)}(x) = x$. So now let $h = f^{(3)}(x)$, then we know that $(h \circ f)(x) = (f \circ h)(x) = x$ and so f is invertible.

So now assume, to the contrary, that $f(x)$ is continuous, and so, by the previous exercise, we know it is strictly increasing everywhere or strictly decreasing

everywhere.

- If f is increasing, then whenever $x < y$ we know $f(x) < f(y)$, and so (applying the increasing f again), we know $f^{(2)}(x) < f^{(2)}(y)$.
- Similarly, if f is decreasing, then when $x < y$ we know that $f(x) > f(y)$, and so (applying the decreasing f again) we know that $f^{(2)}(x) < f^{(2)}(y)$ as well.

However this gives a contradiction since if $x < y$ we know that

$$f(f(x)) = -x > -y = f(f(y)).$$

This f cannot be continuous.

3.3 · The extreme value theorem

· Exercises

3.3.1. Does it have a min or max?

We will use the fact a couple times that if an increasing (resp. decreasing) function is defined on an open interval (a, b), then it must not achieve a maximum or minimum on this interval. This is because if $c \in (a, b)$ is such a potential point, then one can find x, y such that $a < x < c < y < b$ and so

$$f(x) < f(c) < f(y) \quad (\text{resp. } f(x) > f(c) > f(y)).$$

But this is inconsistent with $x = c$ begin the location of a max or min.

- x^2 is continuous on the interval $[0, 2]$, so the EVT 3.3.5 implies it achieves a maximum and a minimum on this interval.
- x^2 is increasing on $(0, 2)$ because

  $$x^2 < xy < y^2$$

  for $0 < x < y$. So it does not achieve a maximum or minimum on this interval.
- $\frac{1}{x}$ is continuous on the interval $[1, 5]$, so the EVT 3.3.5 implies it achieves a maximum and a minimum on this interval. Indeed, a little work shows that the minimum is $f(5) = 1/5$ and the maximum is $f(1) = 1$.
- $\frac{1}{x}$ is decreasing on the open interval $(0, \infty)$, so it does not achieve a maximum or minimum on this interval.

3.3.2. Bounded function on an open set.

The answer is *No*. For instance, we can take a function that oscillates wildly as $x \to a^+$ such as

$$f(x) = \sin(1/(x - a)).$$

3.3.3. Stay away from zero.

For any such $K \subseteq \mathbb{R}$, since it is bounded, we may take $a, b \in \mathbb{R}$ such that $K \subseteq [a, b]$. Then the restriction $f|_{[a,b]} : [a, b] \to \mathbb{R}$ is a continuous function on a closed bounded interval, so it achieves its minimum at some $c \in [a, b]$ by the EVT 3.3.5. But $f(c) > 0$ by assumption and so we have that

$$f(x) \geq f(c) > 0$$

for every $x \in [a, b]$.

If K is not bounded, then the result may not be true. The function $f(x) = \frac{1}{x^2+1}$ is a counterexample. No matter what $\alpha > 0$ we choose (though it should also be $\alpha < 2$), we can take $c = \pm\sqrt{\frac{2-\alpha}{\alpha}}$ and then $f(c) = \alpha/2 < \alpha$. Any x-value further from the origin will also work. Hence in any unbounded interval of the real line we can find a point so that $f(x) < \alpha/2 < \alpha$.

3.3.4. Always hotter (by a lot).

This problem operates under the assumption that the temperature function at any given point on the earth is continuous with respect to time (which is a reasonable assumption!). So if $V(t)$ and $F(t)$ denote the temperatures of Vancouver and Furnace Creek respectively, then we know that

$$f(t) = F(t) - V(t)$$

is a positive function. Moreover, the period of time corresponding to one calendar year is a closed and bounded interval. Then by the previous problem, we know that there is some $c > 0$ such that $f(t) \geq c$ for all t in this year.

3.3.5. Justifying the "obvious".

By the assumption of the limits, we can find some $\delta_1 > 0$ and $\delta_2 > 0$ so that whenever $x \in (a, a + \delta_1)$ or $x \in (b - \delta_2, b)$, we have

$$f(x) \leq f\left(\frac{a+b}{2}\right).$$

Now if we restrict to the remaining interval $[a + \delta_1, b - \delta_2]$, we may apply the EVT 3.3.5 to obtain some point $c \in [a + \delta_1, b - \delta_2]$ that achieves the maximum of f. That is, $f(x) \leq f(c)$ for all $x \in [a + \delta_1, b - \delta_2]$. And for the remaining values of $x \in (a, b) \setminus [a + \delta_1, b - \delta_2]$, we have already seen that

$$f(x) \leq f\left(\frac{a+b}{2}\right) \leq f(c).$$

3.3.6. You can map open to closed continuously.

There are many such functions, for instance $f(x) = \frac{1}{2}(\sin(2\pi x) + 1)$.

3.3.7. You can map bounded to unbounded continuously.

Consider the function $f(x) = \frac{1}{x}$. This maps $(0, 1)$ to $(0, \infty)$ which is unbounded.

You can also see this at the level of sequences. Take the sequence $(\frac{1}{n+1})$, then f maps this to the sequence $(n+1)$ which clearly diverges.

3.3.8. You can map closed to open (with some infinity).

3.3.9. The importance of closed and bounded together.

See Exercise 2.5.7.

3.3.10. Continuous bijection from open to closed.

As per the hint, assume such a function exists. Then there are some $a, b \in (0,1)$ with $f(a) = 0$ and $f(b) = 1$. WLOG assume $a < b$. Then by the IVT, we can say that $f([a,b]) = [0,1]$.

So now consider the point $x = a/2$. We know that $f(a/2) = y \in ([0,1])$, but we also know by the IVT that there exists some $c \in [a,b]$ so that $f(c) = y$. Hence f cannot be injective.

Note that there is nothing special about $x = a/2$, indeed a similar contradiction will be achieved by taking any of the "leftover" points in $(0,a) \cup (b,1)$.

3.3.11. Uniformly continuous is easy if you're closed and bounded.

See Exercise 2.5.8.

4 · Derivatives
4.1 · Derivatives of functions
· Exercises

4.1.1. You need to start somewhere.

We have

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{x+h-x}{h} = \lim_{h\to 0} \frac{h}{h} = 1.$$

4.1.2. Multiplication can smooth out a sharp point.

We have

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{(h)|h| - 0}{h} = \lim_{h\to 0} |h| = 0.$$

4.1.3. The simplest power rule.

For the inductive approach, the base case is established by Exercise 4.1.1. So assume it holds for $n = k \in \mathbb{N}$, and calculate

$$\frac{\mathrm{d}}{\mathrm{d}x} x^{k+1} = x\frac{\mathrm{d}}{\mathrm{d}x} x^k + x^k \frac{\mathrm{d}x}{\mathrm{d}x} = x(kx^{k-1}) + x^k = (k+1)x^k$$

as required. So the base case and inductive steps both hold, and thus by induction the result holds for all $n \in \mathbb{N}$.

For the second approach, we use the hint:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x} x^n &= \lim_{h\to 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h\to 0} \frac{x^n + nx^{n-1}h + h^2 P_n(x,h) - x^n}{h}\end{aligned}$$

$$= nx^{n-1} + \lim_{h\to 0} hP_n(x,h).$$

But since $P_n(x,h)$ is just a polynomial in x, h, it is continuous, so we have

$$\lim_{h\to 0} hP_n(x,h) = 0 \cdot P_n(x,0) = 0.$$

4.1.4. Negative integer powers.

Write $m = -n \in \mathbb{N}$ and use the quotient rule combined with the previous exercise:

$$\frac{\mathrm{d}}{\mathrm{d}x}x^n = \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{x^m} = -\frac{mx^{m-1}}{x^{2m}} = -mx^{-m-1} = nx^{n-1}.$$

4.1.5. Rational powers.

As per the hint, we calculate

$$nx^{\frac{m(n-1)}{n}} f'(x) = nf(x)^{n-1}f'(x) = \frac{d}{dx}f(x)^n = \frac{\mathrm{d}}{\mathrm{d}x}x^m = mx^{m-1}.$$

Rearranging this gives

$$f'(x) = \frac{m}{n}x^{m-1-\frac{m(n-1)}{n}} = \frac{m}{n}x^{\frac{nm-m(n-1)}{n}-1} = \frac{m}{n}x^{\frac{m}{n}-1}$$

4.1.6. Two ways.

- Using algebra:

$$\lim_{h\to 0}\frac{(3+h)^2-9}{h} = \lim_{h\to 0}\frac{9+6h+h^2-9}{h} = \lim_{h\to 0}[6+h] = 6.$$

  Using derivatives: If $f(x) = x^2$, then this limit is the precise definition of $f'(3)$, which we know is 6 from previous work.

- Using algebra:

$$\begin{aligned}\lim_{h\to 0}\frac{\sqrt{4+h}-2}{h} &= \lim_{h\to 0}\frac{4+h-4}{h(\sqrt{4+h}+2)}\\ &= \lim_{h\to 0}\frac{1}{\sqrt{4+h}+2} = \frac{1}{4}.\end{aligned}$$

  Note, in the last equality, we used the continuity of the square root function. Using derivatives: If $f(x) = \sqrt{x}$, then this limit is the precise definition of $f'(4)$, which we know is $\frac{1}{2\sqrt{4}} = \frac{1}{4}$ from previous work.

- Using algebra:

$$\lim_{h\to 0}\frac{\frac{1}{2+h}-\frac{1}{2}}{h} = \frac{2-(2+h)}{2(2+h)h}$$

$$= \lim_{h\to 0} \frac{-h}{2(2+h)h}$$
$$= \lim_{h\to 0} \frac{-1}{2(2+h)} = -\frac{1}{4}.$$

Note, in the last equality, we used the continuity of $1/x$ away from $x = 0$. Using derivatives: If $f(x) = 1/x$, then this limit is the precise definition of $f'(2)$, which we know is $-\frac{1}{4}$ from previous work.

4.1.7. Small steps towards product rule.

Using the definition: We have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(af(x)) &= \lim_{h\to 0} \frac{af(x+h) - af(x)}{h} \\ &= \lim_{h\to 0} \frac{a[f(x+h) - f(x)]}{h} \\ &= a \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = af'(x)\end{aligned}$$

A little algebra: We have

$$\frac{\mathrm{d}}{\mathrm{d}x}(af(x)) = a\frac{\mathrm{d}}{\mathrm{d}x}f(x) + f(x)\frac{\mathrm{d}}{\mathrm{d}x}a = af'(x) + 0 = af'(x),$$

where we have used the fact that the derivative of a constant function is 0.

4.1.8. A pretty classic example.

With all the separate parts of the problem, the only potential issue point is $x = 0$ because away from 0, the function is just the product/composition of infinitely differentiable functions. Also, we will use the fact that the derivative of f for non-zero points is given by

$$f'(x) = ax^{a-1}\sin(x^{-c}) - cx^{a-c-1}\cos(x^{-c}).$$

- f is continuous iff

  $$\lim_{x\to 0^+} x^a \sin(x^{-c}) = \lim_{x\to 0^+} f(x) = f(0) = 0.$$

  Note $\sin(x^{-c})$ is uniformly bounded, but also oscillates wildly near $x = 0$, so the product goes to zero iff the first term does iff $a > 0$.

- $f'(0)$ exists iff the limit

  $$\lim_{h\to 0} \frac{f(h) - f(0)}{h}$$

  exists, which is equivalent to saying that the limit

  $$\lim_{h\to 0^+} h^{a-1}\sin(h^{-c})$$

  exists. This limit exists iff the the limit of the first term in the product is zero (because the second term oscillates wildly as $h \to 0^+$), which is true iff $a - 1 > 0$ iff $a > 1$.

- f' is bounded iff $f'(0)$ exists (so this implies in particular that $a > 1$) and $f'(x)$ is bounded on $\mathbb{R} \setminus \{0\}$. But we have calculated $f'(x)$ for non-zero x-values above, and we can see that the derivative function

$$f'(x) = ax^{a-1}\sin(x^{-c}) - cx^{a-c-1}\cos(x^{-c})$$

is bounded iff $\min\{a-1, a-c-1\} = a-c-1$ is nonnegative, i.e., $a \geq c-1$. Note that this subsumes the condition that $a > 1$ because c is assumed positive. So the derivative is bounded iff $a \geq c-1$.

- f' is continuous iff

$$\lim_{x\to 0}[ax^{a-1}\sin(x^{-c}) - cx^{a-c-1}\cos(x^{-c})] = \lim_{x\to 0} f'(x) = f'(0)$$

In particular, if this condition is to hold, we require that $f'(0)$ exists, which is true iff $a > 1$ and in this case $f'(0) = 0$. So, we can say that f' is continuous iff $a > 1$ and

$$\lim_{x\to 0}[ax^{a-1}\sin(x^{-c}) - cx^{a-c-1}\cos(x^{-c})] = 0.$$

But this limit 0 iff $\min\{a-1, a-c-1\} > 0$ iff $a > c+1$. In particular, this condition subsumes the earlier requirement that $a > 1$. Thus, we have shown f' is continuous iff $a > c+1$.

4.2 · Increasing, decreasing and the derivative

· Exercises

4.2.1. MVT is good for inequalities.

Both inequalities follow from an application of the MVT. Note that both cos is differentiable on its domain, and $\sqrt{\cdot}$ is differentiable away from $x = 0$.

- Applying the MVT on the interval $[x, y]$ (assume $x < y$) gives

$$\frac{\cos(x) - \cos(y)}{x - y} = -\sin(c)$$

for some $c \in (x, y)$. Taking absolute values and using the bound $|-\sin(c)| \leq 1$ gives the result.

- Applying the MVT on the interval $[n, n+1]$ (here we assume $n \in \mathbb{N}$, or at least that $n > 0$) gives

$$\sqrt{n+1} - \sqrt{n} = \frac{\sqrt{n+1} - \sqrt{n}}{n+1-n} = \frac{1}{2\sqrt{c}}$$

for some $c \in (n, n+1)$. Taking absolute values and using the fact that $\sqrt{\cdot}$ is strictly increasing (so that $\sqrt{n} < \sqrt{c} < \sqrt{n+1}$) gives the result.

4.2.2. Zero derivatives and zeros.

This is *True*. It follows from a direct application of Rolle's Theorem 4.2.4. If the function at at least two roots a, b, then we would have

$$0 = \frac{0-0}{b-a} = \frac{f(b)-f(a)}{b-a} = f'(c)$$

for some $c \in (a, b)$.

4.2.3. MVT fights speeding.

The measured speeds at the endpoints here are irrelevant. The important information here is that, if $d(t) =$ distance at time t, then

$$\frac{20 \text{ km} - 11 \text{ km}}{17 \text{ min} - 11 \text{ min}} = \frac{9 \text{ km}}{6 \text{ min}} = 90 \text{ km/h}.$$

But we can also assume that a cars speed is a differentiable function, and so the MVT says that there is some $t_0 \in (11, 17)$ such that $v(t_0) = d'(t_0) = 90$ km/h.

4.2.4. Bounds on f' give bounds on f.

By the MVT, we obtain that

$$\frac{f(7)-f(-2)}{7-(-2)} = f'(c)$$

for some $c \in (-2, 7)$. Then using the bound $f'(c) \leq 2$, and rearranging, we obtain

$$f(7) \leq 2(7-(-2)) + f(-2) = 21.$$

So this is an upper bound for $f(7)$. It is also a *sharp* bound in the sense that it can be achieved by the linear function

$$f(x) = 2x + 7.$$

4.2.5. Prove there is a solution.

Define

$$f(x) = C_0 x + \frac{C_1}{2}x^2 + \cdots + \frac{C_{n-1}}{n}x^n + \frac{C_n}{n+1}x^{n+1}.$$

Then $f(0) = 0$ because there is a common factor of x throughout, and $f(1) = 0$ by assumption. f is also continuous on $[a, b]$ and differentiable on (a, b), so Rolle's Theorem 4.2.4 implies the existence of $c \in (0, 1)$ such that $f'(c) = 0$. This is precisely what we wanted to show.

4.2.6. One fixed point.

Assume there are at least two points $a, b \in [0, 1]$ such that $f(a) = f(b)$, and assume $a < b$. Then the MVT implies there is some $c \in (a, b)$ such that

$$1 = \frac{b-a}{b-a} = \frac{f(b)-f(a)}{b-a} = f'(c) < 1,$$

which is absurd. Therefore, there can be at most one such point. By Exer-

cise 3.2.5.9 there exists at least one such point. Hence there is exactly one fixed point.

4.2.7. A good application of the MVT.

(a) The limit $\lim_{x\to a^-} g(x) = L$ means that for all $\varepsilon > 0$, there is some $\delta > 0$ such that whenever $x \in (a-\delta, a)$, we have

$$|g(x) - L| < \varepsilon.$$

In particular, this holds for c_x, whenever $x > a - \delta$. That is,

$$|g(c_x) - L| < \varepsilon.$$

whenever $x \in (a-\delta, a)$. This proves the desired limit.

(b) The limit $\lim_{x\to\infty} g(x) = L$ means that for all $\varepsilon > 0$, there is some $R > 0$ such that whenever $x \in [R, \infty)$, we have

$$|g(x) - L| < \varepsilon.$$

In particular, this holds for c_x, whenever $x \geq R$. That is,

$$|g(c_x) - L| < \varepsilon.$$

whenever $x \in [R, \infty)$. This proves the desired limit.

(c) The limit $\lim_{x\to a^-} g(x) = \infty$ means that for all $M > 0$, there is some $\delta > 0$ such that whenever $x \in (a-\delta, a)$, we have

$$g(x) \geq M.$$

In particular, this holds for c_x, whenever $x > a - \delta$. That is,

$$g(c_x) \geq M.$$

whenever $x \in (a-\delta, a)$. This proves the desired limit.

(d) The limit $\lim_{x\to\infty} g(x) = \infty$ means that for all $M > 0$, there is some $R > 0$ such that whenever $x \in [R, \infty)$, we have

$$g(x) \geq M.$$

In particular, this holds for c_x, whenever $x \geq R$. That is,

$$g(c_x) \geq M$$

whenever $x \in [R, \infty)$. This proves the desired limit.

4.2.8. It's gotta flatten out.

As per the hint, we apply MVT to $f(x+1)-f(x)$ to obtain that there is some $c_x \in (x, x+1)$ with the property that

$$f(x+1)-f(x) = \frac{f(x+1)-f(x)}{x+1-x} = f'(c_x).$$

Taking the limit as $x \to \infty$, we obtain

$$0 = L - L = \lim_{x\to\infty} f(x+1)-f(x) = \lim_{x\to\infty} f'(c_x).$$

Now we know that the limit

$$\lim_{x\to\infty} f'(x)$$

exists as a real number (which we will call l). So by the previous exercise 4.2.7, we must have

$$\lim_{x\to\infty} f'(c_x) = l$$

as well. But we have already shown the latter limit is 0, which proves the result.

4.2.9. Prove L'Hôptial's rule.

Let $x \in (a,b)$ and apply Cauchy's Mean Value Theorem 4.2.6 to the closed interval $[a,x]$ which implies the existence of some $c_x \in (a,x)$ with

$$f'(c_x)g(x) = f'(c_x)[g(x)-g(a)] = g'(c_x)[f(x)-f(a)] = g'(c_x)f(x).$$

Then whenever $g(x) \neq 0$, we can rearrange to obtain

$$\frac{f'(c_x)}{g'(c_x)} = \frac{f(x)}{g(x)}.$$

But the limit

$$\lim_{x\to 0} \frac{f'(c_x)}{g'(c_x)}$$

is assumed to exist and equal L (by Exercise 4.2.7). So there exists some $\delta > 0$ so that for all $x \in (a, a+\delta)$, it must be true that

$$\left|\frac{f'(c_x)}{g'(c_x)}\right| \leq 2L.$$

Thus on this interval, it is impossible for $g(x) = 0$, and thus the identity

$$\frac{f'(c_x)}{g'(c_x)} = \frac{f(x)}{g(x)}$$

is true over this whole interval $(a, a+\delta)$. In particular, we may now just take $x \to a^+$ on both sides, to obtain

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(c_x)}{g'(c_x)} = L.$$

4.2.10. Darboux's theorem — an IVT for f'.

An excellent homework problem!

4.2.11. f' can be discontinuous, but only in certain ways.

A removable discontinuity of f' at c is one where

$$L := \lim_{x \to c} f'(x) \neq f'(c)$$

though both LHS and RHS exist. WLOG, assume $L < f'(c)$. Let $\varepsilon = f'(c) - L$. Then since the limit exists, we can take $a < c$ close enough to c such that

$$|f'(x) - L| < \varepsilon/2$$

for all $x \in (a, c)$. In particular, we can take

$$\lambda \in \left(\frac{f'(c) + L}{2}, f'(c) \right),$$

then the previous exercise implies there is some $b \in (a, c)$ such that $f'(b) = \lambda$. But this would contradict our work above.

The jump discontinuity is similar. A jump discontinuity of f' at c is one where

$$L := \lim_{x \to c^-} f'(x) \neq \lim_{x \to c^+} f'(x) =: R$$

though both LHS and RHS exist. WLOG, assume $L < R$. Let $\varepsilon = R - L$. Then since the limit exists, we can take $a < c$ close enough to c such that

$$|f'(x) - L| < \varepsilon/4$$

for all $x \in (a, c)$. Similarly we can take $c < b$ close enough to c such that

$$|f'(x) - R| < \varepsilon/4$$

for all $x \in (c, b)$.In particular, we can take

$$\lambda \in (L + \varepsilon/4, R - \varepsilon/4) \setminus f'(c),$$

then the previous exercise implies there is some $d \in (a, b)$ such that $f'(d) = \lambda$. But this would contradict our work above.

4.2.12. An excellent exam question.

We would like to show that the limit

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x}$$

exists and equals 3. For this, we apply the MVT to the interval $[0, x]$ if $x > 0$,

or $[x, 0]$ if $x < 0$, to obtain that

$$\frac{f(x) - f(0)}{x} = f'(c_x)$$

for some c_x in the interval $(0, x)$. The conclusion follows from an application of Exercise 4.2.7.

4.3 · Consequences of the MVT

· Exercises

4.3.1. Too good (bounds) to be true.

Fix some $p \in \mathbb{R}$ and notice that

$$|f'(p)| = \left| \lim_{x \to p} \frac{f(x) - f(p)}{x - p} \right| \leq \lim_{x \to p} |x - p| = 0.$$

The fact that f is constant follows from Corollary 4.3.1.

4.3.2. Strictly monotone is bijective.

By Corollary 4.3.1, $f(x) < f(y)$ whenever $x < y$, so f is injective on (a, b). That the range of f is really contained in (A, B) follows by the increasingness of f as well. The reason why it must also surjective follows from the IVT. In particular, for any $y \in (A < B)$, we can take α, β such that $a < \alpha < y < \beta < b$ and apply IVT to the interval $[\alpha, \beta]$ (note that f is continuous because it is differentiable).

4.3.3. Positive derivative at infinity.

Let $L = \lim_{x \to \infty} f'(x)$, which is positive by assumption. That is to say, for all $\varepsilon > 0$, there exists $R > 0$ large enough so that whenever $x \geq R$, we have

$$|f'(x) - L| < \varepsilon.$$

Applying this with $\varepsilon = L/2$, we can obtain that

$$f'(x) \geq L/2$$

whenever $x \geq R$ for this given R. Now by the MVT, we can say that for any $x > R$, there is some $c \in [R, x]$ such that

$$\frac{f(x) - f(R)}{x - R} = f'(c) \geq L/2.$$

In particular, we have

$$f(x) \geq f(R) + \frac{L}{2}(x - R).$$

So for any $M > 0$ large, we can take x large enough so that

$$f(R) + \frac{L}{2}(x - R) \geq M \iff x \geq \frac{2}{L}(M - f(R)) + R$$

implies that $f(x) \geq M$. That is what it means to say that $\lim_{x \to \infty} f(x) = \infty$.

4.3.4. A small jiggle.

First, $f'(x) = 1 + \varepsilon g'(x)$, and so as long as $\varepsilon \leq M/2$, we have

$$|f'(x)| > 1/2.$$

Thus Exercise 4.3.3 implies that f is a bijection onto its image, which is

$$\left(\lim_{x \to -\infty} f(x), \lim_{x \to \infty} f(x) \right).$$

Moreover, by Exercise 4.3.2 this image must actually be all of $\mathbb{R}$.

4.3.5. Convexity.

First, let's deal with the remark that $f'(x) \leq f'(y)$ if $x \leq y$. This follows from the MVT applied to f', which is valid because f' is continuous on $[x, y]$ and differentiable on (x, y), and the fact that $f'' \geq 0$.

Now that that it out of the way, let's finish the proof. First, note that the inequality is trivial if $s = 0$ or $s = 1$, so let's restrict to $s \in (0, 1)$. As per the hint, let $z = xs + y(1 - s) \in (x, y)$. The MVT applied to the intervals $[x, z]$ and $[z, y]$ give the points z_1, z_2 respectively. Then, using the previously proved fact about increasing derivatives, we have

$$\begin{aligned} \frac{f(xs + y(1-s)) - f(x)}{[xs + y(1-s)] - x} &= \frac{f(z) - f(x)}{z - x} \\ &\leq \frac{f(y) - f(z)}{y - z} \\ &= \frac{f(y) - f(xs + y(1-s))}{y - [xs + y(1-s)]}. \end{aligned}$$

The important inequality here is

$$\frac{f(xs + y(1-s)) - f(x)}{[xs + y(1-s)] - x} \leq \frac{f(y) - f(xs + y(1-s))}{y - [xs + y(1-s)]}.$$

After cross-multiplying and simplifying, this is seen to be equivalent with

$$(y - x)f(xs + y(1-s)) \leq (ys - xs)f(x) + (1 - s)(y - x)f(y).$$

The result follows by dividing through by $(y - x)$.

4.4 · Return to Newton

· Exercises

4.4.1. How many iterations?

Denote $r = \sqrt{2}$. We use the estimates from the notes that on $[1, 2]$, there holds

$$2 \leq |f'| \leq 4, \quad \text{and} \quad f'' >= 2.$$

This gives the recursive formula

$$|r - x_{n+1}| \leq \frac{1}{2}|r - x_n|^2.$$

Now since we're not asked to give a necessarily sharp estimate, there could be many different ways to proceed. But one way would be to notice that

$$|r - x_0| \leq \frac{1}{2}$$

and so

$$|r - x_1| \leq \frac{1}{2}|r - x_0|^2 \leq \frac{1}{8}.$$

Proceeding in this way, we could obtain error estimates of

$$\frac{1}{2^1}, \frac{1}{2^3}, \frac{1}{2^7}, \frac{1}{2^{15}}, \frac{1}{2^{31}}, \frac{1}{2^{63}}, \frac{1}{2^{127}} \ldots.$$

4.4.2. Cube root with quadratic convergence.

As we have seen from the notes, for any function, Newton's Method gives

$$r - x_{n+1} = -\frac{f''(q)}{2f'(x_n)}(r - x_n)^2$$

where $r = \sqrt[3]{2}$, $f = x^3 - 2$, $x_0 = 2$, and q is a number between x_n and r. We will consider the interval $[1, 2]$ where we obtain the bounds

$$|f''| \leq 12 \quad \text{and} \quad |f'| \geq 3.$$

Thus we have

$$|r - x_{n+1}| \leq 2|r - x_n|^2.$$

This is quadratic convergence.

4.4.3. Higher-order zero slows convergence.

The formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

in the case of $f(x) = x^2$ simplifies to

$$x_{n+1} = x_n - \frac{x_n^2}{2x_n} = \frac{1}{2}x_n.$$

Thus the distance to the root is only cut in half with each iteration, and is thus linear convergence (not quadratic). The issue is that f' is not bounded from below - in fact, it goes to zero as $x \to 0$, which cancels part of the bound of ther error.

4.4.4. A recipe for root-p.

Let $p > 0$ be fixed, and note that $f'(x) = 2x$, and $f''(x) = 2$. So f'' always has a uniform upper bound. So we can achieve the inequality

$$|\sqrt{p} - x_{n+1}| \leq \frac{1}{2a}|\sqrt{p} - x_n|^2$$

where $a > 0$ is the left endpoint of the interval that we restrict ourselves to. As long as the initial guess is close enough to $\sqrt{p}$ to guarantee that the successive error is decreasing, this will be quadratic convergence. In particular, we want

$$\frac{1}{2a}|\sqrt{p} - x_0|^2 < |\sqrt{p} - x_0| \;\Rightarrow\; x_0 \in (\sqrt{p} - 2a, \sqrt{p} + 2a).$$

4.5 · Getting closer

· Exercises

4.5.1. Taylor polynomials to compute limits.

First, notice that the claim that

$$\lim_{x\to a} \frac{f(x) - P_n(x)}{(x-a)^n} = 0$$

follows from the fact that $f(x) - P_n(x) = O((x-a)^{n+1})$ as $x \to a$. In particular, we have

$$\left|\frac{f(x) - P_n(x)}{(x-a)^n}\right| \leq \frac{K|x-a|^{n+1}}{|x-a|^n} = K|x-a|$$

which goes to 0 as $x \to a$.

Now, we can use this to compute the given limits.

- We know that the degree-4 Taylor polynomial for $\cos(x)$ is

$$T_4(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!}.$$

Also, all derivative of cos are uniformly bounded by 1, so the result applies. Then using the above limit, we can obtain

$$\begin{aligned}
&\lim_{x\to 0} \frac{\cos(x) - 1 + \frac{x^2}{4}}{11x^4} \\
&= \lim_{x\to 0} \frac{\cos(x) - 1 + \frac{x^2}{4}}{11x^4} - 0 \\
&= \lim_{x\to 0} \frac{\cos(x) - 1 + \frac{x^2}{4}}{11x^4} - \frac{1}{11}\lim_{x\to 0}\frac{\cos(x) - T_4(x)}{x^4} \\
&= \lim_{x\to 0} \frac{\cos(x) - 1 + \frac{x^2}{2} - \cos(x) + 1 - \frac{x^2}{4} + \frac{x^4}{4!}}{11x^4} \\
&= \lim_{x\to 0} \frac{\frac{x^4}{4!}}{11x^4} = \frac{1}{4! \cdot 11}
\end{aligned}$$

- We know that the degree-3 Taylor polynomial for e^{5x} is

$$T_3(x) = 1 + (5x) + \frac{(5x)^2}{2} + \frac{(5x)^3}{3!}$$

Also, all derivatives of e^{5x} are bounded on the interval $[-1, 1]$ (by different bounds, but that is OK), so the result applies. Then using the above limit, we can obtain

$$\begin{aligned}
&\lim_{x\to 0} \frac{e^{5x} - 1 - 5x - \frac{25x^2}{2}}{x^3} \\
&\quad = \lim_{x\to 0} \frac{e^{5x} - 1 - 5x - \frac{25x^2}{2}}{x^3} - 0 \\
&\quad = \lim_{x\to 0} \frac{e^{5x} - 1 - 5x - \frac{25x^2}{2}}{x^3} - \lim_{x\to 0} \frac{e^{5x} - T_3(x)}{x^3} \\
&\quad = \lim_{x\to 0} \frac{e^{5x} - 1 - 5x - \frac{25x^2}{2} - e^{5x} + 1 + 5x + \frac{25x^2}{2} + \frac{(5x)^3}{3!}}{x^3} \\
&\quad = \lim_{x\to 0} \frac{\frac{(5x)^3}{3!}}{x^3} = \frac{5^3}{3!}
\end{aligned}$$

- We know that the degree-4 Taylor polynomial for $\sin(x)$ is

$$T_4(x) = x - \frac{x^3}{3!}$$

Also, all derivative of cos are uniformly bounded by 1, so the result applies. Then using the above limit, we can obtain

$$\begin{aligned}
&\lim_{x\to 0} \frac{\sin(x) - x + \frac{x^3}{6}}{12x^4} \\
&\quad = \lim_{x\to 0} \frac{\sin(x) - x + \frac{x^3}{6}}{12x^4} - 0 \\
&\quad = \lim_{x\to 0} \frac{\sin(x) - x + \frac{x^3}{6}}{12x^4} - \frac{1}{12} \lim_{x\to 0} \frac{\sin(x) - x + \frac{x^3}{6}}{x^4} \\
&\quad = \lim_{x\to 0} \frac{\sin(x) - x + \frac{x^3}{6} + \sin(x) + x - \frac{x^3}{6}}{x^4} \\
&\quad = \lim_{x\to 0} \frac{0}{x^4} = 0
\end{aligned}$$

- This one is a bit of a mess But the technique is the same as the previous parts. You have to determine the degree 8 polynomial for $e^{x^3}\log(1+x^5) + \cos(x^{95}) - \sin(x^5) - 1$. Again, the derivatives of this function are each bounded on $[-1, 1]$, so the result applies. Note we'll be discarding terms of higher order than 8 in the work below.

$$e^{x^3}\log(1 + x^5) + \cos(x^{95}) - \sin(x^5) - 1$$

$$\begin{aligned}
&\approx \left(1 + x^3 + \frac{1}{2}x^6\right)\left(x^5\right) + (1) - \left(x^5\right) - 1 \\
&\approx 1\left(x^5\right) + x^3\left(x^5\right) - \left(x^5\right) \\
&= x^8
\end{aligned}$$

That is,

$$T_8(x) = x^8$$

Then using the above limit, we can obtain

$$\begin{aligned}
&\lim_{x\to 0} \frac{e^{x^3}\log(x^5) + \cos(x^{95}) - \sin(x^5) - 1}{x^8} \\
&\quad = \lim_{x\to 0} \frac{e^{x^3}\log(x^5) + \cos(x^{95}) - \sin(x^5) - 1}{x^8} - 0 \\
&\quad = \lim_{x\to 0} \frac{e^{x^3}\log(x^5) + \cos(x^{95}) - \sin(x^5) - 1}{x^8} \\
&\qquad - \lim_{x\to 0} \frac{e^{x^3}\log(x^5) + \cos(x^{95}) - \sin(x^5) - 1 - x^8}{x^8} \\
&\quad = \lim_{x\to 0} \frac{x^8}{x^8} = 1
\end{aligned}$$

4.5.2. Big-O and zero derivative.

We can write

$$\left|\frac{f(x) - f(c)}{x - c}\right| = \frac{|f(x) - f(c)|}{|x - c|} \leq |x - c|^{\alpha - 1}$$

which goes to zero as $x \to c$. That is to say, $f'(c) = 0$.

4.5.3. Big-O and not differentiable.

This means that for any $M_i \to \infty$, there exists $x_i \in \mathbb{R}$ with $|x_i - c| \to 0$ such that

$$|f(x) - f(c)| \geq M_i |x_i - c|.$$

In particular this implies that

$$f'(c) = \lim_{x\to c} \frac{f(x) - f(c)}{x - c}$$

does not exist.

4.5.4. Big-O and maybe differentiable?

Take $c = 0$ and $f(x) = Lx$ for a range of constants $L \in \mathbb{L}$. Then $f'(0) = L$ and $f = O(x)$ as $x \to 0$. On the other hand,

$$f(x) = \begin{cases} x\sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is not differentiable at $x = 0$ by Exercise 4.1.8, but still $f = O(x)$ as $x \to 0$.

4.5.5. Big-O does not play nicely with derivatives.

This is simple calculation. For $x \neq 0$, we have

$$f'(x) = 2x\sin(1/x) - \cos(1/x).$$

Thus $f' \neq O(x)$ as $x \to 0$ because - in particular - $\lim_{x\to 0} f'(x)$ does not even exist.

4.5.6. Not everything can be Taylored.

- The base case $n = 0$ is already established. So assume that the result is true for $n \in \mathbb{N} \cup \{0\}$. We can readily calculate

$$\begin{aligned} f^{(n+1)}(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(P_n(1/x)e^{-1/x^2}\right) \\ &= -\frac{1}{x^2}P_n'(1/x)e^{-1/x^2} + \frac{2}{x^3}P_n(1/x)e^{-1/x^2} \\ &= \left(-\frac{1}{x^2}P_n'(1/x) + \frac{2}{x^3}P_n(1/x)\right)e^{-1/x^2}. \end{aligned}$$

  The latter bracketed quantity is a polynomial in $1/x$.

- Again, by induction with the case $n = 0$ already having been established. For the inductive step, we have

$$f^{(n+1)}(0) = \lim_{x\to 0}\frac{f^{(n)}(x) - f^{(n)}(0)}{x} = \lim_{x\to 0}\frac{1}{x}P_n(1/x)e^{-1/x^2}.$$

  This limit must be zero because exponentials grow faster as $y \to \infty$ than polynomials (make a substitution $y = |x|$).

4.5.7. Best approximation.

By Taylor's Theorem, for $j \leq n$, we have

$$f(x) = P_j(x) + O(x^{j+1}),$$

which implies

$$\lim_{x\to 0}\frac{f(x) - P_j(x)}{x^j} = 0 = \lim_{x\to 0}\frac{f(x) - Q_j(x)}{x^j}$$

for all $j \leq n$. Here we have denoted Q_j to mean Q_n truncated to degree j.

In particular, this implies that the j'th coefficient of P_n and Q_n are equal, for each such j. Thus $P_n = Q_n$.

4.5.8. Big-O arithmetic.

- WLOG assume $p \leq q$. Then for $x \in [-1, 1]$, $|x^p| \geq |x^q|$. So if M_f, M_g are

the constants in the definition of big-O, we have

$$|f(x)+g(x)| \leq M_f|x|^p + M_g|x|^q \leq \max\{M_f, M_g\}|x|^p.$$

- With M_f, M_g as above, we have

$$|f(x)g(x)| \leq M_f|x|^p M_g x^q = (M_f M_g)x^{p+q}.$$

- The function $F(x) = x^r$ is increasing, so

$$|f(x)| \leq M_f|x|^p \Rightarrow |f(x)|^r \leq M_f^r|x|^{pr}.$$

4.5.9. Shortcuts.

(a) Let's write a_j to be the coefficients of P_n, and b_j for Q_n. By the hinted formula (which comes from n applications of the product rule), we can write

$$\begin{aligned}\frac{\mathrm{d}^n}{\mathrm{d}x^n}(fg)(0) &= \sum_{j=0}^{n} \binom{n}{j} f^{(j)}(0)g^{(n-j)}(0) \\ &= \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} f^{(j)}(0)g^{(n-j)}(0) \\ &= \sum_{j=0}^{n} n! a_j b_{n-j}.\end{aligned}$$

In particular, the m'th coefficient for the Taylor polynomial of the product fg is given by

$$\sum_{j=0}^{n} a_j b_{n-j}.$$

But this is precisely the m'th coefficient of x^n in the product

$$P_n(f)Q_n(g).$$

(b) We know that

$$f(X) = P_n(X) + O(X^{n+1})$$

as $X \to 0$ and

$$g(x) = Q_n(x) + O(x^{n+1})$$

as $x \to 0$. In particular, since $g(x) = 0$, we can make the substitution $X = g(x)$ and write

$$f(g(x)) = P_n(Q_n(x) + O(x^{n+1})) + O([Q_n(x) + O(x^{n+1})]^{n+1}).$$

First, notice that

$$([Q_n(x) + O(x^{n+1})]^{n+1}) = O(x^{n+1})$$

because Q_n is at least degree 1. Also

$$P_n(Q_n(x) + O(x^{n+1})) = P_n(Q_n(x)) + O(x^{n+1})$$

because any expansion of $(Q_n(x) + O(x^{n+1}))^m$ will be precisely $Q_n(x)^m$ plus sometime at least degree $n + 1$.

(c) The degree-12 Taylor polynomial for e^{x^3} is

$$Q_{12}(x) = 1 + x^3 + \frac{x^6}{2} + \frac{x^9}{3!} + \frac{x^{12}}{4!}.$$

The degree-12 Taylor polynomial for $1 - \cos(x^2)$ is

$$P_{12}(x) = \frac{x^4}{2} - \frac{x^8}{4!} + \frac{x^{12}}{6!}.$$

So the degree-12 Taylor polynomial for the product is

$$T(x) = \frac{x^4}{2} + \frac{x^7}{2} - \frac{x^8}{4!} + \frac{x^{10}}{4} - \frac{x^{11}}{4!} + \frac{x^{12}}{6!}$$

This would indeed be very difficult and horribly messy to find using product- and chain-rules 12 times!

5 · Metric spaces

5.1 · Metrics and metric spaces

· Exercises

5.1.1. You'll never forget the properties of a metric space after this.

Proof.

(a) Not a metric because it fails the triangle inequality

$$d(0, 1) = 1 > \frac{1}{2^3} + \frac{1}{2^3} = d(0, 1/2) + d(1/2, 1)$$

(b) Yes, this is a metric. Positivity and Symmetry are easy to check. The triangle inequality follows from the more basic inequality

$$(\sqrt{a} + \sqrt{b})^2 \geq a + b$$

for $a, b \geq 0$.

(c) Not a metric because it is not symmetric

$$d(0,1) = 2 \neq 1 = d(1,0)$$

$X = \mathbb{R}, \quad d(x,y) = |x-y| \cdot |x+y|$

(d) Not a metric because it fails positivity. There are distinct points that are distance zero from each-other:

$$d(-1,1) = |1-(-1)^2| = 0$$

(e) Yes, this is a metric. It is the restriction of the Euclidean metric to the subset $\mathbb{Z} \subset \mathbb{R}$.

(f) Yes, this is a metric. Positivity and symmetry are easy to see. For the triangle inequality, we follow the hint and consider the difference:

$$\begin{aligned} f(a) + f(b) - f(a+b) &= \frac{a}{1+a} + \frac{b}{1+b} - \frac{a+b}{1+a+b} \\ &= \frac{(1+a)(1+b)(1+a+b)}{ab(2+a+b)} > 0 \end{aligned}$$

because $a, b \geq 0$.

(g) Not a metric because it is not zero for $x = y$ for nonzero quantities.

(h) Yes, this is a metric. Positivity and symmetry is easy to see. The triangle inequality is satisfied because for $x, y \in \mathbb{R}$ (assume $x < y$), either $|x-y| < 1$ or $|x-y| \geq 1$ and take any $z \in \mathbb{R}$ without loss of generality between x and y (why?). In the former case, this is just the standard Euclidean triangle inequality. In the latter case, there is no way that z can cut $[x,y]$ into two pieces that sum to less than 1, because otherwise $|x-y| < 1$ in this case as well.

(i) Not a metric because it is not symmetric:

$$d(0,1) = 2 \neq 1 = d(1,0)$$

(j) Not a metric because it fails positivity. There are distinct points that are distance zero from each-other:

$$d(0,\pi) = \sin(\pi) = 0$$

■

5.1.2. Are we out of the woods?[3].

No, this is not a metric. The energy you exert going uphill along a path in one

direction is more than the amount you exert going downhill back along the path. Hence ρ is not symmetric.

5.1.3. A strange metric.

Yes, this is a metric.

- We know $d(A, A) = 0$ because all letters are the same, and no changes are needed.
- The positivity for $A \neq B$ follows, since we know that A, B are different, they either have different lengths (and so letters must be added), and/or differ at at least one letter.
- Symmetry is true because you can just revert the changes one at a time (in reverse order).
- The triangle inequality must also be true because in order to transform A to B requires some positive minimal number of changes, and any set of changes that has a string C as an intermediate will be at least that minimal number of changes.

This distance is an example of a string-metric. A famous example is the Levinshtein metric. There is also the Hamming distance (but that only compares words of equal length). The interested reader should search-engine their way to more on this topic.

5.1.4. A strange metric on positive reals.

- The symmetry of the metric follows immediately from the definition. Then notice that $d(x, x) = 0$, and then if $d(x, y) = 0$ then we must have $|x-y| = 0$ and $|1/x - 1/y| = 0$ and hence $x = y$ as required. The non-negativity also follows quite directly from the definition.

  The triangle inequality follows essentially from the triangle inequality on each absolute-value term:

$$|x - y| + \left|\frac{1}{x} - \frac{1}{y}\right| \leq |x - z| + \left|\frac{1}{x} - \frac{1}{z}\right| + |z - y| + \left|\frac{1}{z} - \frac{1}{y}\right|$$

- Not necessarily. Actually, what is true is that if $(x_n)_n$ is a sequence of positive real numbers that converges to a *positive* real number $x > 0$, then this sequence also converges to x in with respect to the metric d (we will leave the justification of this to the reader). The problem is that you can have a sequence of positive real numbers that converges (with respect to the standard Euclidean metric) to a limiting point that is not contained in the set $\mathbb{R}^+$. For example, consider $x_n = 1/n$, which converges to 0 in the standard Euclidean metric, but $0 \notin \mathbb{R}^+$. In fact, we will see later that this sequence is not even Cauchy with respect to d. In fact, this metric space is **complete**.

5.1.5. Draw the "circles".

Checking that these are metric is pretty standard. The sets look like (in order), a circle, a diamond shape and a square.

5.1.6. Isometric (indistinguishable) metrics.

- The function f is bijection on $\mathbb{R}^2$ because it is given by an invertible matrix

$$f(x,y) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

- To show that f preserves distances, let $(a,b),(c,d) \in \mathbb{R}^2$ and notice that

$$\begin{aligned} &d_3(f(x_1,y_1), f(x_2,y_2)) \\ &\quad = \max\{|(x_1+y_1)-(x_2+y_2)|, |(x_1-y_1)-(x_2-y_2)|\} \\ &\quad = \max\{|(x_1-x_2)+(y_1-y_2)|, |(x_1-x_2)-(y_1-y_2)|\} \\ &\quad = |x_1-x_2|+|y_1-y_2|, \end{aligned}$$

  where the last equality can be shown by breaking into cases of whether or not (x_1-x_2) and (y_1-y_2) have the same sign.

5.1.7. Extended real numbers.

(a) The fact that $d(x,x) = 0$ is clear. If $x \neq y$, then $\varphi(x) \neq \varphi(y)$ because arctan is injective, and never equals $\pm\pi/2$ (it is an increasing function). Symmetry and the triangle inequality follow from the symmetry and triangle inequality for the Euclidean metric.

(b) This preserves the standard ordering on $\mathbb{R}$ because arctan is an increasing function. The upper bound/lower bound claim follows by the fact that the range of arctan is $(-\pi/2, \pi/2)$. The basic properties that $<$ is an ordering are left to the reader.

(c) Consider a set $A \subset X$. If $A \cap \mathbb{R}$ is bounded above or below by an element of $\mathbb{R}$, then the LUB or GLB are the same as those from the standard ordering on $\mathbb{R}$ because the ordering is preserved. Otherwise, the set is either unbounded from above (in which case $+\infty$ is the LUB) or from below (in which case $-\infty$ is the GLB).

5.2 · Open and closed sets

5.2.4 · Exercises

5.2.4.1. No limit points.

Any example where all elements are isolated will do, for instance

$$A = \emptyset,\ A = \mathbb{N},\ A = \{\pi, e, 10^{100}\}, \ldots$$

5.2.4.2. Limit points of a pedagogical example.

The only limit point of A is 0 because 0 is the $|1/n| \to 0$ (this shows 0 is a limit point) and all the other points $1/n$ are isolated because

$$\left|\frac{1}{n} - \frac{1}{m}\right| = \frac{|n-m|}{nm}$$

which is a quantity that is uniformly bounded below if $m \neq m$ (this shows that none of the $1/n$ are limit points of A).

5.2.4.3. A contrived construction.

Let

$$E = \left\{ \frac{1}{n} + k \;\middle|\; n \in \mathbb{N},\, k \in \{0,1,2\} \right\}.$$

Then $E \subseteq [0,3]$, so it's bounded. The sequences $\left\{\frac{1}{n}\right\}$, $\left\{\frac{1}{n}+1\right\}$, and $\left\{\frac{1}{n}+2\right\}$ converge to 0, 1, and 2 respectively. Hence, the limit points of E are exactly 0, 1, and 2.

5.2.4.4. Did you pay attention?

False. The classical recurring example is

$$A = \{1/n \mid n \in \mathbb{N}\},$$

which has 0 as a limit point, but $0 \notin A$.

5.2.4.5. Finite sets are all isolated points.

For any finite set $\{x_j\}_{j=1}^N \subseteq X$, the quantity

$$\min\{d(x_i, x_j) : i \neq j\}$$

is a fixed *positive* number. So each x_j is isolated from the other points. Note that we're using the the *finiteness* of the set critically here when we take the minimum of a finite set of numbers. If the set were infinite, then that minimum would have to be replaced by an infimum, which could, of course, be zero.

5.2.4.6. Limit points of limit points.

This is *False*. For example, take $A = \{1/n : n \in \mathbb{N}\}$ which has $A' = \{0\}$ as we have already seen. But then $(A')' = \emptyset$.

5.2.4.7. Closures and unions.

(a) The $\supset$ containment can be seen to follow from (b) (take the rest of the $A_m = \emptyset$), so it suffices to show that $\subset$ containment. Let $x \in \mathrm{cl}\,(B_n)$. Then either $x \in B_n$ (in which case we would be done) or else x is a limit point. In this case, there exists a sequence of points $x_m \in B_n$ with $d(x_m, x) \to 0$. Then infinitely many of these x_m must be in a single one of the A_i, and thus $x \in \mathrm{cl}\,(A_i)$.

(b) If $x \in \operatorname{cl}(A_i)$, then either $x \in A_i$ (in which case we would be done) or else x is a limit point of A_i. Then there exists a sequence $x_m \in A_i$ with $d(x_m, x) \to 0$, and thus $x \in \operatorname{cl}(B_n)$ as well.

(c) Let $A_i = \{1/i\}$ so that $\operatorname{cl}(A_i) = A_i$ for all $i \in \mathbb{N}$. But

$$\operatorname{cl}(B) = \operatorname{cl}(\{1/i : i \in \mathbb{N}\}) = \{1/i : i \in \mathbb{N}\} \cup \{0\}.$$

5.2.4.8. Limit points in $\mathbb{R}^2$.

For open sets, the answer is "Yes". if E is open, then every point is an interior point, so every neighborhood intersects E at other points. So every point is a limit point.

For closed sets, the answer is "No". A closed set may contain isolated points. For example, $\{(0,0)\} \subseteq \mathbb{R}^2$ is closed, but $(0,0)$ is not a limit point.

5.2.4.9. Both open and closed.

Recall that if d is the discrete metric, this means that $d(x,y) = 1$ whenever $x \neq y$. In particular, $B_{1/2}(x) = \{x\}$ for any $x \in X$. Thus we can write

$$A = \bigcup_{x \in A} B_{1/2}(x),$$

which is a union of open sets, and is thus open. Similarly we could do the same with $X \setminus A$ showing that $X \setminus A$ is also open (and thus A is closed).

5.2.4.10. More properties of interiors.

Notice that things can be a bit subtle due to boundary points and isolated points.

(a) Any point in G has an open ball inside $G \subseteq E$, so it's in $\operatorname{int}(E)$.

(b) No. Example: $\mathbb{Q} \subseteq \mathbb{R}$ has $\operatorname{int}(\operatorname{cl}(\mathbb{Q})) = \operatorname{int}(\mathbb{R}) = \mathbb{R}$, but $\operatorname{int}(\mathbb{Q}) = \emptyset$.

(c) No. Example: $E = (0,1) \cup \{2\}$. $\operatorname{int}(E) = (0,1)$, whose closure is $[0,1]$, but $\operatorname{cl}(E) = [0,1] \cup \{2\}$.

5.2.4.11. Parsing the words can be hard.

Let $E \subseteq X$. The interior $\operatorname{int}(E)$ is open so $B = X \setminus \operatorname{int}(E)$ is closed. Notice that B contains $X \setminus E$, so it contains $\operatorname{cl}(X \setminus E)$. That is, we've shown

$$\operatorname{cl}(X \setminus E) \subseteq X \setminus \operatorname{int}(E).$$

Conversely, let $x \in X \setminus \operatorname{int}(E)$. Then there is no $r > 0$ such that $B_r(x) \subseteq E$. That is, $B_r(x) \cap \operatorname{cl}(E) \neq \emptyset$. In other words, $x \in \operatorname{cl}(X \setminus E)$.

5.2.4.12. Limit points and open sets (this is useful later!).

Let $U \ni x$ be open so that there is some $r_1 > 0$ with $B_{r_1}(x) \subseteq U$. Then there exists $x_1 \in A \setminus \{x\}$ with $x_1 \in B_{r_1}(x)$. Now given $\{x_1, \ldots, x_n\} \subseteq A$, define

$$r_{n+1} = \min\{d(x, x_j) : j \in \{1, \ldots, n\}\},$$

so that $B_{r_{n+1}}(x)$ is an open set containing x. Thus we can find some $x_{n+1} \in A \setminus \{x\}$ with $x_{n+1} \in B_{r_{n+1}}(x)$.

5.2.4.13. Limit points is closed.

Let $x \in (A')'$ (or this set may be empty, in which case, we're done). Then for any $\varepsilon > 0$, there exists $x_1 \in A'$ with $d(x_1, x) < \varepsilon/2$. Likewise there exists $x_2 \in A$ with $d(x_2, x_1) < \varepsilon/2$. But then

$$d(x, x_2) \leq d(x, x_1) + d(x_1, x_2) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

This shows that x is a limit point of A, i.e., that $x \in A'$.

5.2.4.14. Density Equivalences.

- (a) implies (b): Saying "E is dense in X" means that $\mathrm{cl}\,(E) = X$, and $\mathrm{cl}\,(E)$ is the smallest closed subset of X containing E, which is X.
- (b) implies (c): If U is an open set disjoint from E, then $X \setminus U$ is a closed set which contains X, and hence is X itself. That is, $U = \emptyset$.
- (c) implies (d): Let U be a non-empty open set, which must be disjoint from E, i.e., $E \cap U \neq \emptyset$.
- (d) implies (e): Let $U = B_r(x)$ be a ball in X, which is open, so it intersects E.
- (e) implies (a): Let $x \in X$. We will show that $x \in \mathrm{cl}\,(E)$, which, by the arbitrariness of x, will imply that E is dense in X. Either $x \in E$, in which case we are done, or else $x \notin E$. In the latter case, we know that $B_r(x) \cap E \neq \emptyset$ for any $r > 0$. That is, $x \in E'$ and hence in $\mathrm{cl}\,(E)$.

5.3 · Limits in metric spaces

· Exercises

5.3.1. Constant sequences.

For any metric space (X, d) and $x \in X$, if $x_n = x$ for all $n \in \mathbb{N}$, then

$$d(x_n, x) = d(x, x) = 0$$

for all $n \in \mathbb{N}$. In particular, $x_n \to x$.

5.3.2. Meaning of "non-convergence".

That the sequence converges means

$$\exists x \in X,\ \forall \varepsilon >,\ \exists N \in \mathbb{N},\ \text{ such that } n \geq N \ \text{ implies } \ d(x_n, x) < \varepsilon.$$

The negation is

$$\forall x \in X,\ \exists \varepsilon >,\ \forall N \in \mathbb{N},\ \exists n \geq N \ \mid \ d(x_n, x) \geq \varepsilon.$$

5.3.3. Non-convergence and infinitely many points.

We use the previous exercise. Since $x_n \not\to x$, there exists $\varepsilon > 0$ such that for any $N \in \mathbb{N}$, there exists $n \geq N$ with $d(x_n, x) \geq \varepsilon$. In particular, we can inductively find such a required subsequence by letting x_{n_1} be as above with $N = 1$. Now given $x_{n_1}, x_{n_2}, \ldots x_{n_k}$, choose $x_{n_{k+1}}$ as above with the choice of $N = n_k + 1$. This guarantees that each x_{n_k} is a unique member of the original sequence.

5.3.4. A strange metric on $\mathbb{N}$.

First, we show that d is a metric on $\mathbb{N}$.

- Notice that if $n \neq k$ then $d(n,k) > 0$, and if $d(n,k) > 0$ we must have $\frac{1}{n} \neq \frac{1}{k}$ and so $n \neq k$. Similarly, $d(n,n) = 0$ and if $d(n,k) = 0$ we have $1/n = 1/k$ giving $n = k$ as required.
- Then symmetry follows quite directly from the definition: $d(n,k) = |1/n - 1/k| = |1/k - 1/n| = d(k,n)$.
- Finally the triangle inequality follows fairly directly

$$\begin{aligned} d(n,k) &= \left|\frac{1}{n} - \frac{1}{k}\right| \\ &= \left|\frac{1}{n} - \frac{1}{\ell} + \frac{1}{\ell} - \frac{1}{k}\right| \\ &\leq \left|\frac{1}{n} - \frac{1}{\ell}\right| + \left|\frac{1}{\ell} - \frac{1}{k}\right| \\ &= d(n,\ell) + d(\ell,k) \end{aligned}$$

Let (a_n) be a convergent sequence of natural numbers in this metric space, with limit L. Then for any $\varepsilon > 0$ there must some N so that if $n > N$ we have

$$d(a_n, L) = \left|\frac{1}{a_n} - \frac{1}{L}\right| < \varepsilon$$

Notice that if L is not a natural number, then set $L^+ = \lceil L \rceil$ and choose $\varepsilon < 1/L^+$, to get a contradiction (no integer sequence can approach closer to L).

So now assume that $L \in \mathbb{N}$, then for sufficiently large n we have

$$\frac{1 - L\varepsilon}{L} = \frac{1}{L} - \varepsilon < \frac{1}{a_n} < \varepsilon + \frac{1}{L} = \frac{1 + L\varepsilon}{L}$$

Hence

$$\frac{L}{1 + L\varepsilon} < a_n < \frac{L}{1 - L\varepsilon}$$

Now since ε is arbitrary, we can choose ε sufficiently small so that

$$\frac{L}{1 + L\varepsilon} > L - 1$$

and also

$$\frac{L}{1-L\varepsilon} < L+1$$

and hence we must have (for sufficiently large n)

$$L-1 < a_n < L+1$$

Since L, a_n are integers, we must have $a_n = L$.

Hence only *eventually* constant sequences converge.

5.3.5. Limits are unique.

Suppose that $x_n \to x$ and $x_n \to y$, and let $\varepsilon > 0$ be an arbitrarily small positive number. We will show that $d(x,y) < \varepsilon$, which, by the arbitrariness of ε, implies that $d(x,y) = 0$, and subsequently $x = y$ (hence, the limit is unique).

Since the sequence converges, there exists $N \in \mathbb{N}$ large enough so that both $d(x_n, x) < \varepsilon/2$ and $d(x_n, y) < \varepsilon/2$ for all $n \geq N$. Then by the triangle inequality, we have

$$d(x,y) \leq d(x_n, y) + d(x_n, x) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

As outlined above, this completes the proof.

5.3.6. Double sequences and convergence.

By the triangle inequality,

$$d(x_n, y_n) - d(x,y) \leq d(x_n, x) + d(y_n, y).$$

Likewise,

$$d(x,y) - d(x_n, y_n) \leq d(x_n, x) + d(y_n, y).$$

These together imply

$$|d(x_n, y_n) - d(x,y)| \leq d(x_n, x) + d(y_n, y),$$

and the result follows because $x_n \to x$ and $y_n \to y$.

5.3.7. Convergence of subsequences.

Let $\varepsilon > 0$ and take $N \in \mathbb{N}$ large enough to guarantee (by the convergence $x_n \to x$) that $d(x_n, x) < \varepsilon$ for all $n \geq N$. Then for any subsequence $(x_{n_k})_k$, we may assume that $K \in \mathbb{N}$ is large enough so that $n_k \geq N$ for all $k \geq K$. Then

$$d(x_{n_k}, x) < \varepsilon$$

for all $k \geq K$, and so $x_{n_k} \to x$.

5.3.8. Limit points can be the original set.

The set $\mathbb{R}$ satisfies the condition, because for any $x \in \mathbb{R}$, the sequence

$$x_n = x - \frac{1}{10^n}$$

has $x_n \to x$, and $x_n \neq x$ for any n. Therefore, $x \in \mathbb{R}'$, i.e., is a limit point.

5.3.9. Convergent sequences are bounded.

Let L be the limiting value, let $\varepsilon > 0$, and let $N \in \mathbb{N}$ be such that $d(x_n, L) \leq 1$ for all $n \geq N$. Also, define

$$R := \max\{x_1, x_2, \ldots, x_{N-1}, 1\}.$$

Then

$$d(L, x_n) \leq R$$

for every $n \in \mathbb{N}$. Now set $M = \max\{1, R\}$, then we have (as required)

$$\forall n, \quad d(L, x_n) \leq M$$

5.3.10. Discrete metric has boring converging subsequences.

If $d(x_n, x) \to 0$ where d is the discrete metric, then this means in particular that $d(x_n, x) < 1$ whenever $n \geq N$ (here $\varepsilon = 1$). Since the metric is discrete, this forces $x_n = x$ for all such x, i.e., the sequence is eventually constant.

5.3.11. Different metrics change convergence.

The previous exercise makes this straight-forward. Let d_1 be the standard Euclidean metric on $\mathbb{R}$, d_2 the discrete metric. Then set $a_n = 1/n$, which converges in $(\mathbb{R}, d_1)$. However, since $d_2(a_n, a_k) = 1$ for any $n \neq k$, we know that the sequence does not converge in $(\mathbb{R}, d_2)$.

Indeed, one can take any convergent sequence in $(\mathbb{R}, d_1)$ that is not eventually constant, then it will not converge in $(\mathbb{R}, d_2)$.

5.3.12. A strange, but equivalent, subsequence condition.

If x_n does *not* converge to y, then this means there exists some $\varepsilon > 0$ and infinitely many x_{n_k} (that is, a *subsequence*), such that

$$d(x_{n_k}, x) \geq \varepsilon.$$

But, by assumption, a further subsequence of the x_{n_k}, which we denote $x_{n_{k_l}}$, converges to y. So for the given ε above, we can take $L \in \mathbb{N}$ large enough so that when $l \geq L$, there holds

$$d(x_{n_{k_l}}, y) < \varepsilon.$$

This is a contradiction.

5.3.13. Does it still work without the fixed point y?

This is *False*. For example, consider $x_n = (-1)^n$. Then every subsequence has a further subsequence, either a subsequence of (x_{2n}) or (x_{2n+1}), which converges (in fact, is constant), but the overall sequence does not converge.

5.4 · Completeness in metric spaces

· Exercises

5.4.1. Discrete metric is complete.

Any Cauchy sequence in a discrete metric space is necessarily eventually constant, and thus convergent (the precise justification for these claims is left to the reader).

5.4.2. A complete metric on $\mathbb{Q}$.

The answer is "Yes", as demonstrated by the previous exercise.

5.4.3. An unusual metric on $\mathbb{N}$, revisited.

Let $(x_n) \subseteq X$ be a Cauchy sequence. Then since

$$|x_m - x_n| \leq |x_m - x_n| + \left| \frac{1}{x_m} - \frac{1}{x_n} \right|,$$

and so the sequence (x_n) is Cauchy as well with respect to the Euclidean metric. Hence it must converge to some limit $y \in [0, \infty)$ under the Euclidean metric. Now, either $y = 0$ or $y > 0$.

We first show that y cannot be zero. To see this, notice that convergence to zero under the usual metric tells us that for any $\varepsilon > 0$, we must have $|x_n| < \varepsilon$ and so $1/x_n > 1/\varepsilon$. That is, we can make $1/x_n$ arbitrarily large. But now fix any k then $d(x_n, x_k) = |x_n - x_k| + |1/x_n - 1/x_k| > |1/x_n - 1/x_k|$. Then we can make n sufficiently large so that $1/x_n$ is huge (compared to the fixed $1/x_k$), and so $d(x_n, x_k) > 1$. This contradicts the assumption that (x_n) is Cauchy under d.

So we must have$y > 0$. So now fix $\varepsilon > 0$.

- By convergence under the usual metric, there is $N_1 > 0$ so that for any $n > N_1$ we have $|x_n - y| < \varepsilon/2$.
- Then since $x_n \to y > 0$ under the usual metric, we know that $1/x_n \to 1/y$, and so there is N_2 so that for any $n > N_2$ we have $|1/x_n - 1/y| < \varepsilon$.

Hence we can make n sufficiently large so that

$$d(x_n, y) = |x_n - y| + |1/x_n - 1/y| < \varepsilon$$

Thus our sequence converges under d. Hence $(\mathbb{R}^+, d)$ is complete.

5.4.4. If a group of you stick together and some of you are going to Cleveland, then you're all going to Cleveland[4].

Let (x_n) be such a Cauchy sequence, with a convergent subsequent $(x_{n_k})_k$. Write x to be the limit of the subsequence and let $\varepsilon > 0$ be arbitrary. By the subsequence convergence, we can find $K \in \mathbb{N}$ large enough so that whenever $k \geq K$, we have

$$d(x, x_{n_k}) < \varepsilon/2.$$

Similarly, let $N \in \mathbb{N}$ be large enough so that whenever $n, m \geq N$, we have

$$d(x_n, x_m) < \varepsilon/2.$$

Let $M = \max\{n_K, N\}$ and let $n_{k'}$ be some member of the subsequence larger than M. Then by the triangle inequality, we have

$$d(x, x_n) \leq d(x, x_{n_{k'}}) + d(x_{n_{k'}}, x_M) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

5.4.5. Union of complete sets.

The answer is *True*. Let X, Y be complete subsets of a metric space Z. Let $(x_n) \subseteq X \cup Y$ be a Cauchy sequence in the union. Then infinitely many of the x_n lie in (say)X and this subsequence is still Cauchy, so it has a converging subsequence. But we have seen in Exercise 5.4.4 that this is sufficient for the entire sequence to be convergent.

5.4.6. Completeness and isolated points.

The metric d here is fixed, so we will suppress it from our notation. First, let $Y = X \setminus \{c\}$ is complete iff every Cauchy sequence (x_n) in Y converges to a point L in Y.

So now assume that c is not isolated. Then there is a sequence $x_n \to c$ with the property that $x_n \neq c$. Since it converges, it is Cauchy, and it is also a sequence in Y. However this implies that Y is not complete, because it contains a Cauchy sequence that does not converge to a limit in Y.

Now assume that c is isolated, and let (x_n) be any Cauchy sequence in Y. Hence (x_n) is Cauchy in X and so converges to a limit $L \in X$. Since $x_n \in Y$ and c is isolated, we must have $L \neq c$ and so the Cauchy sequence in Y converges to a limit in Y. Thus Y is complete.

5.4.7. THE COMPLETION.

(a) $\mathcal{C}$ is not empty because it contains constant sequences, or more generally, any sequence which is *eventually* constant.

(b) Almost immediately we have Clearly $(x_n) \sim (x_n)$ (reflexive) and $(x_n) \sim (y_n)$ implies $(y_n) \sim (x_n)$ (symmetric) because $d(x, y) = d(y, x)$.

Transitivity follows from the triangle inequality for d:

$$\begin{aligned}\lim_{n\to\infty} d(x_n, z_n) &\leq \lim_{n\to\infty} [d(x_n, y_n) + d(y_n, z_n)] \\ &= \lim_{n\to\infty} d(x_n, y_n) + \lim_{n\to\infty} d(y_n, z_n) = 0\end{aligned}$$

(c) If $(x_n) \sim (a_n)$ and $(y_n) \sim (b_n)$, then

$$\begin{aligned}\lim_{n\to\infty} d(x_n, y_n) &\leq \lim_{n\to\infty} [d(x_n, a_n) + d(a_n, b_n) + d(b_n, y_n)] \\ &= \lim_{n\to\infty} d(a_n, b_n).\end{aligned}$$

The reverse inequality is identical, which shows that $\widehat{d}$ is well-defined. Now we need to show it is a metric. Clearly $\widehat{d} \geq 0$ and

$$\widehat{d}\,([(x_n)], [(y_n)]) = 0$$

$$\begin{aligned} &\Rightarrow \quad \lim_{n\to\infty} d(x_n, y_n) = 0 \\ &\Rightarrow \quad (x_n) \sim (y_n) \\ &\Rightarrow \quad [(x_n)] = [(y_n)]. \end{aligned}$$

The fact that $\widehat{d}$ satisfies the triangle inequality follows from the triangle inequality for d.

(d) We have

$$\widehat{d}(\varphi(x), \varphi(y)) = \lim_{n\to\infty} d(x, y) = d(x, y).$$

(e) Let $[(x_n)] \in \widehat{X}$ and let $\varepsilon > 0$. Since (x_n) is d-Cauchy, we may take $N \in \mathbb{N}$ large enough so that whenever $n, m \geq N$, then $d(x_n, x_m) < \varepsilon$. Then notice that

$$\widehat{d}(\varphi(x_N), [(x_n)]) = \lim_{n\to\infty} d(x_N, x_n) < \varepsilon.$$

This shows that $\varphi(X)$ is dense in $\widehat{X}$

(f) Prove that $(\widehat{X}, \widehat{d})$ is a complete metric space by following the outline described below.

(i) By the hint, we write

$$d(z_n, z_m) = d(x_n^{(n)}, x_m^{(m)}) \leq d(x_n^{(n)}, x_n^{(m)}) + d(x_n^{(m)}, x_m^{(m)}).$$

Since $[(x_n^{(m)})]_{m=1}^{\infty}$ is Cauchy, this means that

$$\widehat{d}([(x_n^{(i)})], [(x_n^{(j)})]) = \lim_{n\to\infty} d(x_n^{(i)}, x_n^{(j)}) < \varepsilon/4$$

whenever i, j are large enough. In particular, we know that whenever $n \geq N$ for some $N \in \mathbb{N}$, there holds

$$d(x_n^{(i)}, x_n^{(j)}) < \varepsilon/2.$$

This is the same as the first term in the RHS of the hint. Also since $(x_n^{(m)})$ is Cauchy for each $m \in \mathbb{N}$, we can make the second term in the RHS small as well, for n large enough.

(ii) This follows from similar technique in the previous part and the hint

$$d(x_n^{(m)}, z_n) \leq d(x_n^{(m)}, x_n^{(n)}) + d(x_n^{(n)}, z_n).$$

Here things are actually easier because the second term in RHS is actually just 0, and the first term is small by the fact that $[(x_n^{(m)})]_{m=1}^{\infty}$ is Cauchy in $\widehat{d}$.

5.5 · Continuity in metric spaces

· Exercises

5.5.1. Is it continuous?

It does *not* imply that f is continuous. For example, take

$$f(x) = \begin{cases} 0 & x = 0 \\ 1 & \text{otherwise.} \end{cases}$$

Then the limit condition above is satisfied for this f, but it is not continuous at $x = 0$.

5.5.2. Distance function is continuous.

By the triangle inequality, we can write

$$d(x, y) - d(z, y) \leq d(x, z)$$

and

$$-d(x, y) + d(z, y) \leq d(x, z).$$

This is equivalent with saying that

$$|f(x) - f(z)| = |d(x, y) - d(z, y)| \leq d(x, z).$$

So for any $\varepsilon > 0$, taking $\delta = \varepsilon$ satisfies the ε-δ definition of continuity.

5.5.3. Continuity and closures.

- Fix some $E \subseteq X$ and let $y \in f(\mathrm{cl}\,(E))$. We need to show also that $y \in \mathrm{cl}\,(f(E))$. Because $y \in f(\mathrm{cl}\,(E))$, there exists some $x \in \mathrm{cl}\,(E)$ such that $f(x) = y$. Then since $x \in \mathrm{cl}\,(E)$, this means that either $x \in E$ or else that x is a limit point of E. The desired containment is trivial in the former case, so we only need to consider the possibility that x is a limit point of E. In other words, we can find $x_n \in E$ with the property $d_X(x_n, x) \to 0$ as $n \to \infty$. But, by Theorem 5.5.3, the continuity of f implies that $d_Y(f(x_n), f(x))$, i.e., that $f(y) \in \mathrm{cl}\,(f(E))$.
- Let $X = Y = \mathbb{R}$, both with the Euclidean metric. Take $f(x) = \arctan(x)$, which is continuous and $E = X$. Then E is closed and so $E = \mathrm{cl}\,(E)$ but we have

$$f(\mathrm{cl}\,(E)) = (-\pi/2, \pi/2) \neq [-\pi/2, \pi/2] = \mathrm{cl}\,(f(E))\,.$$

5.5.4. Continuity and density.

- As per the hint and the density assumption, we have

$$f(X) = f(\mathrm{cl}\,(E)) \subseteq \mathrm{cl}\,(f(E))\,.$$

  This proves the result.

- For any $p \in X$, we can find a sequence $x_n \in E$ with $d_X(x_n, p) \to 0$. Since both f, g are continuous, we have

$$d_Y(f(p), g(p)) \leq d_Y(f(p), f(x_n)) + d_Y(g(p), g(x_n)) \to 0.$$

  Thus, $d_Y(f(p), g(p)) = 0$, which implies $f(p) = g(p)$.

- This is just a special case of part (b).

5.5.5. Zero set is closed.

By Exercise 5.5.3, we have

$$f(\mathrm{cl}\,(Z)) \subseteq \mathrm{cl}\,(f(Z)) = \mathrm{cl}\,(\{0\}) = \{0\}\,.$$

So we can say from this that for every $x \in \mathrm{cl}\,(Z)$, we have that $f(x) = 0$, i.e., that $x \in \mathrm{cl}\,(Z)$. This shows that $Z = \mathrm{cl}\,(Z)$, which is the same as saying that Z is closed.

5.5.6. Identity continuous?

First, consider $\mathrm{id} : (\mathbb{R}, d) \to (\mathbb{R}, d_E)$ from Euclidean to discrete. For any $\varepsilon > 0$, let $\delta \in (0, 1/2]$. Then given a fixed $x \in \mathbb{R}$, the only $y \in \mathbb{R}$ that satisfy

$$d(x, y) < \delta \leq 1/2$$

must be $x = y$, for which

$$d_E(\mathrm{id}(x), \mathrm{id}(y)) = |\mathrm{id}(x) - \mathrm{id}(y)| = 0 < \varepsilon.$$

In other words, id works as a *shrinking* or *compressing* function in this direction and the function is continuous.

For the reverse direction, from discrete to Euclidean, the opposite phenomenon occurs: points that are arbitrarily close with respect to d_E become very stretched out with respect to d, so the identity map is not continuous in this case. We leave the details to the reader.

5.5.7. Popcorn[5].

- Let $x \in \mathbb{Q}$ and take $\varepsilon = f(x) > 0$. As per the hint, regardless of what $\delta > 0$ we take, we can find some $y \in \mathbb{R} \setminus \mathbb{Q}$ with $|x - y| < \delta$. For this y, we have

$$|f(x) - f(y)| = |f(x)| = \varepsilon.$$

  Since $\delta > 0$ was arbitrary, this shows that f cannot be continuous at x.

- Let $x \in \mathbb{R} \setminus \mathbb{Q}$ and let $\varepsilon > 0$. As hinted, let

$$S = \{y \in \mathbb{Q} : |y - x| \leq 1 \ \text{ and } \ n_y \leq 1/\varepsilon\}.$$

  This set is finite, in fact, the cardinality of S is at most $1/\varepsilon^2$ (it is actually

less than this, but this bound will do). So the minimum

$$\delta := \min\{|x - y| : y \in S\}$$

is a strictly positive number. Thus we can take this δ in the definition of continuity. So we can say that whenever $|x - z| < \delta$, either $z \notin \mathbb{Q}$ (so that $f(z) = 0$), or else $f(z) = 1/n_z < \varepsilon$. In either case,

$$|f(z) - f(x)| < \varepsilon.$$

So f is continuous at x.

5.6 · Metric space of continuous bounded functions

· Exercises

5.6.1. Continuity preserved by pointwise limits?

The answer is "No". Consider, for example,

$$f_n(x) = \begin{cases} 0 & x < 0 \\ x^n & 0 \le x \le 1 \\ 1 & x > 1. \end{cases}$$

Each f_n is continuous, but f_n converges pointwise to the function

$$f_n(x) = \begin{cases} 0 & x < 1 \\ 1 & x \ge 1, \end{cases}$$

which is not continuous.

5.6.2. Pointwise vs uniform.

- When $x = 0$, then $f_n(x) = 0$ for every $n \in \mathbb{N}$. For $x \in (0, 1]$, the $(1 - nx)^2$ term grows without bound (with respect to n), and so $f_n(x) \to 0$ as $n \to \infty$ for any such x.
- The maximum of f_n is 1, which is achieved at $x = 1/n$ by standard calculus techniques. Therefore
$$\|f_n\| = 1 \not\to 0.$$

5.6.3. Pointwise convergence can have "bad" limits.

If $x \in \mathbb{Z}$, then $f_n(x) = 1$ for every $n \in \mathbb{N}$, and so $f_n(x) \to 1$ for these values. But when $x \notin \mathbb{Z}$, then $\cos(n\pi x)^2$ is just a number in $(0, 1)$, and so when we raise this to very large m, the result converges to zero. Therefore, the limiting function is

$$f(x) = \begin{cases} 1 & x \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

5.6.4. Converges uniformly equivalent condition.

- If $x = 0$, then $f_n(x) = 0$ for every $n \in \mathbb{N}$, so we only have to concern ourselves with nonzero points. Along those lines, for any $x \neq 0$ and $\varepsilon > 0$, take $N > 1/(|x|\varepsilon)$. Then for any $n \geq N$, there holds

$$|f_n(x)| = \left| \frac{nx}{1+n^2x^2} \right| \leq \frac{1}{|nx|} < \varepsilon.$$

- If $0 \notin A'$, then as per the hint, we may obtain some $\delta > 0$ such that any $x \in A \setminus \{0\}$ satisfies $|x| \geq \delta$. Then similarly as in part (a), for any $\varepsilon > 0$, we may take $N > 1/(\delta\varepsilon)$ to obtain that for any $x \in A \setminus \{0\}$, we have

$$|f_n(x)| = \left| \frac{nx}{1+n^2x^2} \right| \leq \frac{1}{|nx|} < \varepsilon.$$

  This estimate holds uniformly regardless of x, so the convergence is uniform on this set.

- As per the hint, we assume that $0 \in A'$ and so we can find a sequence $x_k \in A$ with $|x_k| \to 0$ and use the notation of n_k as in the hint. Consider $f_{n_k}(x_k)$ to see that

$$|f_{n_k}(x_k)| = \left| \frac{n_k x_k}{1+n_k^2 x_k^2} \right| \geq \frac{\frac{1}{2}}{1+1} = \frac{1}{4}.$$

  But this is in contradiction to our assumption that f_n converges uniformly to the zero function.

5.6.5. Uniform convergence is convenient.

Let $\varepsilon > 0$. As per the hint, we first write

$$|f_n(x_n) - f(x)| \leq |f_n(x_n) - f(x_n)| + |f(x_n) - f(x)|.$$

Since $f \in \mathcal{C}(X)$, it it is continuous. So we may assume n is large enough so that $d(x_n, x)$ is small enough to guarantee that $|f(x_n) - f(x)| < \varepsilon/2$. Likewise, using the uniform convergence $f_n \to f$, we may assume that n is large enough to ensure that $\|f_n - f\|$, and hence $|f_n(x_n) - f(x_n)|$, is also less than $\varepsilon/2$.

5.6.6. Uniform convergence implies integrals converge.

By some basic properties of integrals, we have

$$\begin{aligned} \left| \int_a^b f_n(x)\,\mathrm{d}x - \int_a^b f(x)\,\mathrm{d}x \right| &= \left| \int_a^b f_n(x) - f(x)\,\mathrm{d}x \right| \\ &\leq \int_a^b |f_n(x) - f(x)|\,\mathrm{d}x \end{aligned}$$

$$\leq (b-a)\|f_n - f\|.$$

The latter quantity goes to zero as $n \to \infty$ because $(b-a)$ is just some fixed positive number and $\|f_n - f\| \to 0$ as $n \to \infty$.

5.6.7. Integrals converging does not imply uniform convergence.

(a) For $x = 0$ or $x = 1$, we can see that $f_n(x) = 0$ for all $n \in \mathbb{N}$, so the convergence for these cases is trivial. For $x \in (0,1)$, the quantity $(1-x^2)$ is just a number in $(0,1)$. In particular,

$$n(1-x^2)^n = \frac{n}{((1-x^2)^{-1})^n}$$

goes to 0 as $n \to \infty$ because exponentials grow faster than polynomials.

(b) We can calculate the integral using substitution $u = 1 - x^2$:

$$\int_0^1 f_n(x)\,\mathrm{d}x = \frac{n}{2}\int_0^1 u^n\,\mathrm{d}u = \frac{n}{2(n+1)}.$$

As $n \to \infty$, this quantity converges to $1/2$, not 0 which would be the integral of the limiting function. This is not inconsistent with the previous problem because the convergence $f_n \to 0$ is not uniform in this case, so we should not necessarily expect that

$$\int f_n \to \int f.$$

5.6.8. Converges in one metric, but not the other.

- Showing that $d_2(f,f) = 0$, that $d_2(f,g) \geq 0$, and the triangle are relatively straight-forward. It remains to show that if $f \neq g$, then $d_2(f,g) > 0$. Along those lines, consider the new function $h = |f - g|$, which is still in $\mathcal{C}([0,1])$. If $h(x) > 0$ for some $x_0 \in [0,1]$, then - by the continuity of h - there is some small interval $[a,b] \subseteq [0,1]$ containing x_0 such that

$$h(x) \geq \frac{1}{2}h(x_0) \text{ for all } x \in [a,b].$$

In particular, we can see from this that

$$d_2(f,g) = \int_0^1 h(x)\,\mathrm{d}x \geq \frac{1}{2}(b-a)h(x_0) > 0.$$

This shows that d_2 is a metric.

- For any $x \in [0,1)$, x^n converges pointwise to 0. On the other hand, $f_n(1) = 1$ for all $n \in \mathbb{N}$. We also know that if a function converges

uniformly, then it converges pointwise as well, and so the limiting function must be the one that we have found above. But this function is not continuous and the uniform limit of continuous functions is always continuous by Theorem 5.6.5, so the f_n must not converge with respect to d_1.

On the other hand, if we denote $z(x)$ to be the zero function, then we can see that

$$d_2(f_n, z) = \int_0^1 |x^n| \, \mathrm{d}x = \frac{1}{n+1}.$$

So $f_n \to z$ with respect to d_2.

5.6.9. Back to polynomials.

- As per the hint, given any $\varepsilon > 0$, we can take $g(x) = f(x) + \varepsilon/2 \sin(x)$, which satisfies
$$d(f, g) = \sup_{t \in [a,b]} |\varepsilon/2 \sin(t)| \leq \varepsilon/2.$$
So $g \in B_\varepsilon(f)$. Since $\varepsilon > 0$ was arbitrary and g is not a polynomial, this shows that $\mathcal{P}$ is not open in this metric space.

- By Taylor's Theorem 4.5.1, we may obtain a polynomial $f_n \in \mathcal{P}$ with the property that
$$\sup_{t \in [a,b]} |\sin(t) - f_n(t)| \leq \frac{(b-a)^{n+1}}{(n+1)!}.$$
Since we can take the RHS as small as we like, this shows that $f(x) = \sin(x)$ is a limit point of $\mathcal{P}$, i.e., that $f \in \mathrm{cl}\,(\mathcal{P})$. Since $f \notin \mathcal{P}$, this shows that $\mathcal{P}$ is not closed.

5.7 · Contraction mappings

· Exercises

5.7.1. Completeness is important.

Let $X = (0, 1)$ with the standard Euclidean metric $d(x, y) = |x - y|$. This metric space is not complete by Remark 2.3.22. Let $f : X \to X$ be defined as $f(x) = x/2$, which is a contraction because $|f'(x)| = 1/2 < 1$ uniformly in x. There is no fixed point for this function because $x/2 = x$ implies that $x = 0$, which is not in the domain of f.

5.7.2. A way to calculate square roots.

(a) The fixed point of f satisfies:

$$f(x) = x \Rightarrow \frac{1}{2}\left(x + \frac{c}{x}\right) = x \Rightarrow \frac{c}{x} = x \Rightarrow x^2 = c \Rightarrow x = \sqrt{c}.$$

(b) Calculate

$$f'(x) = \frac{1}{2} - \frac{c}{2x^2}.$$

Since both f and f' are continuous functions, and because $f(\sqrt{c}) = \sqrt{c}$ and $f'(\sqrt{c}) = 0$, we can positive numbers a, b such that $a < \sqrt{c} < b$ and

$$|f'(x)| \leq \frac{1}{2}$$

on $[a, b]$. We can conclude from this that $f([a, b]) \subseteq [a, b]$ and that f is a contraction there.

(c) This follows directly from the Contraction Mapping Theorem 5.7.4.

(d) To compute $\sqrt{2}$, one may check that a suitable interval is $[1, 2]$. Taking any initial guess will do, so let $x_0 = 1$:

$$\begin{aligned} x_1 &= \frac{1}{2}(1 + \frac{2}{1}) = \frac{3}{2} = 1.5, \\ x_2 &= \frac{1}{2}\left(1.5 + \frac{2}{1.5}\right) = \frac{1}{2}(1.5 + \frac{4}{3}) = \frac{1}{2}\left(\frac{13}{6}\right) = \frac{13}{12} \\ x_3 &= \frac{1}{2}\left(\frac{13}{12} + \frac{2}{\frac{13}{12}}\right) = \frac{1}{2}\left(\frac{13}{12} + \frac{24}{13}\right) \approx 1.4142. \end{aligned}$$

5.7.3. Maps and fixed points.

Yes, there is necessarily at least one such point though we have to make some assumptions. We take X to be the set that is the country of Canada on the map, including the coastline with the standard way that we measure distances. This is a complete metric space (it is a closed subset of $\mathbb{R}^2$). Let the function $f : X \to X$ take a point on X to the point on the map directly above which it lays. This will be a contraction because the map is much smaller than the country the map represents. Therefore, as long as this function is continuous, it is guaranteed to have a unique fixed point by Contraction Mapping Theorem 5.7.4.

Crumpling the piece of paper is a continuous action so we are guaranteed to have a fixed point in this setting. However, ripping the map is discontinuous, and so fixed point may not be guaranteed.

5.7.4. Read the theorem carefully!

Consider the function $f(x) = x + (1 + e^x)^{-1}$. It has

$$f'(x) = 1 - \frac{e^x}{(1 + e^x)^2}$$

and so (with a little work) $\frac{3}{4} \leq f'(x) < 1$. Then notice that if we attempt to solve $f(x) = x$ then we have $x = x + (1 + e^x)^{-1}$, which has no solution.

Now, this does not contradict the contraction mapping theorem, since there is no *uniform* constant $q < 1$ so that $|f'(x)| \leq q < 1$ *for all* x. Indeed, $f'(x) \to 1$ (from below) as $x \to \pm\infty$.

5.7.5. Some trickery involved.

The answer is *yes*. To prove it, first note that $h = f \circ \cdots \circ f$ is guaranteed to have a *unique* fixed point by Theorem 5.7.4. Call that fixed point x^*. We know that

$$h(x^*) = (f \circ \cdots \circ f)(x^*) = x^*.$$

Then apply f to both sides, and use some clever trickery:

$$f(x^*) = (f \circ h)(x^*) = \underbrace{(f \circ \cdots \circ f)}_{n+1}(x^*) = h(f(x^*))$$

Now let $y = f(x^*)$, then we have $h(y) = y$. But this means that y is also a fixed point of h. But since h has a unique fixed point, we must have that $y = x^*$. But $y = f(x^*)$, and so we have

$$f(x^*) = x^*.$$

making x^* a fixed point of f. Now, notice that x^* must be a unique fixed point of f, because if f were to have multiple fixed points, then each would be a fixed point of h.

Colophon

This book was authored in PreTeXt.